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NUTRITIONAL PROTOCOLS



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Executive Summary: Compendium of Nutritional Protocols

The human body's capacity to function, repair, and thrive is entirely contingent on the quality and composition of its nutritional inputs. This compendium provides a rigorous, evidence-based analysis of nutritional science across four distinct but interconnected domains: the biochemical foundations of nutrient function, the clinical and behavioural dimensions of dietary protocols, the physiological systems that govern nutrient bioavailability, and the public health implications of nutritional policy in the Indian context. Together, these themes form a complete picture of nutrition as a science, a practice, and a policy challenge.

The Nutritional Foundation: Building Blocks of Health

At the core of nutrition lies a taxonomy of six nutrient classes organised into two tiers. Macronutrients, comprising carbohydrates, proteins, and fats, serve as the primary energy substrates and structural materials for the body, delivering 4, 4, and 9 kilocalories per gram, respectively. Each macronutrient operates through distinct metabolic pathways: carbohydrates fuel glycolysis and the Krebs cycle, proteins supply amino acids for enzymatic and hormonal synthesis, and fats support membrane integrity, hormone production, and the absorption of fat-soluble vitamins. Micronutrients, encompassing 13 vitamins and 16 minerals, act as cofactors and coenzymes that enable virtually every biochemical reaction in the body. Water, often underappreciated, underpins thermoregulation, cellular homeostasis, kidney function, and cardiovascular stability. The central insight is that no nutrient operates in isolation. Deficiency in any single micronutrient creates cascading dysfunctions across energy metabolism, immune competence, cognitive performance, and tissue repair. A diet that is calorically adequate but micronutrient-poor remains fundamentally inadequate.

Dietary Protocols and Their Metabolic Trade-offs

The compendium evaluates a spectrum of dietary models, organised by their principal mechanism of action. Time-restricted protocols, including intermittent fasting and alternate-day fasting, leverage metabolic switching: the shift from glucose to ketone-based fuel that occurs 12 to 24 hours after caloric restriction begins. These approaches improve insulin sensitivity, reduce visceral fat, and activate cellular repair pathways such as autophagy, but carry meaningful risks around muscle catabolism, micronutrient depletion, and hormonal disruption, particularly for women. Therapeutic diets such as the ketogenic diet and the low-FODMAP diet address specific pathological conditions, managing drug-resistant epilepsy and irritable bowel syndrome, respectively, and require clinical supervision given the risk of nutrient gaps. The DASH and Mediterranean diets represent the most broadly evidence-supported models for long-term cardiometabolic and cognitive health. The PREDIMED trial demonstrated a 30 per cent reduction in major cardiovascular events on the Mediterranean diet, while DASH produces clinically meaningful blood pressure reductions within two to three weeks. The core strategic insight is that dietary protocol selection must be individualised. No single diet is universally optimal; effectiveness depends on physiological profile, disease burden, lifestyle sustainability, and nutritional completeness across all essential micronutrients.

Physiological Enablers: Enzymes and the Microbiome

Nutritional status is not determined by intake alone but by absorption and utilisation, both of which are critically mediated by digestive enzymes and the gut microbiome. The body deploys a coordinated relay of hydrolytic enzymes across the oral, gastric, pancreatic, and brush-border phases of digestion. Pancreatic lipase accounts for 70 to 80 per cent of dietary fat hydrolysis; lactase deficiency impairs carbohydrate absorption in a substantial proportion of the global adult population; and pepsin deficiency, most commonly secondary to hypochlorhydria, compromises the release of vitamin B12, iron, and zinc from food matrices. These enzymatic bottlenecks illustrate that absorption capacity is a biological variable, not a constant. Simultaneously, the gut microbiome,

particularly species such as *Lactobacillus acidophilus*, *Bifidobacterium longum*, and *Akkermansia muciniphila*, governs gut barrier integrity, immune modulation, and short-chain fatty acid production, all of which influence systemic metabolic health. Higher *A. muciniphila* abundance at baseline predicts greater metabolic improvement from caloric restriction, positioning microbiome composition as a clinically relevant biomarker. The practical implication is that optimising nutritional outcomes requires attending to the conditions that enable absorption, not merely the quality of foods consumed.

Public Health and Policy: Fortification and the Indian Imperative

India faces a compound public health crisis of micronutrient deficiency. NFHS-5 data document increasing anaemia prevalence across all age and gender groups, alongside population-level shortfalls of 54 percent for iron, 53 percent for vitamin B12, 37 percent for folate, and 61 percent for vitamin D. The current FSSAI fortification standards for folic acid and vitamin B12 in wheat flour and rice are set at levels approximately 90 percent below WHO recommendations, rendering them ineffective at preventing neural tube defects and nutritional anaemia at scale. The regulatory framework governing dietary supplements under Section 22 of the Food Safety and Standards Act, 2006, has created a structural gap: companies routinely make drug-like health claims while classified as food products, with no pre-market efficacy verification and penalties insufficient to deter misleading labelling. The Costa Rican model provides a clear proof point: aligning fortification standards with WHO guidance, alongside mandatory inclusion in public distribution systems, reduced NTD prevalence by 36 per cent within two years of implementation. Three actions are warranted for India. First, revise folic acid fortification standards to 1,300 micrograms per kilogram of wheat flour and rice, in line with WHO guidance. Second, scale biofortification of staple crops, building on established programmes for high-iron pearl millet and zinc-rich wheat. Third, introduce pre-market scientific scrutiny for health claims on nutraceuticals, closing the regulatory gap that currently places the entire burden of consumer protection on post-market complaints.

Conclusion

Nutritional science, dietary protocol selection, physiological absorption mechanisms, and public health policy are four interconnected levers that collectively determine population health outcomes. Advancing health at scale requires interventions that operate across all four: evidence-informed dietary guidance at the individual level, rigorous claim standards at the industry level, and fortification calibrated to actual deficiency burden at the policy level. The cost of inaction, measured in preventable birth defects, chronic disease burden, and cognitive impairment, is demonstrably higher than the cost of coordinated reform.

Introduction to Nutrients

Nutrients are chemical substances required by the body to sustain basic functions and are optimally obtained by eating a balanced diet.¹ Nutrients must be obtained from diet or edible substances, since the human body does not synthesise or produce them. Nutrients have one or more of the three basic functions: providing energy, contributing to body structure, and/or regulating chemical processes in the body.

Foods also contain non-nutrients that may be harmful (such as natural toxins common in plant foods and additives like some dyes and preservatives) or beneficial (such as antioxidants).² However, the beneficial nutrients are classified under six major classes, which are essential for human health.

- Carbohydrates, lipids, and proteins are macronutrients and serve as the major source of energy.
- Water, which is required in large amounts, does not yield energy.
- Vitamins and minerals, which are considered micronutrients and play essential roles in metabolism.³

The two essential umbrella categories here are macro and micro nutrients.

- **Macronutrients:** Nutrients that are needed in large amounts are called macronutrients. There are three classes of macronutrients: carbohydrates, lipids, and proteins. The cellular energy of macronutrients, which is metabolically processed in the body, comes from their chemical bonds. Water is also a macronutrient in the sense that it is required in large amounts, but unlike the other macronutrients, it does not yield calories.
 - **Carbohydrates:** They are an important dietary energy source and provide 4 kcal of energy per gram. Carbohydrate intake raises blood glucose levels and stimulates insulin secretion, promoting glucose uptake into tissues and glucose storage as glycogen. Additionally, carbohydrates play an important role in gut health and immune function. Fibre, a type of non-digestible carbohydrate with multiple subtypes, is important in promoting satiety, improving gastrointestinal function, and reducing cholesterol levels.
 - **Proteins:** These are large molecules comprising varying amounts and combinations of amino acids linked via peptide bonds. The most important function of dietary proteins is to supply amino acids, which provide nitrogen, hydrocarbon skeletons, and sulfur. They contain 4 kcal of energy per gram and are less efficient energy sources than lipids or carbohydrates. In the human body, amino acids are used for mechanical and structural purposes and help synthesise enzymes, hormones, antibodies, cytokines, transporters and neurotransmitters.
 - **Lipids:** Also known as dietary fats, lipids are the most energy-dense macronutrient and provide 9 kcal of energy per gram. In human physiology, lipids are essential for the production of reproductive hormones, maintenance of cellular structure, energy storage as body fat, regulation of body temperature, protection from physical trauma, and the absorption of fat-soluble vitamins A, D, E, and K. Additionally, fats enhance the taste, texture, and palatability of foods. Dietary fats are further distinguished into triglycerides (fats and oils), phospholipids, sterols (cholesterol), and fatty acids. Fatty acids in the diet can be further distinguished according to the presence of double bonds: saturated fats and unsaturated fats (monounsaturated fats and polyunsaturated fats). Different groups of dietary lipids have been known to have distinct physiological properties and health effects.
- **Micronutrients:** Micronutrients are nutrients required by the body in lesser amounts, but are still essential for carrying out bodily functions. Micronutrients include all the essential minerals and

¹ Morris, A. L., & Mohiuddin, S. S. (2023, May 1). *Biochemistry, nutrients*. StatPearls - NCBI Bookshelf.

² Langara College, Nutrition and Food Service Management Program. (n.d.). Introduction to nutrients. Human Nutrition.

³ Morris, A. L., & Mohiuddin, S. S. (2023, May 1). *Biochemistry, nutrients*. StatPearls - NCBI Bookshelf.

vitamins. There are sixteen essential minerals and thirteen vitamins. In contrast to carbohydrates, lipids, and proteins, micronutrients are not sources of energy, but they assist in the process as cofactors or components of enzymes (i.e., coenzymes). Enzymes are proteins that catalyse chemical reactions in the body and are involved in all aspects of body functions, from producing energy to digesting nutrients, to building macromolecules; thus, playing many essential roles in the body.²

- **Minerals:** These are solid inorganic substances that form crystals and are classified depending on how much of them is needed in the body. Many minerals are critical for enzyme function, others are used to maintain fluid balance, build bone tissue, synthesise hormones, transmit nerve impulses, contract and relax muscles, and protect against harmful free radicals in the body that can cause health problems such as cancer.
 - **Trace minerals**, such as molybdenum, selenium, zinc, iron, and iodine, are only required in a few milligrams or less.
 - **Macrominerals**, such as calcium, magnesium, potassium, sodium, and phosphorus, are required in hundreds of milligrams.
- **Vitamins:** Vitamins are required to perform many functions in the body, such as making red blood cells, synthesising bone tissue, and playing a role in normal vision, nervous system function, and immune system function. There are thirteen vitamins, which are categorised as either water-soluble or fat-soluble.
 - **Water-soluble vitamins:** Vitamin C and all the B vitamins are water-soluble, which include thiamine, riboflavin, niacin, pantothenic acid, pyridoxine, biotin, folate and cobalamin. These are dissolved in water, washed out and not easily stored in the body.
 - **Fat-soluble vitamins:** Vitamins A, D, E, and K are fat-soluble, which are absorbed along with dietary fat and easily stored in fat upon absorption.

Introduction to Carbohydrates

Carbohydrates are one of the three macronutrients in the human diet, along with proteins and fats. These molecules contain carbon, hydrogen, and oxygen atoms. By nature, they are sugar molecules, which are the main energy source in the human diet (50% or more total calories).⁴

Carbohydrates act as an energy source, help control blood glucose and insulin metabolism, participate in cholesterol and triglyceride metabolism, and help with fermentation. Carbohydrates are an umbrella term encompassing sugar, fruits, vegetables, fibres, and legumes. While there are numerous divisions of carbohydrates, the human diet benefits mostly from a certain subset.

Structures of carbohydrates

Carbohydrates can be divided into 4 major classes based on the degree of polymerisation:

- **Monosaccharides:** They are the most fundamental unit of a carbohydrate. These are simple sugars and include glucose, galactose, and fructose.
- **Disaccharides:** They are compound sugars containing two monosaccharides with the elimination of a water molecule. Examples of this subtype include sucrose (glucose & fructose), maltose (glucose & glucose), and lactose (glucose and galactose).
- **Oligosaccharides:** These polymers contain three to ten monosaccharides, and their examples include maltodextrins and raffinose.
- **Polysaccharides:** They are polymers containing long chains of monosaccharides connected through glycosidic bonds. Examples of this subtype include starch and cellulose.

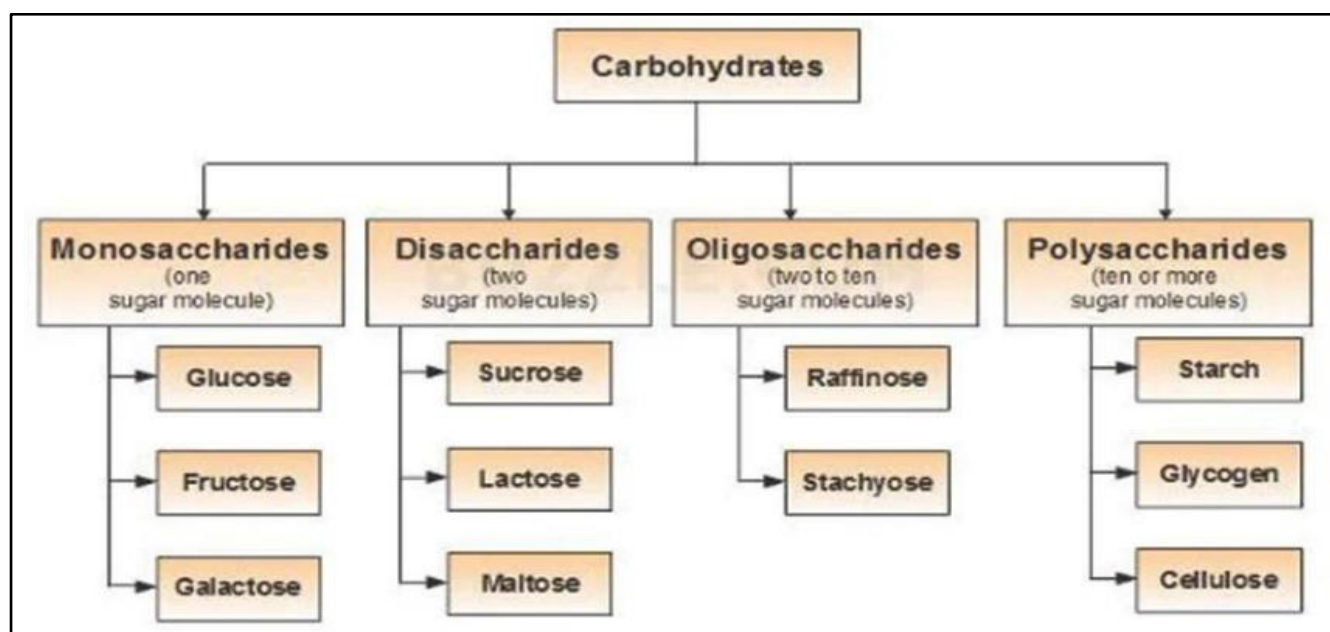


Figure: Structures of carbohydrates

Types of carbohydrates

- **Simple carbohydrates:** These are one or two sugar molecules (monosaccharides or disaccharides) combined in a simple chemical structure. These are easily utilised for energy, causing a rapid rise in blood sugar and insulin secretion from the pancreas. This category includes fructose, lactose, maltose, sucrose, glucose, galactose, and ribose (candy, carbonated beverages, corn syrup, fruit juice, honey, table sugar).
- **Complex carbohydrates:** These are structures involving three or more sugars (oligosaccharides or polysaccharides) bonded together in a complex chemical structure. They take longer to digest and therefore have a more gradual effect on the increase in blood sugar levels. This category

⁴ Holesh, J. E., Aslam, S., & Martin, A. (2023, May 12). Physiology, carbohydrates. StatPearls - NCBI Bookshelf.

includes amylose, cellulose, dextrin (apples, broccoli, lentils, spinach, unrefined whole grains, brown rice).

- **Starches:** They are complex carbohydrates which contain a large number of glucose molecules. Typically, plants produce these polysaccharides. Foods heavy in starches are potatoes, chickpeas, pasta, and wheat.
- **Fibre:** They are non-digestible complex carbohydrates which resist digestion in the small intestine, encourage healthy bacterial growth in the colon and thereby act as a bulking agent, easing defecation. The main components include cellulose, hemicellulose, and pectin.
 - **Insoluble fibre:** They remain in the intestines, softening and bulking the stool, primarily benefitting the body by ensuring regularity of bowel movements. They can be found in foods like beans, seeds, vegetables, brown rice, and potato skins.
 - **Soluble fibre:** They help decrease blood cholesterol and LDL levels, reduce straining with defecation, and blunt postmeal blood glucose levels. They can be found in foods like fleshy fruits, oats, broccoli, and dried beans.

Functions of carbohydrates

- **Energy production:** Primarily, carbohydrates supply energy to the body in the form of glucose. Glucose is the preferred fuel for many tissues and is the main energy source for the central nervous system (brain). Under prolonged fasting or starvation conditions, the brain can partially adapt to using ketone bodies, though glucose remains essential. After absorption, glucose is distributed to tissues for immediate energy needs or metabolic processes. The first stage in the breakdown of glucose is called glycolysis, which occurs in the cytoplasm and involves a series of ten enzyme-mediated reactions that convert glucose into pyruvate while releasing energy. The second stage of glucose breakdown occurs in the mitochondria, where pyruvate is further oxidised. Carbon dioxide is released, and high-energy electrons are captured and transferred within the mitochondria, ultimately producing energy in a form that cells can use.
- **Energy storage:** Carbohydrates provide the body with glucose for immediate energy use, enable short-term energy storage as glycogen in muscle and liver, and support blood glucose regulation. They are essential for sustaining high-intensity physical activity and for maintaining a stable energy supply to glucose-dependent tissues, particularly the brain.
- **Building macromolecules:** Glucose is converted to ribose and deoxyribose, which are essential building blocks of important macromolecules, such as RNA, DNA, and ATP. Glucose is additionally utilised to make the molecule NADPH, which is important for protection against oxidative stress and is used in many other chemical reactions in the body. If any glucose is left post this, it is converted into fat.
- **Molecular recognition and signalling (glycosylation):** Carbohydrates are essential for cell recognition, signalling, and structural integrity via glycoproteins, glycolipids, and proteoglycans. These carbohydrate chains regulate immune recognition, hormone receptor function, cell adhesion, and protein stability.
- **Detoxification and conjugation (glucuronidation):** Glucose-derived compounds are required for glucuronidation, a major liver detox pathway. This process makes hormones, drugs, and toxins water-soluble so they can be excreted.

Digestion of carbohydrates

Carbohydrates, which are generally consumed as starches and sugars, are broken down into single units of glucose, fructose, and/or galactose, both mechanically (chewing) and chemically (enzymes), which are then absorbed and finally transported to different parts of the body for use.

- **Mouth:** Initial starch digestion starts in the mouth by the enzyme α -amylase (ptyalin) present in the saliva, which digests cooked starch to maltose. However, the role of salivary amylase is limited by the short duration of stay of the food in the mouth.
- **Stomach:** The activity of α -amylase, which enters the stomach with food, continues in the stomach for 20-30 minutes until highly acidic gastric juice mixes with the food and makes it

inactive. The optimum pH for the action of salivary amylase is 6-7, and its activity in the stomach completely stops when the pH falls below 4.

- **Small intestine:**

- The carbohydrate digestion is completed in the small intestine (mainly in the jejunum and proximal ileum). The end products of carbohydrates are monosaccharides such as glucose, fructose and galactose.
- Carbohydrates are digested by the amylolytic enzymes present in the pancreatic juice and brush border enzymes of the small intestine.
- Pancreatic amylase is present in the pancreatic juice, which is poured into the duodenum. It acts on starches except cellulose. It hydrolyses almost all the starch within 15 to 30 minutes of the entry of chyme into the duodenum. Its action occurs before the chyme passes beyond the duodenum or upper jejunum. Pancreatic amylase acts in an alkaline medium, converting starch into oligosaccharides such as maltose, maltotriose and dextrin.
- Brush border enzymes of the small intestine include dextrinase, maltase, sucrase, and lactase. They digest the oligosaccharides into monosaccharides on the surface of epithelial cells of villi.
 - α -limiting dextrinase leads to sequential removal of glucose monomers from the dextrans (the breakdown products of starch by the enzyme amylase).

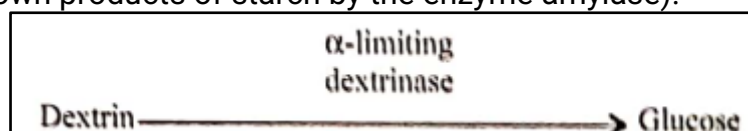


Figure: Conversion of dextrin into glucose

- Sucrase and lactase hydrolyse the corresponding disaccharides into monosaccharides.

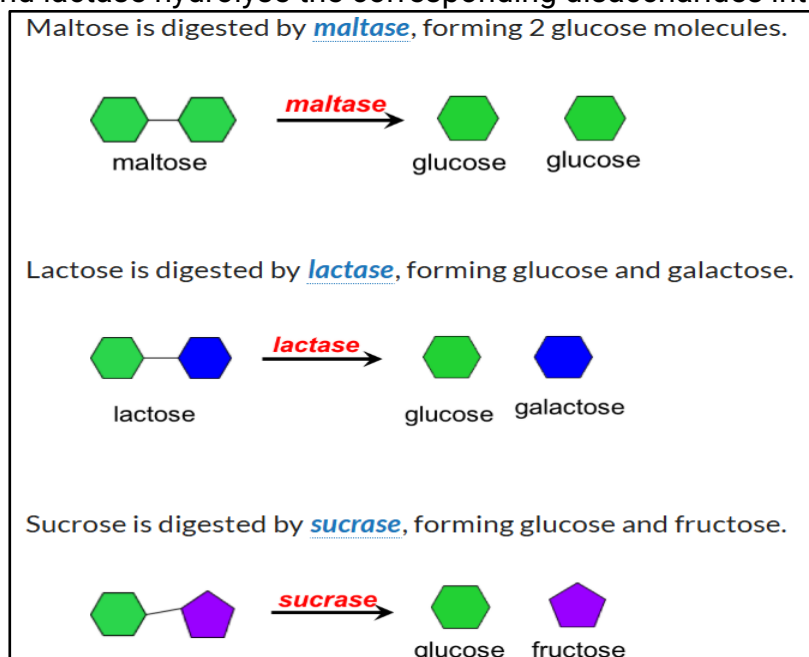


Figure: Conversion of maltose, lactose and sucrose into monosaccharides

Absorption of carbohydrates

Carbohydrates are absorbed in the blood in the form of monosaccharides (both from enzymatic digestion and direct intake through sources like fruits).

- Most of the monosaccharides are absorbed from the mucosal surface of the jejunum and upper ileum almost completely before the remains of the meal reach the terminal ileum. Negligible absorption also occurs in the stomach and colon.
- Mechanism of absorption:
 - Glucose and galactose are absorbed by a common Na^+ dependent active transport system.
 - Fructose is absorbed in the small intestine by facilitated diffusion through the GLUT5

transporter and exits into the bloodstream via GLUT2 on the basolateral membrane.

Absorption of glucose and galactose

Glucose and galactose are absorbed into the epithelial cells (enterocytes) lining the mucous membrane of the small intestine from their brush border surface by an active transport mechanism, called the sodium-coupled transport mechanism.

- **Binding of glucose and Na⁺ to carrier protein:** The carrier protein (present in the cell membrane) has two binding sites, one for sodium and another for glucose, and is called sodium-dependent glucose transporter-1 (SGLT1). The conformational change in the carrier protein occurs only when the binding sites are occupied by sodium and glucose present in the gut lumen, forming the sodium-glucose-carrier complex.
- **Creation of electrochemical gradient across the epithelial cell:** The Na⁺-K⁺ pump pushes sodium (Na⁺) out of the cell and brings potassium (K⁺) in. Because sodium is moved out, the amount of sodium inside the cell becomes low. This difference outside the lining creates an electrochemical gradient, which gives energy that helps the cell move other substances in or out.
- **Movement of sodium and glucose inside the cell:** Because of the electrochemical gradient, sodium moves into the cell (downhill transport). The flow of sodium ions down the gradient is so forceful that the glucose (or galactose) molecule attached to the carrier protein also enters the cell, even against the concentration gradient for glucose (uphill movement). Since two Na are transported down their electrochemical gradient, a large amount of energy is available for transport. Thus, almost all of the glucose and galactose present in the intestine can be absorbed (against the concentration gradient). The energy released is required for Na-K pump activity to maintain the sodium gradient.
- **Transport of glucose into blood capillaries:** From the epithelial cell, the glucose is transported into the interstitial space and then to the blood capillaries of the portal system through facilitated diffusion (transport of substances across a biological membrane from an area of higher concentration to an area of lower concentration with the help of a transport molecule) by glucose transporter-2 (GLUT-2).

Metabolism of carbohydrates

Once the absorbed monosaccharides are transported to the tissues, the process of cellular respiration begins.

1. Glycolysis

Once glucose is transported across the wall of the small intestine and into the circulatory system, it is transported to the liver. In the liver, glucose uses the GLUT2 transporter to move from the bloodstream into hepatocytes because glucose molecules are large and cannot cross the membrane freely. Glucose is phosphorylated using ATP to trap it inside the cell. Hepatocytes either pass the glucose on through the circulatory system or store excess glucose as glycogen.

Cells in the body take up circulating glucose in response to insulin. Through a series of reactions called glycolysis, which takes place in the cytoplasm of all cells and can occur in both aerobic and anaerobic conditions, glucose (a 6-carbon sugar) is broken down to produce energy in the form of ATP. During this process, energy is transferred to ADP to form ATP, and electrons are transferred to molecules that deliver them to the mitochondria, where more ATP will be produced. The last step in glycolysis produces pyruvate, an important three-carbon molecule in cellular respiration.

1.1. Aerobic respiration

In the presence of oxygen, pyruvate can enter the Krebs cycle, where additional energy is extracted

as electrons are transferred from the pyruvate to the receptors NAD^+ , GDP, and FAD, with carbon dioxide being a waste product. The NADH and FADH_2 pass electrons on to the electron transport chain, which uses the transferred energy to produce ATP.

1.1.1. Krebs Cycle/Citric Acid Cycle/Tricarboxylic Acid Cycle

The pyruvate molecules generated during glycolysis are transported across the mitochondrial membrane into the inner mitochondrial matrix, where they are metabolised by enzymes in a pathway called the Krebs cycle. During the cycle, high-energy molecules, including ATP, NADH, and FADH_2 , are created. NADH and FADH_2 then pass electrons through the electron transport chain in the mitochondria to generate more ATP molecules. This cycle starts with the conversion of pyruvate into acetyl-CoA.

1.1.2. Electron transport chain

The electron transport chain (ETC) uses the NADH and FADH_2 produced by the Krebs cycle to generate ATP. Electrons from NADH and FADH_2 are transferred through protein complexes embedded in the inner mitochondrial membrane by a series of enzymatic reactions.

The electron transport chain consists of a series of four enzyme complexes (Complex I – Complex IV) and two coenzymes (ubiquinone and Cytochrome c), which act as electron carriers and proton pumps used to transfer H^+ ions into the space between the inner and outer mitochondrial membranes. The ETC couples the transfer of electrons between a donor (like NADH) and an electron acceptor (like O_2) with the transfer of protons (H^+ ions) across the inner mitochondrial membrane, enabling the process of oxidative phosphorylation. In the presence of oxygen, energy is passed, stepwise, through the electron carriers to gradually collect the energy needed to attach a phosphate to ADP and produce ATP. The role of molecular oxygen, O_2 , is as the terminal electron acceptor for the ETC. This means that once the electrons have passed through the entire ETC, they must be passed to another, separate molecule. These electrons, O_2 , and H^+ ions from the matrix combine to form new water molecules. This is the basis for the need to breathe in oxygen. Without oxygen, electron flow through the ETC ceases.

1.2. Anaerobic respiration

When oxygen is limited or absent, pyruvate enters an anaerobic pathway. In these reactions, pyruvate can be converted into lactic acid. This pathway also serves to keep the pyruvate concentration low so glycolysis continues, and it oxidises NADH into the NAD^+ needed by glycolysis. Anaerobic respiration occurs in most cells of the body when oxygen is limited, or mitochondria are absent or nonfunctional. The lactic acid produced diffuses into the plasma and is carried to the liver, where it is converted back into pyruvate or glucose via the Cori cycle.

2. Gluconeogenesis

Gluconeogenesis is the synthesis of new glucose molecules from pyruvate, lactate, glycerol, or the amino acids alanine or glutamine. This process takes place primarily in the liver during periods of low glucose, that is, under conditions of fasting, starvation, and low-carbohydrate diets. The process is important because some organs only use energy in the form of glucose. Pyruvate is a common starting material for gluconeogenesis. First, the pyruvate is converted into oxaloacetate. Oxaloacetate then serves as a substrate for the enzyme phosphoenolpyruvate carboxykinase (PEPCK), which transforms oxaloacetate into phosphoenolpyruvate (PEP). From this step, gluconeogenesis is nearly the reverse of glycolysis.

Introduction to Fats

Fats are a type of lipid, meaning compounds that are insoluble in water but soluble in organic solvents.⁵ Fats and oils are esters of glycerol and three fatty acids. They are important in the diet as energy sources and as sources of essential fatty acids and fat-soluble vitamins.

Fatty acids generally consist of a straight alkyl chain, terminating with a carboxyl group. The number of carbons in the chain varies, and the compound may be saturated (containing no double bonds) or unsaturated (containing one or more double bonds).

Most fats are made by the body, but some are obtained only from the diet and are known as essential fats. Such essential fats include Omega-3 fats (found in foods such as fish and flaxseed) and Omega-6 fats (found in foods such as nuts, seeds, and corn oil).

Types of fats

- **Monounsaturated fats**

They are unsaturated fats, which have one unsaturated carbon bond in the molecule and typically start to turn solid when chilled. These are found primarily from plant sources. Considered part of a balanced and healthy diet, they have a protective effect on the heart as they lower LDL cholesterol. Oils rich in monounsaturated fats also provide vitamin E, an important antioxidant vitamin. Common sources of monounsaturated fat include avocado, nuts (like almonds, cashews, pecans and peanuts), peanut butter, seeds (such as pumpkin and sesame seeds) and non-tropical vegetable oils (olive, canola, peanut, safflower, sesame)

- **Polyunsaturated fats**

They are unsaturated fats that have more than one unsaturated carbon bond in the molecule. Oils that contain polyunsaturated fats are typically liquid at room temperature but start to turn cloudy or remain in liquid form even when chilled. The oils provide essential fats that the body can't produce on its own, including omega-3 and omega-6 fatty acids. Nontropical vegetable oils high in polyunsaturated fat include canola, corn, soybean and sunflower. Additionally, foods high in polyunsaturated fats include fatty, or oily fish, including anchovies, herring, mackerel, black cod, salmon, sardines, bluefin tuna, whitefish, striped bass and cobia, some nuts and seeds, including walnuts, flaxseeds and sunflower seeds, and tofu/soybeans.

- **Saturated fats**

They are fats in which the fatty acid chains are saturated through single bonds in between them and are no longer available to react with hydrogen. They are typically solid at room temperature. Saturated fats can cause problems with cholesterol levels, which increase the risk of heart disease. Animal products containing saturated fat include lard, fatback and salt pork, high-fat meats like regular ground beef, bologna, hot dogs, sausage, bacon and spareribs, high-fat dairy products such as full-fat cheese, cream, ice cream, whole milk, 2% milk and sour cream, butter, poultry skin (like chicken, turkey etc.) and oils containing saturated fat include palm oil, palm kernel oil and coconut oil.

- **TransFats**

They are made by heating liquid vegetable oils in the presence of hydrogen gas and a catalyst, a process called hydrogenation. They are stable, less likely to become rancid and can stand repeated heating. Naturally occurring trans fats are produced in the gut of some animals, and foods made from these animals (e.g., milk and meat products) may contain small quantities of these fats. Artificial trans fats (or trans fatty acids) are created in an industrial process that adds hydrogen to liquid vegetable oils to make them more solid. Sources of trans fat include processed foods like snacks (crackers and chips) and baked goods (muffins, cookies and cakes) with hydrogenated oil or partially hydrogenated oil, fast food items, such as French fries. Trans

⁵ National Academies Press (US). (1989). *Fats and other lipids*. Diet and Health - NCBI Bookshelf.

fats are the worst type of fat for the heart, blood vessels, and the rest of the body because they raise bad LDL and lower good HDL; create inflammation, which is a reaction related to immunity and has been implicated in heart disease, stroke, diabetes, and other chronic conditions; contribute to insulin resistance; and can have harmful health effects even in small amounts.

Commonly, unsaturated fats are known as good fats, whereas saturated and trans fats are known as bad fats. Saturated fats raise LDL cholesterol, which can increase heart disease risk when consumed in excess. Whereas, trans fats increase LDL (bad) and decrease HDL (good) cholesterol, and cumulatively, are associated with heart risks.

Functions of fats

- **Provision of energy:** Fat is the most concentrated source, providing 9 kcal per 1 gram consumed, which is more than double the energy content of protein or carbohydrate (4 kcal per gram) and more than quadruple the energy content of fibre (2 kcal per gram). It can be stored in body fat tissues, which release fatty acids as and when required.
- **Provides structure:** The membranes around the cell, which separate the inside from the outside of the cell, and control the movement of substances in and out of the cells, are made of phospholipids, triglycerides and cholesterol. Shorter chain fatty acids and unsaturated fatty acids are less stiff and less viscous, which makes membranes more flexible.
- **Carrier of vitamins:** In diets, fat is a carrier for fat-soluble vitamins (A, D, E and K), and supports their absorption in the intestine, thus improving their bioavailability. Consuming sufficient amounts of fatty foods that contain these vitamins is thus essential for adequate intake of these micronutrients.
- **Regulating and signalling:** Fats help the body to produce and regulate hormones. For example, adipose tissue secretes the hormone leptin, which signals the body's energy status and helps to regulate appetite. Omega-3 and omega-6 essential fatty acids help regulate cholesterol and blood clotting and control inflammation in the joints, tissues, and bloodstream. It plays a role in sustaining nerve impulse transmission, memory storage, and tissue structure. Lipids are especially focal to brain activity in structure and in function, helping form nerve cell membranes, insulate neurons, and facilitate the signalling of electrical impulses throughout the brain.
- **Insulation and protection:** Some of the fat is stored within the abdominal cavity (visceral fat), which protects vital organs such as the heart, kidneys, and liver. Some is stored just underneath the skin (subcutaneous fat), which insulates the body from extreme temperatures and helps keep the internal climate under control.

Digestion of fats

In the mouth, an enzyme called lingual lipase is produced by cells on the tongue, which causes enzymatic digestion of some triglycerides, breaking individual fatty acids from the glycerol backbone. Lingual lipase only gets activated in the stomach in low pH provided by hydrochloric acid. Subsequently, in the stomach, the chief cells produce an enzyme called gastric lipase, which begins the digestion of fats in the stomach. But together, these two lipases play only a minor role in fat digestion. The majority of fat digestion takes place in the small intestine.

Mechanism of digestion of fats

The digestion of fat includes three steps:

- **Emulsification of fat by bile salts:** Emulsification, caused by bile salts, is the breaking of large fat drops into smaller droplets, a prerequisite for the action of pancreatic lipase because the pancreatic lipase, being water-soluble, acts only on the oil-water interface of fat. It also increases the surface area for the action of enzymes on fats. Lecithin, a component of bile which has a stabilising action on the emulsions, greatly enhances the emulsifying action of bile salts. The bile salts surround the fine fat droplets in such a way that their lipophilic ends are towards the fat and their hydrophilic ends separate the fat droplets from the aqueous phase.

- **Hydrolysis of fat by pancreatic and intestinal lipolytic enzymes:** When food reaches the small intestine in the form of chyme, a digestive hormone called cholecystokinin (CCK) is released by intestinal cells in the intestinal mucosa. CCK stimulates the release of pancreatic lipase from the pancreas and stimulates the contraction of the gallbladder to release stored bile salts into the intestine. CCK also travels to the brain, where it can act as a hunger suppressant. At the same time, alkaline pancreatic juice (pH 7.8–8.4) neutralises the acidic chyme from the stomach (pH ~6), bringing the intestinal pH close to neutral (~7). This pH environment is optimal for pancreatic lipolytic enzyme activity.
- **Pancreatic digestion of dietary lipids:** Pancreatic lipase is the principal enzyme responsible for fat digestion, hydrolysing emulsified triglycerides into two free fatty acids and one 2-monoglyceride. Its activity is enabled by colipase, which displaces bile salts from the fat droplet surface to allow effective enzyme binding. Other pancreatic enzymes act on non-triglyceride lipids: cholesterol ester hydrolase converts cholesterol esters into free cholesterol and fatty acids, while phospholipase A₂ (activated from its pro-form by trypsin) hydrolyses phospholipids by removing a fatty acid.
- **Limited role of intestinal brush border enzymes:** The epithelial cells lining the intestinal villi contain small amounts of lipase and cholesterol esterase. Their contribution to fat digestion is minor and supplementary, serving only to marginally extend the actions of pancreatic enzymes.
- **Acceleration of fat digestion through micelle formation:** Since triglyceride hydrolysis is reversible, accumulation of monoglycerides and free fatty acids near fat droplets inhibits further digestion. Bile salts resolve this by forming water-soluble mixed micelles, which sequester digestion products away from the lipid-water interface. By continuously removing these products, micelles enable pancreatic lipolysis to proceed efficiently to completion.

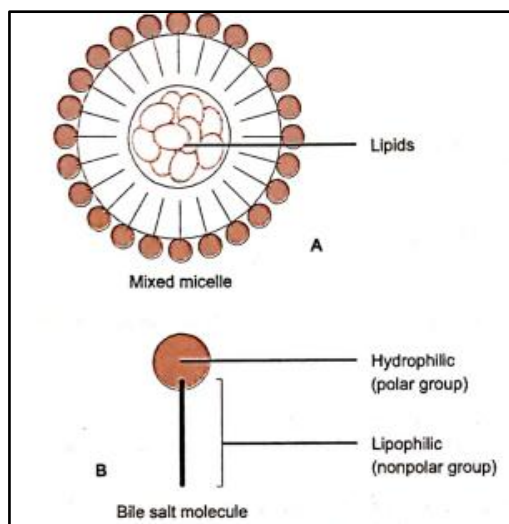


Figure: Micelle formation

Absorption

The majority of the absorption of fats occurs in the duodenum in the following manner:

- **Transportation as micelles to the brush border membrane:** The insolubility of fat globules prevents their diffusion through the aqueous medium of the intestinal lumen to reach the brush border. The outer surface of the micelle is formed by water-soluble ends of bile salts, which help the micelle to diffuse through the aqueous medium to reach the brush border membrane. Thus, the bile salt micelle acts as a transport vehicle for the products of fat digestion.
- **Diffusion of lipids across the enterocyte membrane:** Once the micelle comes in contact with the cell membrane, the monoglycerides, free fatty acids, cholesterol and fat-soluble vitamins diffuse passively at a rapid speed through the enterocyte cell membrane to the interior of the cell, leaving bile salts in the intestinal lumen. Thus, the rate-limiting step in lipid absorption is the formation

and migration of the micelles from the intestinal chyme to the microvilli surface. Bile salts must be present in a certain minimum concentration called the critical micellar concentration before micelles are formed. The bile salts released from micelle after diffusion of their associated lipids are absorbed in the terminal ileum by a Na^+ dependent active transport process.

- **Transport of lipids from inside the enterocytes to the interstitial space:** Once inside the cell, the end products of fat digestion enter the interstitium by two mechanisms:
 - **Diffusion:** The small chain fatty acids with less than 12-14 carbon atoms diffuse across the basal border of enterocytes to enter the interstitium, a supporting or connective tissue or material that lies between the principal cells of an organ, or between structures in the body.
 - **Formation and excretion of chylomicrons from enterocytes:** The large chain fatty acids enter the smooth endoplasmic reticulum, where they are reconstituted. Monoglycerides are combined with fatty acids to produce triglycerides, lysophosphatides are combined with fatty acids to form phospholipids, and cholesterol is re-esterified. The re-formed lipids coalesce to form small lipid droplets called chylomicrons. The chylomicrons are then excreted into the interstitium by exocytosis from the basolateral membrane of the enterocyte. Covering of β -lipoproteins is essential for the exocytosis, a process by which the contents of a cell vacuole are released to the exterior through fusion of the vacuole membrane with the cell membrane, to occur. Therefore, in the absence of B-lipoprotein, exocytosis will not occur, and the enterocytes become engorged with lipids.

After exiting the enterocytes, the chylomicrons merge into larger droplets. From the interstitium, the lipids diffuse into the lacteals, from which they enter the lymphatic circulation and, via the thoracic duct, gain access to the blood circulation.

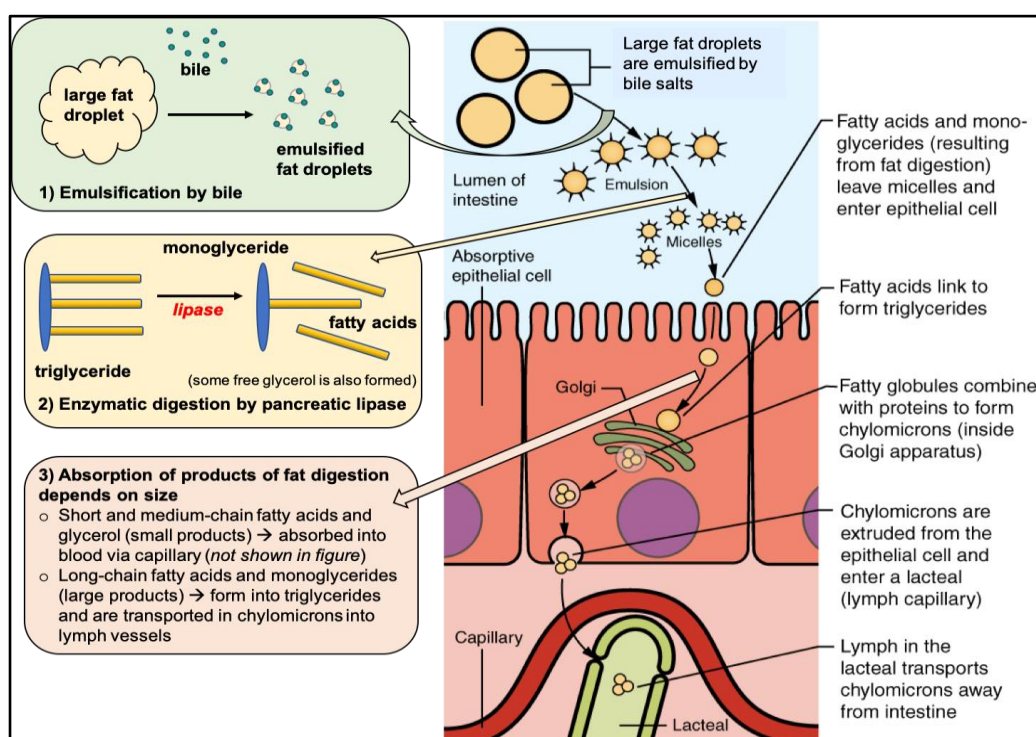


Figure: Absorption of fats

Metabolism of fats

Fats (or triglycerides) within the body are ingested as food or synthesised by adipocytes or hepatocytes from carbohydrate precursors. Lipid metabolism entails the oxidation of fatty acids to either generate energy or synthesise new lipids from smaller constituent molecules. Lipid metabolism is associated with carbohydrate metabolism, as products of glucose (such as acetyl-CoA) can be converted into lipids. Once the fats are digested and enter the bloodstream, they can either go to the liver or be stored in fat cells (adipocytes) that comprise adipose (fat) tissue found throughout the body.

- **Lipolysis**

To obtain energy from fat, triglycerides must first be broken down by hydrolysis into their two principal components, fatty acids and glycerol, through lipolysis, which takes place in the cytoplasm. The resulting fatty acids are oxidised by β -oxidation into acetyl-CoA, which is used by the Krebs cycle. The glycerol released from triglycerides after lipolysis directly enters the glycolysis pathway as DHAP. Because one triglyceride molecule yields three fatty acid molecules with as many as 16 or more carbons in each one, fat molecules yield more energy than carbohydrates and are an important source of energy for the human body. When glucose levels are low, triglycerides can be converted into acetyl-CoA molecules and used to generate ATP through aerobic respiration. The breakdown of fatty acids, called fatty acid oxidation or beta (β)-oxidation, begins in the cytoplasm, where fatty acids are converted into fatty acyl-CoA molecules. This fatty acyl-CoA combines with carnitine to create a fatty acyl carnitine molecule, which helps to transport the fatty acid across the mitochondrial membrane. Once inside the mitochondrial matrix, the fatty acyl carnitine molecule is converted back into fatty acyl CoA and then into acetyl CoA. The newly formed acetyl-CoA enters the Krebs cycle and is used to produce ATP in the same way as acetyl-CoA derived from pyruvate.

- **Ketogenesis**

If excessive acetyl-CoA is created from the oxidation of fatty acids and the Krebs cycle is overloaded and cannot handle it, the acetyl-CoA is diverted to create ketone bodies, because there is no oxaloacetate to continue with the Krebs cycle, as it may have been utilised through gluconeogenesis. These ketone bodies can serve as a fuel source if glucose levels are too low in the body. In both cases, fat stores are liberated to generate energy through the Krebs cycle and generate ketone bodies when too much acetyl-CoA accumulates. In this ketone synthesis reaction, excess acetyl-CoA is converted into hydroxymethylglutaryl-CoA (HMG-CoA). HMG-CoA is a precursor of cholesterol and is an intermediate that is subsequently converted into β -hydroxybutyrate, the primary ketone body in the blood.

- **Ketone oxidation**

When ketones are produced faster than they can be used, they can be broken down into CO₂ and acetone. The acetone is removed by exhalation. The carbon dioxide produced can acidify the blood, leading to diabetic ketoacidosis, a dangerous condition in diabetics.

Ketones oxidise to produce energy for the brain. Beta (β)-hydroxybutyrate is oxidised to acetoacetate, and NADH is released. An HS-CoA molecule is added to acetoacetate, forming acetoacetyl-CoA. The carbon within the acetoacetyl CoA that is not bonded to the CoA then detaches, splitting the molecule in two. This carbon then attaches to another free HS-CoA, resulting in two acetyl-CoA molecules. These two acetyl-CoA molecules are then processed through the Krebs cycle to generate energy. The Krebs cycle here occurs in the cells of the non-liver tissues only, which have enough level of oxaloacetate to initiate the Krebs cycle.

- **Lipogenesis**

When glucose levels are plentiful, the excess acetyl-CoA generated by glycolysis can be converted into fatty acids, triglycerides, cholesterol, steroids, and bile salts. It takes place in the cytoplasm of adipocytes (fat cells) and hepatocytes (liver cells). Lipogenesis begins with acetyl-CoA and advances by the subsequent addition of two carbon atoms from another acetyl-CoA; this process is repeated until fatty acids are the appropriate length, and is an efficient way of storing energy.

Although lipogenesis occurs in the cytoplasm, the necessary acetyl-CoA is created in the mitochondria and cannot be transported across the mitochondrial membrane. For this, pyruvate is converted into both oxaloacetate and acetyl-CoA. Oxaloacetate forms via the action of pyruvate carboxylase, and pyruvate dehydrogenase creates acetyl-CoA. Oxaloacetate and acetyl-CoA combine to form citrate, which crosses the mitochondrial membrane and enters the cytoplasm. In

the cytoplasm, citrate is converted back into oxaloacetate and acetyl-CoA. Oxaloacetate is converted into malate and then into pyruvate. Pyruvate crosses back across the mitochondrial membrane to wait for the next cycle of lipogenesis. The acetyl CoA is converted into malonyl CoA, which is used to synthesise fatty acids.

Introduction to Proteins

Proteins are large, complex molecules that are critical to most of the work done by cells and are required for the structure, function and regulation of the body's tissues and organs. A protein is made up of one or more long, folded chains of amino acids (each called a polypeptide), whose sequences are determined by the DNA sequence of the protein-encoding gene.

Protein is made from twenty-plus basic building blocks, or amino acids. Since the body does not store amino acids, the body makes them in two different ways: either from scratch or by modifying others. Nine amino acids, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine, are known as the essential amino acids and must come from food.

Protein structure

- **Primary structure of protein:** This is the sequence of amino acids in a single polypeptide chain (beads, like amino acids arranged in a specific order, in a beaded necklace). Chemical bonds hold them together. The order changes based on the type of protein and its function in the body.
- **Secondary structure of protein:** This is the shape a polypeptide chain takes due to chemical interactions between its amino acids. The shape may be coiled (alpha helix) or folded in a zig-zag manner (beta sheet).
- **Tertiary structure of protein:** This is the 3D shape that forms when the regions of secondary structure fold together into the final structure of the protein. Water attracts some amino acids and repels others. So, some amino acids stay on the outer surface of the protein, where they can bond with water molecules, while others move toward the interior.
- **Quaternary structure of protein:** This is the fourth and most complex level. It's when a protein has more than one polypeptide chain, and those chains interact and are folded within each other. Chemical bonds hold these chains together in their unique 3D shapes.

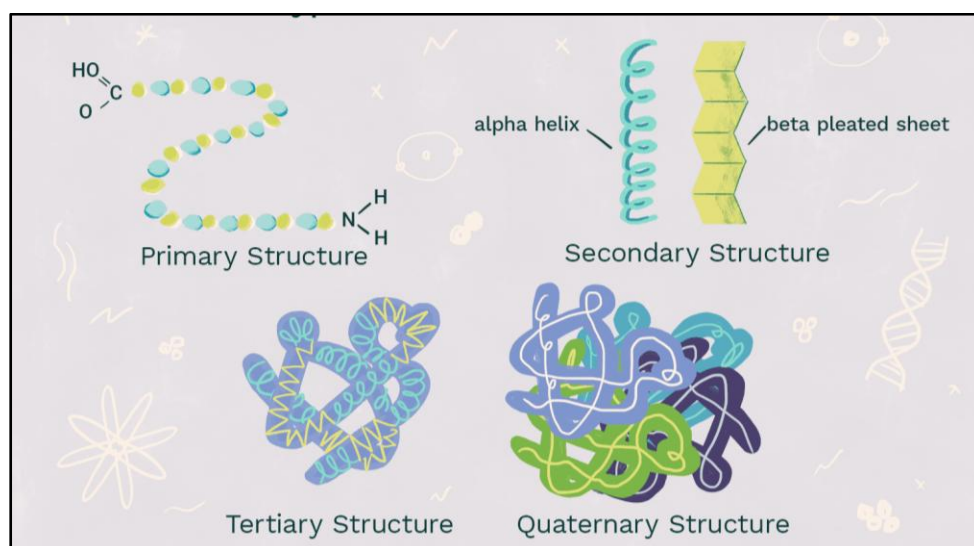


Figure: Types of protein structures

Classes of proteins

- **Non-essential:** They can be obtained from foods and also generated within the body. This category includes alanine, asparagine, aspartic acid, glutamic acid and serine.
- **Conditionally-essential:** Healthy bodies can generate them under normal physiologic conditions. They become essential under certain conditions, like starvation or inborn errors of metabolism. This category includes arginine, cysteine, glutamine, glycine, proline and tyrosine.
- **Essential:** They cannot be generated within the body, and only dietary protein provides them to the body. This category includes histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, valine and tryptophan.

Types of proteins (based on the functions they perform)

- **Antibodies:**
 - They are specialised proteins that defend the body against antigens or foreign invaders.
 - They have the ability to travel through the bloodstream and are utilised by the immune system to identify and defend against bacteria, viruses, and other foreign intruders in the blood. One way antibodies counteract antigens is by immobilising them.
- **Contractile proteins:**
 - They are responsible for muscle contraction and movement.
 - The cytoplasm of cells is a colloidal network of contractile proteins. Actin filaments are the major components of this network.
 - Eukaryotes tend to possess copious amounts of actin, which controls muscle contraction as well as cellular movement and division processes. Myosin powers tasks carried out by actin by supplying it with energy.
- **Enzymes:**
 - All enzymes identified to date are proteins.
 - Enzymes are catalysts of all metabolic reactions, enabling an organism to build the chemical substances necessary for life, i.e., proteins, nucleic acids, carbohydrates, and lipids, to convert them into other substances, and to degrade them.
 - Some proteins function as enzymes, which conduct specific chemical reactions and provide a site for a chemical reaction, lowering the amount of energy and time taken for the chemical reaction.
- **Hormonal proteins:**
 - They are messenger proteins that help coordinate certain bodily functions and are responsible for hormone synthesis.
 - A major function of hormones is to turn enzymes on and off, so some proteins can regulate the actions of other proteins.
 - For example, growth factors are highly specific proteins, a subdivision of cytokines. Growth factors stimulate the division and differentiation of a particular type of cell. In skeletal muscle hypertrophy, growth factors include insulin-like growth factor (IGF). IGF is secreted by skeletal muscle and regulates insulin metabolism and stimulates protein synthesis. Similarly, cortisol is a hormone that is produced in the adrenal cortex of the kidney and is known as the stress hormone.
- **Structural proteins:**
 - They maintain and protect the structure of the animal body. For example, collagen, found in the bones, cells and skin, is a structural protein.
 - They are used to provide internal structure to the cell (the cytoskeleton) and are sometimes involved in cell movement.
- **Storage proteins:**
 - They are proteins responsible for storing and reserving substrates for the body until they are ready for use. For example, ferritin is a storage protein that stores iron.
- **Transport proteins:**
 - They are carrier proteins that move molecules from one place to another in the body.
 - For example, the respiratory protein (haemoglobin) carries oxygen in the blood.

Other functions of proteins

- **Fluid and electrolyte balance:** Fluid balance means maintaining the distribution of water in the body. Water moves toward areas that have higher concentrations of other solutes, such as proteins and glucose. Proteins attract fluid; to keep the water evenly distributed between blood and cells, proteins continuously circulate at high concentrations in the blood. Albumin's presence in the blood makes the protein concentration in the blood similar to that in cells.

- **Wound healing, tissue regeneration, and nerve function:** The healing process begins with proteins that help in dilating blood vessels at the site of injury. Fibrin secures platelets that form a clot to stop the bleeding. In the proliferation phase, cells move in and mend the injured tissue by installing newly made collagen (protein) fibres. In the remodelling phase, more collagen is deposited, forming a scar. During tissue regeneration, an exact structural and functional copy of the old tissue is created. Cells are constantly broken down, repaired, and replaced. When proteins in the cells are broken down, the amino acids are recycled into new proteins. Amino acids can also be used to make neurotransmitters (e.g., epinephrine) that transmit messages from one nerve cell to another.
- **Energy source:** Some amino acids in proteins can be disassembled and used to make energy, through the process of deamination, wherein the amine group is removed from the amino acid, and nitrogen is transported to the kidney for excretion. The remaining components are metabolised for energy.

Digestion of proteins

Proteins are digested by proteolytic enzymes into amino acids and small polypeptides before absorption.

- **Mouth:** No chemical digestion of proteins occurs here, as the required enzymes are not present.
- **Stomach:** Pepsin, secreted by chief cells of the main gastric glands in an inactive form (pepsinogen), digests about 10-15% proteins. Pepsinogen is converted into pepsin (active form) by the action of HCl or preformed pepsin. Pepsin splits proteins into peptones, proteoses and polypeptides. The optimum pH for the action of pepsin is 2.0; therefore, HCl secretion by the stomach is as essential as pepsinogen secretion for the digestion of proteins. Pepsin is unique in its proteolytic action because it digests collagen. Protein digestion within the stomach is important because the protein digestion products stimulate the secretion of proteolytic enzymes of the pancreas.
- **Small intestine:** Proteins are digested here by the pancreatic proteases, brush border peptidases and intracellular peptidases.
 - In order to avoid breaking down the proteins that make up the pancreas and small intestine, pancreatic enzymes are released as inactive proenzymes that are only activated in the small intestine. Pancreatic proteases or proteolytic enzymes of the pancreas include trypsin, chymotrypsin, carboxypeptidase and elastase for the breakdown of proteins.
 - Trypsin, released by the binding of enterokinase to trypsinogen (inactive form), binds to chymotrypsinogen to convert it into the active chymotrypsin. Together, trypsin and chymotrypsin break down large proteins into smaller peptides, a process called proteolysis.
 - Elastase digests the protein elastin, which is found in connective tissues.
 - Carboxypeptidase removes single amino acids from a peptide chain, helping break peptides down into individual amino acids.
 - Pancreatic proteases digest the proteins and split them into dipeptides, tripeptides and small polypeptides, which are further digested by brush border peptidases. Some of the dipeptides and tripeptides are absorbed directly into the epithelial cells of the mucosa of the small intestine and are further digested by intracellular enzymes into amino acids.
 - Brush border peptidases: Peptidases here include aminopeptidases, dipeptidases, tripeptidases, nucleases, and related enzymes. These enzymes continue the digestive process begun by the pancreatic proteases, eventually converting the proteins to small polypeptides and amino acids. Intracellular peptidases are the proteolytic enzymes present in the cytosol (gel-like substance that surrounds the organelles) of epithelial cells of the small intestine. Multiple peptidases present in the enterocytes are specific for linkages between the various amino acids. Within minutes, they digest the last dipeptides and tripeptides into amino acids, which then enter the blood.

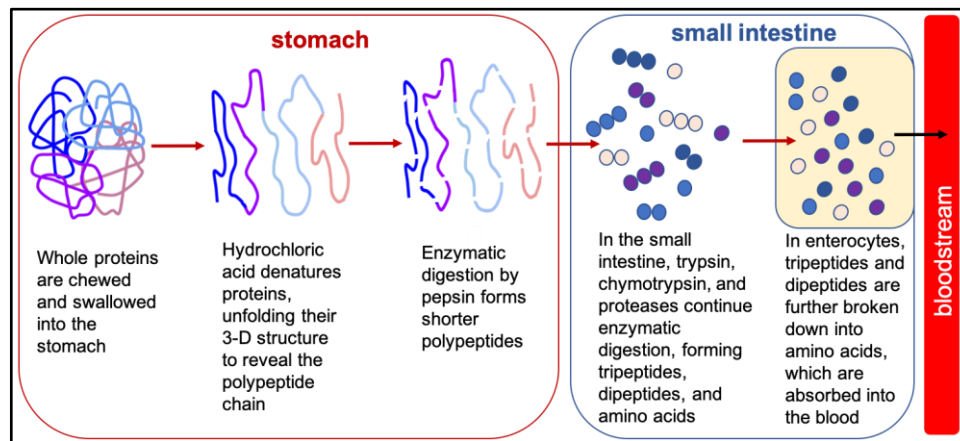


Figure: Digestion of proteins

Absorption of proteins

The end products of protein digestion (amino acids, dipeptides and tripeptides) are absorbed through the epithelial cells of the small intestine. Absorption of amino acids is faster in the duodenum and jejunum and slower in the ileum.

- Mechanisms used for protein absorption into the epithelial cells:
 - **Na⁺ dependent active transport mechanism:** The L-amino acids, dipeptides and tripeptides are absorbed by a Na⁺ dependent active transport mechanism. Separate transporters (carriers) are present for the absorption of basic, acidic and neutral amino acids, and at least two different polypeptide transporters exist. Steps of the sodium-dependent active transport mechanism are similar to those described for glucose absorption.

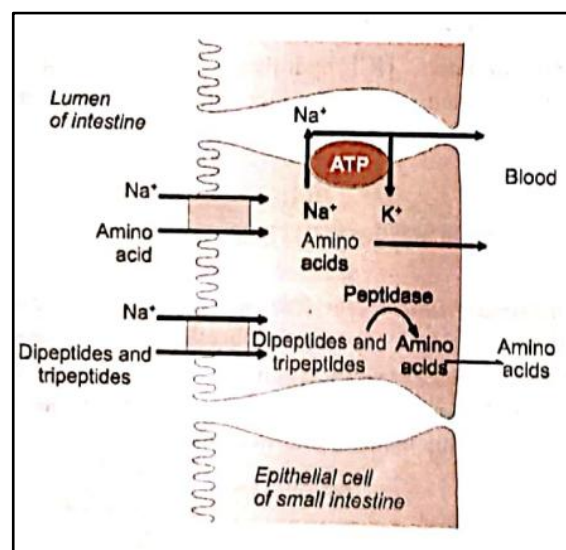


Figure: Na⁺ dependent active transport mechanism of proteins

- **Simple diffusion:** The D-amino acids are absorbed solely by passive diffusion.
- **Endocytosis:** Larger polypeptides cannot be absorbed into the epithelial cells. Endocytosis is the process by which a cell takes in substances from the outside by engulfing them with its cell membrane, which then pinches off to form a vesicle. Proteins absorbed by endocytosis usually elicit immunological/allergic reactions.
- Once amino acids and polypeptides are absorbed into the intestinal epithelial cells, the intracellular peptidases break the remaining linkages of tripeptides and dipeptides, releasing amino acids.
- Transport of amino acids into blood capillaries:

- Once amino acids are inside the intestinal (epithelial) cells, they move out of these cells into the space between the cells (interstitial space). This happens through facilitated diffusion or simple diffusion.
- From there, the amino acids enter tiny blood vessels called capillaries in the villi of the intestine by simple diffusion. These capillaries join to form the portal vein, which carries the absorbed amino acids to the liver, and then they enter the general bloodstream to reach the rest of the body.

Metabolism of proteins

Freely available amino acids are used to create proteins. If amino acids exist in excess, the body has no capacity or mechanism for their storage; thus, they are converted into glucose or ketones (an acid the body produces when it uses fat for energy instead of glucose), or they are decomposed. Amino acid decomposition results in hydrocarbons and nitrogenous waste. However, high concentrations of nitrogen are toxic. The urea cycle processes nitrogen and facilitates its excretion from the body.

Vitamin A (Retinol)

Vitamin A is the name of a group of fat-soluble retinoids, primarily retinol and retinyl esters. The two main forms of vitamin A in the human diet are preformed vitamin A (retinol, retinyl esters) and provitamin A carotenoids such as alpha-carotene and beta-carotene that are converted to retinol. Preformed vitamin A comes from animal products, fortified foods, and vitamin supplements. Carotenoids are found naturally in plant foods. The sources of Vitamin A are leafy green vegetables (kale, spinach, broccoli), orange and yellow vegetables (sweet potatoes, pumpkin and other winter squash, summer squash), carrots, tomatoes, red bell pepper, mango, beef liver, fish oils, milk, eggs, fortified foods, etc.⁶

Absorption of Vitamin A

- Vitamin A is absorbed through the same pathway as dietary fats.
- Upon the consumption of vitamin A, retinyl esters are broken down in the digestive tract into free retinol through the hydrolysis of pancreatic enzymes and intestinal esterase, while carotenoids only partially convert into retinol in the small intestinal mucosal cell walls.
- Both types of vitamin A are emulsified into mixed micelles with bile acids, fats, and other lipids. Mixed micelles are the fundamental structure for the absorption of vitamin A.
- Both free retinol and carotenoids are then absorbed by the enterocytes of the intestinal wall via passive diffusion and transporter-mediated processes. Within the enterocyte cell, retinol is re-esterified to produce retinyl esters, which are then packaged with triglycerides into a structure called chylomicron.
- Chylomicrons are then released from the enterocyte into the lymphatic system and later to the vascular system, providing most of the body's supply of vitamin A to the liver for storage.

Functions of Vitamin A

- **Vision:** It is essential for vision, especially in dark conditions, through its role in forming rhodopsin in the retina. 11-cis-retinal combines with opsin to create rhodopsin, a light-sensitive pigment in rod cells, enabling the eye to detect low light. Its deficiency leads to night blindness and, in severe cases, permanent blindness. It is also used for the prevention or treatment of age-related macular degeneration as it targets oxidative stress in AMD pathophysiology due to its antioxidative properties.
- **Cell differentiation and epithelial integrity:** It controls the development of epithelial cells through its influence on gene expression. If an individual is deficient in Vitamin A, they may experience keratinisation of the mucous covering the epithelial cells of different organs, which results in health problems like xerophthalmia (a vision-related condition) and Bitot's spots (caused by xerophthalmia).
- **Gene regulation:** Retinoic acid, an active form of vitamin A, binds to nuclear receptors (RARs and RXRs) that regulate gene expression. This influences the expression of certain genes which are involved in cell growth, differentiation and tissue development. This mechanism is crucial for embryonic development, including proper limb formation.
- **Immune function:** Vitamin A helps maintain immune competence and epithelial barriers. Its deficiency increases susceptibility to infections by impairing mucosal surfaces, reducing lymphocyte proliferation, and altering T-cell populations.⁷ It also influences cytokine production, which helps in the maintenance of immune balance and supports defence against infections.
- **Embryonic development:** Retinoic acid regulates genes involved in embryogenesis, including limb formation, neural development, and organogenesis. Severe deficiency of this vitamin during pregnancy leads to malformations in the cardiovascular, nervous, and skeletal systems.

⁶ Powell, J. (2024b, November 6). Vitamin A • The Nutrition Source. The Nutrition Source - Harvard Chan School.

⁷ Gerster, H. & F. Hoffmann-La Roche Ltd. (1996). Vitamin A - Functions, requirements, sathe fety. On the Internet. J. Vit. Nutr. Res. (Vol. 67, pp. 71–90)the vitamin.

- **Metabolic regulation:** Retinoids influence metabolic regulation. Dietary sources of vitamin A are directly responsible for signalling through the RAR/RXR receptor systems, impacting lipid and fatty acid metabolism genes, glucose transporter genes, and the genes responsible for insulin resistance and adipocyte differentiation. Additionally, there is emerging evidence on the association of carotenoid consumption with the modification of the risks associated with obesity and metabolic syndrome.⁸
- **Lowers risk of cancer:** Considering its role in regulating cell growth and differentiation, the association between vitamin A and various types of cancers is emerging. Several systematic reviews and meta-analyses of observational studies have shown that higher dietary intakes of retinol, carotenoids, fruits and vegetables, or a combination are associated with a lower risk of lung cancer, pancreatic cancer, oral cavity cancer, oesophageal cancer, ovarian cancer, glioma and bladder cancer.⁹

⁸ Carazo, A., Macáková, K., Matoušová, K., Krčmová, L. K., Protti, M., & Mladěnka, P. (2021). Vitamin A Update: Forms, sources, kinetics, detection, function, deficiency, therapeutic use and toxicity. *Nutrients*, 13(5), 1703.

⁹ Office of Dietary Supplements - Vitamin A and carotenoids. (n.d.).

Vitamin B1 (Thiamine)

Vitamin B1, also known as thiamine or thiamin, is one of the 8 B-complex vitamins and is classified as a water-soluble B vitamin. Thiamine occurs naturally in specific diets, is added to food products, and is available in dietary supplements. Brown rice, whole grains, pork, poultry, soybeans, nuts, peas, dried beans, and fortified or enriched grain products such as bread, cereals, and infant formulas are rich sources of thiamine.¹⁰

Absorption and metabolism of Vitamin B1

- Thiamine occurs in foods in both phosphorylated and non-phosphorylated forms, with animal-derived foods predominantly containing thiamine phosphate esters and plant-derived foods largely providing free thiamine.
- Following ingestion, dietary thiamine phosphates are hydrolysed by intestinal phosphatases, allowing absorption of free thiamine across the intestinal mucosa. In healthy individuals with adequate intake, intestinal absorption efficiency exceeds 95%.
- After absorption, thiamine is transported in the circulation primarily within erythrocytes, where more than 80% of total blood thiamine exists as thiamine diphosphate (TDP) and thiamine triphosphate (TTP), while plasma contains only small amounts of free thiamine, thiamine monophosphate, and protein-bound TDP.
- Tissue distribution of thiamine is highest in metabolically active organs, including skeletal muscle, heart, brain, liver, and kidneys. Within cells, thiamine is phosphorylated by thiamine diphosphokinase to form TDP, the biologically active coenzyme form.
- TDP functions as a coenzyme for transketolase in the pentose phosphate pathway, linking thiamine status to the generation of NADPH and ribose-5-phosphate, which are essential for antioxidant defence, lipid synthesis, and nucleotide biosynthesis.¹¹

Functions of Vitamin B1

- **Energy production:** Thiamine acts as a coenzyme or cofactor in the form of thiamine diphosphate for enzymes critical to carbohydrate metabolism and cellular energy production. These enzymes, such as Pyruvate dehydrogenase (PDH), transketolase, and branched-chain α -ketoacid dehydrogenase, are used to link glycolysis to the TCA cycle, serve the pentose phosphate pathway, which produces ribose for nucleic acids, and amino acid catabolism, respectively.
- **Neural signalling:** It is required for the synthesis of neurotransmitters such as acetylcholine, GABA and glutamate. It also supports myelin formation and neuronal signal transmission. Its deficiency can lead to neurodegenerative symptoms and disorders such as Wernicke's encephalopathy and Korsakoff's syndrome. It prevents depression and has a positive effect on well-being.
- **Acts as an antioxidant:** It is protective against a variety of toxic agents that promote oxidative stress. It protects hepatocytes from iron-catalysed oxidative stress by decreasing lipid peroxidation, mitochondrial and protein damage, and DNA oxidation. Thiamine inhibits intracellular ROS generation and might contribute to the potential regulation of ROS-dependent NETosis (allowing neutrophils to kill extracellular pathogens while minimising damage to the host cells). It also possesses anti-inflammatory and anti-tumor properties by increasing the phagocytic activity of macrophages.¹²

¹⁰ Martel, J. L., Doshi, H., Sina, R. E., & Franklin, D. S. (2024, January 31). Vitamin B1 (Thiamine). StatPearls - NCBI Bookshelf.

¹¹ Kaźmierczak-Barańska, J., Halczuk, K., & Karwowski, B. T. (2025). Thiamine (Vitamin B1)—An Essential Health Regulator. *Nutrients*, 17(13), 2206.

¹² Mrowicka, M., Mrowicki, J., Dragan, G., & Majsterek, I. (2023). The importance of thiamine (vitamin B1) in humans. *Bioscience Reports*, 43(10).

Vitamin B2 (Riboflavin)

What is Vitamin B2?

Vitamin B2, also known as riboflavin, is one of eight B vitamins that make up the B complex. Like all B vitamins, vitamin B2 is water-soluble, and thus its concentration in the body is highly dependent on an individual's intake. Riboflavin is converted to flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN), two cofactors that are crucial for a multitude of reactions within the body. These include reactions crucial for the production of adenosine triphosphate (ATP) in the electron transport chain and elimination of reactive oxygen and nitrogen species.¹³ Foods that are particularly rich in riboflavin include eggs, organ meats (kidneys and liver), lean meats, and milk. Some vegetables also contain riboflavin.¹⁴

Absorption and metabolism of Vitamin B2

Humans are neither capable of in vivo riboflavin synthesis nor efficient in riboflavin storage, requiring a steady supply through dietary intake. Absorption primarily occurs in the proximal small intestine through an active, carrier-mediated, and saturable transport process. The riboflavin status of the individual closely regulates riboflavin metabolism. Conversion of riboflavin to its co-enzymatic forms occurs within the cellular cytoplasm of many tissues, but mostly in the small intestine, liver, heart, and kidney. An ATP-dependent phosphorylation transforms riboflavin into FMN, which is subsequently complexed with certain apoenzymes to various flavoproteins. However, most FMN is phosphorylated to FAD, which is therefore the main form of vitamin B2 in body tissue.¹⁵ Free riboflavin is transported in the plasma, bound both to albumin and to certain immunoglobulins, which will also bind flavin coenzymes. Almost all riboflavin in tissues is enzyme-bound, such as FAD covalently bound to succinic dehydrogenase. Unbound flavins are relatively labile and are rapidly hydrolysed to free riboflavin, which diffuses from cells and is excreted. The intracellular phosphorylation of riboflavin is therefore a form of metabolic trapping key to riboflavin homeostasis.¹⁶

Functions of Vitamin B2

- Has antioxidant properties: Riboflavin contributes to antioxidant defence by enhancing the activity of key antioxidant enzymes such as superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx). With its coenzyme forms FMN and FAD, riboflavin is essential for glutathione reductase (GR) activity, which maintains intracellular reduced glutathione (GSH). These effects collectively reduce reactive oxygen species (ROS), protect against oxidative damage, and support cellular resilience in models of diabetes, cardiac dysfunction, ageing, and migraine-related oxidative stress.
- Malaria infection: Infection with the malaria parasite attacks the plasma membrane of the erythrocyte, compromising its integrity. Riboflavin reduces the level of methaemoglobin content, decreases food vacuole size and inhibits asexual parasite growth in erythrocytes infested with *P. falciparum*. Combined riboflavin and chloroquine treatment shows improvements in

¹³ Donais, J., & Radhakrishnan, K. (2025). Vitamin B2: Riboflavin. In Handbook of Clinical and Practical Pediatric Nutrition (pp. 265–270). https://doi.org/10.1007/978-3-031-97759-6_20

¹⁴ Office of Dietary Supplements - riboflavin. (n.d.). <https://ods.od.nih.gov/factsheets/Riboflavin-HealthProfessional/>

¹⁵ King's College London. (n.d.). Vitamin B2 (riboflavin) | Open-Global | King's College London. <https://www.kcl.ac.uk/open-global/biomarkers/vitamin/vitamin-b2/index>

¹⁶ Hilary J Powers. (n.d.). Riboflavin (vitamin B-2) and health. The American Journal of Clinical Nutrition. <https://doi.org/10.1093/ajcn/77.6.135> the attacks, decrease shows in inhibits

haemoglobin and packed cell volume and reductions in lipid peroxidation, indicating a beneficial effect on cellular integrity during infection.

- Supports the immune system: Riboflavin supports innate immune function by activating phagocytic activity of neutrophils and macrophages and stimulating the multiplication of neutrophils and monocytes. RF down-regulates the NF- κ B activation initiated by ROS, which activate pro-inflammatory cytokines such as interleukin-6 (IL-6), TNF- α , etc., contributing to anti-inflammatory effects. Additionally, riboflavin's cofactor role in NADPH oxidase (Nox2) activity is crucial for ROS generation during phagocytosis, which is important for microbial defence ¹⁷
- Cataracts: Opacity of the lens of the eye increases with age, leading to decreased vision. Sunlight-induced oxidative protein damage in the lens is considered a major cause of such age-related cataracts. Riboflavin is involved in scavenging free radicals, an antioxidant property.
- Intermediate metabolism: It plays a central role in intermediary metabolism by serving as the precursor of the flavin cofactors FMN and FAD, which act as essential electron carriers in numerous redox reactions. Through FAD-dependent enzymes, riboflavin supports key steps in energy production, including succinate dehydrogenase activity in the TCA cycle and electron transport chain, thereby influencing oxidative phosphorylation. Riboflavin is also critical for fatty acid β -oxidation, where flavoproteins function as electron acceptors.
- Iron mobilisation . The mobilisation of iron from the intracellular protein ferritin is a reducing process. Reduced flavins can evidently reduce and thereby mobilise ferritin iron in a variety of tissues, at a physiologically relevant rate.
- Homocysteine metabolism: Vitamin B2 plays an important role in homocysteine metabolism through its function as FAD, a required cofactor for methylenetetrahydrofolate reductase (MTHFR). MTHFR generates the active folate form needed for remethylation of homocysteine to methionine, alongside folate and vitamin B12. Poor riboflavin status can impair this pathway and lead to elevated plasma homocysteine (a cardiovascular risk factor).¹⁸

¹⁷Suwannasom, N., Kao, I., Pruß, A., Georgieva, R., & Bäuml, H. (2020). Riboflavin: the health benefits of a forgotten natural vitamin. *International Journal of Molecular Sciences*, 21(3), 950. <https://doi.org/10.3390/ijms21030950>

¹⁸ Hilary J Powers. (n.d.). Riboflavin (vitamin B-2) and health. *The American Journal of Clinical Nutrition*. <https://doi.org/10.1093/ajcn/77.6.135> the attacks, decrease shows inhibit

Vitamin B3 (Niacin)

What is Vitamin B3?

Vitamin B3, also known as niacin, belongs to the water-soluble B complex group and is found in various foods such as bran, yeast, eggs, peanuts, poultry, red meat, fish, whole-grain cereals, legumes, and seeds.¹⁹ The two most common forms of niacin in food and supplements are nicotinic acid and nicotinamide. The body can also convert tryptophan, an amino acid, to nicotinamide. Niacin works in the body as a coenzyme, with more than 400 enzymes dependent on it for various reactions. Niacin helps to convert nutrients into energy, create cholesterol and fats, create and repair DNA, and exert antioxidant effects.²⁰

Absorption and metabolism of Vitamin B3

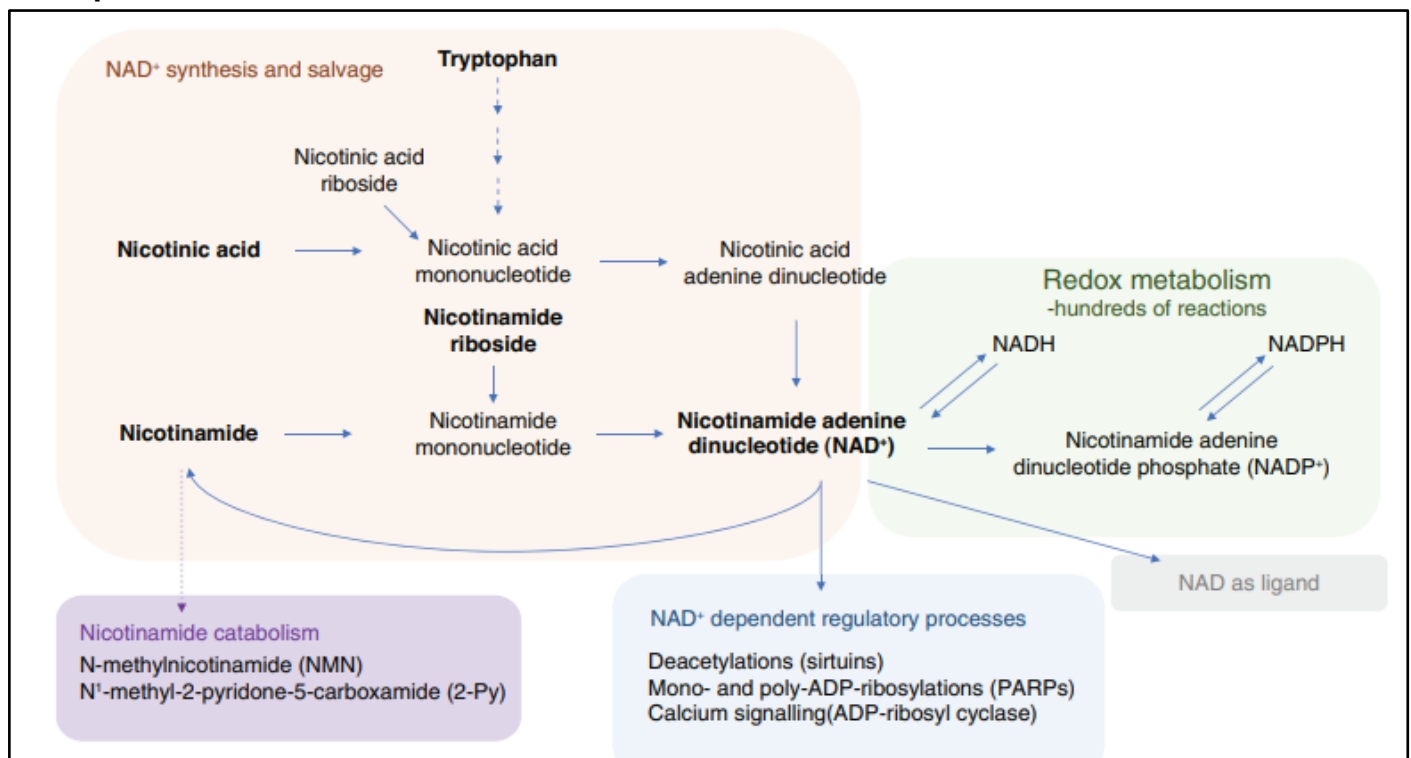


Fig 1: NAD⁺ synthesis and salvage

In meat, niacin is mainly bound to NAD and NADP. The coenzyme forms (NAD, NADP) are hydrolysed by intestinal phosphatases and NAD glycohydrolases to release nicotinamide and nicotinamide riboside, and the latter is further hydrolysed to nicotinamide. In plant products, especially in cereal grains, niacin is found mainly as nicotinic acid and is poorly available. Food processing (alkaline treatment) may increase the bioavailability of the nicotinic acid in cereals. Nicotinic acid and nicotinamide are absorbed by carrier-mediated mechanisms and passive diffusion. Some diffusion takes place in the stomach, but niacin absorption is more effective in the small intestine. The main form of niacin in serum is nicotinamide. Red blood cells contain a circulating pool of pyridine nucleotides. Different NAD precursors are taken into the cells by diffusion or transporter-mediated processes. Liver is the centre of niacin metabolism, and, for instance, a large part of NAD synthesis

¹⁹ Patel, P., & Rout, P. (2024, February 29). Vitamin B3. StatPearls - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK526107/>

²⁰ Powell, J. (2024, November 7). Niacin – Vitamin B3 • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/niacin-vitamin-b3/>

from tryptophan and the conversion to excreted metabolites takes place in the liver. The excretion route of niacin metabolites is urine.²¹

Functions of Vitamin B3

- **Dyslipidemia:** The mechanism of action of niacin in dyslipidemia involves inhibition of lipolysis within adipose tissue, reduction in liver triacylglycerol formation, inhibition of synthesis of Apo B-100 and hepatic VLDL, impaired cholesterol biosynthesis, and lowering of the fractional catabolic rate of Apo A-1 (role in reverse cholesterol transport). In addition to its reductive effects, niacin also increases lipoprotein lipase activity.²²
- **Metabolism:** The coenzymes NAD and NADP actively participate in essential oxidation-reduction reactions, playing key roles in glycolysis, pyruvate metabolism, protein and amino acid metabolism, pentose biosynthesis, glycerol metabolism, synthesis of high-energy phosphate bonds, and fatty acid metabolism.
- **DNA repair and cell signalling:** Niacin-derived NAD⁺ is utilised by non-redox enzymes involved in DNA repair, gene expression, apoptosis, and calcium signalling. Enzymes like poly(ADP-ribose) polymerases (PARPs) and sirtuins utilise NAD⁺ as a substrate, thus establishing the importance of niacin in the regulation of genome stability, stress responses, circadian rhythm, and cell integrity.
- **Inflammation regulation and immune signalling:** NAD⁺ and its derivatives may have an impact on the function of immune cells and the pathways of inflammation, such as the regulation of cytokines, although the complete mechanisms are still being investigated.²³

²¹ Freese, R., & Lysne, V. (2023). Niacin – a scoping review for Nordic Nutrition Recommendations 2023. Food & Nutrition Research, 67.

²² Patel, P., & Rout, P. (2024b, February 29). Vitamin B3. StatPearls - NCBI Bookshelf.

²³ Powell, J. (2024, November 7). Niacin – Vitamin B3 • The Nutrition Source. The Nutrition Source - Harvard Chan School.

Vitamin B5 (Pantothenic Acid)

What is Vitamin B5?

Vitamin B1, also known as Pantothenic acid, is a water-soluble vitamin that has essential functions in the body as a component of coenzyme A (CoA) and acyl carrier protein (ACP). Nutritional deficiency of pantothenic acid is rare, and toxicity is negligible.²⁴ Pantothenic acid is found in a wide variety of foods such as beef, chicken, organ meats, fortified cereals, mushrooms, avocados, dairy milk, yoghurt, potatoes, eggs, etc. Bacteria in the gut can also produce some pantothenic acid, but not enough to meet dietary needs.²⁵

Absorption and metabolism of Vitamin B5

CoA in the diet is hydrolysed in the intestinal lumen to dephospho-CoA, phosphopantetheine, and pantetheine, with the pantetheine subsequently hydrolysed to pantothenic acid. Absorption is by active transport at low concentrations of the vitamin and by passive transport at higher concentrations. Because the active transport system is saturable, absorption will be less efficient at higher concentrations of intake, but the intake levels at which absorptive efficiency decreases in humans are not known. Alongside, the contribution of bacterial synthesis to body pantothenic acid levels or faecal losses in humans has not been quantified.

The synthesis of CoA from pantothenate is regulated primarily by pantothenate kinase, an enzyme that is inhibited by the pathway end products, CoA and acyl-CoA. Thus, CoA production does not reflect the amount of available pantothenate. CoA, in forms such as acetyl CoA and succinyl CoA, plays an important role in the TCA cycle and in the synthesis of fatty acids and membrane phospholipids, amino acids, steroid hormones, vitamins A and D, porphyrin and corrin rings, and neurotransmitters. It is also required for the acetylation and acylation of proteins and the synthesis of α -tubulin.²⁶

Functions of Vitamin B5

- **Hyperlipidemia:** Because of pantothenic acid's role in triglyceride synthesis and lipoprotein metabolism, experts have hypothesised that pantothenic acid supplementation might reduce lipid levels in patients with hyperlipidemia. Several clinical trials have shown that the form of pantothenic acid known as pantethine reduces lipid levels when taken in large amounts (reduction in triglycerides, total cholesterol and non- HDL cholesterol).²⁷
- **Metabolic functions:** As a component of CoA, pantothenic acid is needed in a large number of metabolic reactions. About 4% of cellular enzymes use CoA or its thioester derivatives as a substrate. It is needed in fatty acid synthesis. Also, CoA is a universal cofactor that participates in hundreds of biochemical reactions, including the transfer of acetyl/acyl groups necessary for metabolic pathways across all cells.

²⁴ View of Pantothenic acid – a scoping review for Nordic Nutrition Recommendations 2023. (n.d.). <https://foodandnutritionresearch.net/index.php/fnr/article/view/10255/16572>

²⁵ Otis, B. O. (2025, September 4). Vitamin B5 (Pantothenic Acid) and Health• The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/pantothenic-acid-vitamin-b5/>

²⁶ Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Pantothenic Acid, Biotin, and Choline. (1998). The National Academies Press. <https://doi.org/10.17226/5136> have hypothesised have hypothesised.

²⁷ Office of Dietary Supplements - Pantothenic acid. (n.d.). <https://ods.od.nih.gov/factsheets/PantothenicAcid-HealthProfessional/>

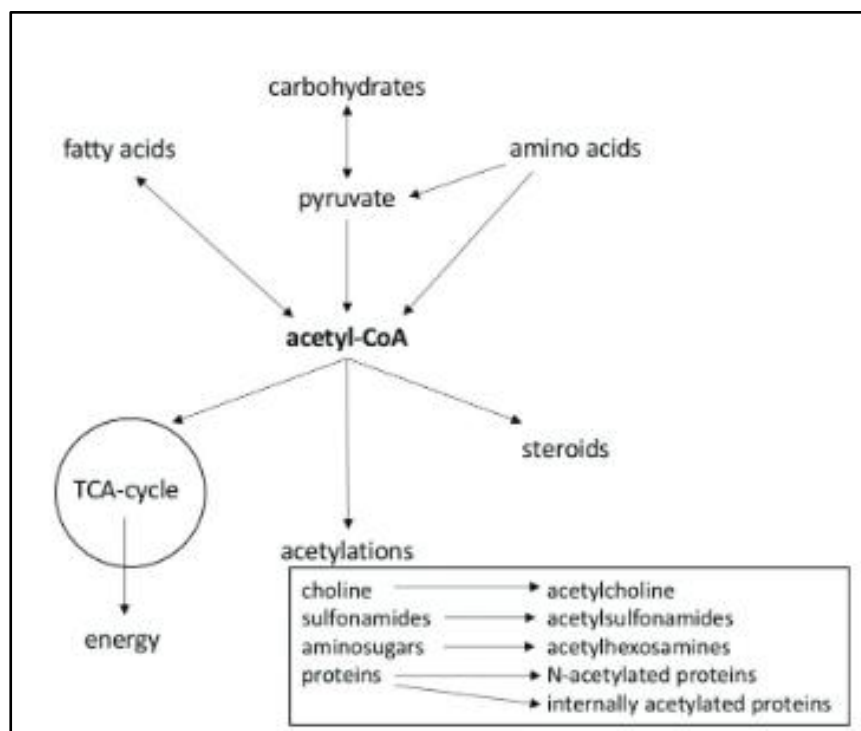


Fig. 1: Role of acetyl-CoA in metabolism²⁸

- Neurotransmitter synthesis: Acetyl-CoA, which is produced from vitamin B5 via CoA synthesis, is a substrate for the synthesis of acetylcholine, which is a prominent neurotransmitter in the central and peripheral nervous systems. It also participates in the synthesis of steroid hormones.

²⁸ View of Pantothenic acid – a scoping review for Nordic Nutrition Recommendations 2023. (n.d.). <https://foodandnutritionresearch.net/index.php/fnr/article/view/10255/16572>

Vitamin B6 (Pyridoxine)

What is Vitamin B6?

Vitamin B1, also known as pyridoxine, is a water-soluble vitamin found naturally in many foods such as beef liver, tuna, salmon, fortified cereals, chickpeas, poultry, some fruits and vegetables like dark leafy greens, bananas, papayas, as well as added to foods and supplements.²⁹ It converts inside the body into essential coenzymes for more than 100 enzymes in the human body.

Vitamin B6 has three natural forms: pyridoxine (PN), pyridoxal (PL), and pyridoxamine (PM), all of which transform into their active forms in the body, which is the coenzyme pyridoxal 5-phosphate (PLP or P5P). PLP, the active molecule in the body, mainly serves as a coenzyme in amino acid, protein, carbohydrate, and lipid metabolism, in addition to neurotransmitter synthesis.³⁰

Absorption of Vitamin B6

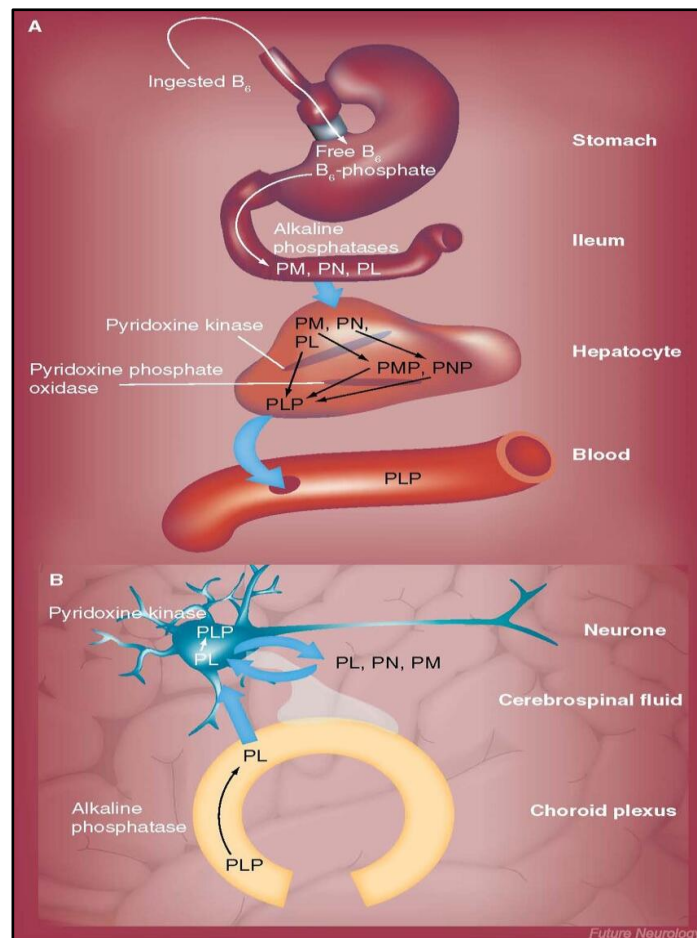


Fig 1: Absorption of Vitamin B6

The absorption has been thought to occur via a passive process in the gut; however, a transport mechanism, i.e., SLC19A2 and SLC19A3, known as thiamine transporter 1 (THTR1) and THTR2, has recently been identified. The body stores of vitamin B6 have been estimated to be approximately 1000 μmol (170 mg), of which 80–90% is found in the muscles, where PLP has a covalent bond to glycogen phosphorylase. Phosphorylation of PLP enzymes plays an important role in mediating

²⁹ Powell, J. (2024b, November 7). Vitamin B6 • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/vitamin-b6/>

³⁰ Abosamak, N. R., & Gupta, V. (2023, August 17). Vitamin B6 (Pyridoxine). StatPearls - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK557436/>

diverse cellular functions, but it is not fully understood. The turnover of the vitamin is relatively fast in plasma, with a half-life of 25–33 days for PLP, but much slower in muscle tissue.³¹

Functions of Vitamin B6

- Protein folding and amino acid biosynthesis: PLP may help to stabilise some PLP-dependent enzymes by encouraging proper folding and preserving the native structure; in other proteins, the binding of PLP locks the enzyme into a stable functional structure. In addition to this structural function, PLP is a key cofactor in amino acid metabolism, facilitating transamination, decarboxylation, racemisation, and other reactions that are crucial for the biosynthesis and catabolism of amino acids. It indirectly controls the biosynthesis of amino acid-derived metabolites, such as hormones and signalling molecules.
- Neurotransmitter biosynthesis: Foods associated with high vitamin B6 are often correlated with better mental health. Vitamin B6-dependent enzymes are needed for the biosynthesis of three important neurotransmitters- epinephrine (adrenaline), dopamine and serotonin.³²
- Haemoglobin synthesis and function: PLP functions as a coenzyme of 5-aminolevulinic acid synthase, which is involved in the synthesis of heme, an iron-containing component of haemoglobin. Haemoglobin is found in red blood cells and is critical to their ability to transport oxygen throughout the body. Both pyridoxal and PLP can bind to the haemoglobin molecule and affect its ability to pick up and release oxygen.
- Hormone function: Steroid hormones, such as estrogen and testosterone, exert their effects in the body by binding to steroid hormone receptors in the nucleus of target cells. The nuclear receptors themselves bind to specific regulatory sequences in DNA and alter the transcription of target genes. Experimental studies have uncovered a mechanism by which PLP may affect the activity of steroid receptors and decrease their effects on gene expression.³³
- Coenzyme in metabolism: PLP serves as an essential coenzyme in over 150 biochemical reactions, including amino acid metabolism, carbohydrate metabolism, lipid metabolism and nucleic acid metabolism.
- Inflammation and immune response: PLP influences the immune system function by regulating cytokine production, modulating inflammasomes such as NLRP3 and supporting immune signalling pathways.³⁴

³¹ Vitamin B6: a scoping review for Nordic Nutrition Recommendations 2023 | Food & Nutrition Research. (n.d.). <https://foodandnutritionresearch.net/index.php/fnr/article/view/10259/16601#toc>

³² Parra, M., Stahl, S., & Hellmann, H. (2018). Vitamin B6 and its role in cell metabolism and physiology. *Cells*, 7(7), 84. <https://doi.org/10.3390/cells7070084>

³³ Vitamin B6. (2026, January 2). Linus Pauling Institute. <https://lpi.oregonstate.edu/mic/vitamins/vitamin-B6#toc-function>

³⁴ Stach, K., Stach, W., & Augoff, K. (2021). Vitamin B6 in health and disease. *Nutrients*, 13(9), 3229. <https://doi.org/10.3390/nu13093229>

Vitamin B7 (Biotin)

What is Vitamin B7?

Vitamin B7, also known as biotin, is a water-soluble B vitamin found naturally in foods such as beef liver, cooked eggs, salmon, avocados, pork, sweet potato, nuts, seeds, etc., and also in supplements. Biotin plays a vital role in assisting enzymes in breaking down fats, carbohydrates, and proteins in food. It also helps to regulate signals sent by cells and the activity of genes.³⁵

Absorption of Vitamin B7

Most biotin in foods is bound to protein, although some dietary biotin is in the free form. Gastrointestinal proteases and peptidases break down the protein-bound forms of ingested biotin into biocytin and biotin-oligopeptides, which undergo further processing by biotinidase, an enzyme, in the intestinal lumen to release free biotin. The free biotin is then absorbed in the small intestine, and most biotin is stored in the liver.³⁶

A biotin carrier located in the intestinal brush border membrane transports biotin against a sodium ion concentration gradient (sodium-dependent multivitamin transporter (SMVT)) and is structurally specific, temperature-dependent, and electroneutral; at pharmacological concentrations, diffusion predominates. Some of the vitamin B7 is also absorbed in the proximal colon, giving credence to the concept that biotin from microbial synthesis within the colon can contribute to meeting the human requirement.³⁷

In the blood, biotin is mainly transported as free biotin (81%) and to a lesser extent, bound to plasma proteins (19%). The uptake of biotin in peripheral tissues and the liver occurs via SMVT and the monocarboxylate transporter (MCT). Inside the cell, biotin is attached to biotin-dependent carboxylases. Limited storage of biotin occurs in the muscles, liver, and brain. Biotin and its metabolites are excreted via the urine, and the non-absorbed biotin synthesised by intestinal bacteria is excreted via faeces.³⁸

Functions of Vitamin B7

- Acts as a cofactor: This water-soluble vitamin is a cofactor for five carboxylases (propionyl-CoA carboxylase, pyruvate carboxylase, methylcrotonyl-CoA carboxylase [MCC], acetyl-CoA carboxylase 1, and acetyl-CoA carboxylase 2) that catalyse critical steps in the metabolism of fatty acids, glucose, and amino acids.³⁹
- Regulation of chromatin structure and gene expression: Vitamin B₇ (biotin) functions as a gene expression regulator by altering histone proteins, which are involved in DNA packaging. Biotin is covalently bound to particular lysine residues of histones, particularly histone H4, via the enzyme holocarboxylase synthetase. Histone biotinylation marks are predominantly observed in densely packed chromatin and are known to be correlated with

³⁵ Office of Dietary Supplements - Biotin. (synthesifaecess,edn.d.). <https://ods.od.nih.gov/factsheets/Biotin-HealthProfessionalsynthesisedfathatcatalyseecescatalyse/>

³⁶ Office of Dietary Supplements - Biotin. (n.d.). <https://ods.od.nih.gov/factsheets/Biotin-HealthProfessional/>

³⁷ Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Pancreatylsetothenic Acisynthesisedd, Biotin, and Choline. (1998). The National Academies Prescan be characteriseds. <https://doi.org/10.17226/5136>.

³⁸ Solvik, B. S., & Strand, T. A. (2024). Biotin: a scoping review for Nordic Nutrition Recommendations 2023. Food & Nutrition Research, 68. <https://doi.org/10.29219/fnr.v68.10256>

³⁹ Office of Dietary Supplements - Biotin. (n.d.). <https://ods.od.nih.gov/factsheets/Biotin-HealthProfessional/>

low gene expression. The extent of histone biotinylation is modulated according to biotin availability, making it possible for biotin status to affect chromatin structure and gene expression, including biotin transport genes.

- Hair, nail and skin health: Signs of biotin deficiency include skin rashes, hair loss, and brittle nails.
- Regulation of fatty acid synthesis and oxidation: Biotin-dependent acetyl-CoA carboxylase determines whether cells will store energy as fat or oxidise fat for energy. ACC1 promotes malonyl-CoA formation, initiating fatty acid synthesis. ACC2 regulates mitochondrial fatty acid entry by controlling malonyl-CoA inhibition of CPT-1.
- Role in glucose homeostasis: Biotin enables pyruvate carboxylase, the enzyme that converts pyruvate into oxaloacetate, which is the first committed step of gluconeogenesis. It supports endogenous production of glucose during fasting periods and maintains glucose stability.
- Cellular signalling and stress response: It does not act as a direct antioxidant; it maintains metabolic pathways that prevent oxidative stress, supports mitochondrial function and reduces the accumulation of stress- inducing metabolites.⁴⁰

⁴⁰ Karalia, S., & Meena, V. K. (2025). Biotin: DNA to diet. *The Journal of Nutritional Biochemistry*, 146, 110081. <https://doi.org/10.1016/j.jnutbio.2025.110081>

Vitamin B9 (Folate)

What is Vitamin B9?

Folate is a generic term that typically refers to a group of water-soluble compounds that play an essential role in DNA biosynthesis. It is also known as vitamin B9. Folic acid is the synthetic form of folate. Folate converts into tetrahydrofolic acid (THF), which undergoes several transfer/methylation reactions that are important for synthesising nitrogenous bases in DNA and RNA and are necessary for the maturation of RBCs.⁴¹

A wide variety of foods naturally contain folate, but the form that is added to foods and supplements, folic acid, is better absorbed. Good sources of folate include dark green leafy vegetables (turnip greens, spinach, romaine lettuce, asparagus, Brussels sprouts, broccoli), beans, peanuts, sunflower seeds, fresh fruits, whole grains, aquatic foods, eggs, liver and fortified foods and supplements.⁴²

Absorption of Vitamin B9

Food folates (polyglutamate derivatives) are hydrolysed to monoglutamate forms in the gut using folate conjugase before absorption across the intestinal mucosa. The monoglutamate form of folate is actively transported across the proximal small intestine by a saturable pH-dependent process.

Monoglutamates, mainly 5-methyl-tetrahydrofolate, are present in the portal circulation. Much of this folate can be taken up by the liver, where it is metabolised to polyglutamate derivatives and retained or released into the blood or bile. Approximately two-thirds of the folate in plasma is protein-bound. A variable proportion of plasma folate is bound to low-affinity protein binders, primarily albumin, which accounts for about 50 per cent of bound folate. Low levels of high-affinity folate binders are also present in plasma.

Cellular transport of folate is mediated by several different folate transport systems, which can be characterised as either membrane carriers or folate-binding protein-mediated systems. These transport systems are not saturated by folate under physiological conditions, and folate influx into tissues would be expected after any elevation in plasma folate after supplementation.

Before being stored in tissue or used as a coenzyme, folate monoglutamate is converted to the polyglutamate form by the enzyme folylpolyglutamate synthetase. When released from tissues into circulation, folate polyglutamates are reconverted to the mono-glutamate form by folate conjugase.⁴³

Functions of Vitamin B9

- Deficiency leads to megaloblastic anaemia: Megaloblastic anaemia (MBA) is a type of anaemia characterised by both reduced numbers and the presence of unusually large and underdeveloped erythrocytes (megaloblasts) and their mature forms (macrocytes and megalocytes). The underlying pathophysiology of this condition includes abnormalities in DNA synthesis and induction of apoptosis in pre-erythroblasts, which ultimately leads to pancytopenia. The induction of apoptosis in cells is associated with increased expression of the

⁴¹ Merrell, B. J., & McMurry, J. P. (2023, August 8). Folic acid. StatPearls - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK554487/>

⁴² Powell, J. (2024a, November 6). Folate (Folic Acid) - Vitamin B9 • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/folic-acid/>

⁴³ Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Pantothenic Acid, Biotin, and Choline. (1998). The National Academies Press. <https://doi.org/10.17226/5136>.

transcription factor, p53 protein, which may be due to low levels of vitamin B12, as well as B9 and high plasma homocysteine levels.

- Supports the nervous system: Folic acid plays an important role in the functioning of the nervous system, by, among other things, participating in the synthesis of neurotransmitters, i.e., epinephrine, norepinephrine and dopamine. Its deficiency in the body can contribute to delayed development of the nervous system, increased risk of demyelination of the brain and peripheral nerves, deterioration of motor and behavioural abilities, as well as neurocognitive disorders. The main dysfunction is associated with high plasma levels of homocysteine, which don't get metabolised properly to methionine due to folate deficiency.
- Prevention of cancer: Folate plays an important role in the prevention of cancer, by influencing the proper course of DNA synthesis, transcription and repair. Its deficiency can promote an increased risk of damage to genetic material, as well as hypomethylation, which can consequently initiate the process of carcinogenesis.⁴⁴
- Neurogenesis: Folate (vitamin B9) has essential properties in the formation of new neurons during neurogenesis. Folic acid enhances the stimulation of neurons by ischemia of notch signalling and hippocampal neurogenesis in the adult brain, and lessens the impairment of cognitive function. Folate is an important regulator of hippocampal neurogenesis, and folic acid is needed prenatally to reduce the risk of neural tube defects.⁴⁵
- Synthesis of methionine: Methionine is a byproduct synthesised as folate reduces homocysteine levels in the blood; 5-10-MTHF donates a methyl group to an enzyme, methyl-tetrahydrofolate reductase (MTHFR), and then becomes 5-methyl THF. 5-methyl THF donates its remaining methyl group to homocysteine via methionine synthase, converting homocysteine to methionine. The presence of vitamin B12 is also important for this process.⁴⁶

⁴⁴ Kurowska, K., Kobylińska, M., & Antosik, K. (2023). Folic acid - importance for human health and its role in COVID-19 therapy. *Roczniki Państwowego Zakładu Higieny*, 74(2), 131–141. <https://doi.org/10.32394/rpzh.2023.0252>

⁴⁵ Godonu, S. S. (2025). Role of vitamin B9 (Folate) in neuronal plasticity. In *International Journal of Scientific Research and Technology* (Vol. 2, Issue 1, pp. 77–82) [Journal-article]. <https://www.ijstjournal.com>

⁴⁶ Merrell, B. J., & McMurry, J. P. (2023b, August 8). Folic acid. *StatPearls* - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK554487/>

Vitamin B12 (Cobalamin)

What is Vitamin B12?

Vitamin B12 is a water-soluble vitamin that contains the mineral cobalt. Compounds with vitamin B12 activity are collectively called cobalamins. Methylcobalamin and 5-deoxyadenosylcobalamin are the metabolically active forms of vitamin B12. However, two other forms, hydroxycobalamin and cyanocobalamin, become biologically active after they are converted to methylcobalamin or 5-deoxyadenosylcobalamin.

Vitamin B12 is present in foods of animal origin, including fish, meat, poultry, eggs, and dairy products. Plant foods do not naturally contain vitamin B12. However, fortified breakfast cereals and fortified nutritional yeasts are readily available sources of vitamin B12 that have high bioavailability.⁴⁷

Absorption of Vitamin B12

Small amounts of B12 are absorbed via an active process that requires an intact stomach, intrinsic factor (a glycoprotein that the parietal cells of the stomach secrete after being stimulated by food), pancreatic sufficiency, and a normally functioning terminal ileum. In the stomach, food-bound B12 is dissociated from proteins in the presence of acid and pepsin. The released B12 then binds to R proteins (haptocorrins) secreted by the salivary glands and the gastric mucosa. In the small intestine, pancreatic proteases partially degrade the R proteins, releasing B12 to bind with intrinsic factor. The resulting complex of intrinsic factor and B12 attaches to specific receptors in the ileal mucosa; after internalisation of the complex, B12 enters the enterocyte. Approximately 3 to 4 hours later, B12 enters the circulation. All circulating B12 is bound to the plasma binding proteins: transcobalamin I, II, or III (TCI, TCII, or TCIII). Although TCI binds approximately 80 per cent of the B12 carried in the blood, TCII is the form that delivers B12 to the tissues through specific receptors for TCII. The liver takes up approximately 50 per cent of the B12,¹² and the remainder is transported to other tissues.

If the circulating B12 exceeds the B12 binding capacity of the blood, the excess is excreted in the urine. The highest losses of B12 ordinarily occur through the faeces. Sources of faecal B12 include unabsorbed B12 from food or bile, desquamated cells, gastric and intestinal secretions, and B12 synthesised by bacteria in the colon.⁴⁸

Functions of Vitamin B12

- Cofactor for methionine synthase: Vitamin B12, in its active methylcobalamin form, serves as an essential cofactor for the enzyme methionine synthase. This enzyme catalyses the transfer of a methyl group from methyltetrahydrofolate to homocysteine to form methionine and tetrahydrofolate, a key reaction in one-carbon metabolism that underpins DNA synthesis and cellular replication. Without adequate B12, this reaction is disrupted, leading to impaired synthesis of purines and pyrimidines, the building blocks of DNA. This disruption is a primary cause of megaloblastic anaemia, where red blood cells cannot divide properly due to defective DNA synthesis.

⁴⁷ Office of Dietary Supplements - Vitamin B12. (n.d.). <https://ods.od.nih.gov/factsheets/VitaminB12-HealthProfessional/>

⁴⁸ Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, influences Vitamin B6, Folate, Vitamanaemia characterised in B12, Pananaemiatothenic Acid, Biotin, and Choline. (1998). The National Academies Press. <https://doi.org/10.17226/5136>.

- Energy and lipid metabolism: Vitamin B12 functions as a cofactor in the mitochondria for methylmalonyl-CoA mutase, which requires the 5-deoxyadenosylcobalamin form of cobalamin. This enzyme converts methylmalonyl-CoA into succinyl-CoA, linking amino acid and odd-chain fatty acid metabolism to the tricarboxylic acid (TCA) cycle.
- Neural tube development and early growth: Vitamin B12 influences neural tube closure during embryonic development. The vitamin's role in one-carbon metabolism, critical for DNA synthesis and methylation, is linked to neural tube defects (NTDs) in human studies, with low maternal B12 status being associated with increased risk of NTD.
- Deficiency leads to megaloblastic anaemia: Megaloblastic anaemia (MBA) is a type of anaemia characterised by both reduced numbers and the presence of unusually large and underdeveloped erythrocytes (megaloblasts) and their mature forms (macrocytes and megalocytes). The underlying pathophysiology of this condition includes abnormalities in DNA synthesis and induction of apoptosis in pre-erythroblasts, which ultimately leads to pancytopenia. The induction of apoptosis in cells is associated with increased expression of the transcription factor, p53 protein, which may be due to low levels of vitamin B12, as well as B9 and high plasma homocysteine levels.⁴⁹
- Inflammation and oxidative stress regulation: A reduced form of vitamin B12 (cob(II)alamin) exhibits antioxidant properties through a reduction in the amount of superoxide ions in the cytosol and mitochondria in the presence of physiological levels of vitamin B12, indicating the direct scavenging of ROS. Cob(II)alamin shows strong ROS-scavenging properties in the nervous system. The antioxidant properties, especially those related to superoxide radical binding, play a key role in its anti-inflammatory and protective effects against oxidative DNA damage. Vitamin B12 can also exert its indirect antioxidant properties by participating in the metabolism of homocysteine.⁵⁰
- Cognition and mental health: Vitamin B12 deficiency is commonly associated with memory problems, cognitive impairment, nominal aphasia, learning problems, headaches and migraines, mood swings, irritability, insomnia, depression, anxiety, hallucinations, delusions, etc.⁵¹

⁴⁹ Kurowska, K., Kobylińska, M., & Antosik, K. (2023). Folic acid - importance for human health and its role in COVID-19 therapy. *Roczniki Państwowego Zakładu Higieny*, 74(2), 131–141. <https://doi.org/10.32394/rpzh.2023.0252>

⁵⁰ Halczuk, K., Kaźmierczak-Barańska, J., Karwowski, B. T., Karmańska, A., & Cieślak, M. (2023). Vitamin B12—Multifaceted In Vivo Functions and In Vitro Applications. *Nutrients*, 15(12), 2734. <https://doi.org/10.3390/nu15122734>

⁵¹ Wolffenbuttel, B. H., Owen, P. J., Ward, M., & Green, R. (2023). Vitamin B 12. *BMJ*, 383, e071725. <https://doi.org/10.1136/bmj-2022-071725>

Vitamin C (L-ascorbic acid)

What is Vitamin C?

Vitamin C is a water-soluble vitamin, antioxidant, and essential cofactor for collagen biosynthesis, carnitine and catecholamine metabolism, and dietary iron absorption. Humans are unable to synthesise vitamin C, so it is strictly obtained through the dietary intake of fruits and vegetables. Citrus fruits, berries, tomatoes, potatoes, and green leafy vegetables are excellent sources of vitamin C

Absorption of Vitamin C

Although most vitamin C is completely absorbed in the small intestine, the percentage of absorbed vitamin C decreases as intraluminal concentrations increase. Absorption is through an energy-dependent process that has two mechanisms: simple diffusion and active transport. Two transporters are involved: SVCTs (sodium-dependent vitamin C transporters) and hexose transporters. The site for absorption is the distal small intestine and is regulated by renal excretion. Usual dietary doses of up to 100 mg/day are almost completely absorbed. The highest ascorbic acid concentrations are in the pituitary gland, the adrenal gland, the brain, leukocytes, and the eyes.⁵²

Functions of Vitamin C

- Modulation of the immune system: Vitamin C has a number of activities that could conceivably contribute to its immune-modulating effects.
 - It is a highly effective antioxidant, due to its ability to readily donate electrons, thus protecting important biomolecules (proteins, lipids, carbohydrates, and nucleic acids) from damage by oxidants generated during normal cell metabolism and through exposure to toxins and pollutants (e.g., cigarette smoke).
 - Vitamin C is also a cofactor for a family of biosynthetic and gene regulatory monooxygenase and dioxygenase enzymes.
 - The vitamin has long been known as a cofactor for the lysyl and prolyl hydroxylases required for stabilisation of the tertiary structure of collagen, and is a cofactor for the two hydroxylases involved in carnitine biosynthesis, a molecule required for transport of fatty acids into mitochondria for the generation of metabolic energy.
 - Vitamin C is also a cofactor for the hydroxylase enzymes involved in the synthesis of catecholamine hormones, e.g., norepinephrine, and amidated peptide hormones, e.g., vasopressin, which are central to the cardiovascular response to severe infection.
 - It regulates gene transcription and cell signalling pathways through the regulation of transcription factor activity and epigenetic marks.
 - It accumulates in immune cells like neutrophils and enhances chemotaxis, phagocytosis, reactive oxygen species generation, and microbial killing.
 - It supports clearance of spent immune cells and may enhance differentiation and proliferation of lymphocytes (B- and T-cells).
- Maintaining barrier integrity and wound healing: Skin contains millimolar concentrations of vitamin C, with higher levels found in the epidermis than the dermis. Vitamin C is actively accumulated into the epidermal and dermal cells via the two sodium-dependent vitamin C

⁵² Abdullah, M., Jamil, R. T., & Attia, F. N. (2023, May 1). Vitamin C (Ascorbic acid). StatPearls - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK499877/>

transporters (SVCT) isoforms 1 and 2, suggesting that the vitamin has crucial functions within the skin.

- Vitamin C is a cofactor for the prolyl and lysyl hydroxylase enzymes that stabilise the tertiary structure of collagen.
- Vitamin C can increase collagen gene expression in fibroblasts.
- Vitamin C is thought to influence several important aspects of neutrophil function: migration in response to inflammatory mediators (chemotaxis), phagocytosis and killing of microbes, and apoptosis and clearance by macrophages, which plays a major role in wound healing.⁵³
- Biosynthesis of collagen: Vitamin C is essential for collagen biosynthesis because it keeps iron in the active (Fe^{2+}) form required by prolyl and lysyl hydroxylases, the enzymes that hydroxylate proline and lysine residues in collagen. It also enhances collagen production by increasing procollagen gene expression, mRNA stability, and secretion into the extracellular matrix.
- Biosynthesis of Norepinephrine: Vitamin C acts as an essential cofactor for dopamine β -hydroxylase, the enzyme that catalyses the final and rate-limiting step in the conversion of dopamine to norepinephrine. Ascorbate provides the reducing equivalents needed to keep the enzyme's copper ions in the active, reduced state, allowing continuous norepinephrine production.⁵⁴
- Reducing and capping agent: Ascorbic acid acts as a reducing and capping agent for the synthesis of metal nanoparticles such as silver, gold, copper, etc. Ascorbic acid molecules can cap or surround the particle and prevent the uncontrolled growth of the particles to micron-sized dimensions.

⁵³ Carr, A. C., & Maggini, S. (2017). Vitamin C and Immune Function. *Nutrients*, 9(11), 1211. <https://doi.org/10.3390/nu9111211>

⁵⁴ Harish Padh, Vitamin C: Newer Insights into Its Biochemical Functions, *Nutrition Reviews*, Volume 49, Issue 3, March 1991, Pages 65–70, <https://doi.org/10.1111/j.1753-4887.1991.tb07407.x>

Vitamin D (Calciferol)

What is Vitamin D?

Vitamin D, also known as calciferol, is a fat-soluble vitamin that is naturally present in a few foods, added to others, and available as a dietary supplement. It is also produced endogenously when ultraviolet (UV) rays from sunlight strike the skin and trigger vitamin D synthesis.⁵⁵

Few foods are naturally rich in vitamin D₃. The best sources are the flesh of fatty fish and fish liver oils. Smaller amounts are found in egg yolks, cheese, and beef liver. Certain mushrooms contain some vitamin D₂; in addition, some commercially sold mushrooms contain higher amounts of D₂ due to being intentionally exposed to high amounts of ultraviolet light.⁵⁶

Absorption of Vitamin D

Owing to its fat-soluble nature, dietary vitamin D (either D₂ or D₃) is absorbed with other dietary fats in the small intestine. The efficient absorption of vitamin D is dependent upon the presence of fat in the lumen, which triggers the release of bile acids and pancreatic lipase. In turn, bile acids initiate the emulsification of lipids, pancreatic lipase hydrolyses the triglycerides into monoglycerides and free fatty acids, and bile acids support the formation of lipid-containing micelles, which diffuse into enterocytes.⁵⁷

Within the intestinal wall, vitamin D, cholesterol, triglycerides, lipoproteins, and other lipids are packaged together into chylomicrons. Importantly, while a fraction of newly absorbed intestinal vitamin D is also transported along with amino acids and carbohydrates into the portal system to reach the liver directly, the main pathway of vitamin D uptake is incorporation into chylomicrons that reach the systemic circulation via the lymphatics. Chylomicron lipids are metabolised in peripheral tissues that express lipoprotein lipase, but particularly in adipose tissue and skeletal muscle, which are rich in this enzyme. During hydrolysis of the chylomicron triglycerides, a fraction of the vitamin D contained in the chylomicron can be taken up by these tissues. What remains of the original chylomicron after lipolysis is a chylomicron remnant, a cholesterol-enriched, triglyceride-depleted particle that still contains a fraction of its vitamin D content.

Metabolism of Vitamin D

Two main processes are required to transform vitamin D, which is inactive by itself, into active hormonal forms⁵⁸:

- Cutaneous synthesis: 7-dehydrocholesterol in the epidermis is transformed into pre-vitamin D₃ and subsequently into vitamin D₃ (cholecalciferol) when skin is exposed to ultraviolet B (UVB) radiation.
- First Hydroxylation (Hepatic Conversion): Proteins are transported to the liver by vitamin D₂ or D₃ from the diet or skin synthesis. 25-hydroxylase (probably CYP2R1) hydroxylates it

⁵⁵ Office of Dietary Supplements - Vitamin D. (n.d.). <https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/>

⁵⁶ Otis, B. O. (2024, October 24). Vitamin D • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/vitamin-d/>

⁵⁷ Ross, A. C., Taylor, C. L., Yaktine, A. L., & Del Valle, H. B. (2011b). Overview of vitamin D. Dietary Reference Intakes for Calcium and Vitamin D - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK56061/>

⁵⁸ Ross, A. C., Taylor, C. L., Yaktine, A. L., & Del Valle, H. B. (2011b). Overview of vitamin D. Dietary Reference Intakes for Calcium and Vitamin D - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK56061/>

in the liver to produce 25-hydroxyvitamin D (25OHD or calcidiol), which is the primary form of vitamin D that is in circulation.

- Second Hydroxylation (Renal Conversion): 1 α -hydroxylase (CYP27B1) then transforms 25OHD in the kidney into the hormonally active form of 1,25-dihydroxyvitamin D (calcitriol). Parathyroid hormone tightly controls this process. The active hormone calcitriol binds to the vitamin D receptor (VDR) in target tissues to exert its effects.

Functions of Vitamin D

- Absorption of calcium and bone homeostasis: Vitamin D promotes calcium absorption in the gut and maintains adequate serum calcium and phosphate concentrations to enable normal bone mineralisation and to prevent hypocalcemic tetany (involuntary contraction of muscles, leading to cramps and spasms). It is also needed for bone growth and bone remodelling by osteoblasts and osteoclasts. Without sufficient vitamin D, bones can become thin, brittle, or misshapen. Vitamin D sufficiency prevents rickets in children and osteomalacia in adults. Together with calcium, vitamin D also helps protect older adults from osteoporosis.⁵⁹
- Systemic functions: The vitamin D receptors (VDR) are present in tissues throughout the body, including antigen-presenting cells, with known direct effects on innate and adaptive immunity. Observational studies demonstrate the association between vitamin D deficiency and the risk of hypertension or cardiovascular events, higher incidence of cancers, more musculoskeletal pain or migraine, and neuropsychiatric disorders such as schizophrenia, dementia or depression.⁶⁰
- Hormonal actions: Once, Vitamin D is converted into its active form 1,25-dihydroxyvitamin D, it acts like a steroid hormone and binds to VDRs present in cells and influences gene expression (approx. 3% of the genome), impacting processes like calcium homeostasis, cellular differentiation, immune activity, and inflammation.
- Regulation of the immune system: By interacting with immune cells, such as T cells, B cells, and macrophages, vitamin D helps regulate the immune response. It may support defence against infections and immune-mediated disorders by inhibiting inflammatory pathways through these mechanisms.⁶¹
- Insulin secretion: Vitamin D impacts insulin release by acting directly on pancreatic β -cells, which express vitamin D receptors and require calcium-dependent signalling for insulin secretion. The active form of vitamin D helps maintain optimal intracellular calcium levels required for glucose-stimulated insulin release, protects β -cells from inflammation-induced dysfunction, and increases the expression of β -cell genes involved in insulin synthesis. Low levels of vitamin D lead to impaired calcium signalling in β -cells, decreased insulin secretion, and an increased risk of type 2 diabetes and β -cell stress.

⁵⁹ Office of Dietary Supplements - Vitamin D. (n.d.). <https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/>

⁶⁰ Chang, S., & Lee, H. (2019). Vitamin D and health - The missing vitamin in humans. *Pediatrics & Neonatology*, 60(3), 237–244. <https://doi.org/10.1016/j.pedneo.2019.04.007>

⁶¹ Mavar, M., Sorić, T., Bagarić, E., Sarić, A., & Sarić, M. M. (2024). The Power of Vitamin D: Is the Future in Precision Nutrition through Personalized Supplementation Plans? *Nutrients*, 16(8), 1176. <https://doi.org/10.3390/nu16081176>

Vitamin E (Tocopherol)

What is Vitamin E?

Vitamin E is defined as α -tocopherol, while tocochromanols describes a family of eight fat-soluble molecules with antioxidant activities: four tocopherol isoforms (α -, β -, γ -, and δ -tocopherol) and four tocotrienol isoforms (α -, β -, γ -, and δ -tocotrienol). Only one form, α -tocopherol, meets human vitamin E requirements. α -Tocopherol is the predominant fat-soluble antioxidant found in the blood and tissues. Moreover, α -tocopherol is the tocochromanol with the greatest nutritional significance.⁶²

Absorption and metabolism of Vitamin E

Vitamin E is absorbed in the small intestine along with dietary fat. Bile acids and pancreatic secretions emulsify fats, allowing vitamin E to be incorporated into mixed micelles. Micellar vitamin E is taken up by enterocytes via transporters such as NPC1L1, SR-BI, and CD36, where it can transiently accumulate in lipid droplets. Enterocytes then assemble chylomicrons, into which vitamin E is packaged and secreted into the lymph. All absorbed tocochromanols are delivered to the liver in chylomicron remnants, but the liver selectively retains and resecreted α -tocopherol via the α -tocopherol transfer protein (α -TTP) into VLDL, and subsequently LDL and HDL. This repeated VLDL cycling enriches plasma lipoproteins with α -tocopherol for tissue delivery. Vitamin E is not stored long-term; it is catabolized and excreted in urine or bile.

Functions of Vitamin E

- Acts as an antioxidant: Tocopherol is uniquely suited to intercept peroxy radicals and thus prevent a chain reaction of lipid peroxidation. When a molecule of α -tocopherol intercepts a free radical, α -tocopherol is oxidised to the α -tocopheroxy radical and must be reduced. Aside from maintaining the integrity of cell membranes throughout the body, α -tocopherol protects the fats in low-density lipoproteins (LDLs) from oxidation, preventing the development of CVDs.
- Cell-mediated immunity: Tocopherol can protect the physiological properties of lipid bilayer membranes and may influence the activity of membrane proteins and enzymes. It improves the formation of the immune synapse between naive T lymphocytes and antigen-presenting cells (APC), which eventually prompts T cell activation and proliferation.
- Regulation of platelet aggregation and Protein Kinase C activation: Vitamin E protects blood vessels by stopping platelet aggregation and promoting prostacyclin release. This release dilates blood vessels and helps prevent clot formation. It reduces blood cell adhesion to the endothelium by lowering ICAM-1 and VCAM-1 levels. It also inhibits protein kinase C and increases the production of nitric oxide and prostacyclin. Natural RRR- α -tocopherol and mixed tocopherols are particularly effective, with mixed forms showing stronger anti-platelet effects.
- Cell membrane integrity: By strengthening membrane lipid structure and preventing oxidative damage to phospholipids, vitamin E supports membrane stability and repair, which is important for normal cell function and muscle maintenance.
- Prevention of diseases: Vitamin E has antioxidant and anti-inflammatory effects, because of which it plays a protective role in the following diseases:

⁶² Vitamin E. (2026, January 2). Linus Pauling Institute. <https://lpi.oregonstate.edu/mic/vitamins/vitamin-E>

- Cardiovascular health: Reducing oxidative modifications of lipids and improving endothelial function
 - Cancer: Influencing gene regulation, inhibiting enzymes that support tumour growth, and inducing apoptosis in some cancer cell types.
 - Eye health (cataracts): Observational evidence suggests slower lens opacification with higher vitamin E status.
 - Neuroprotection: Reducing oxidative stress implicated in Alzheimer's disease and potentially slowing functional decline.⁶³
- Photoprotection: Vitamin E primarily protects the skin from UV-induced damage through its antioxidant properties. As a lipid-soluble antioxidant, it neutralises free radicals and reactive oxygen species generated by UV exposure, preventing lipid peroxidation, DNA damage, and structural changes in skin cells. Better results are seen when vitamin E supplementation is combined with vitamin C or when a mixture of tocopherols and tocotrienols is used, increasing resistance to UV-induced skin damage and DNA damage.⁶⁴

⁶³ Rizvi, S., Raza, S. T., Ahmed, F., Ahmad, A., Abbas, S., & Mahdi, F. (2014, April 7). The role of vitamin E in human health and some diseases. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3997530/>

⁶⁴ Vitamin E and skin health. (2026, January 2). Linus Pauling Institute. <https://lpi.oregonstate.edu/mic/health-disease/skin-health/vitamin-E>

Vitamin K

What is Vitamin K?

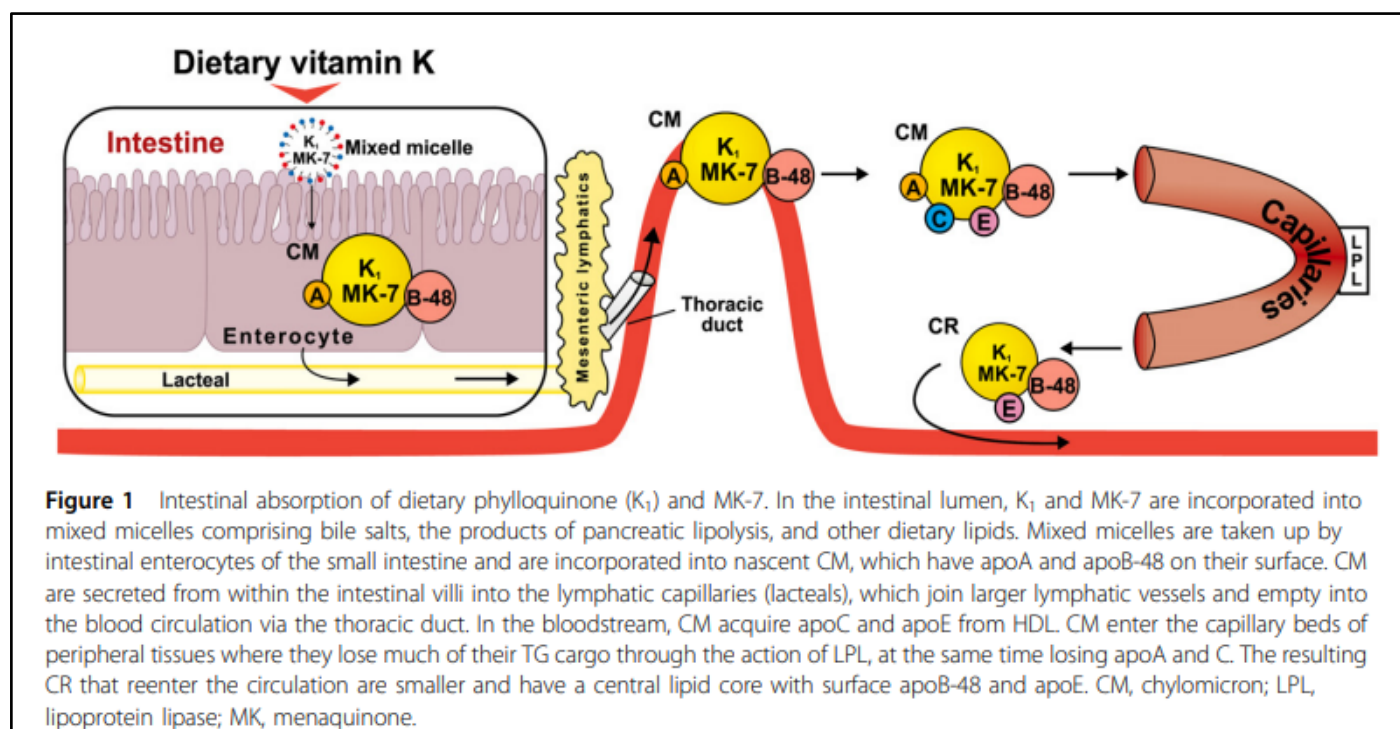
Vitamin K is a fat-soluble vitamin that comes in two forms. The main type is called phyloquinone, found in green leafy vegetables like collard greens, kale, and spinach. The other type, menaquinones, are found in some animal foods and fermented foods such as fermented soybeans, meat, cheese and eggs. Menaquinones can also be produced by bacteria in the human body.

It is found throughout the body, including the liver, brain, heart, pancreas, and bone. It is broken down very quickly and excreted in urine or stool. Because of this, it rarely reaches toxic levels in the body even with high intakes, as may sometimes occur with other fat-soluble vitamins.⁶⁵

Absorption of Vitamin K

Vitamin K, which includes phyloquinone (K_1) and menaquinones (K_2), is mainly absorbed in the small intestine. It needs dietary fat, bile acids, and pancreatic enzymes for proper absorption. In the intestine, vitamin K joins mixed micelles formed from bile salts and products of fat digestion. These micelles carry vitamin K to the intestinal brush border, where enterocytes take it up, likely through passive diffusion and transport mechanisms, similar to other fat-soluble vitamins.

After entering enterocytes, vitamin K gets packed into chylomicrons along with triglycerides, cholesterol, and other fat-soluble vitamins. These chylomicrons are released into the lymphatic system, avoiding the portal circulation, and eventually enter the bloodstream. Chylomicron remnants then transport vitamin K to the liver. There, it can either activate vitamin K-dependent proteins, such as clotting factors, or be redistributed to other tissues through lipoproteins.



⁶⁵ Admin. (2024, November 7). Vitamin K • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/vitamin-k/>

Functions of Vitamin K

- **Blood clotting:** Vitamin K helps to make four of the 13 proteins needed for blood clotting, which stops wounds from continuously bleeding so they can heal. It acts as a cofactor for γ -glutamyl carboxylase, the enzyme that converts specific glutamate residues in clotting proteins into γ -carboxyglutamate (Gla). These γ -carboxylated proteins- prothrombin (factor II) and factors VII, IX, and X, as well as regulatory proteins C and S are required for the normal blood coagulation cascade.
- **Bone health:** Vitamin K is involved with the production of proteins in bone, including osteocalcin, which is needed to prevent the weakening of bones. Some studies have shown that higher vitamin K intakes are associated with a lower incidence of hip fractures and low bone density. In addition, low blood levels of vitamin K have been linked with low bone density.
- **Heart health:** Vitamin K is involved with the production of matrix Gla proteins (MGP), which help to prevent calcification or hardening of heart arteries, a contributor to heart disease.⁶⁷
- **Neuroprotection:** Gas6 (growth arrest-specific 6 protein) is carboxylated in a vitamin K-dependent manner and appears to be involved in brain physiology, including neuroprotection and possibly reduced cognitive decline. Gas6 is also involved in diverse cellular functions, including phagocytosis, cell adhesion, cell proliferation, and protection against apoptosis.⁶⁸
- **Metabolic regulation:** Some research shows that vitamin K may improve insulin sensitivity and is linked to a lower risk of type 2 diabetes. These effects might be due to its impact on metabolic pathways and gene regulation, but the exact mechanisms are still being studied. There is an association between circulating VK and dependent-OC concentration, obesity and T2D risk. Scientific evidence suggests that OC, an osteoblast-derived hormone, is involved in glucose and energy metabolism through multiple mechanisms. It regulates secretion and insulin sensitivity through increased β -cell function and increased adiponectin expression in adipocytes.⁶⁹

⁶⁶ Shearer, M. J., Fu, X., & Booth, S. L. (2012). Vitamin K Nutrition, Metabolism, and Requirements: Current Concepts and Future research. *Advances in Nutrition*, 3(2), 182–195. <https://doi.org/10.3945/an.111.001800>

⁶⁷ Admin. (2024b, November 7). Vitamin K • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/vitamin-k/>

⁶⁸ Vitamin K. (2026, January 2). Linus Pauling Institute. <https://lpi.oregonstate.edu/mic/vitamins/vitamin-K#toc-function>

⁶⁹ Popa, D.-S., Bigman, G., & Rusu, M. E. (2021). The Role of Vitamin K in Humans: Implication in Aging and Age-Associated Diseases. *Antioxidants*, 10(4), 566. <https://doi.org/10.3390/antiox10040566>

Calcium

What is Calcium?

Calcium is the fifth most abundant element in the body, with >99% residing in the skeleton as hydroxyapatite, a complex calcium phosphate molecule. This mineral supplies the strength to bones that support locomotion, but it also serves as a reservoir to maintain serum calcium concentrations. Calcium plays a central role in a wide range of essential functions. Its metabolism is regulated by 3 major transport systems: intestinal absorption, renal reabsorption, and bone turnover.⁷⁰

Natural food sources of calcium are primarily dairy, cheese, yoghurt, winter squash, edamame, almonds, canned sardines and salmon (with bones), leafy greens (collard, mustard, turnip, kale, bok choy, spinach).⁷¹

Absorption of Calcium

Calcium is absorbed from the intestinal lumen by two distinct mechanisms:

- Active, transcellular absorption occurs in the duodenum when calcium intake is low. This process involves the import of calcium into the enterocyte, transport across the cell, and export into the extracellular fluid and blood. Calcium enters the intestinal epithelial cells through voltage-insensitive (TRP) channels and is pumped out of the cell via a calcium-ATPase (energy from ATP is used). The transport across the epithelial cell- carrier protein calbindin, the synthesis of which is totally dependent on vitamin D.
- Passive, paracellular absorption occurs in the jejunum and ileum, and, to a much lesser extent, in the colon when dietary calcium levels are moderate or high. In this case, ionised calcium diffuses through tight junctions into the basolateral spaces around enterocytes, and hence into the blood.

The body works to keep a steady amount of calcium in the blood and tissues. If calcium levels drop too low in the blood, parathyroid hormone (PTH) will signal the bones to release calcium into the bloodstream. This hormone may also activate vitamin D to improve the absorption of calcium in the intestines. At the same time, PTH signals the kidneys to release less calcium in the urine. When the body has enough calcium, a different hormone called calcitonin works to do the opposite: it lowers calcium levels in the blood by stopping the release of calcium from bones and signalling the kidneys to rid more of it in the urine.⁷²

Functions of Calcium

- Regulation of blood pressure: Blood pressure is regulated by intracellular calcium in vascular smooth muscle cells through vasoconstriction and variations in vascular volume. Low calcium intake seems to trigger both mechanisms, raising plasma parathyroid hormone (PTH) levels, which increase intracellular calcium directly or through calcitriol activation, and stimulating the renin-angiotensin-aldosterone signalling pathway that produces sodium and water reabsorption, thus increasing the vascular volume.

⁷⁰ Weaver, C. M., & Peacock, M. (2018). Calcium. *Advances in Nutrition*, 10(3), 546–548. <https://doi.org/10.1093/advances/nmy086>

⁷¹ Powell, J. (2025, September 5). Calcium • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/calcium/>

⁷² Powell, J. (2025, September 5). Calcium • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/calcium/>

- Cholesterol regulation: Calcium (supplementation) leads to a reduction in LDL cholesterol and an increase in HDL cholesterol. This happens as the increase in dietary calcium includes the suppression of calcitrophic hormones that reduce intracellular calcium in adipocytes, thus stimulating lipogenesis and lipid storage. Besides, dietary calcium may decrease serum cholesterol by inhibiting cholesterol and saturated fatty acid absorption.
- Bone health: Calcium is crucial for bone health. Around 99% of the body's calcium is stored in bones and teeth. It combines with phosphate to create hydroxyapatite crystals. These crystals give bones their strength, hardness and help prevent fractures. Calcium is necessary for bone mineralisation during growth. Additionally, it supports bone remodelling, a process where old bone is broken down, and new bone is built. When dietary calcium is low, the body keeps blood calcium levels stable by pulling calcium from bone. Over time, this lowers bone mineral density and raises the risk of osteopenia, osteoporosis, and fractures.⁷³
- Cellular functions: The remaining approximately 1% of calcium moves in the blood and extracellular fluid. It plays important roles in cell signalling, muscle contraction (including the heart), nerve transmission, and hormone secretion. Hormones like parathyroid hormone (PTH) and active vitamin D (1,25-dihydroxyvitamin D) tightly control blood calcium levels to ensure normal body functions.
- Muscular system: Calcium is vital for muscle contraction, which includes skeletal, smooth, and cardiac muscles. In muscle cells, calcium ions serve as a trigger for the actin and myosin interaction that enables contraction. Proper calcium regulation is essential for normal movement, posture, and heart function.
- Nervous signalling: Calcium is important for the nervous system because it acts as a signalling ion. It triggers neurotransmitter release at synapses when it enters nerve terminals. It also regulates neuronal excitability and acts as a second messenger in signalling pathways within cells. Calcium signalling is vital for synaptic plasticity, learning, memory, and proper neuronal development and survival.⁷⁴

⁷³ Cormick, G., & Belizán, J. M. (2019). Calcium intake and health. *Nutrients*, 11(7), 1606. <https://doi.org/10.3390/nu11071606>

⁷⁴ Chen, Y., Forgetta, V., Richards, J. B., & Zhou, S. (2021). Health effects of calcium: Evidence from Mendelian randomization studies. *JBMR Plus*, 5(11), e10542. <https://doi.org/10.1002/jbm4.10542>

Potassium

What is Potassium?

Potassium is the main intracellular cation in the body and is principally involved in membrane potential and electrical excitation of both nerve and muscle cells, as well as acid-base regulation, which is sometimes referred to as an electrolyte because it carries a small electrical charge that activates various cell and nerve functions. Potassium is found naturally in many foods and as a supplement. Its main role in the body is to help maintain normal levels of fluid inside our cells. Sodium, its counterpart, maintains normal fluid levels outside of cells. Potassium is widely available in many foods such as leafy greens, beans, nuts, dried fruits, potatoes, bananas, oranges, coconut water, dairy and plant milks, chicken, salmon, and avocados.⁷⁵

Absorption of Potassium

Potassium is absorbed well in the gastrointestinal tract. About 90% of dietary potassium is absorbed, mostly in the small intestine and to a lesser extent in the colon. Absorption mainly happens through passive diffusion and solvent drag, driven by concentration gradients rather than energy-dependent transport. Once it enters the bloodstream, potassium quickly gets taken up by cells, especially muscle cells. This process is controlled by Na^+/K^+ -ATPase, insulin, and hormones, which help keep normal potassium levels in cells and plasma.⁷⁶

Functions of Potassium

- Blood pressure regulation: Potassium intake is linked to lower blood pressure, especially in people with high sodium intake. More potassium leads to increased sodium excretion in urine and relaxation of blood vessels, which helps lower high blood pressure. High blood pressure is a major risk factor for stroke, heart disease, and kidney disease. Higher potassium levels also relate to a lower risk of stroke and may improve blood vessel health, regardless of blood pressure effects.
- Fluid and electrolyte balance: Potassium is a key electrolyte that helps maintain fluid balance, osmotic equilibrium, and cellular electrical activity.
- Bone health: Potassium, especially from organic salts like potassium citrate or bicarbonate that we find in fruits and vegetables, may help protect bone health. It does this by reducing calcium loss in urine and improving calcium balance. This effect might also relate to neutralising low-grade metabolic acidosis. Without this neutralisation, metabolic acidosis could lead to bone resorption.
- Calcium interaction: Dietary potassium leads to reduced urinary excretion of calcium, supporting a stronger mineral balance in the bones.
- Kidney function: Higher potassium intake is linked to a lower risk of kidney stones returning. This is probably due to its effects on urinary citrate and pH, which prevent calcium oxalate stones

⁷⁵ Powell, J. (2025a, August 5). Potassium • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/potassium/>

⁷⁶ Toft, U., Riis, N. L., & Jula, A. (2024). Potassium – a scoping review for Nordic Nutrition Recommendations 2023. Food & Nutrition Research, 68. <https://doi.org/10.29219/fnr.v68.10365>

from forming. Potassium citrate boosts urinary citrate, which is an important inhibitor of stone formation, and it improves urine pH to levels that make stone formation less likely.⁷⁷

- Signalling processes and nerve transmission: Potassium channels are expressed in virtually all cell types, and their activity is the dominant determinant of cellular membrane potential. As such, potassium flux is a key regulator of many cellular processes, including the regulation of action potentials in excitable cells. Subtle changes in extracellular potassium can initiate signalling processes vital for survival (insulin signalling) while more extreme and chronic changes may lead to pathological states (acid-base disturbances and cardiac arrhythmia).⁷⁸
- Muscle contraction: The potassium gradient is fundamental for skeletal, smooth, and cardiac muscle function. It helps with membrane repolarisation after each contraction, enabling normal muscle relaxation and rhythmic heartbeats. Disruption in potassium levels can affect cardiac rhythm.⁷⁹

⁷⁷ Weaver, C. M. (2013). Potassium and health. *Advances in Nutrition*, 4(3), 368S-377S. <https://doi.org/10.3945/an.112.003533>

⁷⁸ McCormick, J. A., Topf, J., Tomacruz, I. D., & Grimm, P. R. (2023). A new understanding of potassium's influence upon human health and renal physiology. *Advances in Kidney Disease and Health*, 30(2), 137–147. <https://doi.org/10.1053/j.akdh.2023.01.001>

⁷⁹ Office of Dietary Supplements - Potassium. (n.d.). <https://ods.od.nih.gov/factsheets/Potassium-HealthProfessional/>

Sodium

What is Sodium?

Sodium is the major cation of extracellular fluid. The concentration gradient between the ECF and intracellular fluid is maintained by the sodium- potassium pump activity, which transfers sodium and potassium, respectively, from inside to outside the cell and vice versa against the concentration gradient, using the energy supplied by ATP.⁸⁰

Most of the salt in our diets comes from commercially prepared foods, not from salt added to cooking at home or even from salt added at the table before eating. According to The Centres for Disease Control and Prevention, the top 10 sources of sodium in the diets include: breads/rolls; pizza; sandwiches; cold cuts/cured meats; soups; burritos, tacos; savoury snacks (chips, popcorn, pretzels, crackers); chicken; cheese; eggs, omelettes.⁸¹

Absorption of Sodium

The following are the mechanisms of sodium absorption in the body:

- Na^+ glucose, Na^+ -amino acids, and Na^+ -di-or tri-peptide co-transport system account for 30% of Na^+ transport into the cell.
- The Neutral Na-Cl^- co-transport system also transports about 30% of Na^+ : In the small intestine (especially ileum) and colon, there are special transport proteins on the surface of the intestinal cells which let one sodium ion (Na^+) and one chloride ion (Cl^-) move from the intestinal lumen into the cell at the same time. Because they move together, the movement doesn't change the overall electrical charge (hence neutral).
- Na^+-H^+ exchange: On the surface of the intestinal cell, there is a special exchange protein called the Na^+-H^+ exchanger. This exchanger swaps one sodium ion (Na^+) from the intestinal contents into the cell, while it pushes one hydrogen ion (H^+) out of the cell into the intestinal lumen. As sodium keeps moving into the cell through this exchange, its concentration inside starts to rise : but the Na^+-K^+ keeps pumping sodium out into the blood, maintaining a low level of sodium inside the cell. This allows more sodium to keep coming in from the gut, so sodium is continually absorbed from the intestinal contents into the body.

After sodium enters the enterocyte, the Na-K ATPase keeps pumping sodium out of the cell into the interstitial fluid, which diffuses into the capillaries of the intestinal villi and enters the bloodstream.

Functions of Sodium

- Fluid and electrolyte balance: Sodium is the main cation outside cells and an important factor in extracellular fluid (ECF) volume and plasma osmolality. It helps keep water in the bloodstream and tissues, which is crucial for maintaining blood volume and blood pressure. Changes in

⁸⁰ Strazzullo, P., & Leclercq, C. (2014). Sodium. *Advances in Nutrition*, 5(2), 188–190. <https://doi.org/10.3945/an.113.005215>

⁸¹ Admin. (2024a, November 7). Salt and Sodium • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/salt-and-sodium/>

sodium levels cause hormonal responses, such as aldosterone and vasopressin, that alter kidney excretion to keep balance.

- Nerve function and supports nerve impulse transmission: Sodium ions (Na^+) are crucial for generating and spreading electrical signals (action potentials) along nerve cells. When a nerve is stimulated, sodium channels open and let sodium enter the cell. This triggers electrical impulses that allow communication between the brain, nerves, and muscles.
- Muscle contraction: Sodium is essential for muscle excitability and contraction, which includes skeletal, smooth, and cardiac muscle. It plays a role in the electrical changes that trigger muscle fibre activation. Additionally, it works with the sodium-potassium pump to maintain the ion gradients necessary for contraction and relaxation.
- Transportation of nutrients: Sodium gradients drive secondary active transport of key nutrients such as glucose and amino acids. For example, sodium-glucose cotransporters (SGLTs) utilise the sodium gradient to facilitate the absorption of glucose in the intestine and kidney.
- Blood pressure regulation (Hypertension): High intake of sodium plays a significant role in the long-run process of the development of hypertension. Increased intracellular calcium in vascular smooth muscle cells, induced primarily by high sodium intake and sodium retention, may be one of the key factors behind increased contractility of resistance arteries, impaired vascular endothelial function, and increased peripheral resistance in sodium-sensitive hypertension. High sodium intake may induce stiffening of the large arteries and structural changes of the left ventricle, also by pressure-independent mechanisms. Hence, it modulates the fluid volume, vascular tone and pressure.⁸²

⁸² View of Sodium – a systematic review for Nordic Nutrition Recommendations 2023 (n.d.). <https://foodandnutritionresearch.net/index.php/fnr/article/view/10319/16824>

Phosphorus

What is Phosphorus?

Phosphorus is a ubiquitous mineral in nature and the second most abundant mineral in the human body, which represents approximately 1% of total body weight. Less than 1% of unbound inorganic phosphate (PO₄⁻) in the extracellular space is metabolically active and is now considered tightly maintained within a narrow serum concentration range (2.5–4.5 mg/dL) in adults.⁸³

A variety of foods naturally contain phosphorus, and the richest sources are dairy, red meat, poultry, seafood, legumes, and nuts. Phosphorus from these foods is called organic phosphorus. It is absorbed more efficiently from animal foods than from plant foods. Inorganic phosphorus is a processed form added to foods to preserve colour, moisture, and texture. It is found in fast foods, deli meats, canned and bottled beverages, and many other processed foods.⁸⁴

Phosphorus bioavailability varies by food source: it is very high from phosphate additives, moderate from animal foods, and low from plant foods where phosphorus is bound as phytate, which humans cannot digest unless phytate is broken down during food processing (e.g., fermentation).

Absorption of Phosphorus

Dietary phosphorus is absorbed mainly in the duodenum and jejunum of the small intestine through two mechanisms: passive diffusion, which depends on the amount of phosphorus present in the gut, and active, sodium-dependent transport mediated primarily by the NaPi-IIb transporter. Active absorption increases when dietary phosphorus intake is low, because reduced serum phosphate stimulates the production of 1,25-dihydroxyvitamin D (calcitriol), which enhances intestinal phosphate uptake.

After absorption, phosphorus enters the circulation, raising serum phosphate levels, and excess phosphate is primarily handled by the kidneys. About 80–90% of filtered phosphate is reabsorbed in the proximal tubules via sodium-dependent transporters NaPi-IIa, NaPi-IIc, and PiT2. Renal handling is hormonally regulated, mainly by parathyroid hormone (PTH) and fibroblast growth factor-23 (FGF23). When phosphorus intake is high, both hormones are secreted- PTH from the parathyroid glands and FGF23 from bone cells, leading to reduced renal phosphate reabsorption and decreased intestinal absorption by lowering calcitriol levels.⁸⁵

Functions of Phosphorus

- **Bone health:** Phosphorus, mainly as inorganic phosphate (Pi), is a key part of bone mineral (hydroxyapatite) along with calcium. It gives bones and teeth strength and rigidity. Phosphorus also plays a role in forming and maintaining bone by affecting osteoblasts and osteoclasts.
- **Energy metabolism and storage:** Phosphorus is crucial for energy metabolism. Phosphate groups make up key components of ATP, GTP, and other high-energy compounds. These compounds store and transfer energy for almost all cellular activities, such as muscle contraction, nerve function, and biosynthesis.

⁸³ Calvo, M. S., & Lamberg-Allardt, C. J. (2015). Phosphorus. *Advances in Nutrition*, 6(6), 860–862. <https://doi.org/10.3945/an.115.008516>

⁸⁴ Powell, J. (2024d, November 7). Phosphorus • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/phosphorus/>

⁸⁵ Itkonen, S. T., & Lamberg-Allardt, C. (2023). Phosphorus – a scoping review for Nordic Nutrition Recommendations 2023. *Food & Nutrition Research*, 67. <https://doi.org/10.29219/fnr.v67.10318>

- Structure of cell: Phosphorus makes up phospholipids that make biological cell membranes, which ensure membrane integrity and cell compartmentalisation. They also form part of the DNA and RNA backbone structures, which are critical for genetic information storage and transmission.
- Intracellular signalling and regulation: Phosphate plays a role in cell signalling pathways and regulatory networks. Phosphorylation and dephosphorylation of proteins, which involve adding or removing phosphate groups, are a common way to activate or deactivate enzymes and signalling proteins. This process influences metabolism and cellular responses.
- Acid- base balance: Phosphate ions help the buffer system in blood and urine. They keep pH levels stable in body fluids. This buffering action is important for enzyme function and overall metabolic stability.
- Delivery of oxygen in RBCs: Phosphorus affects the oxygen-releasing capacity of haemoglobin through its role in synthesising 2,3-diphosphoglycerate (2,3-DPG) in red blood cells. Higher levels of 2,3-DPG reduce haemoglobin's affinity for oxygen, enhancing tissue oxygenation.

Magnesium

What is Magnesium?

Magnesium, an abundant mineral in the body, is naturally present in many foods, added to other food products, available as a dietary supplement, and present in some medicines (e.g., antacids, laxatives). Magnesium is a cofactor in more than 300 enzyme systems that regulate diverse biochemical reactions in the body, including protein synthesis, muscle and nerve function, blood glucose control, and blood pressure regulation.

An adult body contains approximately 25 grams of magnesium, with 50% to 60% present in the bones and most of the rest in soft tissues. Less than 1% of total magnesium is in blood serum, and these levels are kept under tight control.⁸⁶

Absorption of Magnesium

Magnesium is absorbed in the body in its ionised form (Mg^{2+}), primarily in the ileum of the small intestine. Absorption occurs mainly through a paracellular pathway, meaning magnesium passes between intestinal cells rather than through them. The efficiency of absorption is inversely related to intake: when dietary magnesium is low, a higher percentage is absorbed, and at normal intakes, about 40–50% is absorbed. Organic magnesium salts are generally more bioavailable than inorganic forms. Diets high in phytic acid or phosphate can reduce magnesium absorption by forming insoluble complexes, though the clinical impact is usually limited. After absorption, plasma magnesium is tightly regulated by the kidneys, which decrease excretion when levels are low and increase excretion when levels are high to maintain normal magnesium balance.⁸⁷

Functions of Magnesium

- Cofactor for enzymatic reactions: The complex $MgATP^{2-}$ is required for the activity of many enzymes (approximately over 600 enzymes). In general, Mg^{2+} acts as a cofactor in all reactions involving the utilisation and transfer of ATP, including cellular responses to growth factors and cell proliferation, being thus implicated in virtually every process in the cells. Mg^{2+} availability is a critical issue for carbohydrate metabolism, which may explain its role in diabetes mellitus type 2.
- Nucleic acids: Mg^{2+} is necessary for the correct structure and activity of DNA and RNA polymerases. In addition, topoisomerases, helicases, exonucleases, and large groups of ATPases require Mg^{2+} for their activity. Mg^{2+} is essential in DNA replication, RNA transcription, and protein formation, being thus involved in the control of cell proliferation. Moreover, Mg^{2+} is crucial to maintain genomic and genetic stability, stabilising the natural DNA conformation and acting as a cofactor for almost every enzyme involved in nucleotide and base excision repair and mismatch repair. Given these effects, low Mg^{2+} availability can be involved in the development of cancer.

⁸⁶ Office of Dietary Supplements - magnesium. (n.d.). <https://ods.od.nih.gov/factsheets/Magnesium-HealthProfessional/>

⁸⁷ Henriksen, C., & Aaseth, J. O. (2023). Magnesium: a scoping review for Nordic Nutrition Recommendations 2023. Food & Nutrition Research, 67. <https://doi.org/10.29219/fnr.v67.10314>

- Bone metabolism: Serum Mg^{2+} concentrations are strongly related to bone metabolism; bone surface Mg^{2+} is constantly exchanged with blood Mg^{2+} . Furthermore, Mg^{2+} induces osteoblast proliferation; therefore, the consequences of Mg^{2+} deficiency are accelerated bone loss and a decline in bone formation.
- Calcium homeostasis: Mg^{2+} participates in controlling the activity of some ionic channels in many tissues. Its mechanism of action relies on either direct interaction with the channel, or an indirect modification of channel function through other proteins (e.g., enzymes or G proteins), or via membrane surface charges and phospholipids. Furthermore, Mg^{2+} acts as a physiological Ca^{2+} antagonist within cells, since it can compete with Ca^{2+} for binding sites in proteins and Ca^{2+} transporters. These abilities are involved in the observed effect of magnesium on the cardiovascular system, muscle, and brain.
- Cognitive function: Neuronal magnesium concentrations downregulate the excitability of the N-methyl-D-aspartate (NMDA) receptor, which is essential for excitatory synaptic transmission and neuronal plasticity in learning and memory. Magnesium blocks the calcium channel in the NMDA receptor and must be removed for glutamatergic excitatory signalling. Low serum Mg^{2+} levels increase NMDA receptor activity, thus enhancing Ca^{2+} and Na^{+} influx and neuronal excitability. For these reasons, a deficiency of Mg^{2+} has been hypothesised in many neurological disorders, such as migraine, chronic pain, epilepsy, Alzheimer's, Parkinson's, and stroke, as well as anxiety and depression.
- General physiological support: Through its widespread biochemical roles, magnesium supports muscle contraction and relaxation, nerve impulse transmission, cardiovascular function, protein synthesis, and overall cellular energy metabolism.⁸⁸

⁸⁸ Fiorentini, D., Cappadone, C., Farruggia, G., & Prata, C. (2021). Magnesium: Biochemistry, Nutrition, Detection, and Social Impact of Diseases Linked to Its Deficiency. *Nutrients*, 13(4), 1136. <https://doi.org/10.3390/nu13041136>

Chloride

What is Chloride?

Chloride (Cl^-) is the most abundant ion in humans after sodium and accounts for 70% of the total anions in extracellular fluid. Cl^- is vital for maintaining osmotic pressure, muscle movement, and acid-base balance in the body. Cl^- does not follow the electrochemical equilibrium in most mammalian cells. In several cells, including primary sensory neurons, leukocytes, epithelial cells, sympathetic ganglion, and muscle cells, intracellular Cl^- is maintained above equilibrium levels. Cl^- is a component of a daily diet in the form of sodium chloride (NaCl). It is classified as an electrolyte as it carries a negative charge along with its positive counterparts, K^+ and Na^+ .

Cl^- is mainly found in a diet consisting of seaweed, rye, vegetables such as lettuce, tomatoes, olives, celery, fruits such as apples, melons, berries, and bananas, as well as red meats. Most of the Cl^- also comes from added salt in several food preparations.⁸⁹

Absorption of Chloride

Chloride is absorbed in the small intestine and remains in the body's fluids and blood. Any excess amount is excreted in urine. Chloride is usually bound to sodium, and therefore the amount in blood tends to coincide with sodium levels.⁹⁰

Functions of Chloride

- **Ionic Homeostasis and Electrical Neutrality:** Chloride is the most common extracellular anion in the human body. It plays a crucial role in maintaining ionic homeostasis and electrical neutrality. Because it has a negative charge, chloride balances the positive charges of cations such as sodium and potassium in body fluids. By helping to properly distribute charges, chloride stabilises the resting membrane potential of cells. This stability is vital for the normal function of electrically excitable tissues like nerves and muscles.⁹¹
- **Osmotic Pressure and Fluid Balance:** Osmotic pressure arises from differences in ion concentrations inside and outside cells. Chloride's high concentrations in the extracellular fluid draw water toward it. This maintains proper fluid volumes in the blood and interstitial spaces, influencing blood pressure and nutrient delivery throughout the body. When chloride levels drop abnormally, water distribution can change, potentially leading to dehydration of cells or an imbalance in extracellular fluid.
- **Acid-Base Balance:** Chloride interacts with bicarbonate (HCO_3^-), another key ion in pH regulation, through transport systems like the chloride/bicarbonate exchanger. When bicarbonate concentrations rise, chloride shifts to maintain charge balance, and vice versa. These movements help buffer changes in pH.
- **Digestion and Hydrochloric Acid Secretion:** One of chloride's most direct roles is in stomach acid production. Specialised cells in the stomach lining, known as parietal cells, secrete chloride ions

⁸⁹ Raut, S. K., Singh, K., Sanghvi, S., Loyo-Celis, V., Varghese, L., Singh, E. R., Rao, S. G., & Singh, H. (2024). Chloride ions in health and disease. *Bioscience Reports*, 44(5). <https://doi.org/10.1042/bsr20240029>

⁹⁰ Powell, J. (2024f, November 15). Chloride • The Nutrition Source. The Nutrition Source - Harvard Chan School. <https://nutritionsource.hsph.harvard.edu/chloride/>

⁹¹ Raut, S. K., Singh, K., Sanghvi, S., Loyo-Celis, V., Varghese, L., Singh, E. R., Rao, S. G., & Singh, H. (2024). Chloride ions in health and disease. *Bioscience Reports*, 44(5). <https://doi.org/10.1042/bsr20240029>

into the gastric lumen, where they combine with hydrogen ions to form hydrochloric acid (HCl). Hydrochloric acid lowers the stomach's pH, creating an acidic environment that activates digestive enzymes like pepsinogen into pepsin, which is vital for protein digestion. The low pH from HCl also helps kill harmful microbes that enter with food, serving as a first line of defence in the gut.

- **Muscle Function and Electrical Excitability:** Chloride is important for muscle function by helping control electrical excitability in muscle cells. In skeletal muscle, chloride conductance across the cell membrane stabilises the resting membrane potential. This prevents excessive or prolonged electrical activity that could lead to muscle stiffness or involuntary contractions. In cardiac muscle, chloride movements also influence the shape and duration of electrical signals that control heartbeats, highlighting its role in rhythm and contractility.
- **Cellular Processes and Signalling:** Changes in intracellular chloride concentration can influence gene expression by affecting transcription and translation. Chloride is involved in controlling the cell cycle, cell growth, and protein modifications after translation. Additionally, chloride movements help cells adjust their internal pH and volume in response to metabolic needs or external stresses.
- **Organelle Function and Intracellular Compartmentalisation:** Chloride ions are essential within specific organelles, the cellular compartments where localised chemical conditions support specialised functions.
 - In lysosomes, for example, proper acidification needed for enzyme activity relies on chloride movement.
 - In other organelles like mitochondria and the endoplasmic reticulum, chloride affects ionic balance, influencing energy production, protein synthesis, and calcium storage.
 - In endosomes and phagosomes, chloride aids in vesicle acidification and maturation, which are critical for processes like receptor-mediated endocytosis and pathogen destruction.
- **Charge Compensation for Cation Movements:** Chloride often acts as a compensatory ion when major cations such as sodium, potassium, or calcium move across membranes. As cations are transported into or out of cells or organelles during nutrient absorption, hormone responses, or electrical signalling, chloride shifts accordingly to maintain electrochemical balance. This charge compensation helps avoid dangerous voltage imbalances that could disrupt cellular integrity or signalling. By balancing the movement of other ions, chloride supports a wide range of physiological activities, from nerve conduction to epithelial transport.

Iron

What is Iron?

Iron is a mineral that is naturally present in many foods, added to some food products, and available as a dietary supplement. The trace element iron is essential for optimal physiological functioning and overall health and must be derived from dietary food sources and supplements. The body cannot synthesise iron, but macrophages can recycle and reuse it from senescent erythrocytes.

Dietary iron has two main forms: heme and nonheme. Plants and iron-fortified foods contain nonheme iron only, whereas meat, seafood, and poultry contain both heme and nonheme iron.⁹² The average amount of iron in an adult male is approximately 3 to 4 grams, of which three-quarters are heme proteins, such as haemoglobin, myoglobin, cytochromes, and peroxidases. About 20% to 30% of total body iron exists in the form of storage proteins, including ferritin and hemosiderin.⁹³

Absorption of Iron

Iron is absorbed by villus enterocytes in the duodenum. Efficient absorption requires an acidic environment, and antacids or other conditions that interfere with gastric acid secretion can interfere with iron absorption. Ferric iron (Fe^{+++}) in the duodenal lumen is reduced to its ferrous form, which is easy to absorb. Then, it enters the enterocyte using a transporter called DMT-1 (divalent metal transporter 1).

Once inside the enterocyte, iron follows one of two major pathways.

- Iron abundance states: Iron within the enterocyte is trapped by incorporation into ferritin.
- Iron-limited states: Iron is exported out of the enterocyte via a transporter (ferroportin). It then binds to the iron-carrier transferrin for transport throughout the body.

Heme iron is absorbed by a different process wherein the whole heme group enters the cell, and the iron is released inside.

Functions of Iron

- Oxygen transport and storage: Iron is a central component making up heme in haemoglobin in RBCs and myoglobin in muscles. Iron facilitates the binding of oxygen to haemoglobin and eventually its transportation from the lungs to body tissues. Whereas myoglobin stores oxygen in the muscles for use during activity.
- Cellular respiration and energy production: Iron is essential for the electron transport chain in mitochondria. It is part of cytochromes and iron-sulfur (Fe-S) proteins that help transfer electrons during ATP synthesis. This function supports oxidative phosphorylation, which is the main process cells use to produce energy. Iron deficiency can lead to a reduction in ATP production, which compromises the functionality of high-energy tissues such as muscles and the brain.
- Immune function: Iron also plays an essential role in immune modulation. Its deficiency has been linked with impaired B-cell proliferation, poor T-lymphocyte function and disrupted antibody responses. Iron plays a dual role in host defence and pathogen proliferation. In disease states, the body reduces the amount of iron available to pathogens, reducing the spread of diseases.

⁹² Office of Dietary Supplements - Iron. (n.d.). <https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/>

⁹³ Moustarah, F., & Daley, S. F. (2024, January 8). Dietary iron. StatPearls - NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK540969and_myoglobinwhich_compromisesproductionthe_Gram-negafacilitateactive_can/

This is regulated through inflammatory cytokines, which increase hepcidin production, leading to increased iron sequestration.

- DNA synthesis: Iron serves as a cofactor for enzymes that are crucial to DNA metabolism. It is needed for ribonucleotide reductase, the enzyme that changes ribonucleotides into deoxyribonucleotides, the building blocks of DNA. Iron-dependent enzymes also play a role in DNA repair and cell cycle progression.
- Red blood cell production: Iron plays a crucial role in erythropoiesis or red blood cell production. It is required for haemoglobin synthesis in developing erythrocytes. Iron availability regulates erythroid precursor proliferation and maturation in the bone marrow.⁹⁴
- Cancer fighting and antimicrobial resistance: Mixed iron (III) complexes exhibit both anticancer and antibacterial properties. They contain amino acids and isonitrosoacetophenone, both of which are potent anticancer compounds and can be used in the development of anticancer drugs. The same compounds also depict antimicrobial activity against both Gram-positive and Gram-negative bacteria.
- Muscle health maintenance: Iron can bind free radicals, thereby reducing oxidative stress. This helps reduce muscle damage and lower the risk of sarcopenia, particularly in old age. Additionally, iron contributes to energy generation in skeletal muscles via oxygen-reducing systems. Iron-containing enzymes are required by the skeletal muscle mitochondria to generate energy. Furthermore, oxidative metabolism in these muscles is facilitated by hemoproteins in complexes III and IV, and iron–sulfur clusters (ISC) proteins in complexes I, II and III.
- Prenatal development: Iron plays a critical role in supporting erythropoiesis and fetal growth. It is responsible for cell proliferation and differentiation within the embryo. A continuous supply of iron is required to build fetal liver iron stores. Iron regulates placental transport between the visceral endoderm (VE), early placenta, and specialised placenta to prevent reverse iron transfer via the placenta back into the mother's circulation. Iron metabolic genes have a role in normal fetal development as their disruption can cause developmental issues and fetal mortality.⁹⁵

⁹⁴ Beard, J. L. (2001). Iron biology in immune function, muscle metabolism and neuronal functioning. *Journal of Nutrition*, 131(2), 568S-580S. <https://doi.org/10.1093/jn/131.2.568s>

⁹⁵ Malik, Z. I., Ghafoor, M. U., Shah, S. H. B. U., Abid, J., Farooq, U., & Ahmad, A. M. R. (2025). Unlocking iron: nutritional origins, metabolic pathways, and systemic significance. *Frontiers in Nutrition*, 12, 1637316. <https://doi.org/10.3389/fnut.2025.1637316>

Copper

What is Copper?

Copper, an essential mineral, is naturally present in some foods and is available as a dietary supplement. It is a cofactor for several enzymes (known as cuproenzymes) involved in energy production, iron metabolism, neuropeptide activation, connective tissue synthesis, and neurotransmitter synthesis. Ceruloplasmin, which is a cuproenzyme, carries more than 95% of the total copper in healthy human plasma. Almost two-thirds of the body's copper is located in the skeleton and muscles. Only small amounts of copper are typically stored in the body, and the average adult has a total body content of 50–120 mg of copper. The richest dietary copper sources include shellfish, seeds and nuts, organ meats, wheat-bran cereals, whole-grain products, and chocolate.⁹⁶

Absorption of Copper

Copper is absorbed mainly in the duodenum through an active and tightly regulated transport process. Most dietary copper is present in the oxidised Cu^{2+} form, which cannot be directly taken up by intestinal cells. Before absorption can occur, Cu^{2+} must be reduced to Cu^+ by a brush-border metalloredutase on the surface of enterocytes. The reduced Cu^+ is then transported into the intestinal epithelial cell via copper transporter 1 (CTR1). Once inside the enterocyte, copper is immediately bound to intracellular proteins to prevent toxicity. It is either attached to specific copper chaperones or sequestered by metallothionein, a metal-binding protein whose synthesis is strongly induced by zinc. Copper that is not trapped by metallothionein is guided by chaperone proteins toward export pathways.

Copper exits the enterocyte through the basolateral membrane via the copper-transporting ATPase ATP7A, which enables its controlled release into the circulation. In the bloodstream, copper is transported to the liver bound to carrier molecules such as albumin, transcuprein, and low-molecular-weight copper–histidine complexes. These carriers ensure safe transport and prevent free copper from causing oxidative damage. After hepatic uptake, copper is stored primarily bound to metallothionein, forming a readily mobilisable reserve. When required, copper is released from storage and redistributed within the cell by intracellular copper chaperones, which deliver it specifically to copper-dependent enzymes while preventing non-specific interactions that could damage cellular components. For distribution to peripheral tissues, copper is incorporated mainly into ceruloplasmin, which carries about 70-90% of circulating copper, with the remainder bound to albumin.⁹⁷

Functions of Copper

- Cuproenzyme for cellular energy: Cytochrome oxidase plays an essential role in cellular energy. As catalysing the reduction of molecular oxygen (O_2) to water (H_2O), cytochrome oxidase generates an electrical gradient, which is used by mitochondria to create vital energy for the organism and stored in molecules of ATP.

⁹⁶ Office of Dietary Supplements - Copper. (n.d.). <https://ods.od.nih.gov/factsheets/Copper-HealthProfessional/>

⁹⁷ Henriksen, C., & Aaseth, J. O. (2023b). Magnesium: a scoping review for Nordic Nutrition Recommendations 2023. Food & Nutrition Research, 67. <https://doi.org/10.29219/fnr.v67.10314>

- Iron homeostasis: Two copper-containing enzymes, ceruloplasmin (ferroxidase I) and (ferroxidase II), can oxidise iron (Fe^{2+}) to iron (Fe^{3+}), which are connected to the protein transferrin for transportation to the red blood cells and blood formation.
- Cognitive and nerve function: Many enzymatic reactions, which are essential for the proper functioning of the brain and the nervous system, are catalysed by copper enzymes. Dopamine-beta-mono oxidase catalyses the conversion of the neurotransmitter dopamine into norepinephrine. Monoamine oxidase (MAO) plays a role in the metabolism of the neurotransmitters norepinephrine, epinephrine and dopamine. MAO functions in the breakdown of the neurotransmitter serotonin, which justifies the use of MAO inhibitors as antidepressants. Myelin sheath is made of phospholipids whose synthesis depends on the activity of the cytochrome c oxidase copper enzyme.
- Antioxidant: Superoxide dismutase (SOD) functions as an antioxidant, which catalyses the conversion of free radicals into hydrogen peroxide that can subsequently be reduced to water by other antioxidant enzymes. Two forms of SOD contain copper:
 - Copper/zinc SOD is found in most cells of the organism, including red blood cells, and
 - Extracellular SOD is a copper-containing enzyme, located in large quantities in the lungs and in low levels in plasma
- Gene expression: Copper-dependent enzymes are involved in the regulation of gene expression. Cellular copper levels may affect the synthesis of proteins in the organism by enhancing or inhibiting the transcription of specific genes. Enzymes, which are involved in the regulation of gene expression, are copper/ zinc superoxide dismutase (Cu/Zn SOD), catalase (another antioxidant enzyme).⁹⁸

⁹⁸ COPPER IN THE HUMAN ORGANISM. (2011). In Trakia Journal of Sciences (Vols. 9–9, Issue 1, pp. 88–98) [Mini-Review]. <http://www.uni-sz.bg>

Zinc

What is Zinc?

Zinc is an essential mineral which is naturally present in some foods, added to others, and available as a dietary supplement. Zinc is also found in some cold lozenges, over-the-counter drugs sold as cold remedies, and some denture adhesive creams.

The total amount of zinc in the body is approximately 1.5 g in women and 2.5 g in men. Most of this zinc is stored in skeletal muscle and bone. Major dietary sources of zinc include meat, fish and other seafood, eggs, dairy products, oysters, beans, nuts and whole grains.⁹⁹

Absorption of Zinc

Zinc absorption/uptake primarily occurs in the proximal part of the small intestine, in the distal duodenum and proximal jejunum. Zn^{2+} ion entry into the enterocyte is mediated via ZIP4 and/or ZnT5B in the duodenum and jejunum. Another transmembrane ion transporter potentially involved in zinc uptake into the enterocyte is Divalent Metal Transporter-1 (DMT-1). Regardless of the transporter used, once Zn^{2+} ions are in the enterocyte, ZnT1 transports zinc from the enterocyte into the portal blood. From here, these ions travel through the hepatic portal vein, most likely bound to ligands such as amino acids and citrate. At the liver, the portal vein branches, and the blood drains through sinusoids. Some zinc enters hepatocytes, most likely via ZIP14. After the blood passes through the sinusoids, it is directed to the central vein, then the hepatic veins and eventually the systemic circulation. From the systemic circulation, zinc ions are transported to various body tissues such as the brain, muscle, and bone.

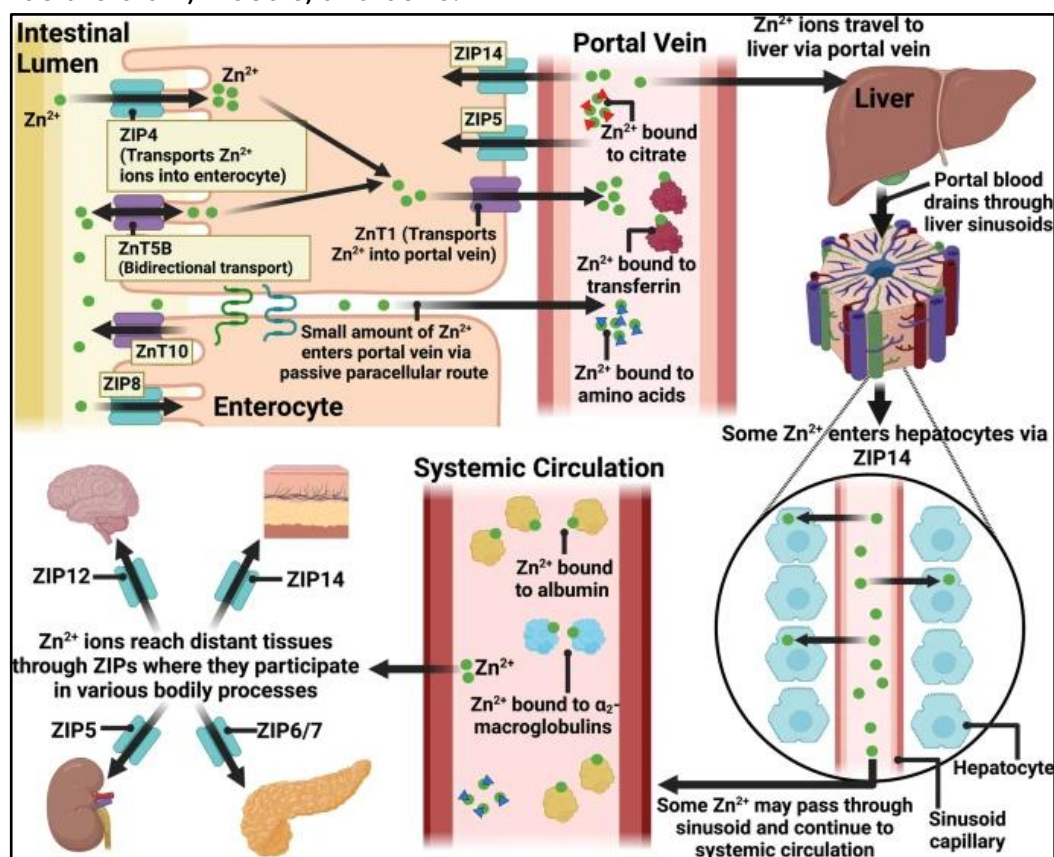


Fig 1: Absorption of zinc¹⁰⁰

Functions of Zinc

⁹⁹ Office of Dietary Supplements - Zinc. (n.d.). <https://ods.od.nih.gov/factsheets/Zinc-HealthProfessional/>

¹⁰⁰ Stiles, L. I., Ferrao, K., & Mehta, K. J. (2024). Role of zinc in health and disease. *Clinical and Experimental Medicine*, 24(1), 38. <https://doi.org/10.1007/s10238-024-01302-6>

- **Zinc as a Structural Component of Proteins and Enzymes:** Zinc has a structural role in many proteins. It stabilises protein structure by forming zinc fingers, zinc clusters, and zinc ribbons. Zinc finger motifs help proteins, especially transcription factors, bind to DNA, RNA, and other proteins. It is estimated that about 10% of all human proteins bind zinc. This structural role allows zinc to maintain protein integrity, ensure correct folding, and support gene regulation.
- **Zinc as a Catalytic Cofactor:** Zinc is an essential catalytic cofactor for over 300 enzymes. It takes part in hydrolytic reactions, like those in carbonic anhydrase and alkaline phosphatase. Zinc also stabilises negative charges during reactions and activates substrates. Unlike iron or copper, zinc does not undergo redox reactions, making it ideal for stable enzymatic activity. Zinc-dependent enzymes are critical for digestion, metabolism, acid-base balance, and biosynthetic pathways.
- **Regulation of Gene Expression and Transcription:** Zinc regulates gene expression mainly through zinc-dependent transcription factors. Zinc finger proteins bind to specific DNA sequences to activate or repress transcription. Zinc also affects epigenetic processes by influencing chromatin structure and transcription machinery.
- **Immune System Function:** Zinc is vital for both innate and adaptive immunity. It supports the development and function of T cells, B cells, macrophages, neutrophils, and NK cells. Zinc regulates cytokine production and inflammatory signalling pathways. A deficiency can lead to weakened immune responses and a higher risk of infections.
- **Cell Growth, Division, and Differentiation:** Zinc is essential for cell cycle progression. It regulates DNA synthesis, mitosis, and apoptosis. Zinc influences signalling pathways involved in stem cell maintenance and tissue regeneration. Sufficient zinc is necessary for normal growth, wound healing, and tissue repair.
- **Antioxidant Defence and Redox Homeostasis:** Zinc contributes indirectly to antioxidant protection. It helps stabilise cell membranes and proteins against oxidative damage. Zinc is a structural component of Cu/Zn superoxide dismutase (SOD1). It protects sulfhydryl groups in proteins from oxidation. Zinc reduces oxidative stress and limits cellular damage without entering harmful redox cycles.
- **Nervous System Function:** Zinc serves both structural and signalling roles in the nervous system. It is involved in synaptic transmission, particularly in glutamatergic neurons. Zinc modulates neurotransmitter receptors and neuronal plasticity. It supports neurodevelopment and cognitive function.
- **Hormone Synthesis and Signalling:** Zinc influences endocrine function. It is necessary for insulin synthesis, storage, and secretion in pancreatic β -cells. Zinc modulates hormone receptor signalling and intracellular signal pathways.
- **Apoptosis and Cellular Stress Responses:** Zinc regulates programmed cell death. It inhibits caspase activation under normal conditions. Changes in intracellular zinc levels can trigger apoptosis during cellular stress.¹⁰¹

¹⁰¹ Maywald, M., & Rink, L. (2022). Zinc in human health and infectious diseases. *Biomolecules*. <https://doi.org/10.3390/biom12121748>

Iodine

What is Iodine?

Iodine is a trace element that is naturally present in some foods, is added to some types of salt, and is available as a dietary supplement. It is an essential component of thyroid hormones thyroxine (T4) and triiodothyronine (T3), which are critical for liver, kidney, muscle, brain and central nervous system function.¹⁰²

Iodine in food and iodised salt is present in several chemical forms, including sodium and potassium salts, inorganic iodine (I₂), iodate, and iodide, the reduced form of iodine. Iodine rarely occurs as the element but rather as a salt; for this reason, it is referred to as iodide and not iodine. The iodine-replete healthy adult has about 15–20 mg of iodine, 70%–80% of which is contained in the thyroid. Seaweed, fish, other seafood, eggs, human breast milk, dairy products, oysters, iodised table salt, etc., are some major sources of iodine.¹⁰³

Absorption of Iodine

In food, iodine is present mainly as iodide (I⁻) or iodate (IO₃⁻). Iodide is almost completely absorbed in the gastrointestinal tract, whereas iodate must first be reduced to iodide before absorption can occur. The uptake of iodide across the intestinal epithelium is mediated by the sodium–iodide symporter (NIS), an active transport system that couples iodide movement to sodium gradients.

Once absorbed into the bloodstream, iodide is distributed throughout the body, but it is selectively concentrated in the thyroid gland and several extrathyroidal tissues such as the salivary glands, mammary glands, stomach, and intestine. This uptake also occurs via NIS, with the thyroid being the primary site of iodide utilisation. The expression and activity of NIS in thyroid cells are tightly regulated by thyroid-stimulating hormone (TSH), which increases iodide uptake when thyroid hormone levels are low. At the same time, iodide itself exerts autoregulation, as excess iodide inhibits its own transport into thyroid cells, preventing overproduction of thyroid hormones.¹⁰⁴

Functions of Iodine

- Regulation of thyroid function: Within the thyroid follicular cells, iodide undergoes oxidation by the enzyme thyroid peroxidase. The oxidised iodine is then incorporated into tyrosine residues of thyroglobulin (TG), a large protein synthesised by thyroid cells. This process, known as organification, leads to the formation of iodinated tyrosines, which subsequently couple to form the thyroid hormones thyroxine (T4) and triiodothyronine (T3). These hormones remain stored within the thyroglobulin molecule in the thyroid follicle lumen until the body signals a need for their release. They regulate gene transcription, thereby controlling basal metabolic rate, growth, development, and energy metabolism. In addition to these genomic actions, thyroid hormones also exert rapid nongenomic effects through other cellular pathways.¹⁰⁵
- Brain development: Iodine is essential for healthy brain development, particularly during fetal life and early childhood. Proper levels of thyroid hormones, which rely on enough iodine, are vital for

¹⁰² Hatch-McChesney, A., & Lieberman, H. R. (2022). Iodine and Iodine Deficiency: A Comprehensive Review of a Re-Emerging Issue. *Nutrients*, 14(17), 3474. <https://doi.org/10.3390/nu14173474>

¹⁰³ Office of Dietary Supplements - iodine. (n.d.). <https://ods.od.nih.gov/factsheets/Iodine-HealthProfessional/>

¹⁰⁴ Nedić, O. (2023). Iodine: Physiological importance and food sources. *eFood*, 4(1). <https://doi.org/10.1002/efd2.63>

¹⁰⁵ Nedić, O. (2023). Iodine: Physiological importance and food sources. *eFood*, 4(1). <https://doi.org/10.1002/efd2.63>

neuronal differentiation, myelination, and synaptogenesis. Severe iodine deficiency during pregnancy can cause intellectual disability and cretinism.

- Fetal growth: Iodine needs are greater during pregnancy and lactation. This is because iodine passes to the fetus through the placenta and to the baby through breast milk. Enough iodine helps with normal fetal growth, brain development, and the baby's neurological health. A lack of iodine during these stages raises the risk of poor growth and developmental issues.¹⁰⁶
- Antioxidative properties: Iodine acts as a scavenger of reactive oxygen species (ROS) and helps neutralise harmful free radicals. These free radicals can damage cellular components like lipids, proteins, and DNA. This process contributes to cellular protection and overall oxidative balance.
- Antimicrobial effects: Iodine shows strong antimicrobial activity when it is oxidised to hypoiodite (IO^-), a reactive form of iodine that effectively kills bacteria, viruses, and fungi. The oxidation helps protect tissues from harmful organisms. Simultaneously, iodine may help protect mucosal surfaces in organs like the salivary glands, stomach, and intestine, adding to the immune function of the body.¹⁰⁷

¹⁰⁶ Hatch-McChesney, A., & Lieberman, H. R. (2022). Iodine and Iodine Deficiency: A Comprehensive Review of a Re-Emerging Issue. *Nutrients*, 14(17), 3474. <https://doi.org/10.3390/nu14173474>

¹⁰⁷ Sorrenti, S., Baldini, E., Pironi, D., Lauro, A., D'Orazi, V., Tartaglia, F., Tripodi, D., Lori, E., Gagliardi, F., Praticò, M., Illuminati, G., D'Andrea, V., Palumbo, P., & Ulisse, S. (2021). Iodine: Its role in thyroid hormone biosynthesis and beyond. *Nutrients*, 13(12), 4469. <https://doi.org/10.3390/nu13124469>

Water

Water is essential for life and comprises 75% body weight in infants to 55% in the elderly and is essential for cellular homeostasis and life. We get water not only directly as a beverage but also from food and, to a very small extent, also from oxidation of macronutrients (metabolic water). The proportion of water that comes from beverages and food varies with the proportion of fruits and vegetables in the diet. In the United States, it is estimated that about 22% of water comes from our food intake, while it would be much higher in European countries, particularly a country like Greece with its higher intake of fruits and vegetables or South Korea. The fine intrinsic regulation of hydration and water intake in individuals mitigates against prevalent underhydration in populations and its effects on function and disease.¹⁰⁸

Infants 0–6 months*	0.7 litres
Infants 7–12 months#	0.8 litres total (with 0.6 litres as fluids)
Girls and boys 1–3 years	1 litre (about 4 cups)
Girls and boys 4–8 years	1.2 litres (about 5 cups)
Boys 9–13 years	1.6 litres (about 6 cups)
Boys 14–18 years	1.9 litres (about 7–8 cups)
Girls 9–13 years	1.4 litres (about 5–6 cups)
Girls 14–18 years	1.6 litres (about 6 cups)
Men 19 years+	2.6 litres (about 10 cups)
Women 19 years+	2.1 litres (about 8 cups)
Pregnant girls 14–18 years	1.8 litres (about 7 cups)
Pregnant women 19 years+	2.3 litres (about 9 cups)
Lactating girls 14–18 years	2.3 litres (about 9 cups)
Lactating women 19 years+	2.6 litres (about 10 cups)

Fig 1: Water requirements according to age groups

Absorption of water

The GI tract absorbs 99% of ingested water and ions. Absorption is determined by transporters and diffusion, with a transcellular pathway across cell membranes or a paracellular pathway through tight junctions. The tightness of tight junctions determines water and ion permeability. Tight

¹⁰⁸ Popkin, B. M., D'Anci, K. E., & Rosenberg, I. H. (2010). Water, hydration, and health. *Nutrition Reviews* to diin the faecesarrhoea, 68(8), 439–458. <https://doi.org/10.1111/j.1753-4887.2010.00304.x>

junctions are loosest in the duodenum and tightest in the colon and stomach. The GI tract receives 2 L/day from ingestion and 7 L/day from GI secretions. Only 50 to 100 mL/day is lost in the faeces. The jejunum is the primary site of water absorption. The colon absorbs 400 mL/day, against an apparent osmotic gradient.

The jejunum is also the primary site of Na⁺ absorption, secondary to amino acid and carbohydrate absorption. Apical entry occurs by diffusion, and basolateral surface secretion occurs by the primary active transporter Na⁺/K⁺-ATPase. The colon has active, amiloride-sensitive Na⁺ absorption. Net K⁺ balance is determined in the colon. Potassium is reabsorbed in the jejunum and ileum. In these regions, K⁺ levels are tied to water movements. Transport in the colon determines the final K⁺ balance. There is reabsorption if luminal K⁺ concentration is high, and secretion if luminal K⁺ concentration is low. Aldosterone enhances Na⁺/K⁺ exchange and causes K⁺ secretion. A significant K⁺ loss can accompany water loss due to diarrhoea.¹⁰⁹

Regulation of water balance

The regulation of water balance is based on a feedback mechanism involving the hypothalamus, the neurohypophysis, and the kidneys. Arginine vasopressin (AVP), also denoted anti-diuretic hormone, is the body's primary water-regulating hormone. It functions to maintain body water balance by keeping plasma osmolality within narrow limits and allowing the kidneys to alter water excretion in response to the body's needs, in conjunction with thirst.

Osmoreceptors in the hypothalamus sense plasma osmolality. When water loss exceeds intake, blood volume decreases and plasma osmolality increases. An increase in osmolality above a physiologic threshold (290 to 295 mOsm per kilogram of water in most persons) leads to an increased secretion of AVP that binds to the vasopressin 2(V₂) receptors in the kidney. This increases the permeability to water in the last portion of the nephron, leading to a reabsorption of solute-free water and consequently a decreased water output in urine. An increase in the osmolality also activates the renin-angiotensin system, leading to increases in renin and angiotensin II concentrations. The latter, along with aldosterone, promotes sodium and chloride reabsorption in the kidneys and thus water via osmosis, and thereby decreases urine output. Increased renal reabsorption of water in response to AVP lowers plasma osmolality, thereby reducing the stimulus for AVP secretion and thirst and completing the feedback loop. The physiology is reversed in the case of an excess of extracellular fluid in the body: osmolality decreases, leading to decreased renal water reabsorption and excretion of diluted urine.¹¹⁰

Functions of water

- **Thermoregulation:** Water is an important component of thermoregulation as it facilitates the efficient dissipation of heat, maintenance of cardiovascular stability, and buffering of changes in body temperature. Sufficient body water is necessary for sweat secretion, and the evaporation of sweat from the skin's surface is the main mechanism for heat dissipation during heat stress and exercise. Water also maintains plasma volume and normal plasma osmolality, which are vital for the maintenance of skin blood flow and sweat secretion. Also, the high specific heat of

¹⁰⁹ Carroll, R. G. (2007). Gastrointestinal system. In Elsevier eBooks (pp. 139–156). <https://doi.org/10.1016/b978-0-323-04318-2.50018-in-the-faeces2>

¹¹⁰ Fluid and water balance: a scoping review for the Nordic Nutrition Recommendations 2023 | Food & Nutrition Research. (n.d.). <https://foodandnutritionresearch.net/index.php/fnr/article/view/9975/16389>

water facilitates the absorption and redistribution of heat by total body water, thereby preventing abrupt changes in body temperature.

- Cellular homeostasis: Water is crucial for survival, and humans cannot survive without water for long periods of time. It constitutes a large proportion of body weight ($\approx 75\%$ in infants to 55% in the elderly) and is indispensable for cells to maintain balance and function.
- Gastrointestinal function: Water ingested with meals is absorbed by the gastrointestinal tract (small intestine and colon). Adequate fluid intake helps maintain normal bowel function and prevents constipation; inadequate fluid intake contributes to constipation, particularly in the elderly.
- Kidney function and waste excretion: The kidneys require water to filter blood, regulate plasma osmolality, maintain blood pressure, and eliminate waste. Water influences the kidney's ability to concentrate or dilute urine and to excrete solutes effectively.
- Heart function and hemodynamic effects: Water intake influences blood volume, blood pressure, and heart rate. Drinking water can trigger increases in blood pressure and decreases in heart rate, and helps to prevent passing out (such as orthostatic hypotension) by maintaining cardiovascular function.
- Headache prevention/relief: Water deprivation and dehydration can cause headaches, including migraine or tension headache, probably because of decreased plasma volume and intracranial fluid volume; water ingestion relieves these symptoms in dehydrated patients.
- Skin hydration: Water makes up approximately 30% of skin hydration. Severe dehydration can decrease skin turgor (manifested as "tenting" when pinched). Adequate water intake may increase skin thickness and oppose transepidermal water loss.
- Hydration and chronic disease modulation: Hydration is linked to a decreased risk of a number of conditions, most strongly for kidney stones. There is also evidence, although less robust, that hydration can decrease the risk of constipation, exercise-induced asthma, and some metabolic disorders, as well as the risk of fatal coronary heart disease and other vascular events.¹¹¹

¹¹¹ Popkin, B. M., D'Anci, K. E., & Rosenberg, I. H. (2010). Water, hydration, and health. *Nutrition Reviews*, 68(8), 439–458. <https://doi.org/10.1111/j.1753-4887.2010.00304.x>

Primer on Types of Diets

A diet is fundamentally the pattern of foods and beverages habitually consumed by an individual or group.¹¹² It may also be a prescribed eating plan with specific restrictions or goals, like for weight loss, or as a therapeutic regimen. They can be structured around health conditions or risks, or cultural and regional food traditions. Irrespective, it must be aligned with individuals' circadian rhythm and ought to provide adequate calories and nutrients to meet physiological needs without excess, ensuring a balance between major macronutrients (proteins, fats, carbohydrates) and all essential vitamins and minerals (micronutrients).

Types of Diets

Dietary patterns play a crucial role in shaping human health, influencing metabolic processes, disease risk, and overall well-being. In recent years, a wide range of dietary approaches have emerged based on differences in meal timing, nutrient composition, therapeutic purpose, and lifestyle preferences. These include fasting or time-restricted diets such as intermittent fasting (IF) and alternate-day fasting (ADF); nutrient-focused approaches like the ketogenic diet and low-FODMAP diet; clinical dietary models including the DASH (Dietary Approaches to Stop Hypertension) diet and the gluten-free diet (GFD); and plant-based or traditional patterns such as vegetarian, vegan, and Mediterranean diet (MedDiet).

1. Fasting or Time-Restricted Diets

Fasting refers to the intentional and voluntary avoidance of foods and beverages that provide calories for a defined period of time. In this approach, individuals follow specific eating windows and fasting periods, thereby regulating when energy intake occurs rather than modifying the nutritional composition of the diet itself. Its physiological significance lies in its capacity to induce metabolic switching, enhance cellular repair mechanisms and regulate systemic homeostasis. This shift is commonly referred to as the 'glucose-to-ketone switch', wherein the body's metabolism shifts from using glucose (sugar) as its primary energy source to using ketone bodies derived from fat. In the fed state, the body mainly relies on glucose from carbohydrates for energy. However, during prolonged fasting, carbohydrate restriction, or starvation, blood glucose and insulin levels fall, which triggers the breakdown of stored fat. The liver then converts fatty acids into ketone bodies (such as β -hydroxybutyrate), which serve as an alternative fuel for the brain and other tissues. This typically occurs approximately 12-24 hours after food deprivation and represents a fundamental adaptive response to energy scarcity.

From a social and cultural standpoint, cultural and religious practices promote the concept of fasting. The practice of Hinduism comprises numerous fasting days (like *Ekadashi*, *Navratri*) that involve restrictions such as avoiding grains, eating only plant foods, or one meal per day, which modify energy and nutrient intake cyclically. Similarly, Islamic followers observe fasting through the celebration of *ramzan/ramadan*, which introduces a daily fasting-feasting rhythm. However, the temporary benefits of these practices can affect metabolic health either positively or negatively, depending on food choices. Beyond cultural reasons, fasting is also practised by individuals for health and weight-loss purposes - usually either in an alternate-day fasting model or the popular intermittent fasting regime.

1.1. Intermittent fasting (IF)

- It is a dietary regimen characterised by recurring cycles of fasting and feeding within defined time windows, and generally does not prescribe any specific nutrient composition or caloric restriction during eating periods. Common models include specific windows, like 16:8 or

¹¹² Cambridge Dictionary. (2019, November 20). DIET | definition in the Cambridge English Dictionary. Cambridge.org.

14:10, with hour fasting-to-feeding ratios.¹¹³

- Intermittent fasting operates through cyclical metabolic and hormonal adaptations that occur during repeated fasting intervals. During fasting phases, declining insulin levels and rising glucagon stimulate glycogenolysis followed by lipolysis, resulting in the mobilisation of free fatty acids and the production of ketone bodies as alternative energy substrates. This periodic metabolic switching enhances fat oxidation and reduces reliance on continuous glucose metabolism.
- At the cellular level, intermittent fasting activates regulatory pathways including AMP-activated protein kinase (AMPK), sirtuin signalling, and autophagy processes, which collectively promote cellular maintenance, stress resistance, and metabolic efficiency. Repeated cycles of fasting also improve circadian rhythm alignment by synchronising feeding patterns with endogenous metabolic clocks.
- Through the above mechanics, IF has been linked to reductions in systemic inflammation, oxidative stress, and adiposity. From a cardiometabolic standpoint, intermittent fasting may contribute to improving lipid profiles, reducing visceral fat accumulation, and lowering other risk markers associated with metabolic syndrome (MetS).
- However, IF is generally not recommended for persons with type-2 diabetes as it may cause severe hypoglycemia, especially when combined with insulin or sulfonylureas, increasing risks of dizziness, fainting, and dangerous fluctuations.

1.2. Alternate-day fasting (ADF)

- It is a strict form of IF that alternates between 24-hour fasts and 24-hour eating days. On fasting days, caloric intake is either completely absent or significantly restricted, typically to about 20-25% of daily energy requirements, while feeding days allow normal or ad libitum consumption.
- While a daily 16-hour fast only allows the body to begin utilising fat oxidation, ADF pushes the body to rely on fat as its primary fuel source for significantly longer, typically up to 24 hours per fasting cycle.
- Fasting may contribute to metabolic changes that trigger a surge in growth hormone secretion.
 - **Growth hormone levels** increase during fasting periods. Research shows that a two-day fast in healthy individuals results in augmented GH secretory burst frequency and amplitude, leading to significantly enhanced total GH secretion.
 - The primary explained benefit of this surge is the **maintenance of muscle mass**. While the body is in a catabolic state (breaking down fat for energy), the elevated GH helps prevent muscle catabolism, ensuring that fat is lost rather than lean body mass.
 - This rise in GH occurs as the body flips the metabolic switch from using glucose to using fatty acids and ketones for fuel, typically beginning 12 to 36 hours after the last meal.
 - Despite the rise in growth hormone, **levels of IGF-1 (insulin-like growth factor 1) decrease** during fasting. This reduction in IGF-1 is linked to lowered mTOR pathway activity, which **promotes cellular repair and autophagy**.

¹¹³ de Cabo, R., & Mattson, M. P. (2019). Effects of Intermittent Fasting on Health, Aging, and Disease. *New England Journal of Medicine*, 381(26), 2541–2551.

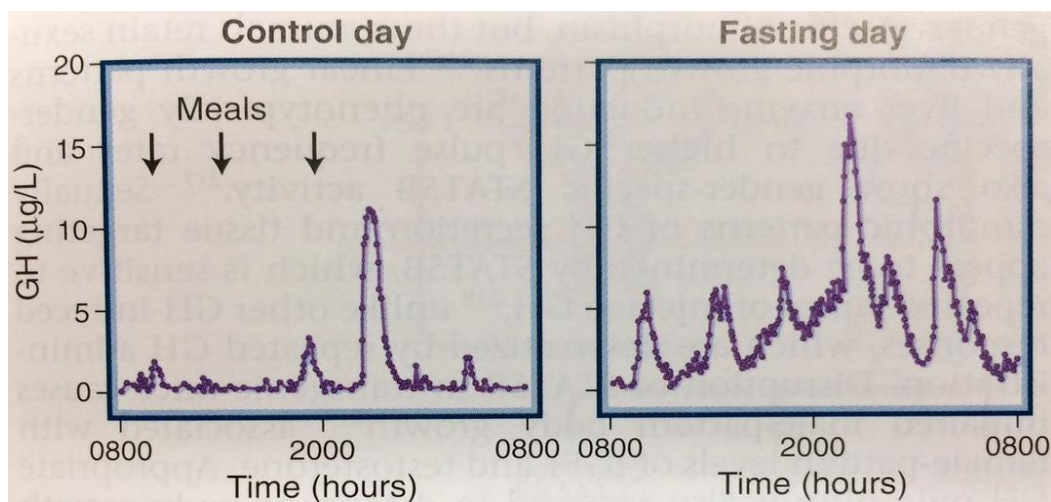


Figure: Graphs showing the level of growth hormone while having meals and while fasting¹¹⁴

2. Nutrient-reductionist diets

Often also known as therapeutic or specialised diets, these diets target specific macronutrients or micronutrients, for health goals, like weight loss, allergy management, or chronic disease control, like irritable bowel syndrome (IBS), diabetes and hypertension. The focus on the isolated nutrient can either be to maximise or minimise its consumption in the daily diet.

2.1. Ketogenic diet

- It is a very low-carbohydrate, moderate-protein, high-fat diet designed to induce ketosis, where the body primarily utilises fat instead of glucose for energy. Thus, carbohydrates are typically restricted to about 5-10% of total energy (often <50 g/day).¹¹⁵
- In a ketogenic diet, typical foods include meat, fish, eggs, cheese, butter, oils (specifically medium-chain triglyceride (MCT) oils like coconut oil), nuts and low-starch vegetables like spinach, kale, arugula, collard greens, and all types of lettuce and cruciferous vegetables like broccoli and cauliflower. Starchy foods, grains, sugars, and most fruits are virtually eliminated.
- During the practice of this diet, the hepatic glycogen stores become depleted, and the liver converts fatty acids into ketone bodies, primarily beta-hydroxybutyrate, acetoacetate, and acetone, which serve as alternative fuels for the brain, muscles, and other tissues.
- It helps in the reduction of the frequency of epilepsy seizures by elevating the seizure threshold through reduced glucose availability and opening ATP-sensitive potassium (KATP) channels to stabilise neuronal synapses. Furthermore, the diet modulates neurotransmitter function by increasing inhibitory GABA levels and decreasing excitatory markers like aspartate and vesicular glutamate transport. An increase in GABA levels promotes neuronal inhibition by reducing excessive electrical activity in the brain and by enhancing inhibitory signalling and reducing excitatory neurotransmission, stabilising neuronal firing and decreasing seizure frequency, which is why ketogenic dietary therapies are widely used in drug-resistant epilepsy. Also, Keto diets reduce systemic inflammation and improve neurotransmitter balance (serotonin and dopamine regulation), both of which are strongly linked to depression.
- Keto has several known risks. In the short term, many people experience 'keto flu', a temporary set of symptoms including headache, fatigue, irritability, nausea, and brain fog that occur as the body adapts to a low-carb, high-fat ketogenic diet. Additionally, the high saturated fat content can raise LDL cholesterol in some, potentially increasing heart disease risk. Kidney stones are relatively common due to an acid load and low citrate.

¹¹⁴ Hartman, M. L., Veldhuis, J. D., Johnson, M. L., Lee, M. M., Alberti, K. G., Samojlik, E., & Thorner, M. O. (1992). Augmented growth hormone (GH) secretory burst frequency and amplitude mediate enhanced GH secretion during a two-day fast in normal men. *The Journal of Clinical Endocrinology and Metabolism*, 74(4), 757–765.

¹¹⁵ D'Andrea Meira, I., Romão, T. T., Pires do Prado, H. J., Krüger, L. T., Pires, M. E. P., & da Conceição, P. O. (2019). Ketogenic Diet and Epilepsy: What We Know So Far. *Frontiers in Neuroscience*, 13(5).

2.2. Gluten-free diet (GFD)

- It is a therapeutic elimination dietary model designed to remove gluten proteins from the diet. GFD is classified as a food-exclusion diet because its central feature is the removal of a specific dietary component rather than altering nutrient timing or macronutrient ratios.
- The diet replaces gluten-containing grains with alternatives such as rice, maize, millets, quinoa, and other naturally gluten-free foods and functions as an immune-modulating nutritional intervention that restores intestinal integrity and metabolic health.
- GFD is used to manage conditions such as celiac disease, non-celiac gluten sensitivity, and wheat allergy. In celiac disease, gluten ingestion provokes an autoimmune reaction that damages the small intestinal mucosa, leading to villous atrophy, impaired nutrient absorption, and subsequent complications such as anaemia, osteoporosis, and chronic systemic inflammation. By allowing regeneration of villi, restoration of gastrointestinal barrier integrity, and normalisation of nutrient absorption. This mucosal healing also lowers systemic inflammatory markers and helps prevent long-term complications associated with chronic malabsorption.
- Gluten-free diets are often deficient in B vitamins (thiamin, riboflavin, niacin, folate, B12), iron, and magnesium. Deficiencies in these areas are linked to symptoms like depression, anxiety, and memory loss¹¹⁶. As these nutrients are essential for neurotransmitter synthesis (such as serotonin and dopamine), oxygen delivery to the brain, regulation of nerve function and neural signalling.

3. Clinical Therapeutic Diets

Clinical therapeutic diets are dietary patterns specifically designed to prevent, manage, or treat particular diseases through targeted nutritional modifications. Unlike general healthy diets, they focus on altering specific nutrients or food components to influence disease-related physiological pathways. The principle behind these diets is that nutrition acts as a non-pharmacological therapeutic tool capable of regulating metabolic functions, reducing symptoms, and improving clinical outcomes.

3.1 Low-FODMAP diet

- The low-FODMAP diet is disease-oriented, primarily focused on reducing digestive symptoms (bloating, gas, pain) in people with irritable bowel syndrome (IBS) by restricting short-chain carbohydrates (fermentable oligosaccharides, disaccharides, monosaccharides), and polyols that are poorly absorbed in the small intestine. These carbohydrates typically ferment quickly in the gut and draw water into the bowel, causing abdominal distension, pain, and diarrhoea.¹¹⁷
- Generally, allowed foods include low-carb sources, rice, oats, corn, and low-lactose dairy. Fibre intake may drop initially due to loss of legumes or wheat, so attention to low-FODMAP fibre sources is recommended.
- While a ketogenic diet focuses on overall health improvement through low-carbohydrate intake, the low-FODMAP diet is implemented for a special or specific purpose. The diet is typically implemented in phases with strict elimination for several weeks, followed by gradual reintroduction.
- High-FODMAP foods can lead to excessive fermentation, and also increase levels of lipopolysaccharides (LPS) that are pro-inflammatory bacterial components producing metabolites like hydrogen and methane, and increasing intestinal permeability (leaky gut). When the gut barrier is compromised, inflammatory cytokines can enter the bloodstream and cross the blood-brain barrier. Thus, having a low-FODMAP diet reduces gut fermentation, lowers

¹¹⁶ Burkhart, A. (2019, October 28). The Mental Toll of the Gluten-Free Diet. Amy Burkhart, MD, RD.

¹¹⁷ Alrasheedi, A. A., Jahlan, E. A., & Bakarman, M. A. (2025). The effect of low-FODMAP diet on patients with irritable bowel syndrome. *Scientific Reports*, 15(1)low-carbohydrate.

systemic inflammation and neuroinflammation, and often results in improved focus and mental energy.

3.2. DASH (Dietary Approaches to Stop Hypertension) diet

- It is a specialised diet aimed at lowering blood pressure by limiting saturated fat and cholesterol intake through minimal consumption of fatty meats, full-fat dairy, and tropical oils. In addition, sweets and sugary beverages are also restricted, and foods rich in potassium, magnesium, calcium, fibre, and protein are encouraged.
- Notably, reducing sodium intake to 1,500 mg can help to lower blood pressure even more effectively.¹¹⁸ Resultingly, reduced sodium intake lowers fluid retention and decreases vascular resistance. Simultaneously, high potassium intake promotes sodium excretion and supports normal muscle and nerve function in blood vessels.
- DASH has been proven to lower blood pressure significantly within 2-3 weeks. It also improves cholesterol profiles by providing soluble fibre, which dissolves in water to form a gel-like substance in the small intestine. This gel binds to bile acids, which are made of cholesterol. Normally, the body recycles bile acids, but when fibre traps them, they are excreted in waste instead. To make more bile, the liver is forced to pull LDL out of the bloodstream; thus lowering LDL and improving the cholesterol profile of the blood.
- DASH provides nutrients that help the endothelium (the inner lining of blood vessels) produce Nitric Oxide (NO). Through the pathway associated with weight and glycemic control, it may also help prevent type 2 diabetes; the diet provides high fibre content, which slows the rate at which glucose enters the bloodstream.¹¹⁹ This prevents glucose excursions.
- Additionally, magnesium is a necessary cofactor for the enzymes involved in glucose metabolism required for the tyrosine kinase activity of the insulin receptor. If an individual is magnesium-deficient (common in diabetic patients), the cells ignore insulin. By restoring magnesium levels, DASH makes cells more responsive to insulin, effectively lowering blood sugar with less effort from the body.
- Higher adherence to the DASH diet is linked to a lower risk of depression. This effect is attributed to its rich supply of folate, magnesium, antioxidants, and omega-3-containing foods, which support neurotransmitter synthesis (serotonin and dopamine) and reduce inflammation.
- The DASH diet improves cognitive outcomes primarily by lowering blood pressure and improving endothelial function; it helps maintain adequate cerebral blood flow and reduces the risk of small-vessel disease linked to cognitive decline.

4. Food source or type-based Diets

Food source or food type-based dietary patterns define the key kinds of foods included or excluded by selecting specific food groups such as plant-based foods, animal-derived products or whole foods, rather than focusing on calorie levels or nutrient ratios. The rationale behind such diets is rooted in the fact that different food sources carry distinct nutrient profiles, bioactive compounds, and physiological effects that are aligned with clinical, ethical, lifestyle or even cultural considerations, with more focus on the latter objective than nutrition.

4.1. Vegetarian Diet

- A vegetarian diet is a primarily plant-based eating pattern that excludes meat, poultry, and seafood, focusing instead on fruits, vegetables, grains, nuts, and seeds. It is primarily promoted due to cultural, ethical, or environmental reasons, with variations including the consumption of

¹¹⁸ Sacks, F. M., Svetkey, L. P., Vollmer, W. M., Appel, L. J., Bray, G. A., Harsha, D., Obarzanek, E., Conlin, P. R., Miller, E. R., Simons-Morton, D. G., Karanja, N., & Lin, P. H. (2001). Effects on blood pressure of reduced dietary sodium and the dietary approaches to stop hypertension (DASH) diet. DASH-Sodium Collaborative Research Group. The New England Journal of Medicine, 344(1), 3–10.

¹¹⁹ National Heart, Lung, and Blood Institute. (2025, January 10). DASH eating plan. National Heart, Lung, and Blood Institute.

dairy and/or eggs. The variations of a vegetarian diet are as follows:

- **Lacto-ovo vegetarian diet:** The lacto-ovo vegetarian diet is characterised by the exclusion of all flesh foods (meat, poultry, and fish) while permitting the consumption of both dairy products and eggs. From a nutritional science perspective, this pattern provides access to high-biological-value proteins, essential amino acids, vitamin B12, calcium, and fat-soluble vitamins through dairy and egg intake, while maintaining high dietary fibre, phytonutrients, and antioxidant intake from plant foods. It is therefore considered a nutritionally balanced plant-forward dietary model when appropriately planned.
- **Lacto-vegetarian diet:** The lacto-vegetarian diet excludes all animal flesh and eggs but includes dairy products such as milk, yoghurt, and cheese. This dietary pattern supplies calcium, riboflavin, and vitamin B12 through dairy consumption while relying on plant sources for protein, carbohydrates, and micronutrients.
- **Ovo-vegetarian diet:** The ovo-vegetarian diet excludes meat, fish, and dairy products but allows egg consumption as the primary animal-derived food source. Eggs provide complete proteins with high digestibility, essential fatty acids, choline, and fat-soluble vitamins, thereby compensating for the absence of dairy and flesh foods.
- **Pescatarian diet:** The pescatarian diet is often considered a semi-vegetarian or flexitarian dietary pattern that excludes terrestrial animal meats like red meat and poultry but includes fish and seafood along with plant foods, and sometimes dairy and eggs. This diet provides high-quality protein, long-chain omega-3 fatty acids (EPA and DHA), iodine, and vitamin D from marine sources.
- High intake of plant foods contributes to better glycemic control, improved lipid profiles, and reduced systemic inflammation due to abundant fibre and phytochemicals.

4.2 Vegan diet

- It **excludes all animal products**, including meat, fish, poultry, eggs, milk products, cheese, yoghurt, and all ingredients derived from animals like gelatin, rennet, casein and whey. It is an entirely plant-based diet with fruits, vegetables, whole grains, legumes, nuts, seeds, plant oils, and often plant milks and meat substitutes like mock meat made from soya.
- The typical nutrient profile of such a diet is higher in carbohydrates, contributing 50-65% of daily calories.
- High fibre intake under the vegan diet improves glycemic regulation, gut microbiota diversity, and lipid metabolism, while abundant antioxidants and polyphenols reduce oxidative stress and systemic inflammation.
- A 2019 review study conducted by the Diabetes and Nutrition Study Group (DNSG) of the European Association for the Study of Diabetes (EASD) associated that both the vegetarian and vegan eating patterns helps in an approximately 28% reduction in the incidence of coronary heart disease, and 22% drop in mortality from such conditions because vegan diet lowers LDL cholesterol levels, lowers blood pressure, reduces body weight, and lowers levels of fasting glucose and triglycerides which has a combined positive effect on the health. It happens because low saturated fat intake decreases hepatic cholesterol production by reducing the activity of HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis. When hepatic cholesterol falls, the liver upregulates LDL receptors with increased clearance of circulating LDL particles. Also, plant sterols compete with cholesterol for absorption in the intestine, reducing cholesterol uptake.
- From a cautionary view, vegans may have lower bone mineral density and higher fracture rates, likely due to a combination of lower calcium intake and, even when calcium-rich plant foods are consumed, their bioavailability is often limited by organic acids like phytic acid and oxalic acid, which bind to minerals and inhibit their absorption. Deficiency in vitamin B-12 is the most critical risk, as plant foods naturally contain no reliable B12. Deficiency can cause irreversible neurological damage, megaloblastic anaemia, and cognitive changes like depression and memory loss. Apart from these, there also remains a risk of iron, zinc and omega-3, as these are

scarce in plant-based diets.

5. Region-based diets

Regional diets are traditional, location-based eating patterns shaped by geography, climate, and culture, often focusing on locally available staples. Apart from geography, socio-economic factors, primarily income, education, occupation, and environment, significantly shape dietary habits, with lower status often linked to higher consumption of energy-dense, processed foods and lower intake of nutrient-rich foods. Among all such diets, the Mediterranean diet is globally the most renowned and followed dietary pattern.

5.1 Mediterranean diet (MedDiet)

- It is a plant-rich, whole-food dietary pattern based on traditional cuisines originating in countries bordering the Mediterranean Sea, like Greece, Italy and Turkey. Its hallmarks include abundant fruits, vegetables, legumes, nuts, and whole grains, with olive oil as the primary fat. Red meat is allowed occasionally, along with wine.
- Its high content of unsaturated fats may improve lipid profiles by lowering LDL cholesterol, while maintaining or increasing high-density lipoprotein (HDL) cholesterol. The abundance of plant foods provides antioxidants and anti-inflammatory compounds that help in reducing oxidative stress and vascular inflammation. Additionally, its low glycemic load and high fibre content support glycemic control, making it beneficial in preventing metabolic syndrome and type 2 diabetes.
- The PREDIMED study, involving over 7,400 participants at high risk for cardiovascular disease, showed a 30% relative risk reduction for major cardiovascular events, with particularly significant results for stroke prevention.¹²⁰ In obese and overweight subjects, the MedDiet was shown to decrease inflammatory markers (such as TNF- α , IL-6, and myeloperoxidase), while increasing anti-inflammatory markers like adiponectin and IL-10.¹²¹
- MedDiet benefits brain health, and higher adherence slows cognitive decline and lowers the risk of mild cognitive impairment, as observed in cognitive impairments like Alzheimer's disease and vascular dementia. By reducing vascular risk factors like hypertension and diabetes, they also protect against vascular dementia. This is because polyphenols such as hydroxytyrosol and resveratrol cross the blood-brain barrier and exert antioxidant activity, preventing lipid peroxidation and protecting neuronal membranes. Chronic inflammation and oxidative damage are known to accelerate amyloid- β aggregation and tau pathology in Alzheimer's disease; thus, anti-inflammatory dietary patterns can slow these pathological processes.
- MedDiet's folate and other B-vitamins support one-carbon metabolism and the synthesis of neurotransmitters such as serotonin, dopamine, and norepinephrine; deficiency can elevate homocysteine levels, which is neurotoxic and associated with depression. Antioxidants reduce oxidative stress in brain regions involved in emotional regulation, such as the hippocampus and prefrontal cortex. Furthermore, polyphenols can modulate the gut microbiome, increasing beneficial bacteria that produce short-chain fatty acids and influence the gut-brain axis, thereby enhancing stress resilience and emotional stability.
- However, high-fat intake, associated with the Med-diet, even from healthy fats, may aggravate symptoms in individuals with gallbladder disease as it triggers gallbladder contraction through the release of the hormone cholecystikinin to aid digestion. In diseased gallbladders with stones or sludge, this contraction forces stones against the ducts, causing pain, nausea, and severe discomfort. Additionally, high intake of potassium-rich foods (fruits, vegetables) may be problematic in advanced kidney disease because damaged kidneys cannot effectively remove excess potassium from the blood, leading to a dangerous condition called hyperkalemia.

¹²⁰ Tosti, V., Bertozzi, B., & Fontana, L. (2018). Health Benefits of the Mediterranean Diet: Metabolic and Molecular Mechanisms. *The Journals of Gerontology. Series A, Biological Sciences and Medical Sciences*, 73(3), 318–326.

¹²¹ Finicelli, M., Di Salle, A., Galderisi, U., & Peluso, G. (2022). The Mediterranean Diet: An Update of the Clinical Trials. *Nutrients*, 14(14), 2956.

Elevated potassium levels can cause severe muscle weakness, irregular heartbeats, or even fatal heart attacks.

Conclusion

The various types of diets, ranging from Mediterranean and DASH diets to therapeutic approaches like ketogenic, low-glycemic, vegetarian, and culturally specific dietary models, demonstrate that no single diet is universally optimal for all populations. Instead, their effectiveness depends on individual physiological needs, genetic predisposition, lifestyle factors, disease status, socioeconomic context, and long-term sustainability.

Diets emphasising whole, minimally processed foods, balanced macronutrient distribution, adequate micronutrient density, and high intake of plant-based components are associated with reduced risks of non-communicable diseases, improved metabolic health, and enhanced cognitive and psychological outcomes. Conversely, dietary patterns characterised by high levels of refined sugars, saturated fats, sodium, and ultra-processed foods contribute significantly to the global burden of obesity, cardiovascular disease, diabetes, and inflammatory disorders.

Primer on the Differential Impact of Complete and Partial Fasting

Fasting is the deliberate restriction of food intake over a defined period and has been practised across cultures for millennia. Complete fasting, also called water fasting or absolute fasting, refers to the total elimination of caloric intake for a defined period of time. Only water and sometimes non-caloric beverages such as black coffee or plain tea are permitted. The body receives zero exogenous energy and must meet all its energetic demands through internal reserves, and is managed with electrolyte supplementation. Partial fasting encompasses a broad family of practices that share the trait of reducing rather than eliminating caloric intake. Key examples are Intermittent Fasting (IF), which is a time-restricted eating patterns, alternate Day Fasting (ADF) and dietary restriction fastings such as the Daniel Fast, which completely focuses on plant-based, whole foods, and protein-sparing modified fasting (PSMF) consisting of low calorie intake (~400-800 kcal/day) primarily from lean protein sources, designed to preserve lean body mass.

Energy source shifting: Complete vs partial fasting

The metabolic sequence during complete fasting is more aggressive and higher than the gradual shifts seen in partial fasting. This difference carries downstream consequences, from cognitive function to lean mass.

1. Glycogen Bridge

In both complete and partial fasting, the body's first response to diminishing food intake is to draw on hepatic glycogen to maintain blood glucose, while muscle glycogen continues to fuel muscle activity locally. The liver holds approximately 75-100 gm of glycogen at any time, providing roughly 300-400 kcal of readily accessible energy. Muscle glycogen is also used, although muscle glycogen cannot be directly exported into the bloodstream; it fuels the muscle itself.

Under complete fasting, hepatic glycogen is typically exhausted within 12-18 hours. At this point, blood glucose cannot be sustained by stored carbohydrates alone, and the body must urgently activate two compensatory systems: gluconeogenesis (synthesising glucose from non-carbohydrate sources) and ketogenesis (producing ketone bodies from fatty acids). Under partial fasting, if some carbohydrate is still consumed, glycogen depletion is delayed, or the impact is partial, meaning the urgency of these switches is blunted, and the hormonal signals that drive them are less extreme.¹²² Thus, complete fasting drives a sharper and faster shift toward fat oxidation. Partial fasting can achieve a similar direction of effect more gradually and sustainably without the physiological stress of total fuel withdrawal.

2. Ketosis

Ketosis is the metabolic state in which the liver produces ketone bodies, primarily beta-hydroxybutyrate (BHB), acetoacetate, and acetone from the breakdown of fatty acids. This occurs because, in the absence of sufficient glucose, acetyl-CoA, the product of fatty acid oxidation, cannot be fully funnelled into the TCA cycle (Krebs cycle), which oxidises acetyl-CoA derived from carbs, fats, and proteins, to generate energy and metabolic intermediates. Acetyl-CoA is instead rerouted into ketone synthesis.

In complete fasting, nutritional ketosis typically develops within 24-48 hours in lean individuals and can reach therapeutic levels by 48-72 hours. In partial fasting, the carbohydrate content of meals can suppress ketone production by maintaining insulin levels above the threshold needed to inhibit adipose lipolysis (metabolic breakdown of stored triacylglycerols into glycerol and free fatty acids). Time-restricted eating without explicit carbohydrate control rarely sustains significant ketosis. Partial fasting with very low carbohydrate and calorie restriction is the exception, where ketosis may

¹²² Cahill, G. F. (2006). Fuel metabolism in starvation. *Annual Review of Nutrition*, 26.

be achieved even with some caloric intake.¹²³

When complete fasting forces a rapid transition from glycogen use to gluconeogenesis and then to ketone-driven metabolism, reliance on lipolysis increases the release of glycerol and free fatty acids processed by the liver. At the cellular level, the metabolic switch encourages the removal of damaged metabolic intermediates and promotes the turnover of dysfunctional proteins and organelles, while in the brain, ketones provide a cleaner, more efficient fuel that can reduce oxidative substrate stress and may indirectly favour glymphatic/brain-waste clearance through metabolic and sleep-related mechanisms.

For women, an abrupt, energy-deficient state from prolonged complete fasting is more likely to suppress hypothalamic-pituitary-gonadal signalling, a hormonal communication pathway that regulates reproductive function, risking menstrual irregularity or temporary amenorrhea, because the body down-prioritises reproductive function when energy availability is low.

Protein integrity and lean mass preservation - Complete vs Partial fasting

One of the most practically significant differences between complete and partial fasting is the risk to skeletal muscle, a risk that escalates sharply with complete fasting as duration extends. The body maintains a circulating amino acid pool sourced from dietary protein, and when dietary input ceases, it shifts to catabolism of structural proteins, primarily skeletal muscle. During the early phase of complete fasting (first 24-48 hours), gluconeogenic demand is largely met through alanine and glutamine released from muscle. Alanine is particularly important as it is transaminated in the liver to form pyruvate, which is then converted to glucose via gluconeogenesis. Thus, even in the absence of any dietary protein, glucose production continues but at the cost of muscle mass. During complete fasting, the body is lacking dietary amino acids and increasingly turns to muscle protein to maintain blood sugar and support vital functions.

Partial fasting, especially when it includes adequate dietary protein (even in modest amounts), substantially preserves muscle mass. Even 600-800 kcal per day with a protein focus can provide enough amino acids to limit the body's need to cannibalise muscle tissue. This is why modified fasting approaches such as protein-sparing modified fasts were developed to capture fat loss benefits while protecting lean body mass. For most people concerned with body composition, partial fasting with a protein emphasis is significantly more protective of muscle than complete fasting.¹²⁴

In total fasting, the accelerated gluconeogenic demand increases proteolysis, which is the enzymatic breakdown of proteins into smaller polypeptides or amino acids through the hydrolysis of peptide bonds. The resulting amino acids are further metabolised. When they are broken down for energy or due to excess protein, the amino group is removed in a process called deamination. The deamination process releases nitrogen in the form of ammonia, imposing a whole-body burden of nitrogenous waste (urea production) that the liver and kidneys clear.

Partial fasting protocols that include dietary protein blunt the waste (nitrogen) production, and cellular proteostasis is better maintained, inflammatory markers tend to fall without the same proteolytic stress, mitochondria retain protein turnover substrates, and women are far less likely to experience disruptions to the menstrual cycle when sufficient calories and protein are consumed during feeding windows.

¹²³ Westman, E. C., Feinman, R. D., Mavropoulos, J. C., Vernon, M. C., Volek, J. S., Wortman, J. A., Yancy, W. S., & Phinney, S. D. (2007). Low-carbohydrate nutrition and metabolism. *The American Journal of Clinical Nutrition*, 86(2), 276–284.

¹²⁴ Stiegler, P., & Cunliffe, A. (2006). The Role of Diet and Exercise for the Maintenance of Fat-Free Mass and Resting Metabolic Rate During Weight Loss. *Sports Medicine*.

Hormonal Signalling - Complete vs Partial fasting

The hormonal landscape of fasting is not uniform; the degree to which insulin falls and glucagon rises determines the extent of metabolic shift. This is where complete and partial fasting diverge most sharply. Women's reproductive hormones are particularly sensitive to chronic alterations in insulin and GH signalling: prolonged elevation of the catabolic hormone profile can suppress ovulatory cycles, whereas intermittent or partial fasting that avoids sustained hormonal extremes usually preserves menstrual function.

1. Insulin

Complete fasting causes a sharper drop in insulin levels, which is a key driver of fat burning. It also triggers a more pronounced rise in stress hormones (cortisol and adrenaline), which mobilise energy stores but also contribute to the muscle breakdown mentioned above. Hunger hormones such as ghrelin tend to spike significantly, which is why extended complete fasting is harder to adhere to.

Partial fasting lowers insulin and promotes fat burning as well, but more modestly, and the cortisol response is typically less intense. Hunger is more manageable, particularly once the body adapts to the eating pattern over a few weeks. Time-restricted eating in particular appears to improve insulin sensitivity over time with relatively low physiological stress. Both approaches improve insulin sensitivity and metabolic markers, but partial fasting achieves this with less hormonal disruption, making it more sustainable for regular, long-term use.

2. Glucagon

Glucagon, the counter-regulatory hormone to insulin released by the pancreatic alpha cells, rises in response to falling blood glucose and acts on the liver to stimulate glycogenolysis and gluconeogenesis. Glycogenolysis is the rapid breakdown of stored glycogen into glucose, acting as the primary, immediate response to low blood sugar. Conversely, gluconeogenesis is the slower, long-term synthesis of new glucose from non-carbohydrate sources like amino acids, lactate, and glycerol. Complete fasting produces a persistently high glucagon-to-insulin ratio; partial fasting produces a fluctuating ratio that may or may not cross the threshold for sustained fat mobilisation.

During complete fasting, no food enters the body, meaning there is no dietary glucose or amino acid stimulation of insulin secretion. As blood glucose levels gradually fall, pancreatic β -cells reduce insulin secretion, while α -cells increase glucagon release to maintain blood glucose for vital organs such as the brain. Because there are no intermittent nutrient inputs, insulin remains consistently low, while glucagon remains consistently elevated. This creates a persistently high glucagon-to-insulin ratio in complete fasting, which signals the body that it must rely on stored fuels and thus, increase in fat mobilisation for fulfilling energy needs.

3. Growth Hormone (GH)

Fasting strongly stimulates GH secretion, particularly during sleep, and GH promotes lipolysis while serving a protein-sparing function by reducing peripheral amino acid oxidation. This effect is observed in both complete and significant partial fasting, though it is most pronounced during complete fasting. The GH response is one reason why some degree of lean mass protection occurs even during extended complete fasting.¹²⁵

Micronutrient Intake

Immune cells depend on micronutrients (zinc, selenium, B-vitamins) for normal function. Therefore, prolonged micronutrient deficits from complete fasting can impair immune responses in humans; partial fasting with adequate food windows maintains micronutrient supply and supports normal immune surveillance.

¹²⁵ Patterson, R. E., & Sears, D. D. (2017). Metabolic Effects of Intermittent Fasting. *Annual Review of Nutrition*, 37(1).

During complete fasting, the supply of all micronutrients, like vitamins, minerals, and trace elements, is effectively suspended. Water-soluble vitamins (B-complex, vitamin C) have limited tissue storage and begin to deplete within days. The B vitamins are of particular concern because they serve as coenzymes in energy metabolism itself. Thiamine (B1) is essential for pyruvate dehydrogenase activity, which is a vital enzyme complex that converts pyruvate into Acetyl-CoA, acting as the key link between sugar breakdown (glycolysis) and energy production (Krebs cycle). Riboflavin (B2) and niacin (B3) are central to the electron transport chain.

Electrolyte management is perhaps the most clinically urgent micronutrient concern in complete fasting. As insulin decreases, the kidneys, which are stimulated by insulin to retain sodium, begin excreting sodium more rapidly. This renal sodium loss drags water with it, contributing to the initial rapid weight loss seen in complete fasting. Electrolytes such as sodium, potassium, and magnesium can also decline if losses are not replaced, potentially causing muscle cramps or arrhythmias. This is why unsupervised extended water fasting is medically discouraged, and supervised fasting protocols always include electrolyte supplementation.

Partial fasting maintains some micronutrient intake. The quality and completeness of that intake, however, depend entirely on food choices during the eating window or non-fasting days. A person doing 16:8 intermittent fasting who eats ultra-processed food predominantly may be in a worse micronutrient position than someone doing a disciplined, supervised, complete fast (5:2 cycle) that prioritises nutrient-dense, whole foods.

Thus, partial fasting is not automatically micronutrient-safe; it simply provides the opportunity to maintain micronutrient sufficiency, which complete fasting does not. Thus, partial fasting is far less likely to cause micronutrient depletion, provided food choices are thoughtful. Electrolyte loss through urine and sweat during complete fasting warrants attention even in short-term protocols.¹²⁶

Gut-Microbiome modulation

The gut microbiome derives its sustenance primarily from dietary substrates, particularly fermentable fibres and complex carbohydrates that reach the colon undigested. During complete fasting, the absence of dietary substrates can alter the relative abundance of several gut microbial taxa, including fibre-fermenting species such as *Lactobacillus* and *Bifidobacterium*. Complete fasting is not entirely detrimental to gut microbiome dynamics as post-fasting refeeding can reset and partially diversify the microbiome if refeeding is done with fibre-rich, plant-forward foods. The fasting period appears to function as a form of microbial ecology reset during substrate scarcity. However, if refeeding returns to poor dietary patterns, the recovery of beneficial taxa is limited.¹²⁷

Partial fasting, especially protocols that maintain fibre intake, produces a more controlled and sustainable modulation of the microbiome. Time-restricted eating has been associated in some studies with increases in taxa such as *Akkermansia muciniphila* (linked to gut barrier integrity and reduced metabolic inflammation) and *Faecalibacterium prausnitzii* (a major butyrate producer) as it creates a daily, rhythmic fasting-feeding cycle. This selectively favours mucin-degrading and fibre-fermenting bacteria without depriving them of substrates for prolonged periods. Therefore, short-term complete fasting may offer gut rest benefits, but partial fasting with high dietary fibre is better for maintaining a healthy and diverse gut microbiome over time.¹²⁸

¹²⁶ Hanssen, N. M. J., & Nieuwdorp, M. (2021). The gut microbiome, a hidden endocrine organ with widespread metabolic implications? *Best Practice & Research Clinical Endocrinology & Metabolism*, 35(3), 101543.

¹²⁷ Obesity Updates: Understanding Obesity as Disease and Intermittent Fasting. (2022). *The Singapore Family Physician*, 48(1).

¹²⁸ Stevens, H. C. (2012). Fructose impacts on gut microbiota and obesity by A. N. Payne et al. *Obesity Reviews*, 13(12), 1182–1183.

For women, microbiome-driven metabolism of sex-hormone precursors can be sensitive to large dietary swings; thus, complete fasting may more readily perturb hormone recycling and menstrual regularity, while thoughtfully structured partial fasting tends to preserve microbiome functions that support reproductive hormonal balance.

Conclusion

Complete fasting delivers a more total and abrupt metabolic shift: glycogen is rapidly depleted, ketosis sets in within 24-48 hours, insulin bottoms out, and the body enters a deeply catabolic state that is protein-sparing only once ketosis is fully established. The micronutrient supply is completely suspended, and the gut microbiome undergoes significant, if partially reversible, restructuring. Partial fasting induces the same directional changes, like lower insulin, mobilisation of fat stores, and reduced inflammatory tone, but more gradually and with greater nutritional continuity. For most individuals, partial fasting represents a more sustainable and safer way to achieve the metabolic benefits associated with fasting.

Primer on Nutritional Intake Benchmarks

The measurement of nutrition helps in understanding the intake (what is consumed), requirement (what is needed) and utilisation (what is being used by the body). Typically, individuals measure nutritional intake by its source, which is the consumed food; this forms the starting point for nutrient measurement. However, food is only the vehicle for nutrients and energy, which finally interacts with the human body.

The following is the general hierarchy of food measurement:

- **Food:** The siloed measurement of food items (grains, fruits, vegetables, dairy, etc.) in grams, portions or servings, though relevant, conveys little information about the nutritional and energy content.
- **Nutrients:** They constitute the chemical composition of food in the form of macronutrients (carbohydrates, proteins and fats) and micronutrients (vitamins and minerals). They translate food intake into biological inputs, generally in grams or milligrams, making them comparable to the standard RDA and other nutrition intake requirements.
- **Energy:** It is the fuel derived from nutrients, expressed in kilocalories or kilojoules. It helps in regulating body weight and metabolism.

Human energy needs and energy balance

Energy requirement is the amount of food energy needed to balance energy expenditure for maintaining body composition and a level of physical activity consistent with long-term good health and requirements. Beyond physical activity, this energy is also needed to fuel metabolic processes and bodily functions, including growth and development and deposition of tissues.¹²⁹ The concept of **energy balance** is based on the fundamental thermodynamic principle that energy cannot be created or destroyed, only transformed. Literally, it means the state achieved when the energy intake equals energy expenditure. Humans take in energy through food and other edibles, and expend energy by maintaining homeostasis (measured through resting metabolic rate), metabolising nutrients (known as the thermic effect of food) or performing activities like exercising (thermic effect of physical activity) or non-exercise movements like walking (non-exercise activity thermogenesis). The energy balance can be positive, negative or neutral, depending upon the intake and expenditure of energy.

- **Positive energy balance:** In this situation, energy intake exceeds energy expenditure, which causes body mass gain, mostly attributed to body fat.
- **Negative energy balance:** In negative energy balance, energy expenditure exceeds energy intake, resulting in a loss of body mass, mainly body fat.
- **Neutral energy balance:** In this situation, the energy expenditure equals the energy intake, maintaining equilibrium.

Metrics for the measurement of energy

Energy expenditure (EE) is the total amount of energy the body converts from chemical substrates (carbohydrates, fats, proteins, alcohol) into ATP and heat over a given period of time, usually expressed in kilocalories per day (kcal/day) or kilojoules per day (kJ/day).

Total energy expenditure (TEE) is the energy burned by an organism daily, and directionally guides energy requirements to maintain equilibrium or achieve a positive balance, depending upon the individual goal. It is determined by the sum of the following 3 components:

- **Basal metabolic rate (BMR):** It is the rate of energy expenditure of a person at rest, eliminating the variable effect of physical activity. BMR accounts for approximately 60% of the daily energy

expenditure. Thus, it includes energy used for normal body cellular homeostasis, cardiac function, and the function of other organs and tissues. It is related to body weight and approximately calculated as $\text{BMR (kcal/day)} = 24 \times \text{body weight (kg)}$ in the case of males.¹³⁰

- **Thermic effect of food (TEF):** The thermic effect of food represents the energy used for the ingestion and digestion of food, along with the absorption, transport, and storage of nutrients. The contribution of each process to the TEF depends on food characteristics, including the type, macronutrient composition, and energy content. Other factors, such as the autonomic nervous balance (sympathetic versus parasympathetic), may also play a role.¹³¹ Overall, it represents 5% to 15% of TEE.
 - It can be divided into two distinct phases: the cephalic and the gastrointestinal phases. The cephalic phase is related to sympathetic nervous system action, which is activated by food sensory properties, while the gastrointestinal phase is characterised by ATP consumption during the absorption and utilisation of nutrients.¹³²
- **Energy expenditure of activity (activity thermogenesis):** The energy expenditure from activity refers to the energy required by the body to move (i.e., perform muscular work) during non-exercise activity thermogenesis and voluntary activity (e.g., exercise, sports). It varies significantly as a proportion of TEE, and typically ranges from 15% (sedentary individuals) to 50% (physically active individuals). It is also impacted by variables like age, sex, body size and composition, genetic traits, etc.¹³³

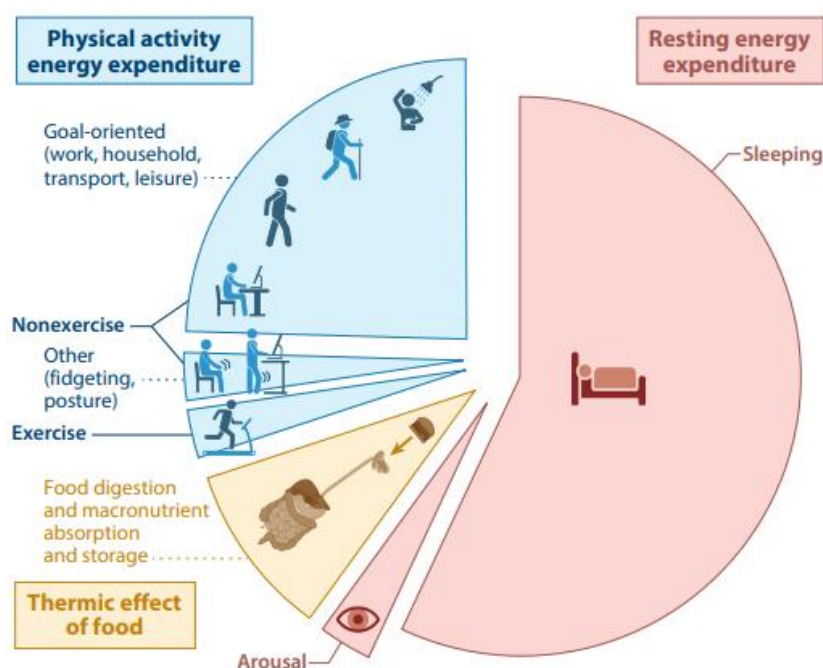


Figure: Components of energy expenditure

Major nutrient intake benchmark standards

• Estimated Average Requirement (EAR)

It is the daily intake value that is estimated to meet the requirement, as defined by the specified indicator or criterion of adequacy, of half of the apparently healthy individuals in a life stage or

¹³⁰ Elsevier's Integrated Review Biochemistry (2nd ed.). (n.d.).

¹³¹ Energy expenditure in humans: principles, methods, and changes throughout the life course. (2024). Annual Review of Nutrition, 51–76.

¹³² Pinheiro Volp, A. C., Esteves De Oliveira, F. C., Duarte Moreira Alves, R., Esteves, E. A., & Bressan, J. (2011). Energy expenditure: components and evaluation methods. Nutr Hosp, 26(3), 430–440.

¹³³ National Academies Press (US). (2023). *Factors affecting energy expenditure and requirements*. Dietary Reference Intakes for Energy - NCBI Bookshelf.

gender group.¹³⁴ It represents the median of the distribution of the nutrient requirements in all people.¹³⁵

- **Recommended Dietary Allowances (RDA)**

It is the average daily dietary intake level that is sufficient to meet the nutrient requirements of nearly all healthy individuals in a particular life stage and gender group. The RDA is intended to be used as a goal for daily intake by individuals.⁷ It represents the 97.5th percentile of the distribution of the nutrient requirements in all healthy individuals.⁸

The process for determining RDA depends on EAR. Before setting the EAR, a specific criterion of adequacy is selected. When selecting the criterion, sometimes the reduction of disease risk is also considered, along with other health parameters. The RDA is set at the EAR plus twice the standard deviation (SD), if known [$RDA = EAR + 2 SD$]. If data about variability in requirements are insufficient to calculate an SD, a coefficient of variation for the EAR of 10% is ordinarily assumed [$RDA = 1.2 \times EAR$].¹³⁶

- **Adequate Intake (AI)**

If sufficient scientific evidence is not available to calculate an EAR, a reference intake called an adequate intake (AI) is provided instead of an RDA. AI is based on experimentally derived intake levels by a group (or groups) of healthy people. The AIs for children and adults are expected to meet or exceed the amount needed to maintain a defined nutritional state or criterion of adequacy in essentially all members of a specific, apparently healthy population. Examples of defined nutritional states include normal growth, maintenance of normal circulating nutrient values, or other aspects of nutritional well-being or general health.⁷

The methods for deriving AI differ substantially among nutrients and life stages for the same nutrients.

- In some cases, AI is based on the observed mean intakes of groups maintaining health and nutritional status consistent with an apparent low incidence of inadequacy.
- In other cases, AI can be derived from the lowest level of intake at which all subjects in an experimental study met the criterion of adequacy. This is different from (and generally lower than) the group mean intake that is consistent with all subjects meeting the criterion of adequacy.
- AI can also be estimated as an approximation of intake in a group with knowledge of the actual requirements of only a few individuals.⁷

However, there is much less certainty about the AI value than about the RDA value.

- **Tolerable Upper Intake Level (UL)**

It is the highest level of daily nutrient intake, which is likely to pose no risk of adverse health effects for almost all apparently healthy individuals in the specified life stage group. As intake increases above the UL, the potential risk of adverse effects may increase.⁷

- **Daily Values (DV)**

Instead of an intake value determinant, DV is a communication-oriented reference number, which helps consumers determine if a food contains a lot or a little of a specific nutrient, based on the RDA or AI for that nutrient. DVs are used on Nutrition Facts panels, but don't take into account age, sex, or other factors affecting one's daily calorie needs. They're presented as

¹³⁴ DIETARY REFERENCE INTAKES FOR Water, Potassium, Sodium, Chloride, and Sulfate. (2005).

¹³⁵ ICMR - NIN. (2020). A brief note on nutrient requirements for Indians, the recommended Dietary Allowances (RDA) and the Estimated Average Requirements (EAR).

¹³⁶ National Academies Press (US). (1998). *What are Dietary Reference Intakes?* Dietary Reference Intakes - NCBI Bookshelf.

percentages of total daily intake; for example, in the USA, it is calculated for an average person eating 2,000 calories a day.¹³⁷

- **Energy Percentage (% of Total Energy Intake)**

Often denoted as %E, %TEI (Total Energy Intake) or E%, it is a standardised metric used to describe the contribution of a specific nutrient, food group, or meal to an individual's or group's total daily calorie consumption. It is calculated as: $\text{Energy\%} = (\text{Calories from nutrient} / \text{Total daily calories}) * 100$.⁷

¹³⁷ Harvard Health. (2023). What are RDAs, DRIs, and DVs?

Note on Important Enzymes For The Body

The chemical digestion of food is fundamentally an enzyme-driven process. Macronutrients arrive in the GI tract as large complex polymers with thousands of glucose residues, proteins composed of polypeptide chains containing hundreds of amino acid residues, and triglycerides packed into lipid droplets, most of which can cross the intestinal epithelium in their native form. Thus, the body deploys a relay of hydrolytic enzymes across anatomically distinct compartments, each optimised to the pH, substrate milieu, and absorptive need of its location. At each stage, products of one enzyme become the substrate for the next.

A. Oral phase: The first stage

Saliva contains enzymes such as salivary amylase, which begins the breakdown of carbohydrates. The oral microbiota contains bacteria capable of metabolising glycoproteins and sugars present in the oral cavity, one of the steps in food processing.

1. Salivary Alpha-Amylase (Ptyalin)

Salivary alpha-amylase (sAA) is a calcium-dependent digestive metalloenzyme, a type of enzyme that works as a biological catalyst to speed up chemical reactions in the body, and is secreted by the parotid, submandibular, and sublingual glands at concentrations of about 100-300 mg/L of saliva, where it initiates carbohydrate digestion in the oral cavity under near-neutral pH conditions with chloride ions acting as cofactors.

Functionally, sAA hydrolyses glycosidic bonds in starch and glycogen, producing maltose, maltotriose, and branched limit dextrins, which are then further digested by intestinal enzymes. After swallowing, the enzyme continues to act transiently within the food bolus in the stomach until gastric acid lowers the pH below 4, by which point approximately 30-40% of dietary starch may already be hydrolysed, thereby reducing the digestive burden on pancreatic amylase. Notably, the AMY1 gene, responsible for the amount of salivary amylase in humans, exhibits significant copy-number variation across populations, with individuals consuming traditionally high-starch diets possessing more gene copies and consequently higher salivary amylase levels, which accelerates oral starch breakdown and is associated with improved glycemic control, including reduced postprandial glucose spikes due to more efficient early carbohydrate predigestion.

In neonates, sAA is nearly absent (<10% of adult levels), making starch digestion almost entirely dependent on pancreatic amylase and on mammary gland-derived amylase in breast milk.¹³⁸ By age 2-3, sAA increases to near-adult levels as dietary starch increases. Adult activity plateaus and remains relatively stable through middle age. In older adults (>65 years), sAA may decline due to glandular atrophy, reduced autonomic stimulation, and polypharmacy (anticholinergic drugs reduce salivary secretion). Reduced sAA in the elderly shifts greater carbohydrate hydrolysis load to the pancreas and brush border, and may contribute to bloating and carbohydrate malabsorption if compounded by other deficiencies.¹³⁹

2. Lingual Lipase

Lingual lipase, secreted by the glands of von Ebner on the posterior tongue, provides the first step in triglyceride hydrolysis and is uniquely acid-stable, allowing continued activity in the stomach. Unlike most digestive enzymes, which require near-neutral pH, lingual lipase is optimally active at pH 3.5-6.0, positioning it perfectly for gastric action.

¹³⁸ Zoppi, G., Andreotti, G., Pajno-Ferrara, F., Njai, D. M., & Gaburro, D. (1972). Exocrine Pancreas Function in Premature and Full Term Neonates. *Pediatric Research*, 6(12), 880–886.

¹³⁹ Holt, P. R. (2007). Intestinal Malabsorption in the Elderly. *Digestive Diseases*, 25(2), 144–150.

In adults, lingual and gastric lipases together account for approximately 10-30% of triglyceride hydrolysis before pancreatic lipase predominates in the intestine. However, lingual lipase has disproportionate importance in neonatal fat digestion, particularly for medium-chain triglycerides in breast milk. Neonates have markedly reduced pancreatic lipase secretion, approximately 5-10% of adult levels at birth, making lingual and gastric lipases critical for fat absorption in the first months of life.¹⁴⁰

B. Gastric phase: Acid-activated digestion

Upon the food bolus reaching the stomach lumen, HCl initiates the conversion of pepsinogen to pepsin, after which active pepsin catalyses further pepsinogen activation.

1. Pepsin

Pepsin is the prototypic acid protease of the stomach because it is the primary digestive enzyme responsible for initiating protein breakdown in a highly acidic (pH 1.5-2.5) gastric environment. As an aspartic protease, it uses aspartate residues in its active site to hydrolyse peptide bonds. Its role also includes the liberation of peptides that serve as signals for downstream digestive regulation. Protein fragments generated by pepsin stimulate G-cells in the gastric antrum to secrete gastrin, which drives further HCl production and pepsinogen secretion. Pepsin also degrades collagen crosslinks in meat and connective tissue, releasing heme iron and minerals bound within food matrices, indirectly facilitating their absorption. Pepsin becomes irreversibly denatured when gastric pH rises above 6.

Females demonstrate lower basal gastric acid output (and therefore lower pepsin activity) compared to males, a difference with clinical and nutritional implications. Basal acid output (BAO) in males averages 2-4 mEq/hour (milliequivalents per hour) compared to 1-2 mEq/hour in females. In terms of gastric juice production, this represents the rate of hydrochloric acid (HCl) secretion by the stomach, measuring the total amount of hydrogen ions produced over one hour. Estrogen inhibits parietal cell acid secretion via prostaglandin-mediated mechanisms, while testosterone appears to upregulate acid output. Practically, females may have somewhat less efficient peptic protein digestion, though this is compensated for by pancreatic proteases.

Pepsin deficiency clinically manifests as protein maldigestion but rarely as an isolated deficiency; it is almost always secondary to hypochlorhydria. Consequences include vitamin B12 deficiency because pepsin is required to dissociate food-bound B12 from its protein carrier (crystalline B12 supplements bypass this step); iron deficiency, as pepsin-released non-heme iron remains oxidised and poorly absorbable; and reduced release of zinc and magnesium from food matrices. Pepsin excess per se does not cause GI toxicity but contributes to laryngopharyngeal reflux injury when regurgitated.

2. Gastric Lipase

Gastric lipase is an acid-stable preduodenal lipase secreted by chief cells of the gastric fundus, acting synergistically with lingual lipase to accomplish 20-30% of total fat digestion before contents reach the duodenum. Gastric lipase preferentially hydrolyses short- and medium-chain triglycerides, with optimal activity at pH 3-6. Unlike pancreatic lipase, it does not require bile salts or colipase, making it particularly important in settings where these cofactors are unavailable, like the neonatal period, cholestatic disease, and cystic fibrosis, where pancreatic lipase secretion is severely impaired.

The products of gastric lipase - 1- and 2-diacylglycerols and free fatty acids serve as potent

¹⁴⁰ Laugier, R., Bernard, J.-P., Berthezene, P., & Dupuy, P. (1991). Changes in Pancreatic Exocrine Secretion with Age: Pancreatic Exocrine Secretion Does Decrease in the Elderly. *Digestion*, 50(3-4), 202–211.

emulsifiers that facilitate subsequent pancreatic lipase action in the duodenum. As the gastric chyme enters the duodenum, the diacylglycerols produced by gastric and lingual lipase enhance emulsification of the remaining triglyceride bulk, increasing the surface area available to pancreatic lipase. This mechanistic synergy is known as sequential enzymatic synergy.

Gastric lipase output is proportionally higher in neonates relative to adult output, making it critical for neonatal fat digestion. At birth, pancreatic lipase is virtually absent (<1% of adults), while gastric lipase is functional. In combination with bile salt-stimulated lipase (BSSL) present in human breast milk, neonates achieve adequate fat digestion through a pathway entirely distinct from adult pancreatic lipase-dependent digestion.

C. Pancreatic phase: The enzymatic powerhouse

The exocrine pancreas is the most important organ for intraluminal digestion, functioning as the body's primary biochemical digestive factory.' Each day, it secretes approximately 1.5-2 litres of pancreatic juice into the duodenum through the pancreatic duct. This secretion contains two critical components: bicarbonate ions, which neutralise acidic chyme arriving from the stomach and raise duodenal pH from ~2 to a near-neutral range of 6-7; and a highly concentrated mixture of digestive hydrolases, which break down carbohydrates, proteins, lipids, and nucleic acids. Pancreatic secretion is tightly regulated hormonally: secretin, released by S-cells in response to acid, stimulates bicarbonate secretion, while cholecystikinin (CCK), released from I-cells in response to fats and peptides, stimulates enzyme secretion. To prevent autodigestion of pancreatic tissue, most enzymes are secreted as inactive enzymes (zymogens), which are activated sequentially only after entering the duodenal lumen, forming a precisely controlled enzymatic cascade essential for safe digestion.

Pancreatic protease secretion exhibits modest sex-based differences influenced primarily by hormonal regulation. Males often demonstrate somewhat higher CCK-stimulated protease secretion, potentially linked to testosterone-associated increases in pancreatic acinar cell mass. However, findings vary across populations.¹⁴¹

1. Pancreatic Alpha-Amylase

Pancreatic alpha-amylase is the principal enzyme responsible for carbohydrate digestion in the small intestine and is secreted in an active form, unlike most pancreatic enzymes. Produced from the AMY2A and AMY2B genes, it is structurally stabilised by calcium ions and activated allosterically by chloride ions. Its optimal pH (6.7-7.0) precisely matches the bicarbonate-buffered environment of the duodenum, ensuring maximal catalytic efficiency. The magnitude of its secretion far exceeds that of salivary amylase: a single meal can stimulate pancreatic amylase output 10-100 times greater than total salivary contribution, making it the dominant enzyme in starch digestion despite acting later in the digestive sequence. Importantly, its products are not directly absorbable and require further hydrolysis by brush-border disaccharidases before glucose absorption can occur.¹⁴²

2. Pancreatic Lipase and Colipase

Pancreatic lipase is the most critical enzyme for lipid digestion, responsible for approximately 70-80% of dietary triglyceride hydrolysis in healthy individuals. Its activity is uniquely dependent on a protein cofactor called colipase, which is secreted as procolipase and activated by trypsin.¹⁴³ Colipase anchors lipase at the lipid-water interface of bile salt-stabilised emulsions, preventing bile

¹⁴¹ Feldman, M., & Richardson, C. T. (1986). Role of thought, sight, smell, and taste of food in the cephalic phase of gastric acid secretion in humans. *Gastroenterology*, 90(2), 428-433.

¹⁴² Amann, S. T., Bishop, M., Curington, C., & Toskes, P. P. (1996). Fecal Pancreatic Elastase Is Inaccurate in the Diagnosis of Chronic Pancreatitis. *Pancreas*, 13(3), 226-230.

¹⁴³ DeNigris, S. J., Hamosh, M., Kasbekar, D. K., Lee, T. C., & Hamosh, P. (1988). Lingual and gastric lipases: species differences in the origin of prepancreatic digestive lipases and in the localization of gastric lipase. *Biochimica et Biophysica Acta (BBA) - Lipids and Lipid Metabolism*, 959(1), 38-45.

salts from displacing lipase from its substrate. Thus, fat digestion depends on a tri-component system: pancreatic lipase, colipase, and bile salts. Failure of any component, whether due to liver disease (reduced bile salts), pancreatic disease (reduced enzyme secretion), or ileal disease (bile salt loss), leads to fat malabsorption and excretion of excess fat in the stool.

Significant sex-based differences exist in fat digestion efficiency. Females absorb dietary lipids more efficiently due to hormonal regulation of bile secretion and pancreatic responsiveness. Also, estrogen increases hepatic cholesterol secretion into bile, expanding the bile salt pool and enhancing micelle formation.¹⁴⁴

3. Pancreatic Proteases: Trypsin, Chymotrypsin, Elastase, Carboxypeptidases

The pancreatic proteases form an interconnected activation cascade, with trypsin acting as the master regulator that sequentially activates all other pancreatic zymogens. Enterokinase (enteropeptidase) on the duodenal brush border cleaves trypsinogen to trypsin, removing the inhibitory propeptide. Trypsin then autocatalytically activates more trypsinogen and sequentially splits: chymotrypsinogen to chymotrypsin; proelastase to elastase; procarboxypeptidase A/B to carboxypeptidase A/B; procolipase to colipase; and phospholipase A2 to phospholipase A2.

- **Trypsin**

Trypsin is the most abundant pancreatic protease and plays a crucial regulatory role by activating other pancreatic zymogens, making it the central enzyme in the proteolytic cascade. Its specificity produces peptide fragments enriched in lysine and arginine, which act as potent stimulators of CCK release, thereby creating a luminal feedback mechanism that sustains pancreatic enzyme secretion. Trypsin also degrades pancreatic secretory trypsin inhibitor (SPINK1), a protective molecule that prevents premature enzyme activation within the pancreas.

- **Chymotrypsin**

Chymotrypsin complements trypsin by targeting peptide bonds adjacent to aromatic and bulky hydrophobic amino acids, including phenylalanine, tyrosine, tryptophan, and leucine. Together, trypsin and chymotrypsin ensure nearly complete cleavage of internal peptide bonds in dietary proteins, reducing them into oligopeptides typically 2-8 amino acids long, which are optimal substrates for subsequent brush-border peptidases. This coordinated enzymatic action ensures efficient protein digestion regardless of the structural diversity of dietary proteins.

- **Elastase**

Pancreatic elastase is uniquely specialised for digesting elastin, a highly cross-linked structural protein abundant in connective tissue and certain plant cell wall components. It preferentially cleaves peptide bonds adjacent to small neutral amino acids such as alanine, glycine, and valine, matching elastin's repetitive glycine-alanine-rich sequences. Clinically, faecal elastase-1 (FE-1) measurement is widely used to assess exocrine pancreatic function, with low levels indicating pancreatic insufficiency. Unlike many digestive enzymes, elastase remains stable in stool for several days, making it particularly practical for non-invasive testing of pancreatic function.

- **Carboxypeptidases A and B**

Carboxypeptidases A and B are exopeptidases that complete pancreatic protein digestion by sequentially removing single amino acids from the C-terminal ends of peptides generated by endopeptidases. Carboxypeptidase A preferentially releases aromatic and aliphatic amino acids, whereas carboxypeptidase B specifically releases basic residues such as lysine and arginine. Carboxypeptidase A functions as a zinc-dependent metalloenzyme, highlighting the importance of trace minerals in digestive enzymology. The combined activity of these enzymes produces a mixture of free amino acids and small peptides, which are then further processed by brush-border and intracellular peptidases before absorption.

¹⁴⁴ Castiñeira-Alvariño, M., Luaces-Regueira, M., Nieto, L., Iglesias-Garcia, J., Larino-Noia, J., & Dominguez-Munoz, E. (2011). Association of Fat Consumption With Clinical Manifestations, Diagnosis and Severity of Chronic Pancreatitis. *Gastroenterology*, 140(5), S-859.

D. Small Intestinal brush-border phase: Final hydrolysis & absorption

The intestinal brush border is the final site of digestion before nutrients are absorbed into the body. It consists of tiny finger-like projections called microvilli on the surface of intestinal cells (enterocytes). These microvilli contain enzymes attached directly to the cell membrane. Unlike enzymes floating freely in the gut, brush-border enzymes work right next to transporters that immediately absorb the nutrients they produce. From a nutrition perspective, they convert partially digested carbohydrates and proteins into small absorbable units such as glucose, amino acids, and small peptides.

1. Enterokinase (Enteropeptidase)

Enterokinase is not directly involved in digesting food, but it is essential because it activates protein digestion. Located in the upper small intestine, it converts inactive trypsinogen into active trypsin, which then activates other pancreatic proteases. Without enterokinase, protein digestion cannot begin properly, even if the pancreas produces normal enzymes. In nutritional terms, its importance lies in its role as the starter enzyme that switches on the entire protein-digesting system.

2. Lactase-Phlorizin Hydrolase (LPH)

Lactase is the enzyme responsible for digesting lactose, the main sugar in milk. It breaks lactose into glucose and galactose, which can then be absorbed. From a nutrition viewpoint, lactase is clinically important because its levels naturally decline after childhood in many populations (lactase non-persistence).¹⁴⁵ When lactase levels are low, lactose reaches the colon undigested, where bacteria ferment it, causing symptoms like bloating, gas, and diarrhoea. Many individuals with low lactase can still tolerate small amounts of lactose or fermented dairy products like yoghurt, where bacteria have already partially broken down lactose. Celiac disease, Crohn's disease, viral gastroenteritis, tropical sprue, or short bowel syndrome occur due to destruction of the lactase-expressing tip-of-villus enterocytes (the cells with the highest LPH expression). Lactase excess has no known pathological consequence; increased lactase activity simply ensures complete lactose digestion.

3. Sucrase-Isomaltase

Sucrase-isomaltase is a dual-function enzyme that helps digest common dietary sugars. It breaks down sucrose (table sugar) into glucose and fructose and also helps digest branched starch fragments produced during earlier digestion. From a nutritional perspective, this enzyme plays a major role because sucrose is a large part of many diets. Deficiency is rare but can cause diarrhoea and poor tolerance of sugary foods, particularly in infancy.

4. Maltase-Glucoamylase

Maltase-glucoamylase is the main enzyme responsible for converting the final products of starch digestion into glucose. It breaks down small starch fragments like maltose and dextrins into individual glucose molecules. Nutritionally, this enzyme is crucial because it completes carbohydrate digestion, enabling efficient glucose absorption. Some diabetes medications work by slowing this enzyme's activity to reduce post-meal blood sugar spikes.

4.5 Aminopeptidase N (APN, CD13)

Aminopeptidases are brush-border enzymes located on the microvilli membrane of enterocytes (intestinal absorptive cells) in the small intestine. Their primary role is to carry out the final step of protein digestion after pancreatic enzymes have already broken proteins into smaller peptides. During digestion, pancreatic proteases such as trypsin and chymotrypsin convert dietary proteins

¹⁴⁵ Tishkoff, S. A., Reed, F. A., Ranciaro, A., Voight, B. F., Babbitt, C. C., Silverman, J. S., Powell, K., Mortensen, H. M., Hirbo, J. B., Osman, M., Ibrahim, M., Omar, S. A., Lema, G., Nyambo, T. B., Gori, J., Bumpstead, S., Pritchard, J. K., Wray, G. A., & Deloukas, P. (2006). Convergent adaptation of human lactase persistence in Africa and Europe. *Nature Genetics*, 39(1), 31–40.

into oligopeptides (short peptide chains). However, these peptides are still too large to be directly absorbed. At this stage, aminopeptidases act on the brush-border surface and cleave amino acids sequentially from the N-terminal end of peptides, converting oligopeptides into free amino acids and very small dipeptides or tripeptides.

This final hydrolysis is essential because intestinal epithelial cells absorb nutrients mainly in these simpler forms. The resulting free amino acids are transported across the enterocyte membrane through sodium-dependent amino acid transporters, while dipeptides and tripeptides enter via the PEPT1 transporter.

Enzyme excess and toxicity

Enzyme excess in the digestive system produces toxicity primarily through two distinct mechanisms: intrapancreatic autoactivation, causing gland autodigestion, and systemic enzyme spillage, causing multiorgan damage.

- **Trypsin**

Under normal conditions, it is secreted as the inactive precursor trypsinogen, which becomes activated only in the intestinal lumen. However, in acute pancreatitis, premature activation of trypsin occurs within the pancreatic acinar cells. This leads to autodigestion of pancreatic tissue, because trypsin also activates other digestive enzymes such as phospholipase and elastase. The enzyme spillage into the bloodstream can further provoke systemic inflammation.

- **Serum Amylase**

Elevated serum amylase levels are often used as a marker of pancreatic injury. In some cases, excess amylase forms complexes with immunoglobulins, producing a condition called macroamylasemia. These complexes are too large to be filtered by the kidneys, resulting in persistently elevated blood amylase levels. Increased amylase may also occur in parotitis (salivary gland inflammation), pancreatitis, or renal failure, where clearance is reduced.

- **Pancreatic Lipase**

Pancreatic lipase is responsible for the digestion of dietary triglycerides into fatty acids and monoglycerides. In acute pancreatitis, excessive lipase leaks from damaged pancreatic cells into surrounding tissues. This enzyme can cause lipolysis of nearby adipose tissue, releasing free fatty acids (FFAs). These FFAs bind calcium to form calcium soaps, a process known as fat saponification, which contributes to tissue necrosis.

- **Elastase**

Elastase is a proteolytic enzyme released mainly from activated neutrophils during inflammation, and it is also produced by the pancreas. Its physiological role is to degrade elastin, a structural protein found in connective tissues and blood vessels. When released in excessive amounts, elastase can damage vascular walls and lung tissue, contributing to widespread inflammatory injury. The enzyme degrades elastin within the extracellular matrix, weakening tissue structure.

- **DPP-IV**

Dipeptidyl peptidase-IV (DPP-IV) is an enzyme that degrades incretin hormones such as GLP-1 (glucagon-like peptide-1) and GIP (glucose-dependent insulintropic peptide). These hormones enhance insulin secretion and help regulate blood glucose. In diabetes treatment, drugs known as DPP-IV inhibitors (gliptins) block this enzyme to prolong incretin activity, improving glycemic control. While beneficial, excessive pharmacological inhibition may alter immune signalling and incretin balance.

Conclusion

Digestive enzymes function as a tightly coordinated biochemical network that determines how efficiently nutrients are released, processed, and ultimately absorbed by the body, making them central to nutritional status rather than merely mechanical aids to digestion. Nutritionally significant deficiencies, such as reduced lactase activity or pancreatic lipase insufficiency, illustrate how even

a single enzymatic bottleneck can alter dietary tolerance, nutrient bioavailability, and long-term health outcomes. Variation in digestive enzyme expression across individuals and populations may represent an adaptive interface between diet and physiology, microbiome composition, and disease susceptibility over time. Thus, from a nutritional perspective, digestive enzymes are not just digestive tools, but dynamic regulators, linking diet composition, absorption capacity, and metabolic health.

Note on Important Bacteria for the Body

The human body hosts trillions of microorganisms - bacteria, viruses, fungi, and archaea, collectively referred to as the human microbiome. Among these microorganisms, bacteria are the most abundant and functionally significant, forming complex ecological communities that inhabit different regions of the body, including the gastrointestinal tract, skin, oral cavity, respiratory tract, and genitourinary system. Rather than being merely passive residents, these bacteria exist in a symbiotic relationship with the human host, meaning that both the microbes and the body benefit from each other's presence.

Beneficial bacteria also play a crucial role in immune system regulation. The microbiome helps train and modulate the immune response from early life by teaching immune cells to distinguish between harmless substances and harmful pathogens. A balanced microbial community helps maintain immune tolerance, reduces excessive inflammatory responses, and protects the body against infections. Many beneficial bacteria also produce antimicrobial compounds that inhibit the growth of harmful microbes, thereby preventing colonisation by pathogens.

A. Gut Microbiome Bacteria

The gastrointestinal tract houses the most diverse and dense bacterial community in the body. A healthy gut microbiome is essential for metabolic and immunological health.¹⁴⁶ Bacteria are particularly essential for maintaining digestive and metabolic health. In the gastrointestinal tract, especially the colon, these beneficial bacteria help to break down complex carbohydrates, dietary fibres, and resistant starches that the human digestive system cannot digest on its own. Through fermentation processes, these bacteria produce short-chain fatty acids (SCFAs) such as butyrate, acetate, and propionate, which serve as an important energy source for intestinal cells, strengthen the gut lining, and help regulate metabolic processes such as glucose and lipid metabolism.

1. *Lactobacillus Acidophilus*

Lactobacillus acidophilus (Neo-Latin 'acid-loving milk-bacillus') is a rod-shaped, anaerobic microbe, which was first isolated from infant faeces in the year 1900. The species is commonly found in humans, specifically in the gastrointestinal tract and oral cavity. The species most readily grows at low pH levels (below 5.0), and has an optimum growth temperature of 37 °C. Certain strains of *L. acidophilus* show strong probiotic effects and are commercially used in dairy production. *L. acidophilus* has antagonistic effects on the growth of *Staphylococcus aureus*, *Escherichia coli*, *Salmonella typhimurium*, and *Clostridium perfringens*.¹⁴⁷ Out of the four organisms, *Staphylococcus aureus*, which commonly causes skin infections like boils, cellulitis, impetigo, abscesses and food poisoning, is the most affected.

L. acidophilus is one of the most extensively studied probiotic bacteria in humans. It ferments dietary carbohydrates to produce lactic acid, acidifying the gut environment and inhibiting harmful pathogens, while supporting the integrity of the intestinal lining. Its colonisation of both the small intestine and vaginal epithelium makes it uniquely important for both digestive and reproductive health.¹⁴⁸

● Key Functions:

- Supports lactose digestion through enzymatic lactase production, reducing symptoms of lactose intolerance. This bacterium produces the enzyme β -galactosidase (functionally

¹⁴⁶ Bisanz, J. E., Upadhyay, V., Turnbaugh, J. A., Ly, K., & Turnbaugh, P. (2019). Diet Induces Reproducible Alterations in the Mouse and Human Gut Microbiome. SSRN Electronic Journal.

¹⁴⁷ Gilliland SE, Speck ML (December 1977). "Antagonistic Action of *Lactobacillus acidophilus* Toward Intestinal and Foodborne Pathogens in Associative Cultures 1". *Journal of Food Protection*. 40 (12): 820-823.

¹⁴⁸ Oak, S. J., & Jha, R. (2018). The effects of probiotics in lactose intolerance: A systematic review. *Critical Reviews in Food Science and Nutrition*, 59(11), 1675-1683.

similar to lactase), which helps break lactose into glucose and galactose, two simpler sugars that can be easily absorbed by the intestinal cells. In individuals with lactose intolerance, the small intestine produces insufficient amounts of the enzyme lactase, so lactose remains undigested and reaches the large intestine.

- *L. acidophilus* present in the gut helps restore microbial balance by repopulating beneficial bacteria, competing with harmful microbes for nutrients in attachment sites, and producing antimicrobial substances that inhibit pathogen growth. It also contributes to maintaining the intestinal barrier and normal metabolic activity of the microbiome. Through these mechanisms, the probiotic can limit excessive pathogen overgrowth and stabilise gut function, thereby reducing both the likelihood and severity of antibiotic-associated diarrhoea.¹⁴⁹
- In a healthy vaginal ecosystem, *Lactobacillus* species produce lactic acid, which keeps the vaginal pH mildly acidic (typically around 3.8–4.5). This acidic environment creates unfavourable conditions for pathogenic bacteria to grow. Additionally, *L. acidophilus* competes with harmful microbes such as *Gardnerella vaginalis* for space and nutrients on the vaginal epithelial surface.

● Impact of Imbalance or Deficiency

Depletion of *L. acidophilus*, commonly caused by antibiotic use, poor diet, or chronic stress, is directly linked to antibiotic-associated diarrhoea and increased risk of *Clostridium difficile* infection (a bacterium infection causing severe, watery diarrhoea, fever, and abdominal pain, usually after antibiotic use disrupts normal gut flora). In women, its loss is a primary ecological antecedent of bacterial vaginosis and recurrent vaginal candidiasis (fungal infection causing intense vaginal itching, irritation, and thick white discharge, usually caused by *Candida albicans*). It results from an overgrowth of yeast, often triggered by antibiotics, pregnancy, or diabetes.

● Restoration through Nutrition

Yoghurt containing live active cultures, kefir, cultured buttermilk, and aged cheeses (cheddar, gouda) are direct dietary sources of *L. acidophilus*. Prebiotic foods rich in inulin, like garlic, onions, and leeks, selectively promote *L. acidophilus* growth by providing a fermentable substrate that favours its metabolism over competing organisms.

2. *Bifidobacterium longum*

Bifidobacterium longum is a Gram-positive, rod-shaped bacterium present in the human gastrointestinal tract. *B. longum* is among the first bacteria to colonise the infant gut and remains a dominant and protective species throughout life. It ferments complex carbohydrates that the human body cannot digest independently.

● Key Functions:

- Many complex carbohydrates and fibres are not digested in the small intestine and therefore pass into the colon, where bacteria such as *B. longum* metabolise them. During this fermentation process, the bacterium produces short-chain fatty acids (SCFAs), particularly acetate and lactate. These SCFAs act as energy substrates for colonocytes (cells lining the colon), helping maintain the health and integrity of the intestinal lining. By providing energy to these cells, SCFAs support normal cellular metabolism, epithelial repair, and barrier function, which helps prevent harmful substances or pathogens from crossing into the bloodstream. Additionally, SCFAs contribute to maintaining a slightly acidic colonic environment, which favours beneficial microbes and discourages the growth of pathogenic bacteria.

¹⁴⁹ Mullish, B., McDonald, J., Pechlivanis, A., Rees, D., Williams, H., Marchesi, J., & Thursz, M. (2016). PWE-094 Understanding The Efficacy of Faecal Microbiota Transplantation in *Clostridium Difficile* Infection: Re-Establishment of Gut Microbiota with The Ability to Degrade Bile? *Gut*, 65(Suppl 1), A184.2-A184.

- *B. longum* promotes the activity and development of regulatory T cells (Tregs), a specialised group of immune cells responsible for maintaining immune tolerance and preventing excessive inflammatory reactions. By supporting Treg function, the bacterium helps limit overactive immune responses and maintain immune balance in the gut while still allowing the immune system to respond effectively to infections, thereby contributing to overall intestinal immune stability.
- Additionally, due to the above phenomenon, the body learns to distinguish between harmless environmental substances (such as food proteins and pollen) and harmful pathogens. Infants with a gut microbiome rich in *Bifidobacterium* during early life are often observed to have a lower likelihood of allergic sensitisation, which can reduce the risk of developing atopic conditions such as eczema, allergic rhinitis, and certain food allergies during childhood.

● **Impact of Imbalance or Deficiency**

Low levels of *B. longum* are associated with inflammatory bowel disease, including Crohn's disease and ulcerative colitis. Loss of *B. longum* is also associated with dysregulated gut immune responses and elevated systemic inflammation markers.

● **Restoration through Nutrition**

Chicory root, artichokes, slightly unripe bananas, garlic, and whole grains rich in inulin and fructooligosaccharides (FOS) are the most effective prebiotic foods for supporting *B. longum* populations. Diet composition rapidly and reproducibly shifts *Bifidobacterium* abundance, with fibre-rich plant-based diets consistently associated with higher colonocyte-nourishing SCFA output. Fermented dairy products (yoghurt, kefir) also deliver live *Bifidobacterium* cultures.

3. *Akkermansia muciniphila*

A. muciniphila resides specifically in the mucus layer of the colon and plays a vital role in maintaining gut barrier integrity. It represents 1-4% of total gut bacteria in healthy individuals. It colonises the gastrointestinal tract of humans and animals. It can be found within the intestinal mucosal layer of the epithelial crypts as well as in the cecum. It specifically resides at the oxic-anoxic interface. *A. muciniphila* is found in about 90% of healthy humans, makes up about 1% to 3% of the faecal microbiota and colonises the gut during the first year of life. Its prevalence can decrease with age or in disease states.¹⁵⁰

● **Key functions:**

- *Akkermansia muciniphila* is a beneficial gut bacterium that lives in close contact with the intestinal mucus layer, where it uses mucin (a glycoprotein that forms mucus) as a nutrient source. By metabolising mucin in a controlled manner, the bacterium stimulates the renewal and maintenance of the mucus layer, which strengthens the intestinal barrier. A healthier intestinal barrier reduces the leakage of bacterial components such as endotoxins into the bloodstream, a process that can otherwise trigger low-grade systemic inflammation commonly observed in obesity and metabolic disorders. Lower inflammation improves the body's responsiveness to insulin, which contributes to better insulin sensitivity and improved fasting glucose control.¹⁵¹
- The bacterium also alters the microbial ecosystem in a way that supports healthier lipid metabolism, contributing to reductions in circulating triglyceride levels. Through combined effects of strengthening the gut barrier, reducing inflammatory signalling, and improving metabolic communication between the gut and the host, *Akkermansia muciniphila* is

¹⁵⁰ O'Callaghan, A., & van Sinderen, D. (2016). *Bifidobacteria and Their Role as Members of the Human Gut Microbiota*. *Frontiers in Microbiology*, 7(925).

¹⁵¹ Messer, D. J., & et al. (1986). Relation between mastery behavior in infancy and competence in early childhood. *Developmental Psychology*, 22(3), 366-372.

associated with improvements in key metabolic health markers in overweight and obese individuals.¹⁵²

- **Impact of imbalance or deficiency**

A thinned gut mucus layer leads to increased bacterial translocation across the intestinal wall, promoting systemic low-grade inflammation and insulin resistance. Importantly, higher *A. muciniphila* abundance at baseline in humans undergoing caloric restriction predicted greater metabolic improvement, establishing it as a clinically relevant biomarker.

- **Restoration through Nutrition**

Polyphenol-rich foods are the strongest dietary modulators of *A. muciniphila* in human populations, with pomegranate, cranberries, red grapes, green tea, and dark chocolate shown to support its growth.

B. Oral Microbiome Bacteria

The human oral microbiome is one of the earliest microbial species to colonise the mouth shortly after birth. As a dominant resident of the tongue and oral mucosal surfaces, it plays an important role in maintaining microbial balance within the oral cavity. ***Streptococcus salivarius*** is a primary commensal that occupies receptor sites on oral mucosal cells, physically preventing pathogenic bacteria from gaining a foothold.¹⁵³

- **Key functions**

- BLIS molecules produced by *S. salivarius* work by interfering with the cell membranes and metabolic processes of competing bacteria, preventing them from multiplying or colonising the throat lining. One of the key pathogens they inhibit is *Streptococcus pyogenes*, the bacterium responsible for strep throat and some other upper respiratory infections. By occupying adhesion sites on the mucosal surfaces and simultaneously releasing these inhibitory substances, *S. salivarius* creates a competitive environment that limits the ability of *S. pyogenes* to establish infection.
- *Streptococcus salivarius* contributes to the regulation of oral and upper respiratory immune responses by interacting with immune cells present in the mucosal lining of the mouth and throat. As an early coloniser of the oral cavity, it helps maintain a balanced microbial environment that allows the immune system to respond appropriately without triggering excessive inflammation. One way it supports mucosal immunity is by promoting the production of secretory Immunoglobulin A (sIgA), an important antibody found in saliva and respiratory secretions. sIgA plays a key defensive role by binding to microbes and preventing them from attaching to mucosal surfaces, thereby limiting colonisation by potential pathogens.¹⁵⁴

- **Impact of Imbalance or Deficiency**

Disruption of *S. salivarius* populations from antibiotic use and mouthwash overuse is associated with recurrent tonsillitis, ear infections, and increased dental caries risk.

- **Restoration through Nutrition**

¹⁵² Depommier, C., Everard, A., Druart, C., Plovier, H., Van Hul, M., Vieira-Silva, S., Falony, G., Raes, J., Maiter, D., Delzenne, N. M., de Barse, M., Loumave, A., Hermans, M. P., Thissen, J.-P., de Vos, W. M., & Cani, P. D. (2019). Supplementation with *Akkermansia muciniphila* in overweight and obese human volunteers: a proof-of-concept exploratory study. *Nature Medicine*, 25(7), 1096–1103.

¹⁵³ Marco ML et al. (2021). Health benefits of fermented foods. *Nature Reviews Gastroenterology & Hepatology*

¹⁵⁴ Patras, K. A., Wescombe, P. A., Rösler, B., Hale, J. D., Tagg, J. R., & Doran, K. S. (2015). *Streptococcus salivarius* K12 Limits Group B *Streptococcus* Vaginal Colonization. *Infection and Immunity*, 83(9), 3438–3444.

Naturally fermented foods such as plain yoghurt and kefir contribute to mucosal *Lactobacillus* colonisation throughout the upper gastrointestinal and oral tracts. Polyphenol-rich foods, including green tea, berries, and apples, have demonstrated selective inhibition of pathogenic oral bacteria in human oral microbiome studies while being more favourable to commensal strains, creating an ecological environment conducive to *S. salivarius* dominance.

C. Vaginal microbiome bacteria

Unlike the gut, a healthy vaginal microbiome is characterised by low diversity and the dominance of *Lactobacillus* species. ***Lactobacillus crispatus*** is a species of lactobacilli that is frequently found in the vaginal flora of healthy women, where it serves as a predominant microorganism contributing to the maintenance of vaginal health. *Lactobacillus crispatus* is considered the most protective of all vaginal *Lactobacillus* species. Its dominance of the vaginal microbiome in healthy, reproductive-age women is the ecological benchmark against which vaginal dysbiosis is assessed. It is associated with the lowest rates of bacterial vaginosis, sexually transmitted infections, urinary tract infections, and adverse pregnancy outcomes.¹⁵⁵

- **Key functions**

- *Lactobacillus crispatus* produces bacteriocins, antimicrobial peptides that inhibit the growth of competing pathogenic bacteria. These bacteriocins can directly suppress *Gardnerella vaginalis*, a key organism involved in the development of bacterial vaginosis (BV). By interfering with the growth and colonisation of *G. vaginalis*, *L. crispatus* helps prevent the microbial imbalance or dysbiotic shift that characterises BV, thereby supporting the stability of the vaginal microbial environment.
- Produces both D- and L-lactic acid, maintaining vaginal pH below 4.5, an environment hostile to sexually transmitted pathogens and anaerobic bacteria that cause bacterial vaginosis.
- Generates hydrogen peroxide (H₂O₂), which has direct antimicrobial properties against sexually transmitted pathogens.

- **Impact of Imbalance or Deficiency**

Displacement of *L. crispatus* creates conditions for bacterial vaginosis, yeast infections (*Candida*), and increased susceptibility to sexually transmitted infections.

- **Restoration through nutrition**

A diet rich in folate (leafy greens, legumes), antioxidants, and low in refined sugars supports hormonal balance and immune defences that maintain vaginal *Lactobacillus* dominance. Fermented dairy products contribute broadly supportive *Lactobacillus* species through mucosal pathways.

D. Skin Microbiome bacteria

Staphylococcus epidermidis is a common commensal bacterium that resides on healthy human skin and plays an important protective role when the skin microbiome is balanced. As part of the normal skin ecosystem, it contributes to defence and immune regulation by helping limit the colonisation of harmful microbes, particularly *Staphylococcus aureus*, a pathogen associated with skin infections, abscesses, and MRSA (Methicillin-Resistant *Staphylococcus aureus*, a strain of the bacterium *Staphylococcus aureus* that has developed resistance to many commonly used beta-lactam antibiotics, including methicillin, penicillin, and related drugs).¹⁵⁶

- **Key Functions**

¹⁵⁵ O'Hanlon, D. E., Moench, T. R., & Cone, R. A. (2013). Vaginal pH and Microbicidal Lactic Acid When Lactobacilli Dominate the Microbiota. *PLoS ONE*, 8(11), e80074.

¹⁵⁶ Cogen AL et al., (2010). *Journal of Investigative Dermatology*, Selective antimicrobial action is provided by phenol-soluble modulins derived from *Staphylococcus epidermidis*.

- Staphylococcus epidermidis protects the skin by producing several antimicrobial molecules, most notably phenol-soluble modulins (PSMs) and the serine protease Esp. These substances act as targeted biochemical defences that suppress the growth of competing pathogens on the skin surface. Phenol-soluble modulins are small peptides capable of disrupting bacterial membranes and interfering with essential cellular processes in susceptible microbes. By damaging the structural integrity of competing bacteria, they prevent those organisms from establishing stable colonies on the skin.¹⁵⁷
- The serine protease Esp (epidermidis serine protease) provides an additional protective mechanism. This enzyme degrades proteins required by pathogens to form biofilm-structured microbial communities that allow bacteria to attach firmly to skin or medical devices. By breaking down these adhesion and biofilm components, Esp limits the ability of pathogens such as Staphylococcus aureus and Group A Streptococcus to persist on the skin surface. Through these antimicrobial strategies, S. epidermidis functions as a biological defence system that reduces the likelihood of pathogenic colonisation and infection.
- Beyond direct antimicrobial activity, S. epidermidis also strengthens the skin's defence system by interacting with keratinocytes, the primary cells that form the outermost layer of the skin. Keratinocytes are not only structural cells but also important participants in innate immunity, capable of sensing microbial signals and producing immune mediators. When S. epidermidis interacts with these cells, it can stimulate them to produce protective cytokines and antimicrobial peptides that enhance the skin's ability to resist infection. These signalling molecules help recruit immune cells, regulate inflammatory responses, and reinforce the integrity of the skin barrier. Through this interaction, the bacterium helps maintain a state of immune readiness at the skin surface.

● Impact of Imbalance or Deficiency

In conditions such as atopic dermatitis (Eczema), this protective mechanism is often impaired. Individuals with this inflammatory skin disorder frequently exhibit reduced levels of beneficial commensal microbes, including S. epidermidis, and therefore produce fewer antimicrobial compounds that normally suppress pathogen growth.¹⁵⁸

● Restoration through Nutrition

Skin microbiome health is influenced by diet through systemic pathways. Omega-3 fatty acids (oily fish: salmon, sardines, mackerel; walnuts; flaxseed) reduce skin inflammation by shifting eicosanoid balance. Fermented foods may positively influence skin via the gut-skin axis. Adequate intake of zinc (pumpkin seeds, legumes, whole grains) and vitamin A (sweet potato, carrots, leafy greens) is foundational for maintaining skin barrier integrity, upon which the skin microbiome ecology depends.

Conclusion

Maintaining a healthy microbiome largely depends on consistent dietary and lifestyle choices that support beneficial microbial communities. Consuming a wide variety of plant foods, including those rich in fibre and prebiotics, provides the substrates that beneficial bacteria need to thrive, while fermented foods introduce helpful microbes that can enhance microbial diversity and support immune balance. Polyphenol-rich foods further shape the microbial environment by encouraging the growth of protective species and discouraging harmful ones. Because the microbiome can respond rapidly to dietary changes, everyday food choices play a significant role in influencing gut microbial composition and metabolic activity. At the same time, unnecessary antibiotic use should be avoided to prevent disruption of microbial ecosystems, and when antibiotics are required, supportive strategies such as probiotics may help restore balance. For women, targeted probiotic

¹⁵⁷ Nakatsuji, T., Fenical, W., & Gallo, R. L. (2019). Response to Comment on "A commensal strain of Staphylococcus epidermidis protects against skin neoplasia".

¹⁵⁸ Cargill, J. S., & Upton, M. (2007). P618 The effect of Staphylococcus epidermidis culture supernatants on the biofilm density of other S. epidermidis strains. International Journal of Antimicrobial Agents, 29.

supplementation can also contribute to maintaining a stable vaginal microbiome. Together, these practices help sustain a diverse and resilient microbial community that supports digestive health, immune function, and overall physiological stability.

Note on Fortification Mechanisms and Current Guidelines in India

Nutritionally, micronutrients are required in small quantities for human functioning, but their insufficiency is associated with adverse health consequences among all. In terms of health, these consequences span poor physical and cognitive development, congenital disabilities, increased risk and severity of infections due to decreased immune function, and increased risk of child and maternal mortality; in terms of other outcomes, it may contribute to reduced productivity, indicating all-encompassing effects on human life.¹⁵⁹

Despite their severity, micronutrient deficiency continues to be one of the top health risk factors and concerns for decades across countries.

The most recent data from the National Family Health Survey (NFHS-5) indicate limited improvement in nutrition outcomes since 2015-2016 (NFHS-4) and an actual increase in the prevalence of anaemia across all age and gender groups through the income quintiles, most notably women of reproductive age (WRA), and preschool children. The main cause of anaemia remains iron deficiency, and the deficiencies of other micronutrients such as folic acid, vitamin B6, and vitamin B12, which are also widely prevalent in India, especially in children, as indicated by the Comprehensive National Nutrition Survey (CNNS) 2016-2018 report. 50%-70% of anaemia is caused by iron, folate, or vitamin B12 deficiency or a combination of these micronutrients. A recent systematic review highlights the high burden of micronutrient deficiencies in India, particularly that of folic acid (37%), iron (54%), vitamin B12 (53%), and vitamin D (61%).

Analyses for the Copenhagen Consensus have consistently ranked micronutrient interventions such as salt iodisation and industrial fortification of wheat and maize flour with iron among the most cost-effective development investments, with very high benefit–cost ratios. Best practice papers from Horton and colleagues emphasise universal salt iodisation, scaling iron fortification of flour from large mills to reach at least half of the population, and expanding fortification to other widely consumed vehicles like condiments and rice so that vulnerable groups are covered even where industrial flour consumption is low.¹⁶⁰

However, reviews such as the Copenhagen Consensus have consistently ranked micronutrient interventions as the most cost-effective development interventions that provide significant returns for a low cost.¹⁶¹

Current scenario of fortification in India

In India, the fortification of Vanaspathi was first introduced in 1953. India was one of the first countries in the world to initiate a Universal Salt Iodisation program in 1962. The CNNS survey demonstrated optimal iodine status among children, confirming the success of the USI program, while the deficiencies of other micronutrients are still high.

The National Nutrition Policy, 9th, 11th and 12th 5-year Plan, as well as the National Nutrition Strategy on “Kuposhan Mukta Bharat” all recommend fortification of staples with micronutrients as one of the key strategies for tackling malnutrition. Directives for inclusion of fortified staples in social safety net programs (SSNP), programs like public distribution system (PDS), integrated child development scheme (ICDS), and mid-day meal (MDM) also exist. The fortified staples include rice and wheat (fortified with iron, folic acid and vitamin B12), oil and milk (fortified with vitamin A and

¹⁵⁹ World Health Organisation. Malnutrition - Fact Sheet; 2021

¹⁶⁰ GtG: Micronutrient Fortification and Biofortification Research. (2026). Copenhagen Consensus.

¹⁶¹ Horton S, Mannar V, Wesley A. Micronutrient Fortification (Iron and Salt Iodization): Best Practice Paper. Copenhagen Consensus Center; 2008.

D) and double fortified salt (DFS) with iron and iodine. The directive for its scale-up through the ICDS and MDM to reduce the prevalence of nutritional deficiency anaemia was issued in 2021.¹⁶²

A literature review of 47 studies highlights improvements in haemoglobin or iron markers when fortifying foods with either iron alone or iron combined with other nutrients. The study concluded that fortification with multiple nutrients had better health impacts than using a single fortificant. A recent case-control study from Gujarat showed increased mean haemoglobin by 0.4 g/dL, an 11.3-point improvement in cognitive scores and a reduced anaemia prevalence of 10% among school-aged children after receiving fortified rice over a period of 8 months through the MDM program. Similarly, in a few other Indian studies conducted among pregnant women and school children, a positive impact of providing DFS was observed on haemoglobin and urinary iodine levels as well as the status of other micronutrients.¹⁶³

The current national regulations for folic acid and vitamin B12 fortification in staple foods are significantly below what the World Health Organisation (WHO) recommends for the Indian context, potentially rendering them ineffective in preventing widespread micronutrient deficiencies and severe, life-threatening birth defects. This regulatory decision yields unintended health consequences and presents a significant challenge to enhancing the health of some of the most vulnerable populations nationwide.¹⁶⁴

There is a high prevalence of neural tube defects (NTDs) and nutritional anaemias that increase the risk of birth defects in India. The current staple food fortification regulations for folic acid and vitamin B12 are ineffective in the country. India allows food producers to voluntarily fortify wheat flour and rice sold on the open market, and the government requires rice distributed through safety-net programs to be fortified.

Food fortification regulation, recommended in 2018 by the Food Safety and Standards Authority of India, allows adding 75 - 125 micrograms (mcg) of folic acid per kilogram (kg), and 0.75-1.25 mcg of vitamin B12 per kg, as fortificants in wheat flour or rice. These levels are 90% lower than what the WHO recommends based on daily consumption of staple foods in India.¹⁶⁵ Micronutrient deficiencies, particularly of folate (vitamin B9) and vitamin B12, are highly prevalent in India. A recent meta-analysis (2015-2020) indicated that 37% of the Indian population is folate-deficient and 53% is vitamin B12-deficient.¹⁶⁶ NTDs, like anencephaly and spina bifida, lead to stillbirths, early mortality, and lifelong disabilities. Vitamin B12 deficiency impacts a large segment of the population in India, resulting in anaemia, neurological complications, and cognitive issues. Most of these deficiencies can be prevented through population-based nutrition interventions such as staple food fortification with adequate levels of micronutrients.¹⁶⁷

The biggest challenge in fortifying foods with iron is to use bioavailable forms of iron that are better absorbed in the body, but do not alter the appearance or taste of the food vehicle to which they are added. For example, white insoluble iron phosphate compounds are stable under a variety of storage conditions but are poorly absorbed. In contrast, soluble iron salts like ferrous sulphate are

¹⁶² Large-scale Staple Food Fortification as a Complementary Strategy to Address Vitamin and Mineral Vulnerabilities in India, NITI Aayog

¹⁶³ Reddy KJ, Nair S. Double fortified salt: An effective measure to control micronutrient deficiencies in Indian pregnant women. *Int J Community Med Public Health* 2016;3:679-86.

¹⁶⁴ Food Fortification Ineffective in Preventing Neural Tube Defects in India due to Regulation Promoting Inadequate Levels of Folic Acid and Vitamin B12 - Kancherla - 2025

¹⁶⁵ New Publication Raises Concerns that India's Food Fortification Standards Are Insufficient, <https://ffinetwork.org/concerns-that-indian-food-fortification-standards-are-insufficient/>

¹⁶⁶ Micronutrient's deficiency in India: a systematic review and meta-analysis | *Journal of Nutritional Science* | Cambridge Core

¹⁶⁷ A global update on the status of prevention of folic acid-preventable spina bifida and anencephaly in year 2022 - PubMed

well absorbed but can discolour fortified foods by reacting with some ingredients under certain conditions. Thus, the form of iron should be selected based on maximum bio-availability, at the lowest cost, without adversely affecting the properties of flour and flour products.

Note: IIT Kharagpur's report and the pause on rice fortification¹⁶⁸

In early 2026, the Union government decided to temporarily discontinue rice fortification under the Pradhan Mantri Garib Kalyan Anna Yojana and allied schemes after a study by IIT Kharagpur raised concerns about the stability of added nutrients in fortified rice kernels over their shelf life and use. Policy notes clarify that this pause is intended to remain in place until a more robust and effective nutrient delivery mechanism is developed, and that beneficiaries continue to receive their full grain entitlements under the PDS, ICDS and school meal schemes, albeit with non-fortified rice during this interim period.

Bio-fortification in India

Biofortification increases the nutrient density of crops through plant breeding, agronomic measures or biotechnology, so that populations obtain more iron, zinc or provitamin A from the same amount of staple foods. Trials with iron biofortified pearl millet in Indian schoolchildren have shown improved iron stores and reversal of iron deficiency, while iron biofortified rice and beans have improved iron status in women in the Philippines and Rwanda, demonstrating that biofortified staples can meaningfully reduce “hidden hunger.” This approach is particularly relevant for India, where over one-third of agricultural land is zinc-deficient and plant-based diets dominate, making zinc and iron-biofortified staples an important complement to industrial fortification and supplementation. Over the last decade, India has released and started scaling several biofortified varieties such as the high iron pearl millet Dhanashakti and related hybrids, zinc rich wheat and high zinc rice varieties like DRR Dhan 45 and DRR Dhan 49, and the Indian Council of Agricultural Research along with HarvestPlus and ICRISAT is now mainstreaming biofortification in its breeding programmes, with state governments like Bihar and Odisha committing to promote zinc biofortified wheat and rice through public systems.¹⁶⁹

Global impact of biofortified varieties

The deployment of biofortified cultivars holds great promise for the health and well-being of the human population. Several studies have demonstrated the positive effects of these biofortified crops on humans. A study conducted on 246 children of 12-16 yr of age in Maharashtra, by feeding them with ‘bhakri’ (round flat unleavened bread) made from iron-rich and conventional pearl millet grains, demonstrated that feeding iron-rich pearl millet was an efficient approach to improve iron status in school-age children²⁸. In another, serum xanthophylls and retinol were significantly improved in Zambian children fed with provitamin-A-rich maize grains. In a study on 679 Zambian children, it was found that consumption of β -carotene-rich maize significantly improved their serum β -carotene concentrations compared with traditional maize. The beneficial effects of QPM are also well-demonstrated worldwide. Porridge made from QPM resulted in fewer sick days among children compared to those who had porridge from normal maize. Infants and young children fed with QPM expressed a 12 per cent higher rate of growth in weight and nine per cent in height compared to the group given only normal maize. Another study showed that consumption of 100 g QPM was sufficient for children to meet the requirement of lysine, resulting in a reduction in maize to the tune of 40 per cent relative to normal maize. The orange-fleshed sweet potatoes (OFSP) were fed to South African school children aged 5-10 yr, and a favourable response in the vitamin-A status of children was observed compared with the children fed with a traditional white variety of sweet

¹⁶⁸ PIB: Recalibration of Nutrition Intervention-Temporary Discontinuation of Rice-biofortified zinc-deficient non-fortified Fortification. (2026).

¹⁶⁹ ISSCA - Biofortified pearl millet Variety (Dhanashakti) and Hybrids to enhance nutrition and help address malnutrition and achieved climate resilience. (2025). Icrisat.org.

potato. In Mozambique, children who consumed OFSP over the two-year intervention possessed significantly higher serum retinol concentrations compared with non-intervention children. The development and promotion of biofortified varieties would help address malnutrition and achieve the SDGs.¹⁷⁰

Point-of-use (home) fortification:

Point-of-use or home fortification uses single-dose sachets of multiple micronutrient powders or small quantity lipid-based supplements that caregivers mix into a child's semi-solid food immediately before consumption, so the taste and appearance of the meal remain acceptable. A global UNICEF–CDC assessment documented rapid expansion of these interventions in low and middle-income countries, with programs using micronutrient powders to prevent anaemia and other deficiencies among infants and young children where routine supplement adherence is low. Cochrane reviews find that home fortification with multiple micronutrient powders in children under two reduces anaemia and iron deficiency compared with no intervention or iron drops, and such powders are now manufactured and supplied at scale in South Asia, including India, for use by governments, UN agencies and NGOs.¹⁷¹

Fortification Strategies across the world and comparison with India

America's fortification of industrially milled wheat flour and corn flour is almost universal. The relatively low level of iron deficiency among school-age children and women of childbearing age in the U.S. and Canada is partly attributed to flour fortification. Chile has an iron deficiency prevalence of less than 1% in children 2-6 (WHO, 2008), which most observers attribute to a strong national flour fortification program (using ferrous sulphate). More recently, all corn and wheat flour in Venezuela is being fortified with iron (a mixture of ferrous fumarate and reduced iron) and a number of vitamins. Fortified flour products currently contribute about 48% of the recommended dietary iron for the average Venezuelan, and the nutritional impact of flour fortification in Venezuela has been both swift and dramatic.

Problems with fortification in India

Prior to 2018, India's fortification standards aligned with WHO recommendations and specified that wheat flour should be fortified with 1300 mcg/kg of folic acid and 10 mcg/kg of vitamin B12. The decision to lower these standards in 2018 was reportedly driven by a sole focus on anaemia prevention, without considering the high prevalence of NTDs and population-wide vitamin B12 levels. There was also a perception of harm associated with higher levels of folic acid, despite extensive evidence confirming the safety of folic acid fortification, an intervention implemented in more than 65 countries.

The potential to reduce child mortality by preventing folic acid-preventable spina bifida and anencephaly (FFASBA is inadequately appreciated. Prevention of FAP SBA would have reduced annually, neonatal, infant, and under-five mortality by 10.2%, 8.9%, and 8.3%, respectively. These reductions would have contributed 18.5% and 17.2% to the reductions in neonatal and under-five mortality, respectively, needed by India to achieve its 2030 Sustainable Development Goal Target 3.2 addressing preventable child mortality.¹⁷²

As indicated by large surveys such as NFHS 5, the compliance with iron folic acid supplementation remains low and challenging in India. Only 26% and 44.1% of pregnant women consume iron folic

¹⁷⁰ Mohapatra, T., Yadava, D., & Hossain, F. (2018). Nutritional security through crop biofortification in India: Status & future prospects. *Indian Journal of Medical Research*, 148(5), 621.

¹⁷¹ Jefferds, M. E., Irizarry, L., Timmer, A., & Tripp, K. (2013). UNICEF–CDC Global Assessment of Home Fortification Interventions 2011: Current Status, New Directions, and Implications for Policy and Programmatic Guidance. *Food and Nutrition Bulletin*, 34(4), 434–443.

¹⁷² Total prevention of folic acid-preventable spina bifida and anencephaly would reduce child mortality in India: Implications in achieving Target 3.2 of the Sustainable Development Goals - PubMed

acid supplements for 180 days and 100 days, respectively. The coverage of supplementation programs for the other vulnerable groups, such as WRA, school children, and adolescent girls, is still a miss (10%-20%).¹⁷³

Vitamin A supplementation alone is not sufficient to provide all Vitamin A needs for preschool children, who are the main target for Vitamin A supplementation programs. To address Vitamin A deficiency, dietary intake alone is not sufficient as it contributes to only 16%-34% of the requirements for Vitamin A across age groups. Furthermore, vitamin A supplementation programs are in place only for preschool children and not for any other age group, nor is there a supplementation program for other nutrients like Vitamin D.¹⁷⁴

Recommendations and way forward

Re-setting fortification standards for folic acid and vitamin B12 to levels recommended by the WHO in India will avert thousands of NTD-associated pregnancies each year, as well as morbidity and mortality associated with them, by mandating 1300 mcg of folic acid and 10 mcg of vitamin B12 per kilogram of wheat flour or rice) to be included in the regulation.

A population-based study in Haryana showed that when atta fortified according to WHO guidelines for folic acid was provided to Public Distribution System beneficiaries, their blood folate levels were raised above the threshold needed to prevent NTDs. This clearly illustrates that raising fortification standards to WHO-recommended levels and implementing large-scale food fortification significantly improve micronutrient levels in the population and prevent NTDs and other consequences of micronutrient deficiencies.¹⁷⁵

At present, of the total fortifiable oil (11.25-11.75 MMT), 65.2% (7.5MMT/annum edible oil) is being fortified. Since a significant amount of oil is produced and marketed by leading oil brands, including small and medium industries, and reaches different parts of the country, fortification of oil to reach the masses is a feasible strategy. Similarly, 416 lakh litres per day (LLPD) is produced by milk cooperatives and the private sector, of which about 150 LLPD gets fortified currently, reaching about 100 million people. If fortified, the total quality of milk can potentially benefit almost 275 million people.¹⁷⁶

Costa Rica case study: Costa Rica has implemented one of the most comprehensive national fortification programmes, mandating fortification of salt, wheat flour, maize flour, sugar, milk and later rice, with iron, folic acid and other micronutrients. Fortification of wheat flour in 1997 and corn flour in 1999 with folic acid, among other micronutrients, was implemented in Costa Rica by means of two decrees, resulting in an effective public health impact. A prevalence of 25% of folic acid serum levels deficiency detected in fertile women in 1996 decreased 87% in urban areas two years later, whereas in rural areas it diminished by 63%. In addition, a significant reduction of neural tube defects at the national level has been reported, dropping from a rate of 9.7 per 1000 live births during the period 1996-1998 to 6.3 per 1000 live births in the period 1999-2000. Finally, there has been a reported 74% reduction in the number of Neural Tube Defects at Birth (NTB) at the National Children's Hospital, resulting in 105 cases treated in 1995 to 26 cases in 2001.¹⁷⁷

¹⁷³WHO. Guideline: Daily Iron and Folic Acid Supplementation in Pregnant Women. Geneva: World Health Organization; 2012.

¹⁷⁴ Reddy GB, Pullakhandam R, Ghosh S, Boiroju NK, Tattari S, Laxmaiah A, et al. Vitamin A deficiency among children younger than 5 y in India: An analysis of national data sets to reflect on the need for vitamin A supplementation. *Am J Clin Nutr* 2021;113:939-47.

¹⁷⁵ India: Fortified Atta Proves to Strengthen Health in Haryana – Food Fortification Initiative, <https://ffinetwork.org/india-haryana-resultsprevent-NTDs/>

¹⁷⁶ Milk Fortification and Oil Fortification. Tata Trust. Available from: <https://www.tatatrusts.org/our-work/nutrition/food-fortification/milk-fortification>.

¹⁷⁷ Chen, L. T., & Rivera, M. A. (2004). The Costa Rican Experience: Reduction of Neural Tube Defects following Food Fortification Programs. *Nutrition Reviews*, 62, S40-S43.

For India, the Costa Rican experience is important because it shows that when fortification standards are set at appropriate levels, the right fortificants are chosen, and monitoring systems are in place, national programmes can virtually eliminate iron deficiency anaemia in children and sharply reduce anaemia in women within a decade. The large and sustained fall in neural tube defects after folic acid fortification of multiple staples underlines why India needs to align its folic acid standards with WHO guidance and pair regulation with strong surveillance, rather than relying on low-dose fortification or supplementation alone.

Regulatory and Licensing System for Dietary Supplements in India

India's regulatory and licensing framework for dietary supplements is structurally inadequate. Companies systematically avoid the drug approval pathway by exploiting food-law classification under Section 22 of the Food Safety and Standards Act, 2006. Claim verification is entirely post-market and complaint-driven with no pre-market scientific screening. Penalties under the Act are low relative to market profits, and no public register of approved health claims exists. The result is that consumers across a market valued at USD 5 billion in 2019 are routinely exposed to drug-like health claims that carry no independent evidentiary support. This stands in stark contrast to the European Union, the United States, Canada, and Australia, each of which operates a more rigorous framework for verifying the scientific basis of health claims before or alongside market entry.¹⁷⁸

The Classification Framework In India

In India, the question of whether a product is a food or a drug is not just a small detail; it decides the whole path a product must follow. If treated as a drug, the Drugs and Cosmetics Act, 1940 comes into force with its requirements for licences, safety checks, clinical trials, and proof of claims before sale. If treated as food, the product falls under the Food Safety and Standards Act, 2006, where the process is much lighter. Food law does not require clinical trials or pre-market proof of efficacy. This is the main reason why most companies prefer to present their supplements as food while still making strong health claims on their packaging. The statutes are as follows:

- **The Food Safety and Standards Act, 2006**

The FSS Act gives a very broad meaning to food. Section 3(1)(j) includes any substance intended for human consumption but explicitly excludes drugs. Section 22 lists special categories: foods for special dietary use, foods for special medical purposes, functional foods, nutraceuticals, health supplements, and novel foods. By writing nutraceuticals and supplements into this provision, the legislature gave companies a route to avoid drug law and remain within the food regulatory system. Section 22 further specifies that a qualifying product must not claim to cure or mitigate any specific disease or disorder (except for health benefit or promotion claims as may be permitted by FSSAI regulations); must not include a drug as defined under the Drugs and Cosmetics Act, 1940; must not include a narcotic or psychotropic substance; and must not exceed recommended daily allowances (RDA) for nutrients.

- **The Drugs and Cosmetics Act, 1940**

Section 3(b) uses a wide language, defining drugs to cover all medicines for internal or external use, and any substance used for diagnosis, treatment, mitigation, or prevention of disease. Section 18 prohibits the sale or manufacture of a drug without a licence. By this wording, many products now sold as supplements could be classified as drugs. Companies, however, know that if they use softer language, they can remain under food law. For example, instead of claiming that the supplement 'cures diabetes (T2DM),' the product's label would claim 'controls blood sugar,' 'reduces risks of diabetes,' or 'does not spike insulin, keeping them away from the drug regulator.

- **The Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954**

Section 3 bans advertisements claiming to cure certain diseases. This law was introduced to stop miracle cure advertisements. Companies avoid direct cure claims and use phrases like 'helps in sugar control.' The buyer may still believe the product is almost like medicine, but legally, it is seen as food. The list of prohibited diseases is outdated and does not cover many modern lifestyle disorders, such as obesity, cholesterol imbalance, or muscle growth, which further enlarges the space available for supplement marketing.

Key Judicial Pronouncements

¹⁷⁸ Bansal, R., & Dhiman, A. (2019). Line of Progression: Indian Regulatory Framework for Nutraceuticals and Dietary Supplements. *Applied Clinical Research, Clinical Trials and Regulatory Affairs*, 6(1), 46-61.

- **Common parlance test:** In *Puma Ayurvedic Herbal (P) Ltd. v. Commissioner of Central Excise*, the Supreme Court held that how ordinary people in common parlance perceive a product is a material factor. If ordinary people see it as medicine or a cure, it should be treated as a medicine.
- **Therapeutic use test:** In *Commissioner of Central Excise v. Wockhardt Life Sciences Ltd.*, the Court reinforced that therapeutic or prophylactic use points to classification as a drug.
- **No pre-approval under Section 22:** In *Vital Nutraceuticals Pvt. Ltd. v. Union of India*, the Bombay High Court struck down FSSAI advisories attempting to introduce a product approval system for nutraceuticals, holding that FSSAI lacked statutory authority to issue them without following the procedure under Sections 92 and 93 of the FSS Act, 2006. The Supreme Court later refused to interfere with the interim order. This confirmed that once a product falls under Section 22, no drug-style pre-approval is required.
- **Food licence as shield from prosecution:** In *M/s Truweight Wellness Pvt. Ltd. v. State of Telangana*, the High Court quashed prosecution under the Drugs and Magic Remedies Act for a diet product already holding a food licence, noting that the product label did not claim to cure disease. As long as a company avoids direct words of cure, courts allow the product to remain in the food category.
- **Medicinal perception in IP law:** In *Sun Pharma Laboratories Ltd. v. Ajanta Pharma Ltd.*, the Delhi High Court applied the stricter standard ordinarily reserved for medicinal products when assessing the likelihood of confusion in a passing-off action, even though the product was legally classified as a nutraceutical. The Court reasoned that nutraceuticals are frequently perceived by consumers as therapeutic and are often used on medical advice, blurring the practical distinction between food and drug. This verdict highlighted a structural tension: regulatory law places supplements and nutraceuticals under food law, but judicial assessment in trademark law may treat them closer to medicines due to their functional positioning and consumer perception.¹⁷⁹

Regulatory Architecture under FSS Act, 2006

The Food Safety and Standards Act was enacted in 2006 to establish the statutory body FSSAI, which controls the manufacturing, storage, distribution, sale, and import of food and food products to ensure their availability inside the country. The FSS Act and its Rules and Regulations of 2011 classify nutraceuticals as foods. The FSSAI was formally constituted in 2008. A pre-publication consultation procedure was initiated in 2006, during which several rules and regulations were drafted and forwarded for notification by the end of September 2010.

According to Section 22(1) of the FSSA, 'foods for special dietary uses or functional foods or Nutraceutical or health supplements' are defined as:

- Foods that are specially processed or formulated to meet specific dietary requirements arising from a specific physical or physiological condition or specific diseases and disorders, presented as such, whose composition must differ significantly from ordinary foods of comparable nature, if such ordinary foods exist.
- A product labelled as food for special dietary uses, functional food, nutraceutical, health supplement, or similar food that is not intended for use as a conventional food and may be formulated as powders, granules, tablets, capsules, liquids, jelly, and other dosage forms but not parenteral, and intended for oral administration.

Nutraceuticals are classified based on their origin, which may be natural or synthetic, into two broad categories: traditional foods and non-traditional foods.

1. Traditional Nutraceuticals (Natural Origin)

¹⁷⁹ Kaushik Anand, Legal Classification And Regulation Of Dietary Supplements And Functional Health Claim Foods, Indian Journal Of Legal Review (Ijlr), 6 (2) Of 2026, Pg. 214 228.

Traditional nutraceuticals are primarily obtained from natural sources such as plants, animals, and microorganisms. These include naturally occurring bioactive components, like omega-3 fatty acids in salmon, saponins in soy, and lycopene in tomatoes. Traditional nutraceuticals can be further classified into three sub-groups.

- **Chemical Constituents**, which include nutrients, herbals, and phytochemicals.
 - **Nutrients:** Nutrients such as amino acids, fatty acids, vitamins, and minerals play essential roles in maintaining health. Vitamins, found in most foods, help prevent disorders like stroke, cataracts, osteoporosis, and heart disease. Minerals derived from plant, animal, and dairy sources contribute to strong bones, teeth, and muscles while supporting nerve function and heart rhythm.
 - **Herbals:** Herbal nutraceuticals are known for their role in improving health and preventing chronic diseases, offering properties such as analgesic, anti-inflammatory, antipyretic, and antiarrhythmic effects. Compounds like tannins help manage conditions such as stress, cough, and hypertension, while proanthocyanidins are associated with benefits in cancer, ulcers, and urinary tract infections.
 - **Phytochemicals:** Phytochemicals, also known as phytonutrients, are plant-derived compounds with specific biological activities that support human health by acting as substrates, cofactors, enzyme inhibitors, and agents that enhance nutrient absorption.

- **Nutraceutical Enzymes**

Nutraceutical enzymes are derived from plants, animals, and microbial sources and are essential for proper physiological functioning. Enzyme supplements can aid in managing conditions such as digestive disorders, blood sugar imbalances, and obesity.

- **Probiotic Microorganisms**

Probiotic microorganisms, meaning 'for life,' are live beneficial microbes that improve gut health when consumed in adequate amounts. They enhance digestion, support nutrient absorption, and protect the body by inhibiting harmful pathogens, including bacteria, viruses, and yeasts, while maintaining a healthy symbiotic relationship with the gastrointestinal system.

2. Non-Traditional Nutraceuticals (Biotech-Modified)

Non-traditional nutraceuticals are artificially developed or biotechnologically modified foods designed to enhance health benefits. These include fortified and recombinant nutraceuticals.

- **Fortified:** Fortified nutraceuticals are created by adding nutrients to foods through agricultural practices or direct fortification, such as vitamin-enriched cereals, vitamin-D-fortified milk, and folic acid-enriched flour.
- **Recombinant:** Recombinant nutraceuticals involve advanced techniques, like genetic engineering and fermentation, to enhance or extract bioactive compounds. Genetically modified cows producing human lactoferrin to address deficiencies or () are examples of this category.

3. Other Related Categories

- **Functional foods:** Functional foods are enriched or fortified during processing and provide health benefits beyond basic nutrition. They contain physiologically active components derived from plant or animal sources. Examples include probiotic yoghurts for gut health, fibre- and vitamin-enriched cereals, and omega-3-fortified milk for cardiovascular health.
- **Medicinal foods:** Medicinal foods are specifically formulated for consumption under medical supervision to manage particular diseases or conditions with unique nutritional requirements. Examples include medicated health bars, transgenic plants for oral vaccines, and lactoferrin-enriched products for immunity enhancement.
- **Dietary supplements:** Dietary supplements are products intended to supplement the diet and are consumed orally. They may contain vitamins, minerals, herbs, amino acids, metabolites, or combinations of these. These supplements must be regulated and labelled appropriately,

ensuring they are not presented as conventional foods. Examples include minerals, coenzyme Q, carnitine, and herbal products such as ginseng, ginkgo biloba, and saw palmetto.

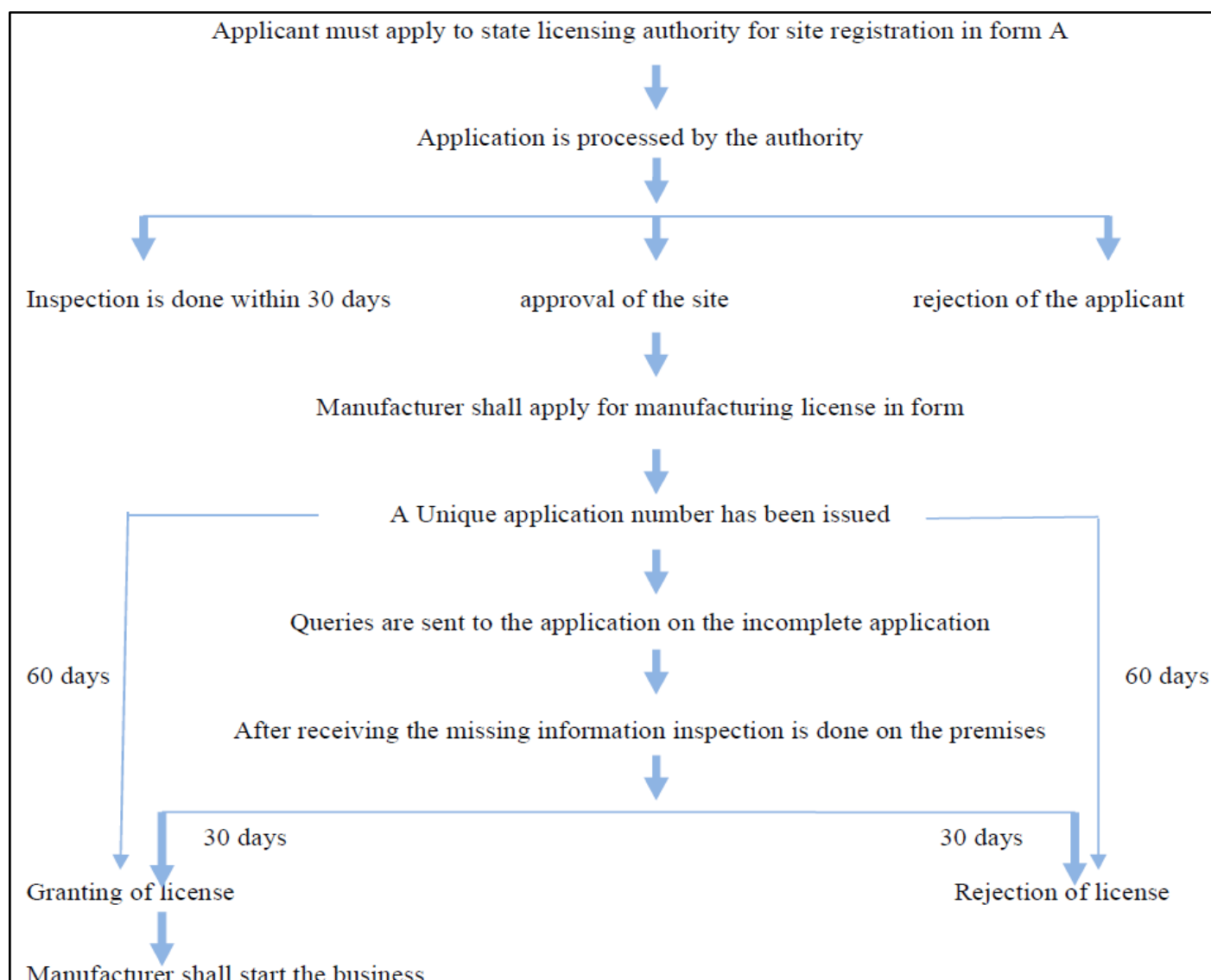


Figure: Registration process of nutraceuticals in India¹⁸⁰

Labelling and claims in nutraceuticals: Common practices around the world

India and China, among several other countries, have a rich traditional practice in the administration of plant and herbal remedies for health and medicinal purposes. While traditional medicine is well recognised and regulated, health supplements are yet to achieve similar acceptance in the regulatory space, particularly for claims.

The WHO acknowledges traditional knowledge in its definition: 'traditional and complementary medicine is the sum total of the knowledge, skills, and practices based on the theories, beliefs, and experiences indigenous to different cultures, whether explicable or not, used in the maintenance of health as well as in the prevention, diagnosis, improvement or treatment of physical and mental illness.'¹⁸¹

¹⁸⁰ Bansal, R., & Dhiman, A. (2020). Nutraceuticals: A Comparative Analysis of ReModernisation regulatory Framework in Different Countries of the World. *Endocrine, Metabolic & Immune Disorders - Drug Targets*, 20(10), 1654-1663.

¹⁸¹ Bansal, R., & Dhiman, A. (2020). Nutraceuticals: A Comparative Analysis of Regulatory Framework in Different Countries of the World. *Endocrine, Metabolic & Immune Disorders - Drug Targets*, 20(10), 1654-1663.

While the classical methodology of clinical evaluations and dose-response studies forms the core of evaluating the safety of substances added to foods (e.g., food additives), a regulatory approach of 'history of safe use' and 'qualified presumption of safety' needs to be relied upon where traditional knowledge exists about a food or food component (botanicals and their extracts), in the country or elsewhere.

The Scientific Committee of the European Food Safety Authority (EFSA) used the 'qualified presumption of safety' approach for microorganisms in food to propose criteria for presuming a botanical or botanical preparation to be safe. Most products still do not need stringent regulation over formulas and branding. Health claims on nutraceuticals serve to inform customers that, when combined with a good diet, they may lessen the risk of certain diseases.

As part of the Nutrition Labelling and Education Act (NLEA) of 1990, the FDA first authorised seven health claims in 1993. Six further claims have been approved by the FDA since 1993. To help consumers get this information faster, the Food and Drug Administration Modernisation Act of 1997 contained a provision to expedite the process of establishing the scientific foundation for health claims. Despite this, food producers may make health claims to advertise their products, offering information on healthy eating habits that may help lower the risk of heart disease, cancer, osteoporosis, high blood pressure, dental cavities, or certain birth abnormalities. Structure and function statements are not the same as health claims: unlike health claims, they do not address illness risk reduction, and the FDA neither pre-approves nor authorises structure or function claims. When a manufacturer makes a structure or function claim, the firm is responsible for ensuring that the claim is accurate and not misleading.¹⁸²

Many academic, scientific, and regulatory organisations are examining how to establish the scientific foundation for claims for nutraceutical functional components. The five categories of health-related statements that can appear on food and dietary supplement labels are as follows:

- **Nutrient-content claims** to show the existence of a certain nutrient at a given level.
- **Structure and function claims** to explain the impact of dietary components on the body's normal structure or function. Dietary guidance **claims** to address the health advantages of certain food groups.
- **Qualified health claims** to reflect a growing link between dietary components and illness risk, as recognised by the FDA and substantiated by a substantial body of reliable scientific data.
- **Health claims** for demonstrating a link between dietary components and the risk of illness or health condition, as approved by the FDA and backed by considerable scientific consensus.

Weak Regulation Of Claims In India

Once a product is placed under food law, the main control on what a company can say comes from the Food Safety and Standards Act, 2006 and the regulations made under it. Section 24 of the Act says that no person can make false or misleading claims on the label or in advertising. These rules speak of general principles, like truthful labelling, but in practice, they leave a lot of scope for interpretation. They do not create a system like drug law, where efficacy has to be proved before a product is launched.¹⁸³

The Structural Deficit in Claim Verification

The 2011 Food Safety and Standards (Packaging and Labelling) Regulations and the 2016 Food Safety and Standards (Health Supplements, Nutraceuticals, Food for Special Dietary Use, Food for Special Medical Purpose, Functional Food and Novel Food) Regulations also try to control the way

¹⁸² Patel, H., Shah D, Maheshwari D. A Short Review on a Comparative Study of Regulation of Nutraceuticals in the USA and India. International Journal for Pharmaceutical Research Scholars(IJPRS), 2014, vol 3, issue -1, 736-741.

¹⁸³ Boindala, S., & Lewis, J. I. (2019). The grand challenge of regulating health foods in India. The Indian Journal of Medical Research, 150(3), 248-253.

claims are made. Many products in the Indian market use phrases that sound scientific but are not tested by an independent regulator. Words like 'clinically tested' or 'backed by research' appear often, but there is no system where the Food Safety Authority checks the research before allowing the claim. The authority mostly acts when someone complains or during inspections. Even then, the penalties are weak compared to the profit companies make.

A good example of the weakness is the way muscle-building supplements, protein powders, and immunity boosters advertise themselves. They regularly use numbers like “50% better absorption” or 'two times faster recovery.' These claims sound like hard scientific results, but often the only basis is a small self-funded study or a report in a journal that does not have strong peer review.

- **Rahul Shekhawat v. Big Muscle Nutrition:** In *Rahul Shekhawat v. Big Muscle Nutrition*, a case filed before the consumer forum, the complaint was that a protein supplement was mislabelled and did not meet the nutritional claims printed on the box. The consumer forum looked at lab reports and found that the claims were not fully supported. The case is important because it showed that individual consumers often have to fight misleading labels themselves instead of the regulator taking strong action.

Case Study: MuscleBlaze’s big claims with thin evidence¹⁸⁴

A market leader can be seen turning a small, in-house, open-label, surrogate-endpoint study published in a discredited journal into a nationwide '50% absorption' headline. This is exactly how the food classification route, plus weak scrutiny of scientific claims, allows drug-like marketing without drug-level proof. The net effect is that the Indian framework puts too much responsibility on the consumer to question or complain. The case shows that MuscleBlaze promoted its Biozyme Whey Protein with a '50% higher absorption' claim based on a small, in-house, open-label study with only 24 participants and no blinding, where surrogate markers like amino acid levels and nitrogen balance were used instead of direct absorption or clinical outcomes. Overall, the claim was found to be an overstatement of weak evidence, highlighting how limited scrutiny and low penalties in India allow exaggerated, drug-like marketing claims, placing the burden on consumers to question them.

Moreover, the penalties under Sections 53 and 54 of the Food Safety Act are mostly fines, often less than the profit from a single advertisement campaign. As a result, the risk of making a false or inflated claim is low.

International Regulatory Frameworks

1. Nutraceutical Regulation in the United States

In the United States, nutraceuticals are generally referred to as dietary supplements. The U.S. Food and Drug Administration (USFDA) defines nutraceuticals as dietary supplements or ingredients. The regulatory framework for these products was established under the Dietary Supplement Health and Education Act of 1994 (DSHEA), which classifies dietary supplements as a separate category of food rather than drugs. The DSHEA stripped the FDA of its jurisdiction to oversee 'dietary supplements,' classifying them as a 'separate regulatory category of food.' They do not fall under either the food or medication categories.

DSHEA describes a dietary supplement as 'a substance intended to complement a diet that carries or includes one or more of the following dietary ingredients: vitamins, minerals, amino acids, herbs or other botanical ingredients; a concentrate, metabolite, component, extract, or mixture of the above-listed ingredients.' It must also be used in pills, capsules, tablets, powders, or liquids; not be

¹⁸⁴ Anupam Trehan, Anjali Chawla, Perna Sharma & Vipin Sharma, An Open Label Clinical Study to Determine the Effect of Enhanced Absorption Formula (MB EnzymePro) on the Bioavailability of Whey Protein in Healthy Male Subjects, 13 J. Food Processing & Tech. 820 2022

used as a conventional food or as a single meal or diet product; and be marked as a dietary supplement. Nutraceuticals, as dietary supplements, are typically derived from food sources but contain concentrated amounts of bioactive substances in non-food forms to enhance health benefits and are often consumed in higher doses than those obtained from regular diets. For example, ipriflavone, a synthetic derivative of the isoflavone daidzein found in soy protein, is considered a nutraceutical.

DSHEA notes that the product safety evaluation is the responsibility of the manufacturer. If a dietary supplement contains a new ingredient, the manufacturer must notify the FDA within 75 days of launch that the new ingredient 'can reasonably be expected to be healthy.' This is a less rigorous process than that needed for food additives (articles used to improve a food's scent, colour, structure, or taste), for which a standardised health assessment process is performed.

FDA and FTC Roles

- Under DSHEA, manufacturers are responsible for ensuring the safety of dietary supplements before they are marketed. Once the product is available in the market, the FDA monitors its safety and can take action against products found to be unsafe. According to FDA regulations, all domestic and international companies involved in the manufacturing, packaging, labelling, holding, or distribution of dietary supplements in the United States must comply with Current Good Manufacturing Practices (CGMPs) to ensure product quality and safety. Manufacturers, packers, or distributors are required to report any serious adverse events associated with dietary supplement use to the FDA, particularly when their name appears on the product label. The FDA also oversees product-related information, including labelling, claims, package inserts, and supporting literature. Advertising of dietary supplements is regulated by the Federal Trade Commission (FTC), ensuring that promotional claims are truthful and not misleading. In recent years, the FTC has actually taken companies to court for making unsupported claims about weight loss or muscle growth products.

The registration process for nutraceuticals in the US requires the applicant to file an application to the FDA along with GMP requirements. An inspection is carried out by an FDA member. The applicant must update their registration within 60 days of any change in information. The applicant must review their registration every two years. If the applicant fails to renew, FSMA authorises the FDA to suspend the registration. If any contamination is found in the food or supplement, the application is rejected.

Health claims approved under the Nutrition Labelling and Education Act of 1990 (NLEA) identify a relationship between a food product, food ingredient, or dietary supplement and reduced risk of a disease or health-related condition. Health claims are categorised into the following types.¹⁸⁵

- **Significant Scientific Agreement (SSA) claims:** These claims can be used for traditional foods and dietary supplements. The generic SSA standard is used to assess the well-established relationship between a nutrient and a disease.
- **FDA Modernisation Act (FDAMA) claims:** These statements may only be used in conventional food and may not be used in dietary supplements. The FDA authorises the use of an FDAMA statement as a result of a stakeholder notice.
- **Qualified health claims:** These may be used for traditional foods and dietary supplements. Any interested party can ask the FDA to issue a health statement regulation. The application is reviewed by the FDA according to the SSA standard.

¹⁸⁵ Rani, A., & Shukla, V. K. (2020). Study of Nutraceutical regulations around the Globe. *International Journal of Drug Regulatory Affairs*, 8(1), 15-24.

- **Nutrient content claims:** It includes the amount of certain nutrients or substances in a food, such as 'low in fat' or 'a good calcium source,' and are used to define the percentage of a nutrient in a product compared to the daily value.
- **Structure or function claims:** Approved under DSHEA 1994, these arguments entail a positive contribution to health, the enhancement of a function, or the alteration or protection of health. However, such claims must not imply the diagnosis, cure, mitigation, treatment, or prevention of any disease, as this would classify the product as a drug and require separate regulatory approval.

2. European Union Regulations

In European Union food law, a specific regulatory framework for 'functional foods' or 'nutraceuticals' as a single category does not exist. The rules are numerous and depend on the nature of the foodstuff. The rules of the General Food Law Regulation, including responsibility for food safety, traceability, recall, and notification, apply to all foods. If a claim implies a medicinal benefit for a nutraceutical product, the product would need to comply with the regulatory requirements for medicinal products in respect of safety, efficacy, quality testing, and marketing authorisation procedures.

According to EU regulations, claims regarding the beneficial effects of nutraceuticals can only be 'health claims' and not 'medicinal claims.' Claims must not state that by eating or taking the nutraceutical, a disease will be prevented or cured; only that it may help to improve health, possibly assisting in the avoidance of the onset of illness. Many functional foods may be subject to the regulatory frameworks of PARNUTS (foods prepared for particular nutritional purposes) or dietary foods and supplements. The Novel Food Regulation applies to fresh functional foods.

The EU's central instrument on nutrition and health claims says that no health claim can be made unless it has been approved after a scientific review by EFSA. The authority runs a strict pre-market authorisation process. Claims are published in a public register so both consumers and competitors can check which claims are authorised and which are rejected. The standard of proof is also high: EFSA requires human studies with proper controls, and looks for consistency with the wider body of science. Because of this, many claims are actually rejected. Hundreds of probiotic health claims were refused because they were not backed by solid evidence.

More specifically, nutraceuticals are regulated as 'food supplements' in compliance with Directive 2002/46/EC of the European Parliament and of the Council of 2002 on the interpretation of Member States' legislation on food supplements. It lays down harmonised guidelines for the labelling of food supplements and provides specific rules for vitamins and minerals in food supplements. The European Food Safety Authority (EFSA) controls the legislation.

Article 5 of the regulation, the requirements for a claim include the availability of the active ingredient in sufficient quantities to achieve the reported nutritional or physiological effect; the nutritional information must be given in compliance with the Directive.

Article 10 special conditions prevail and state that a varied and healthy diet is important. Appropriate warnings must be given for products that, if consumed excessively, are likely to present a health risk.

Article 6 sets out the general substantiation criteria. Health claims would only be accepted for use on food labelling, display, and advertisement on the EU market after the highest possible scientific assessment.

Article 4 provides for the establishment of nutrient profiles, i.e., the total amounts of fat, salt, and sugar that products must contain to make acceptable nutritional and safety claims. Nutrient profiles must be based on scientific knowledge of diet and nutrition and their relationship to health and must be developed within two years of the entry into force of the Regulation.

Claim Types in the EU

- Nutrition claims are only allowed when they match the approved statements and quantitative conditions listed in the Regulation's Annex and the EC's accepted and rejected claims database. They are generally not permitted on alcoholic products above 1.2% alcohol by volume, unless the claim specifically concerns ethanol content or reduced calories.
- Health claims based on generally accepted scientific evidence had to be submitted through Member States so the European Commission could compile the approved claims lists within 12 months of the Regulation taking effect. In simple terms, only evidence-backed claims that passed this formal process could be used.
- Article 18 applies to claims relying on newly generated scientific evidence or on proprietary data that needs protection. Such claims cannot be used freely and must first receive specific authorisation under the Article 18 request process.
- Reduction of Disease Risk Claims and children's growth, development, and health claims fall under Article 14 and require authorisation under Article 15 before use. RDRC statements must also carry a specific warning that the disease mentioned has multiple risk factors and that changing one risk factor may or may not help, while all authorised health claims must be used only within their approved terms, except where proprietary data protection applies.

The above comparison outlines that the regulations in the EU and the USA are not perfect; they still face aggressive marketing and sometimes weak oversight. However, they do have institutional checks that India lacks. EFSA uses a transparent and rigorous scientific process with human trials. The FDA and FTC combine disclosure with active enforcement. India, on the other hand, has laws on paper but depends on post-market complaints and minor inspections.

Recommendations for India

In India, the real issue is not the absence of law but the way the existing law is designed and enforced. The Food Safety and Standards Act, 2006, the Drugs and Cosmetics Act, 1940, and the Consumer Protection Act, 2019 all create duties, but none of them fully cover the grey area under which nutraceuticals and functional foods operate. The gap lets companies make drug-like claims without drug-level proof.

The following five recommendations address each identified gap.¹⁸⁶

• **Recommendation 1: Create a Separate Category for Nutraceuticals with Stricter and Clearer Rules**

Section 22 of the Food Safety and Standards Act already recognises food for special dietary uses and functional foods, but it is too broad. There should be a new chapter or rules under the Act that only focus on health claims. Every product that carries a health benefit claim should have to submit proof before entering the market, the same way the European Union uses pre-market authorisation.

• **Recommendation 2: Make Disclaimers Mandatory for All Functional Foods and Supplements**

The United States already requires a statement that the product is not intended to diagnose, treat, cure, or prevent any disease. Adding a similar rule in India would make it clear to buyers that these products are not equal to drugs, even when the marketing tries to suggest otherwise.

• **Recommendation 3: Create a Public Register of Approved Claims**

The European Food Safety Authority keeps a database that anyone can check. If India adopted something like this, companies could not hide behind vague scientific terms, and consumers, doctors, and lawyers could all see which claims are backed and which are not.

¹⁸⁶ Lewis JI. Good regulatory practice: Health supplements & nutraceuticals: PFNDAI Bulletin; March 2017. p. 8, 10-12

- **Recommendation 4: Strengthen Enforcement**

The Authorities should put more effort into the evaluation of the scientific claims. The Central Consumer Protection Authority also has the power under Section 18 of the Consumer Protection Act, 2019, to stop misleading advertisements. It should use this more actively in the health food market. The existing lab infrastructure of ICMR-NIN can be used for the pre-market evaluations. The Right to Information reply from the Food Safety and Standards Authority of India shows that most violations are only about misbranding or labelling. The authority should put more effort into evaluating the scientific claims.

- **Recommendation 5: Increase Penalties**

At present, the fines under the Food Safety and Standards Act are small compared with the money companies earn from these products. Raising the penalties and including imprisonment would make it less attractive to stretch the truth in advertisements. Such procedural pressure can be created upon the companies by making specific amendments to the Companies Act, 2013.

Comparative Note on Indian Nutrition Guidelines: FSSAI v. ICMR

In a country of 1.4 billion people, the government simultaneously battles severe undernutrition in rural populations, addresses rising obesity and non-communicable diseases (NCDs) especially in urban centres, and corrects widespread micronutrient deficiencies that cut across both geographies. This convergence of three distinct forms of nutritional crisis, often referred to as the 'triple burden of malnutrition', requires a layered policy response.¹⁸⁷ The nutritional guidelines released by governments or their responsible arms are one of the first steps in this direction.

India has two major advisories, the Dietary Guidelines for Indians (DGI) by ICMR-NIN and the Eat Right India Handbook (ERI Handbook) by FSSAI, that direct populations and policymakers in matters of nutrition intake.

- **The ICMR-NIN DGI 2024** is the third edition of India's national food-based dietary guidelines. Its purpose is to translate scientific evidence on nutrient requirements into practical, population-level dietary advice. It addresses India's triple burden of malnutrition, provides quantitative food-based guidance across the life course, and aims to prevent major diet-related conditions, including type 2 diabetes, cardiovascular diseases, and hypertension. The scientific foundation is the ICMR-NIN Expert Group's updated Recommended Dietary Allowances (RDA) report, first published in 2020 and updated in 2024. While the RDA report is a scientific evaluation and assessment output, the DGI is the food-level guidance based on the report.
- **The ERI Handbook's** purpose is different at its root. It is the operational manual for the Eat Right India (ERI) movement, launched by FSSAI in October 2018. The movement was born out of FSSAI's mandate under the Food Safety and Standards Act 2006 to ensure 'safe and wholesome food' for the country, reinterpreted in light of the National Health Policy 2017's emphasis on preventive and promotive healthcare. The Handbook's stated aim is to enable State Food Safety Commissioners, District Commissioners, Food Safety Officers, NetProFaN (Network of Professionals in Food and Nutrition) members, and CSR-funded corporates to implement ERI initiatives and scale them up. The Handbook positions itself explicitly as a 'one-stop shop' on all matters related to Eat Right India in action. It is anchored in three themes: eat safe (personal and surrounding hygiene, combating adulteration, reducing toxins and contaminants), eat healthy (promoting diet-diversity and balanced diets, eliminating industrial trans fats, reducing salt/sugar/saturated fats, large-scale fortification to address micronutrient deficiencies), and eat sustainable (local and seasonal foods, preventing food waste, conserving water, reducing chemicals in food production, safe food packaging). These themes are operationalised through six strategic priorities: robust food standards and codes of practice; credible food testing and effective surveillance; strengthened compliance, enforcement, and emergency response; a strong culture of self-compliance; promoting healthy diets and sustainability; and empowered consumers.¹⁸⁸

1. End-User

The two documents are aimed at fundamentally different audiences: the DGI 2024 is designed for health and nutrition professionals and scientifically literate citizens, while the ERI Handbook primarily reaches administrators and institutional implementers, making the ERI's reach broader in terms of institutional scope and the DGI's authority deeper in scientific specificity.

- **The DGI 2024** is written for health and nutrition professionals, including clinicians, dietitians, nutritionists, public health practitioners, and policymakers. Its technical language and

¹⁸⁷ Singh, A., Agrawal, D., & Bhardwaj, A. (2025). Changes in Dietary Guidelines by National Institute of Nutrition (NIN) and ICMR: From 2011 to 2024. *Indian Journal of Community Medicine*, 50(6), 1088-1089.

¹⁸⁸ FSSAI (2018, updated). Eat Right India Handbook. Food Safety and Standards Authority of India, Ministry of Health and Family Welfare, New Delhi,.

quantitative specificity make it most accessible to this professional audience, with the expectation that guidance cascades to citizens through intermediaries and national programmes. It also speaks to scientifically literate citizens, particularly in its chapters on food labels, food safety, and healthy lifestyles.

- **The ERI Handbook's** primary end-users are administrative and implementation stakeholders, including State Food Safety Commissioners, District Commissioners, Food Safety Officers, NetProFaN members, and corporates deploying CSR funds. A secondary channel is frontline health workers (Anganwadi Workers, ASHA workers, ASHA Sangini, Multi-Purpose Worker-Females, Community Health Officers, Mid-Level Health Providers, and Medical Officers at Health and Wellness Centres) who receive the ERI Toolkit, co-developed by FSSAI, the National Health Systems Resource Centre (NHSRC), and the Voluntary Health Association of India (VHAI), to guide community-level dietary and food safety messaging.

2. Methodology

The methodologies adopted for framing these guidelines are divergent, reflecting the difference in their purpose: the DGI 2024 is grounded in systematic scientific review and quantitative calibration, while the ERI Handbook is built on stakeholder consultation and communication design.

- **The DGI 2024** draws on systematic reviews and meta-analyses, data from the National Family Health Survey-5 (NFHS-5) and the Comprehensive National Nutrition Survey 2016-18 (CNNS), Indian-specific research, and global nutrition science advances. The expert group reviewed existing nutrient requirement data, translated them into food-based guidelines calibrated to Indian dietary patterns, and arrived at 17 dietary guidelines covering food groups, population segments, food safety, physical activity, and sustainable diets. The reference person for quantitative recommendations is a sedentary adult man weighing 65 kg with a 2000 kcal per day energy requirement, with adjustments specified for age, sex, physiological status, and activity level.
- **The ERI Handbook** was developed through stakeholder consultations using a flexible, adaptive design approach comparable to a lean start-up model. It draws on theories of behavioural economics (particularly nudge theory), communication science, systems change frameworks (NHS Large-Scale Change model, CLEAR Framework for Leading Systems Change), and a review of advanced food regulatory systems from countries including Australia, New Zealand, Canada, the EU, and the US. Notably, it is not the product of a nutrition science review process. Wherever it carries dietary content, the recommendations are derived from existing guidelines and FSSAI's regulatory standards rather than from independent scientific synthesis.

3. Broad Framework and Key Areas

3.1. Overview of Frameworks

The DGI 2024 and the ERI Handbook are organised along fundamentally different architectures, the former as a scientific reference document structured around nutrients, food groups, and population segments, and the latter as an operational implementation guide structured around themes and institutional actors, making a direct structural comparison instructive for understanding how each seeks to influence nutritional outcomes.

- **The DGI 2024** is organised into 17 dietary guidelines covering food diversity, caloric and energy balance, ten food groups (cereals and millets, pulses, vegetables, fruits, dairy, eggs, fish and meat, nuts and oilseeds, fats and oils, and herbs and spices), population-specific needs (infants, children, adolescents, adults, pregnant and lactating women, elderly), food safety and hygiene, physical activity, sustainable dietary choices, and food label literacy. On caloric intake, the DGI 2024 uses an Estimated Energy Requirement (EER) framework rather than a single fixed figure, stratified by age, sex, physiological status, and activity level, anchored to a reference sedentary adult male of 65 kg requiring approximately 2000 kcal per day.

- **The ERI Handbook** is structured in three parts: Part 1 (The Vision) establishes the challenge and the three-pillar framework (eat safe, eat healthy, eat sustainable); Part 2 (Strategy and Action Plans) covers supply-side initiatives, demand-side initiatives, mechanisms for enabling healthy choices, and sustainability initiatives; and Part 3 covers partners, coordination, rewards, recognition, and networks. The Handbook provides no caloric targets for any population group, consistent with its role as a communication and implementation manual rather than a clinical or scientific reference.

3.2. Overall Daily Diet Principles and Recommendations

Both documents approach daily dietary guidance through different lenses, quantitative and food-group-specific in the case of the DGI 2024, and behavioural and qualitative in the case of the ERI Handbook, reflecting their distinct purposes and audiences.

- **The DGI 2024** introduces the 'My Plate for the Day' as its central dietary visual. Its approach emphasises dietary diversity, a shift from refined carbohydrates to whole grains and millets, and a reduction of cereal intake from the current 50-70% of total energy to no more than 45%. The ten food groups (an expansion from four in the 2011 edition) include a separate category for herbs and spices, recognising their micronutrient and phytochemical significance. Quantitative food-group targets are detailed in the nutrient-specific sections that follow.
- **The ERI Handbook's** dietary approach under the Eat Healthy pillar is explicitly behaviour-oriented and qualitative. It promotes dietary diversity and balanced diets and communicates via simplified behavioural nudges. It does not specify food group proportions, serving sizes, or daily quantities. The Eat Right Toolkit, which is the ground-level implementation of the Eat Healthy message and is integrated into the training modules of frontline health workers at Aarogya Mandirs (Health and Wellness Centres), is the closest the ERI ecosystem comes to dietary guidance at the citizen level.

4. Assessing Nutrient-Specific Guidelines

Across all macronutrients and micronutrients, the DGI 2024 consistently provides quantitatively specific, evidence-grounded targets calibrated to Indian dietary patterns, while the ERI Handbook engages with nutrition primarily through regulatory action, behaviour change campaigns, and food safety instruments rather than through nutrient-level guidance.

4.1. Carbohydrates

DGI guidelines outline that total carbohydrate intake should constitute 45-60% of total energy, with cereals specifically capped at 45% of total energy and a quantitative emphasis on shifting from refined carbohydrates (polished white rice, refined wheat flour) to whole grains and millets. At least 50% of all cereal intake should come from whole grains or millets, and added sugar intake should be capped at less than 5% of total calories, specifically to tackle the rising prevalence of type 2 diabetes and obesity. Dietary fibre is recommended at 25-30 grams per day with food-source guidance, linking to reduced digestive disease risk and improved insulin sensitivity.

The ERI Handbook does not provide carbohydrate intake targets of any kind. Its dietary campaigns address salt, sugar, and fat broadly rather than carbohydrate composition specifically.

The DGI 2024, despite its quantitative targets, provides insufficient guidance on the glycaemic load and glycaemic index of Indian staple foods, which is especially critical given India's epidemic of type 2 diabetes. Additionally, the distinction between refined carbohydrates and ultra-processed carbohydrate-rich foods (biscuits, instant noodles, ready-to-eat snacks) is underemphasised.

4.2. Proteins

The DGI 2024 addresses protein with particular emphasis on quality. The guidelines stress complementary protein combinations for predominantly plant-based diets, specifying combining

cereals with pulses and oilseeds to improve the overall amino acid profile. For example, a rice-dal or roti-dal combination provides better essential amino acid coverage than either food alone. Dairy, eggs, and fish are consistently recognised as high biological value protein sources providing all essential amino acids in appropriate proportions. Fish is specifically recommended at 80 grams per day, or 250-300 grams twice weekly, for its combined protein and omega-3 EPA/DHA benefits. Protein requirements are also adjusted for physiological status: the second trimester of pregnancy requires an additional approximately 8 grams per day, and the third trimester approximately 18 grams per day. A significant 2024 addition is an explicit advisory against protein supplements, noting that adequate protein can be obtained through appropriate food combinations and that supplements carry risks of overconsumption and micronutrient imbalances.

The ERI Handbook's engagement with protein is limited to the 1000-day nutrition concept (gestation through the child's second birthday), which recognises the critical window for adequate protein in early life but does not specify quantities, quality considerations, or food-source guidance. There is no discussion of complementary amino acid combining, protein supplements, or recommendations for any other population segment.

Notably, neither document provides guidance on protein timing and pacing across meals, a dimension increasingly recognised as important for muscle protein synthesis, satiety, and metabolic health. Amino acid optimisation for special populations such as the elderly (at risk of sarcopenia), adolescent girls, and pregnant women with predominantly vegetarian diets also warrants more detailed guidance than either document currently provides.

4.3. Fats

The DGI 2024 is most technically detailed on fats, reflecting the quality-over-quantity shift in global evidence on dietary fat and cardiovascular risk. Total visible fat intake is recommended at approximately 27 grams per day, with saturated fat intake to be kept below 7% of total caloric intake. Industrial trans fat intake limits are set at zero. Omega-3 fatty acids (EPA and DHA from marine sources) are emphasised. The DGI 2024 recommends diversifying fat sources through oilseeds, nuts, millets, and legumes and rotating between different cooking oils (groundnut, mustard, sesame, rice bran, and coconut in regional contexts) to ensure a balance of saturated, monounsaturated, and polyunsaturated fatty acids. Ghee and butter are recognised as traditional fat sources, but should be limited given their saturated fat content. Invisible fat (naturally present in nuts, seeds, milk, and fish) should be counted toward total daily fat intake. Additionally, the advisory on reused cooking oil notes that repeated heating generates harmful compounds, including trans fats.

The ERI Handbook addresses fats through three specific instruments:

- The *Aaj Se Thoda Kam* campaign targets a gradual reduction in all HFSS (high fat, salt, and sugar) foods through behaviour change communication via celebrity-fronted short videos, social media, television channels, including Doordarshan and Food Food, cinema halls, including PVR screens, and occasional events such as the Swasth Bharat Yatra cyclothon. Fifteen major food companies, including Britannia and Marico, took voluntary pledges to reformulate packaged foods to reduce fat, sugar, and salt levels; three food service establishments (Bikanervala, Haldiram's, and Jubilant Foods) pledged to provide healthier food options and introduce menu labelling.
- The RUCO (Repurpose Used Cooking Oil) initiative, run in partnership with the Ministries of Health and Family Welfare and Petroleum and Natural Gas, collects used cooking oil and converts it into biodiesel, addressing both the health risk of repeated oil use and generating an environmental benefit through biodiesel production.
- Most substantively, the Trans-fat Free India@75 initiative establishes a regulatory timeline for industrial trans fat elimination. FSSAI regulations have progressively reduced the permissible level of trans fatty acids (TFA) in partially hydrogenated vegetable oils (PHVOs), including

vanaspati, margarine, bakery shortening, and industrial margarine, reaching a current limit of not more than 2% of total fats from January 2022 (reduced from an earlier limit of 5%).

This is the most regulatory and quantitatively specified nutritional content in the ERI Handbook, and also the area where the Handbook's content is most closely aligned with DGI's zero trans fat recommendation.

Despite emphasis on this macronutrient, both documents provide weak guidance on omega-3 fatty acid balance, particularly the ratio of omega-6 to omega-3 in the average Indian diet, which is estimated to be heavily skewed towards omega-6 due to the predominance of sunflower and soybean oils. Neither document addresses processed fat sources comprehensively, including the hydrogenated or interesterified fats present in commercial baked goods, fried snacks, and margarine-based products that are increasingly prevalent in Indian diets.

4.4. Micronutrients

The DGI 2024 addresses micronutrients with depth, particularly given India's documented burden across iron, folate, vitamin B12, vitamin D, vitamin A, iodine, zinc, and calcium.

For **iron**, the primary concern is the low bioavailability of non-haem iron from plant sources, which constitutes the bulk of iron intake in predominantly vegetarian Indian diets. Green leafy vegetables, legumes, and fortified foods are recommended as iron sources, and consuming vitamin C-rich foods (citrus fruits, amla) alongside iron-rich plant foods enhances absorption. Supplementation protocols for pregnant women specify 30-60 mg of elemental iron and 400 micrograms of folic acid daily.

Vitamin B12 deficiency is flagged as pervasive among vegetarians; dairy, eggs, and fish are recommended as sources, or supplements where dietary sources are inadequate. **Vitamin D** deficiency is also acknowledged; safe sun exposure, fatty fish, eggs, and fortified milk are recommended as the primary sources. For **calcium**, the DGI 2024 sets a population-level requirement of 600 mg per day for general adults, rising to 1200 mg per day for pregnant and lactating women, with supplementation in pregnancy set at 1 gram per day from 14 weeks onward. **Iodine** requirements are addressed through universal iodised salt. The 2024 edition introduces microgreens as nutrient-dense sources of amino acids, fatty acids, micronutrients, antioxidants, and bioactive compounds.

The ERI Handbook's engagement with micronutrients is through food fortification, its most substantive nutritional policy lever. The Handbook positions fortification explicitly as complementary to dietary diversification, while a balanced diet remains the primary source for fulfilling nutritional requirements. This is in line with the FSSAI gazetted Food Safety and Standards (Fortification of Foods) Regulations, which cover five key staples: oil and milk (with Vitamins A and D), wheat flour and rice (with iron, folic acid, and Vitamin B12), and double-fortified salt (with iodine and iron).

Both frameworks are missing operational strategies for addressing micronutrient deficiencies at the individual level. ICMR identifies the deficiencies and their food sources with precision, but does not provide structured protocols for population-level correction. FSSAI leverages its regulatory and fortification infrastructure powerfully, ultimately, but does not articulate a nutrient-specific outcome framework. The absence of a unified bioavailability-aware recommendation system, one that accounts for the effect of phytates, tannins, and calcium on the absorption of iron and zinc from plant foods, is a notable shared gap.

4.5. Key Foods Addressed

Both documents engage with the major food categories consumed across India, though their emphasis, depth, and framing vary considerably; the DGI 2024 approaches these categories through

nutritional science, while the ERI Handbook typically engages through food safety standards and regulatory action.

- **Milk and Dairy**

ICMR-NIN recommends moderate inclusion of milk and milk products as a source of calcium, high-quality protein, riboflavin (B2), and vitamin D. The guidelines note that fermented products such as curd and buttermilk are particularly valuable because fermentation enhances the bioavailability of certain nutrients. FSSAI's guidance devotes significant attention to milk safety, covering proper storage temperatures, pasteurisation, and testing for adulteration. The National Dairy Development Board's milk fortification programme (adding Vitamin D and Vitamin A to milk) is aligned with FSSAI's regulatory framework. Neither document provides sufficiently differentiated guidance on the nutritional differences between full-fat and low-fat dairy, or on plant-based dairy alternatives for lactose-intolerant populations.

- **Oils and Fats**

ICMR-NIN recommends rotating between different cooking oils to balance fatty acid intake, specifically highlighting mustard oil (rich in MUFA and ALA, an omega-3), groundnut oil, sesame oil, and rice bran oil as culturally common and nutritionally acceptable choices. The guidelines recommend limiting total visible fat to 27 grams per day and avoiding reusing heated oils, which generate harmful trans-fatty acids. FSSAI has made industrial trans fat elimination a regulatory priority, with the current limit of 2% trans fat in partially hydrogenated vegetable oils effective from January 2022. The Orange Book, FSSAI's Guide for Safe and Nutritious Food at the Workplace, one of three colour-coded citizen-facing resource books in the ERI ecosystem alongside the Pink Book for homes and the Yellow Book for schools, advises workplace cafeterias to use healthier cooking methods and limit oil usage in menu planning.

- **Cereals and Millets**

This is the area of greatest convergence between the two frameworks, and also the area where the 2024 updates have been most significant. ICMR-NIN's 2024 edition enthusiastically promotes millets, specifically ragi (finger millet), bajra (pearl millet), jowar (sorghum), foxtail millet, kodo millet, and proso millet as nutrient-dense alternatives to polished rice and refined wheat. Millets are highlighted for their fibre content, micronutrient density (iron, calcium, zinc), low glycaemic index, and environmental sustainability. The Eat Right India framework similarly promoted millets strongly, especially in 2023 when India hosted the International Year of Millets (IYM 2023). Both FSSAI's programmes and ICMR's guidelines align with the government's broader Millet Mission, representing one of the few areas of explicit policy convergence.

- **Fruits and Vegetables**

ICMR-NIN recommends consuming at least 400 grams of vegetables per day, divided across three sub-groups: green leafy vegetables, other vegetables, and roots and tubers, with a further 100 grams of fruit recommended daily. Together, vegetables and fruits are intended to form approximately half the plate in the 'My Plate for the Day' model. Seasonal and locally available produce is encouraged. FSSAI's guidance on fruits and vegetables centres on safe handling (washing, storage, avoiding contamination) and adulteration detection (wax coatings, artificial ripening agents). The FSSAI's Eat Right Fruits and Vegetables Market initiative certifies produce markets for hygiene and safety standards. Neither document provides adequate guidance on the differential nutritional value of raw versus cooked vegetables, or on the effect of extended storage on micronutrient content.

- **Emerging and Functional Foods**

The ICMR-NIN 2024 edition introduces microgreens as a category of emerging functional foods, highlighted for their exceptional concentrations of amino acids, fatty acids, micronutrients, and bioactive compounds, including polyphenols and antioxidants. Fortified foods are addressed in the context of national programmes, and the guidelines support food fortification as a

population-level strategy for addressing micronutrient gaps. FSSAI's regulatory framework covers fortified foods under the FSS (Fortification of Foods) Regulations, mandating standards for fortified rice, wheat flour, edible oil, milk, and salt. Plant-based protein foods and novel protein sources receive limited attention in both documents.

4.6. Food Preparation and Cooking

The DGI 2024 and the ERI ecosystem both engage with cooking guidance, but from different vantage points: the DGI through the lens of nutrient retention, and FSSAI through the lens of food safety and hygiene in household and institutional kitchens.

The DGI 2024 provides practical cooking guidance grounded in nutrient preservation:

- Low-oil cooking methods (baking, steaming, boiling, stir-frying) are recommended to reduce total fat intake.
- Parboiling rice preserves B vitamins otherwise lost in milling.
- Germination and fermentation of pulses and cereals improve nutrient bioavailability by reducing antinutrients like phytic acid, thereby increasing the absorption of iron and zinc.
- Cooking green leafy vegetables briefly and in small amounts of water preserves heat-sensitive vitamins.
- Soaking pulses and discarding soak water reduces antinutrient load.
- Consuming vitamin C-rich foods alongside iron-rich plant foods at the same meal enhances iron absorption.

FSSAI's Pink Book is perhaps the most practically detailed of all ERI publications when it comes to cooking guidance. It covers nine domains of household food management, from buying and storing to cooking, serving, and managing leftovers, with specific attention to hygiene, cross-contamination, and temperature safety. The Orange Book extends cooking guidance to workplace cafeterias, covering safe food preparation, vendor management, and healthy menu planning. The Yellow Book introduces cooking as a subject of scientific inquiry for school children, including experiments using the DART Book (Detecting Adulteration with Rapid Tests) – a companion resource providing simple, home-based test protocols for identifying common adulterants in everyday food items such as milk, edible oil, and spices.

However, neither document provides an actionable cooking framework that integrates nutrient retention, culinary tradition, and practical constraints into a single coherent guide. A practical guide connecting regional Indian cooking styles (South Indian steam cooking versus North Indian tadka) to nutritional outcomes would significantly enhance both frameworks.

5. Food Safety and Hygiene

Food safety serves as a point of interface between the two documents, though with markedly different depth and emphasis: the DGI 2024 approaches it as a component of consumer education, while FSSAI – whose founding mandate is food safety – has built a comprehensive programmatic architecture around it.

The DGI 2024 devotes a section to food safety from a consumer education standpoint, covering safe storage, hygiene during food preparation, food adulteration detection, and food label reading. It advises on reading ingredient lists and nutrition labels, understanding use-by dates, and identifying adulterated foods. It flags that processed and packaged foods often contain hidden sources of salt, sugar, and trans fats not apparent from the product name alone, and positions label literacy as a key protective health skill.

Food safety is the founding mandate of FSSAI, and the ERI Handbook is far more developed on this dimension. The Eat Safe pillar is operationalised through FoSTaC (Food Safety Training and Certification, with 224 empanelled training partners and over 279,000 trained Food Safety

Supervisors by the Handbook's publication date), hygiene ratings for restaurants and other food service establishments (on a 1-5 scale, covering restaurants, meat shops, sweet shops and bakeries), clean street food hubs (working with municipal bodies to provide electricity, hot running water, and waste management to clusters of street vendors), Food Safety on Wheels mobile testing vans (54 vans in 32 states and UTs), the DART Book for rapid adulteration testing at home, the BHOG initiative for places of worship, Eat Right Station certification for railway stations, and the Consumer Connect guidance notes and Myth-Busters programme.

Despite FSSAI's stronger coverage of food safety, ICMR guidelines do not cross-reference FSSAI's food safety framework at all, creating a siloed approach to what should be a unified food systems message.

6. Fortification and Supplementation

Fortification and supplementation represent the most direct nutritional policy lever in both documents. While the two frameworks approach these tools differently, they share a common objective of addressing India's persisting micronutrient gaps, which were documented in Section 2.3 above.

Beyond what was noted in the micronutrient section, the DGI 2024 specifies supplementation protocols for vulnerable groups: iron-folic acid supplements for pregnant women, calcium supplementation from 14 weeks of pregnancy, and consideration of multiple micronutrient supplements in research settings. Iodised salt is recommended universally, and protein supplements are explicitly discouraged.

FSSAI's fortification architecture covering the five key staples noted earlier (rice, wheat flour, edible oil, milk, and salt) represents one of the most systematically executed nutrition interventions in the country, directly mapped to India's most common micronutrient deficiencies. The ERI Handbook does not distinguish between population-level fortification as a complementary public health strategy and individual therapeutic supplementation, presenting fortification as the primary lever for addressing micronutrient deficiency at scale. This means the Handbook does not guide practitioners on when fortification is insufficient and individual supplementation is warranted.

There is a significant gap in the integration of India's national fortification programmes with dietary planning guidance. Neither framework provides a clear consumer-facing message explaining how fortified staples contribute to meeting daily micronutrient needs, and what residual gaps must be filled through diet or supplementation. The absence of structured supplementation protocols in the ICMR document is particularly notable, as therapeutic supplementation is an established part of clinical practice.

7. Other Key Recommendations

Beyond macronutrients, micronutrients, and food safety, both documents engage with broader determinants of health, though the DGI 2024 is considerably more comprehensive in this space, addressing physical activity, hydration, and sustainability in quantitative terms that the ERI Handbook largely does not match.

The DGI 2024 recommends at least 30 minutes of moderate-intensity physical activity per day for adults and at least 60 minutes daily for children, noting that healthy diets and regular physical activity together can prevent up to 80% of type 2 diabetes cases and substantially reduce the burden of coronary heart disease and hypertension. Screen time is explicitly identified as a specific NCD risk factor, particularly for children and adolescents. Hydration guidelines are set at 8-10 glasses (approximately 2 to 2.5 litres) per day, varying by climate and activity, with culturally familiar options such as coconut water, buttermilk, and unsweetened fruit juices recommended while sugary drinks and excessive caffeine are cautioned against. Sustainability considerations are also woven into the

guidelines: reduced red meat consumption, millet and plant-based protein diversification, and food waste minimisation are linked explicitly to planetary health alongside personal health.

The ERI Handbook does not address physical activity, screen time, or hydration intake targets. Its Eat Sustainable pillar is more operationally developed than the DGI's sustainability integration, including Jaivik Bharat (organic food promotion through the India Organic logo), Save Food Share Food (food waste prevention), Safe and Sustainable Packaging in the Food and Beverage Sector, and RUCO. The ERI movement also explicitly maps its work to all 17 United Nations Sustainable Development Goals.

Neither document presents an integrated lifestyle-diet model that treats physical activity, sleep, stress management, and dietary behaviour as a unified system. The WHO's 'Best Buys' for NCDs and the evidence base for lifestyle medicine both point to the need for holistic frameworks that go beyond food composition.

8. Life-Stage Specific Guidance

India's nutritional burden is unevenly distributed across the life course, with specific vulnerabilities at each stage from infancy through old age. The DGI 2024 consistently leads with quantitative, evidence-based guidance calibrated to each life stage, while the ERI Handbook provides regulatory and programme-level support, with greater depth for some segments (school-age children) than others (older adults, pregnant women).

● Infants and Young Children (0-5 Years)

ICMR-NIN places extraordinary emphasis on the first 1000 days of life, spanning from conception through the child's second birthday, encompassing approximately 270 days of gestation and 730 days of early childhood as the most critical nutritional window. The guidelines mandate early initiation of breastfeeding (within one hour of birth), exclusive breastfeeding for the first six months, and continuation of breastfeeding alongside complementary feeding until two years and beyond. Colostrum is specifically identified as a critically important food source rich in proteins, minerals, and vitamins, particularly Vitamin A, and must not be discarded. The guidelines are explicit that infants should not be given water, honey, glucose, or formula in the first six months, as these increase diarrhoeal risk and displace breastfeeding. Complementary feeding should begin at six months with nutrient-rich homemade semi-solid foods such as mashed cereals, pulses, fruits, and vegetables. Young children below five years should receive less bulky but energy-dense foods rich in pulses, nuts, edible oil or ghee, milk, and eggs.

FSSAI's food safety regulations cover infant formula standards, and its general food safety guidance is relevant to the preparation of complementary foods, but there is no FSSAI resource book specifically dedicated to infant and young child feeding.

● School-Age Children and Adolescents (6-19 Years)

ICMR-NIN's 2024 edition addresses children and adolescents with a focus on supporting growth, cognitive development, and the prevention of both undernutrition and childhood obesity. Adolescents are noted to require elevated calcium intake to achieve optimal peak bone mass. The guidelines caution against HFSS foods for this age group, noting that they are calorie-dense but low in micronutrients and fibre. The 2024 edition also introduces guidance on screen time and sedentary behaviour as risk factors for adolescent health.

FSSAI's Yellow Book is the most extensive stakeholder-specific resource in the entire Eat Right portfolio for this age group. Developed through consultation with CBSE, NCERT, and state education boards, the Yellow Book is available in eleven Indian languages and is structured in two volumes targeting children from grades 1 to 8 (ages roughly 6 to 14). It covers food safety, personal hygiene, balanced diet, lunchbox planning, prevention of nutritional deficiencies, and making healthy choices, all delivered through age-appropriate language, activities, and illustrations. The FSSAI's Food Safety and Standards (Safe Food and Balanced Diets for Children in School) Regulations 2020 complement

the Yellow Book by mandating minimum nutritional standards for food served in and around schools.

● **Women: Pregnancy and Lactation**

The ICMR-NIN 2024 edition provides the most detailed pregnancy and lactation nutrition guidance thus far. For the first time, it adopts the ‘1000-day nutrition’ concept explicitly, which spans from conception (day one of gestation) through the child’s second birthday – a window of approximately 270 days of pregnancy and 730 days of early childhood. Specific micronutrient supplementation protocols are provided, including calcium supplementation beginning at 14 weeks of pregnancy, iron and folic acid supplementation to prevent anaemia and neural tube defects, and additional attention to iodine and long-chain omega-3 PUFAs (particularly DHA for foetal brain development). Additional caloric allowances are specified for each trimester and for lactation. The guidelines recognise workplace and social factors that affect breastfeeding success, reflecting a more holistic understanding of maternal nutrition than previous editions.

FSSAI’s Eat Right India framework does not have a dedicated resource book for pregnant and lactating women. The general dietary guidance in the Handbook and the Pink Book applies, and the national fortification programmes for rice, wheat, and edible oil indirectly benefit this population. The absence of a trimester-specific, pregnancy-specific nutrition communication tool is a meaningful gap in the FSSAI portfolio.

● **Older Adults (60 Years and Above)**

ICMR-NIN’s guidelines address older adults, recommending nutrient-rich, easily digestible foods that are low in sodium and added sugars, and high in fibre, calcium, and protein to prevent the onset of sarcopenia (age-related muscle loss), osteoporosis, and constipation. Energy requirements decline with age, but micronutrient needs remain high, making nutrient density even more critical for this population. The guidelines emphasise maintaining physical activity to preserve muscle mass and bone density.

FSSAI does not have a dedicated resource book for older adults, and the Eat Right India Handbook’s coverage of this demographic is minimal.

Despite the comprehensive life-cycle framework of the ICMR-NIN 2024 guidelines, several important population groups remain underserved in both documents:

- Women with polycystic ovarian syndrome (PCOS) have distinct dietary needs related to insulin resistance, hormonal balance, and inflammation that are not addressed in either framework.
- Women entering menopause face specific risks of calcium loss, cardiovascular disease, and weight redistribution that are not mentioned.
- Postpartum recovery nutrition, particularly for women managing both lactation and nutritional replenishment after childbirth, receives no standalone guidance.
- The dietary management of sarcopenia in the elderly, which requires specific protein quality, timing, and resistance exercise integration, is mentioned but not operationalised.

9. FSSAI’s Pink Book and Orange Book

The Pink and Orange Books are the closest FSSAI comes to ground-level dietary guidance for citizens and institutional establishments, respectively. They sit one layer closer to the consumer than the ERI Handbook, but still one layer above the scientific rigour of the DGI 2024.

9.1. The Pink Book: Guide for Safe and Nutritious Food at Home

The Pink Book is FSSAI’s guide for Indian home kitchens, designed to cover the full arc of food interaction within a household, organised into nine sections: buying food, storing food, preparing

and cooking food, serving food, eating healthy food, managing leftover food, packaging food, maintaining hygiene and sanitation in the kitchen, and managing waste.

On food buying, the book advises consumers to prioritise seasonal fruits and vegetables free from dark patches. It gives practical quality markers for fish (pink gills, clear eyes, flesh that bounces back when pressed), meat (pink flesh, lean, less cartilage), and milk (preferring pasteurised packaged milk with a valid best-before date). It advises checking expiry dates on frozen foods and looking for the veg logo, the +F fortification mark, and the FSSAI licence number on packaged products before purchasing.

On eating healthy, the book uses the food pyramid as a visual guide to balanced nutrition. It advises buying whole grains and cereals, using two or more types of fats and oils for cooking, and reading nutrition labels to make informed food choices.

On food preparation, the book covers segregation of vegetarian and non-vegetarian products to prevent cross-contamination, thorough washing of raw fruits and vegetables before cooking, avoiding serving food in plastic bags or newspaper, proper storage temperatures for different food categories (room temperature for bananas and cereals, refrigeration for fresh vegetables, milk, and eggs), and safe microwave use (only glass, ceramic, and specifically labelled microwave-safe plastics).

On hygiene and waste, the book covers kitchen sanitation practices and responsible waste management in the household. The mascots Master Sehat and Miss Sehat were created alongside the Pink Book for use in activation, communication, promotion, and training under the SNF campaign.

9.2. The Orange Book: Guide for Safe and Nutritious Food at the Workplace

The Orange Book was launched on 15 May 2018 as part of the SNF@Workplace initiative. Its premise is that most working Indians eat at least one meal during working hours, and that meal is prepared and served in conditions that vary enormously across formal corporate canteens, government office kitchens, and smaller workplace eating spaces. The book is structured for three separate audiences within the same workplace:

- **Part 1 is for HR and administration.** It lays out the responsibilities of workplace management in creating an enabling environment for food safety and nutrition. Key requirements include ensuring the kitchen and canteen follow FSMS guidelines for both in-house and external catering services, appointing a Food Safety Supervisor (FSS) trained and certified through FoSTaC for Catering (Advance), and engaging a Health and Wellness Coordinator (HWC) certified under a dedicated FSSAI training programme. The FSS is responsible for hygiene and sanitary practices, timely testing, audits, and regular training of food handlers, at a ratio of one FSS for every 25 food handlers.
- **Part 2 is for the canteen or cafeteria establishment.** It specifies best practices for ensuring food served at the workplace is safe and wholesome, including valid licensing and registration requirements, Good Hygiene and Manufacturing Practices as per FSSAI schedules, monthly testing of kitchens, and third-party audits.
- **Part 3 is for employees.** It provides dos and don'ts, practical tips, and suggestions to help employees make healthier food choices at the workplace.

FSSAI designed a 15-point Health and Wellness Matrix with 3, 4, and 5-star ratings for workplaces based on the quality of food and canteens and systemic efforts around employee health and wellness, creating a marketplace incentive comparable to the hygiene ratings for restaurants already deployed under the ERI framework. However, the full-scale rollout of this rating mechanism was not completed; this is likely attributable to challenges in operationalising voluntary institutional

compliance at scale and to the disruption caused by the COVID-19 pandemic, which significantly interrupted planned implementation timelines for several ERI initiatives between 2020 and 2022.

9.3. The Books' Contribution to The Broader Framework

Both the Pink and Orange Books would benefit from being updated to reflect DGI 2024 targets, as both predate the 2024 guidelines by several years and carry messaging that is either outdated (the older food pyramid in the Pink Book) or insufficiently specific (the general advice to use two or more types of oils without guidance on the saturated fat limit or the omega-3 emphasis). An update cycle tied to each new edition of the DGI would ensure the citizen-facing materials stay scientifically current.

The food pyramid visual framework in the Pink Book should be replaced with the DGI 2024's My Plate model, making citizen-level messaging coherent with the scientific guidance. The Orange Book's Food Safety Supervisor and Health and Wellness Coordinator roles are an underused template for institutional dietary intervention beyond the workplace, and could be adapted for government hospitals, large educational institutions, and community health centres under Ayushman Bharat. The Orange Book's 15-point Health and Wellness Matrix, if formally scaled up alongside the Eat Right Campus framework, could create measurable, comparable incentives for institutions to improve their food environments. The Pink Book's emphasis on label reading should be expanded in a revised edition to include guidance on reading front-of-pack nutrition information once FSSAI's FOPL regulation is finalised.

10. Key Gaps

The most fundamental divergence is structural. The DGI 2024 provides quantitative targets and scientific evidence; the ERI Handbook provides implementation mechanisms and qualitative behaviour change messaging. Temporally, the Handbook (2018, updated 2020) predates the DGI (2024) by several years; therefore, several important DGI 2024 updates are not reflected in the Handbook at all, including the My Plate concept, the specific added sugar cap of less than 5% of total calories, the 25-30 grams per day fibre target, the protein supplement advisory, the introduction of microgreens, screen time as an NCD risk factor, climate-specific hydration guidance, and the detailed millet promotion framework.

The most fundamental structural gap is the fragmentation between scientific nutrition knowledge and public health communication. ICMR produces rigorous, evidence-grounded nutrient recommendations that are largely inaccessible to the general public, while FSSAI communicates accessible messages that are not always grounded in quantitative nutritional evidence. **A second structural gap is the absence of personalisation.** Both frameworks provide population-level averages that do not account for individual variation in genetics, gut microbiome composition, metabolic rate, food preferences, or chronic disease status. **A third structural gap is the uneven treatment of the urban-rural dietary divide.** India has two nutritional realities coexisting: rural populations facing caloric insufficiency, micronutrient deficiencies, and limited dietary diversity, and urban populations dealing with caloric excess, HFSS food overconsumption, and sedentary lifestyles. Neither document is sufficiently differentiated to address both realities with equal effectiveness.

Moving beyond structural gaps, user-facing, translational gaps also exist. The DGI 2024 is written for professionals, and the ERI Handbook is written for administrators. Between these, there is a missing middle layer, drafted on the ethos of regional contextualisation, cultural relevance, language accessibility, and pictorial guidance. The ERI Toolkit goes some way toward filling this gap at the community level through frontline health workers, and affiliated materials like the Yellow Book for schools and the Pink Book for homes extend this further, but none of these materials provides scientifically rooted quantitative guidance.

Ultra-processed foods are not systematically addressed by either document. The NOVA classification system (or an equivalent) is absent from both. Given that India's packaged food market is growing rapidly and ultra-processed food (UPF) consumption is linked directly to obesity and NCDs, the absence of a UPF-specific advisory or tiered food rating system is a significant gap.

Regional dietary adaptation is acknowledged in the DGI 2024 but not operationalised in either document. India's dietary diversity across north, south, east, and west, between coastal and inland populations, and between tribal and mainstream communities means that a single national My Plate has limited applicability at the ground level without state-level variants. **Mental health and the relationship between diet and mental well-being are not addressed by either document.** Food allergen guidance, present in both the US and EU dietary guidelines as a practical food safety and dietary planning dimension, is also absent from both Indian documents.

Population-specific guidance remains limited. Athletes and physically active populations, people with common diet-related conditions (type 2 diabetes, cardiovascular disease, chronic kidney disease), and older adults receive minimal separate guidance in the DGI and none in the ERI. Social determinants of diet, including income, geographic access, caste-based food restrictions, and urban food deserts, receive more programmatic attention in ERI's food systems approach than in the DGI, but neither document provides structured nutrition equity guidance of the kind found in more advanced national frameworks.

11. Comparison with Global Standards

India's two nutritional frameworks, while well-grounded in the domestic context, can be benchmarked against international guidelines to identify where they align with global best practice and where meaningful gaps remain relative to the highest-performing national dietary guidance systems.

The ICMR-NIN 2024 guidelines are broadly aligned with WHO and FAO principles for sustainable healthy diets, including adequate energy intake, food diversity, minimisation of ultra-processed foods, adequate water intake, and reduced salt and sugar consumption. The 2024 edition also addresses environmental sustainability in dietary choices, which aligns with the FAO's 16 guiding principles for sustainable healthy diets. However, compared to the DASH diet's structured sodium and nutrient targets for hypertension management, or the Mediterranean diet's documented cardiovascular and metabolic benefits, the ICMR guidelines do not provide therapeutic dietary patterns for specific disease management.

The FSSAI's Eat Right framework is philosophically aligned with WHO's 'Best Buys' recommendations for NCD prevention, particularly due to its focus on reducing salt, sugar, and fat consumption. However, it lacks the quantitative specificity of WHO's nutrient intake recommendations and does not formally benchmark its programmatic targets against WHO thresholds.

Key areas of alignment with global standards include: both frameworks' alignment with WHO's recommendation of less than 5g of salt per day; ICMR's alignment with FAO's 10-food-group diversity approach; FSSAI's trans-fat elimination initiative aligning with WHO's REPLACE agenda; both frameworks' emphasis on vegetable and fruit consumption consistent with WHO targets; and environmental sustainability being addressed in DGI 2024 per FAO principles.

Key areas of divergence or gap include: neither framework providing therapeutic dietary patterns (DASH, Mediterranean) for NCD management; glycaemic index and load guidance being absent despite WHO recommendations for diabetic populations; omega-3/omega-6 balance guidance being absent despite WHO's fat quality recommendations; protein quality scoring (PDCAAS, DIAAS) not being incorporated despite FAO guidance; and no formal alignment with the Nordic Nutrition Recommendations 2023 or a comparable contemporary framework.

12. Way Forward

While institution-specific gaps chart a probable route to addressing individual deficiencies, any meaningful improvement in India's nutritional outcomes requires joint efforts between ICMR-NIN and FSSAI. The most transformative step would be the creation of a unified national dietary framework that formally integrates ICMR's scientific rigour with FSSAI's implementation architecture.

This framework could be co-developed by ICMR-NIN and FSSAI, endorsed by the Ministry of Health and Family Welfare, and operationalised through all national nutrition programmes, including the National Nutrition Mission (POSHAN Abhiyaan), the Mid-Day Meal Scheme, the Integrated Child Development Services (ICDS), and the National Rural Health Mission. The framework would include an India-specific dietary model (akin to the DASH or Mediterranean diet) that is rooted in Indian culinary traditions, evidence-based, and applicable across the country's socioeconomic and regional diversity. FSSAI and ICMR-NIN, jointly with state health ministries, would establish a national dietary monitoring framework that tracks actual dietary intake against DGI targets at a higher frequency than the current CNNS cycle allows.

Both institutions would benefit from investing in a joint digital communication platform that provides personalised dietary guidance tools based on ICMR thresholds, FSSAI food safety standards, and user inputs on region, age, physiological state, and health conditions. This platform could leverage FSSAI's Safe and Nutritious Food app infrastructure and ICMR's scientific content to create India's first integrated public nutrition advisory system. Regular joint reviews, perhaps every two to three years, should synchronise the update cycles of both frameworks so that they remain coherent and mutually reinforcing.

FSSAI and ICMR-NIN should jointly update the ERI Handbook and align it explicitly with DGI 2024. A revised Handbook should incorporate a simplified My Plate visual consistent with the DGI 2024, behaviour change messages derived from specific quantitative targets (the 5% added sugar cap, the 25-30g fibre recommendation, the zero trans fat goal, the millet promotion), and updated supplementation protocols for frontline health workers.

Both institutions could jointly issue state-specific seasonal dietary guidelines rather than providing year-round national guidelines, which would reduce volume and improve relevance and readability for the general population.¹⁸⁹

¹⁸⁹ WHO and FAO (2023). Sustainable Healthy Diets: Guiding Principles. World Health Organization and Food and Agriculture Organization of the United Nations, Rome and Geneva.

GLOSSARY

α -Tocopherol Transfer Protein (α -TTP) : A hepatic protein that selectively binds and resecretates α -tocopherol into nascent VLDL particles, establishing it as the predominant vitamin E isoform in plasma.

Acetyl-CoA: A central two-carbon metabolic intermediate produced from pyruvate, fatty acids, or amino acids that feeds the Krebs cycle, lipogenesis, or ketogenesis depending on prevailing energy status.

Adequate Intake (AI) : A reference nutrient intake value used in place of an RDA when insufficient data exist to calculate an EAR, derived from observed intake levels in apparently healthy populations.

Akkermansia muciniphila : A colonic mucus-layer bacterium representing 1–3% of healthy gut flora that maintains intestinal barrier integrity and is inversely correlated with metabolic inflammation and insulin resistance.

Aldosterone : An adrenal mineralocorticoid hormone that promotes renal sodium and water reabsorption to regulate extracellular fluid volume and blood pressure.

AMPK (AMP-activated protein kinase) : An energy-sensing kinase activated during low-ATP states that stimulates catabolic pathways such as fat oxidation while inhibiting anabolic processes like lipogenesis and protein synthesis.

Arginine vasopressin (AVP) : The primary antidiuretic hormone secreted by the hypothalamus in response to elevated plasma osmolality, increasing renal water reabsorption via collecting duct V2 receptors.

Autophagy : A lysosomal self-degradation mechanism activated during fasting or nutrient deprivation that selectively removes damaged organelles and misfolded proteins to restore cellular homeostasis.

Bacteriocins : Ribosomally synthesized antimicrobial peptides produced by commensal bacteria that inhibit or kill competing pathogenic strains by disrupting their membranes or metabolic processes.

Basal Metabolic Rate (BMR) : The rate of energy expenditure by an individual at complete rest in a thermoneutral, post-absorptive state, accounting for approximately 60% of total daily energy expenditure.

Beta-oxidation (β -oxidation) : The mitochondrial catabolic cycle that sequentially cleaves two-carbon units from fatty acyl-CoA molecules, yielding acetyl-CoA, NADH, and FADH₂ for ATP production.

Bile salts : Cholesterol-derived amphipathic molecules secreted by the liver that emulsify dietary fats, enable mixed micelle formation, and are essential for absorption of fat-soluble nutrients.

Bioavailability : The proportion of an ingested nutrient that is absorbed, transported to target tissues, and made available for physiological function.

Biofilm : A structured community of microorganisms encapsulated in a self-produced extracellular matrix that enables surface adhesion and confers resistance to antimicrobial agents.

Biofortification : The process of enhancing the nutrient density of staple food crops through selective plant breeding, agronomic practices, or genetic engineering.

Brush border : The apical surface of intestinal enterocytes formed by dense microvilli that harbor membrane-bound digestive enzymes and nutrient transporters, responsible for final hydrolysis and absorption.

Butyrate : A four-carbon short-chain fatty acid produced by colonic bacterial fermentation of dietary fiber that is the primary energy substrate for colonocytes, reinforces the gut barrier, and exerts anti-inflammatory and epigenetic regulatory effects.

Calbindin : A vitamin D-dependent calcium-binding protein synthesized in intestinal enterocytes that facilitates transcellular calcium transport from the apical to the basolateral membrane.

Calcitriol (1,25-dihydroxyvitamin D) : The hormonally active form of vitamin D produced in the kidney by CYP27B1 that binds VDR to regulate intestinal calcium absorption, bone metabolism, and immune gene expression.

Carnitine : A lysine- and methionine-derived molecule whose hepatic synthesis requires vitamin C, functioning as the essential shuttle that transfers long-chain fatty acyl-CoA groups across the inner mitochondrial membrane for beta-oxidation.

Cephalic phase (of digestion) : The anticipatory, sensory-triggered neural and hormonal response to the sight, smell, or taste of food that initiates gastric acid and digestive enzyme secretion before food is swallowed.

Ceruloplasmin : A copper-containing plasma ferroxidase that oxidizes Fe^{2+} to Fe^{3+} , enabling loading of iron onto transferrin for transport to erythroid precursors; it also carries more than 95% of circulating copper.

Cholecystokinin (CCK) : An intestinal I-cell hormone released in response to dietary fats and peptides that stimulates gallbladder contraction, pancreatic enzyme secretion, and hypothalamic satiety signaling.

Chylomicrons : Large triglyceride-rich lipoprotein particles assembled by intestinal enterocytes that transport dietary lipids, cholesterol, and fat-soluble vitamins through the lymphatic system into systemic circulation.

Chyme : The semi-fluid, acidic mixture of partially digested food and gastric secretions propelled from the stomach into the duodenum for further enzymatic processing.

Circadian rhythm : The endogenous approximately 24-hour biological clock governing sleep, metabolism, and hormonal secretion, which can be synchronized or disrupted by the timing of food intake.

Coenzyme A (CoA) : A pantothenic acid-derived cofactor that carries and transfers acyl groups in over 100 biochemical reactions, including the TCA cycle, fatty acid synthesis, and steroid hormone biosynthesis.

Colipase : A pancreatic cofactor secreted as inactive procolipase and activated by trypsin that anchors pancreatic lipase to the lipid-water interface of bile salt-stabilized fat emulsions, enabling efficient triglyceride hydrolysis.

Collagen : The most abundant structural protein in the body, whose synthesis requires vitamin C for the hydroxylation of proline and lysine residues, forming the fibrous scaffold of connective tissue, bone, and skin.

Cori cycle : The metabolic pathway by which lactate generated during anaerobic glycolysis in peripheral tissues is transported to the liver and reconverted to glucose for recirculation.

Cortisol : An adrenal glucocorticoid stress hormone that promotes hepatic gluconeogenesis, mobilizes amino acids from skeletal muscle, and suppresses immune responses.

Critical micellar concentration : The minimum bile salt concentration required for spontaneous formation of lipid-transporting micelles in the intestinal lumen, below which fat-soluble nutrient absorption is substantially impaired.

Cuproenzymes : A class of copper-dependent metalloenzymes: including ceruloplasmin, cytochrome c oxidase, and superoxide dismutase: that catalyze critical reactions in energy production, iron metabolism, and antioxidant defense.

Cytochrome c oxidase : The copper- and iron-containing Complex IV of the electron transport chain that catalyzes the reduction of molecular oxygen to water, generating the proton gradient used to synthesize ATP.

Cytokines : Small secreted signaling proteins produced by immune and stromal cells that mediate intercellular communication, regulate inflammation, and coordinate adaptive and innate immune responses.

2,3-Diphosphoglycerate (2,3-DPG) : A phosphorus-containing compound in erythrocytes that allosterically reduces hemoglobin's oxygen affinity, promoting oxygen delivery to metabolically active peripheral tissues.

Deamination : The enzymatic removal of an amino group from an amino acid, releasing nitrogen as ammonia that the liver incorporates into urea for safe renal excretion.

DSHEA (Dietary Supplement Health and Education Act, 1994) : The US federal law that classifies dietary supplements as a distinct food subcategory, placing pre-market safety responsibility on manufacturers rather than requiring FDA approval.

Dipeptidyl Peptidase-IV (DPP-IV) : A serine protease expressed on intestinal brush-border and systemic tissues that rapidly degrades incretin hormones GLP-1 and GIP, limiting their insulin-stimulating duration.

DMT-1 (Divalent Metal Transporter-1) : An apical brush-border transporter in duodenal enterocytes that imports ferrous iron (Fe^{2+}) and other divalent cations from the intestinal lumen into the cell.

Dysbiosis : A pathological imbalance in the composition or metabolic activity of the microbiome, associated with inflammatory bowel disease, metabolic disorders, and impaired immune regulation.

Electrochemical gradient : The combined driving force of electrical charge difference and ion concentration difference across a biological membrane that powers secondary active transport of nutrients including glucose and amino acids.

Electron transport chain (ETC) : A series of four protein complexes in the inner mitochondrial membrane that transfer electrons from NADH and FADH_2 to molecular oxygen, generating the proton gradient that drives ATP synthesis.

Emulsification : The bile salt-mediated dispersion of large fat globules into smaller droplets, increasing the lipid-water interfacial surface area and enabling efficient pancreatic lipase-catalyzed hydrolysis.

Energy balance : The thermodynamic relationship between caloric energy intake and total energy expenditure; a positive balance causes weight gain, a negative balance causes weight loss, and equilibrium maintains body composition.

Enterokinase (enteropeptidase) : A duodenal brush-border serine protease that activates trypsinogen by cleaving its inhibitory propeptide, thereby initiating the entire cascade of pancreatic zymogen activation.

Enterocytes : The principal absorptive epithelial cells lining the small intestinal villi that execute final luminal hydrolysis via brush-border enzymes and mediate transcellular transport of nutrients into blood and lymph.

Epigenetics : The study of heritable changes in gene expression that occur without DNA sequence alteration, mediated through histone modifications (biotinylation, methylation, acetylation) and DNA methylation patterns.

Erythropoiesis : The process of red blood cell formation in the bone marrow, dependent on iron for hemoglobin synthesis, vitamin B12 and folate for DNA replication, and erythropoietin as the principal hormonal regulator.

Estimated Average Requirement (EAR) : The daily nutrient intake level estimated to meet the requirement of 50% of healthy individuals in a specified life stage and sex group, serving as the statistical basis for deriving the RDA.

Exocytosis : The cellular process in which intracellular membrane-bound vesicles fuse with the plasma membrane to release their contents: such as chylomicrons: into the extracellular space.

FAD (Flavin Adenine Dinucleotide) : A riboflavin-derived electron carrier coenzyme that accepts electrons during the TCA cycle (succinate dehydrogenase) and fatty acid beta-oxidation, donating them to the ETC at Complex II.

Fat saponification : The binding of calcium ions by free fatty acids released from adipose tissue by excess pancreatic lipase, forming insoluble calcium soaps: a hallmark of peripancreatic fat necrosis in acute pancreatitis.

Ferroportin : The only known mammalian cellular iron export protein on the basolateral membrane of enterocytes and macrophages whose surface expression is post-translationally regulated by hepcidin.

Ferritin : The primary intracellular iron storage protein that sequesters iron in a soluble, non-toxic form; serum ferritin concentration is the standard clinical indicator of total body iron stores.

FGF23 (Fibroblast Growth Factor-23) : An osteocyte-secreted endocrine hormone that reduces renal phosphate reabsorption and suppresses calcitriol synthesis in response to elevated serum phosphate.

FMN (Flavin Mononucleotide) : A riboflavin-derived coenzyme that serves as an electron carrier in oxidoreductase reactions and is the immediate precursor of FAD; it constitutes the prosthetic group of NADH dehydrogenase (Complex I).

FODMAP : An acronym for fermentable oligosaccharides, disaccharides, monosaccharides, and polyols: poorly absorbed short-chain carbohydrates that undergo rapid colonic fermentation and osmotic water influx, triggering IBS symptoms.

Gas6 (Growth Arrest-Specific 6) : A vitamin K-dependent, gamma-carboxylated secreted protein expressed in the nervous system that mediates neuroprotection, phagocytosis, cell adhesion, and inhibition of neuronal apoptosis.

Gastrin : A peptide hormone secreted by gastric G-cells, stimulated by luminal protein fragments, that drives hydrochloric acid secretion from parietal cells and pepsinogen release from chief cells.

GLP-1 (Glucagon-Like Peptide-1) : An incretin hormone secreted by intestinal L-cells postprandially that amplifies glucose-dependent insulin secretion, inhibits glucagon, and promotes satiety; rapidly inactivated by DPP-IV.

GLUT2 : A high-capacity, low-affinity facilitated diffusion transporter on the basolateral membrane of enterocytes and hepatocytes responsible for exporting absorbed glucose and galactose into portal blood.

GLUT5 : A fructose-selective facilitated diffusion transporter on the apical membrane of small intestinal enterocytes that mediates fructose entry from the gut lumen into the cell.

Glucagon : A peptide hormone secreted by pancreatic alpha-cells in response to hypoglycemia that activates hepatic glycogenolysis and gluconeogenesis to restore circulating blood glucose.

Gluconeogenesis : The hepatic synthesis of new glucose molecules from non-carbohydrate precursors: pyruvate, lactate, glycerol, alanine, and glutamine: activated during prolonged fasting, starvation, or carbohydrate restriction.

Glucuronidation : A major hepatic phase II biotransformation reaction in which glucose-derived glucuronic acid is conjugated to hormones, drugs, and environmental toxins to render them water-soluble for biliary or renal excretion.

Glycogen : The highly branched polymeric storage form of glucose maintained in the liver (~75–100 g) and skeletal muscle, mobilized by glycogenolysis during fasting, exercise, or glucagon stimulation.

Glycogenolysis : The enzymatic breakdown of glycogen into glucose-1-phosphate: the immediate hepatic response to hypoglycemia or glucagon signaling: that rapidly restores circulating blood glucose.

Glycolysis : The ten-step cytoplasmic pathway that converts one glucose molecule into two pyruvate molecules, generating a net of 2 ATP and 2 NADH, functional under both aerobic and anaerobic conditions.

Glycosylation : The enzymatic addition of carbohydrate chains to proteins or lipids, forming glycoproteins, glycolipids, and proteoglycans that mediate cell-cell recognition, immune function, and receptor signaling.

Gut-brain axis : The bidirectional communication network between the enteric nervous system, gut microbiota, and the central nervous system, mediated through neural, endocrine, immune, and microbial metabolite signals.

Haptocorrins (R proteins) : Salivary and gastric glycoproteins that rapidly bind dietary vitamin B12 in the acidic stomach to protect it from degradation, transferring it to intrinsic factor in the alkaline duodenum.

HDL (High-Density Lipoprotein) : A lipoprotein that mediates reverse cholesterol transport by scavenging excess cholesterol from peripheral tissues and returning it to the liver for excretion or recycling; inversely associated with cardiovascular disease risk.

Heme iron : Dietary iron incorporated within the porphyrin ring of hemoglobin and myoglobin in animal foods, absorbed intact by intestinal enterocytes and substantially more bioavailable than non-heme iron.

Hemosiderin : An insoluble, protein-iron aggregated complex that accumulates in macrophages and parenchymal cells when ferritin capacity is exceeded; represents a secondary, less mobilizable form of iron storage.

Hepcidin : A liver-derived peptide hormone that is the master systemic regulator of iron homeostasis, binding ferroportin to trigger its internalization and degradation, thereby limiting both intestinal iron absorption and macrophage iron release.

HMG-CoA (3-Hydroxy-3-methylglutaryl-CoA) : A key hepatic metabolic intermediate positioned at the branch point between cholesterol biosynthesis and ketone body production, depending on cellular energy status.

HMG-CoA reductase : The rate-limiting enzyme of the mevalonate (cholesterol biosynthesis) pathway; its inhibition by statins reduces hepatic cholesterol production and lowers circulating LDL.

Homocysteine : A sulfur-containing amino acid metabolic intermediate that accumulates in plasma when folate, vitamin B6, or vitamin B12 is deficient, and whose elevation constitutes a risk factor for cardiovascular disease, stroke, and neural tube defects.

Hydroxyapatite : The crystalline calcium phosphate mineral $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ that constitutes approximately 70% of bone dry weight, conferring the structural hardness, rigidity, and compressive strength of the skeleton.

Hypochlorhydria : A pathological state of insufficient gastric acid secretion that impairs peptic protein digestion, dissociation of food-bound vitamin B12, and solubilization of dietary minerals.

IGF-1 (Insulin-like Growth Factor-1) : A liver-derived anabolic peptide hormone that mediates growth hormone's downstream effects on protein synthesis, cell proliferation, and glucose uptake; plasma levels fall during fasting due to reduced mTOR activity.

Incretin hormones : Gut-derived peptide hormones (GLP-1 and GIP) released postprandially that amplify glucose-dependent pancreatic insulin secretion and are rapidly degraded by DPP-IV.

Intestinal permeability (leaky gut) : A pathological condition in which compromised tight junctions between intestinal epithelial cells allow luminal bacteria, lipopolysaccharides, and antigens to translocate into the bloodstream, driving systemic low-grade inflammation.

Intrinsic factor : A glycoprotein secreted by gastric parietal cells that binds vitamin B12 in the duodenum and delivers it to specific cubilin receptors in the terminal ileal mucosa for receptor-mediated absorption.

Inulin : A prebiotic fructooligosaccharide fiber found in chicory, garlic, and onions that resists digestion and selectively promotes the growth of Bifidobacterium and Lactobacillus species in the colon.

KATP channels (ATP-sensitive potassium channels) : Ion channels in pancreatic beta-cell and neuronal membranes whose closure in response to elevated intracellular ATP triggers membrane depolarization leading to insulin secretion or neurotransmitter release.

Ketogenesis : The hepatic conversion of excess acetyl-CoA (from accelerated fatty acid oxidation) into the ketone bodies beta-hydroxybutyrate, acetoacetate, and acetone when TCA cycle oxaloacetate is depleted.

Ketone bodies : Three water-soluble hepatically produced compounds: beta-hydroxybutyrate, acetoacetate, and acetone: that serve as alternative fuels for the brain, heart, and other tissues during glucose scarcity.

Ketosis : A metabolic state achieved within 24–48 hours of complete fasting or carbohydrate restriction in which hepatic ketone body production exceeds peripheral utilization, elevating blood beta-hydroxybutyrate above 0.5 mmol/L.

Krebs cycle (TCA/Citric Acid Cycle) : A series of eight enzymatic reactions in the mitochondrial matrix that oxidize acetyl-CoA to CO_2 , generating NADH, FADH_2 , and GTP as high-energy substrates for the electron transport chain.

Lacteals : Lymphatic capillaries within intestinal villi that absorb chylomicrons and fat-soluble vitamins from the interstitium, draining into the thoracic duct and then into systemic venous blood.

Lactase-phlorizin hydrolase (LPH) : The brush-border enzyme that hydrolyzes lactose into glucose and galactose; its genetically regulated decline after weaning (lactase non-persistence) causes primary lactose intolerance in the majority of adults worldwide.

LDL (Low-Density Lipoprotein) : The principal cholesterol-carrying lipoprotein in plasma whose oxidized form initiates atherogenesis by depositing in the arterial intima; the primary clinical cardiovascular risk biomarker.

Leptin : An adipocyte-secreted satiety hormone that signals fat mass and energy sufficiency to hypothalamic centers, suppressing appetite and increasing energy expenditure proportionally.

Lipogenesis : The biosynthetic pathway that converts excess acetyl-CoA into fatty acids, triglycerides, and other lipids primarily in hepatocytes and adipocytes, upregulated during caloric surplus and insulin signaling.

Lipolysis : The hydrolysis of stored adipose triglycerides into glycerol and free fatty acids, stimulated by fasting, glucagon, catecholamines, and cortisol when blood glucose and insulin levels fall.

Lipopolysaccharides (LPS) : Structural components of gram-negative bacterial outer membranes that act as potent pro-inflammatory endotoxins when they enter systemic circulation through a compromised intestinal barrier.

Macroamylasemia : A benign condition in which amylase forms large complexes with circulating immunoglobulins that are too large for renal filtration, causing persistently elevated serum amylase in the absence of pancreatic disease.

Malonyl-CoA : A three-carbon intermediate in fatty acid synthesis produced by acetyl-CoA carboxylase that serves simultaneously as the chain-elongation substrate and as an inhibitor of CPT-1, preventing futile concurrent fat synthesis and oxidation.

Matrix Gla Protein (MGP) : A vitamin K-dependent gamma-carboxylated protein synthesized by vascular smooth muscle and chondrocytes that inhibits arterial and cartilage calcification by sequestering free calcium ions.

Megaloblastic anaemia : A haematological disorder characterized by abnormally enlarged, structurally abnormal red blood cell precursors (megaloblasts) caused by impaired DNA synthesis due to deficiency of vitamin B12 or folate.

Metabolic switching : The physiological shift from glucose as the primary cellular energy substrate to fat-derived ketone bodies, occurring after hepatic glycogen depletion during prolonged fasting or carbohydrate restriction.

Metabolic syndrome (MetS) : A cluster of co-occurring metabolic abnormalities: central obesity, insulin resistance, hypertension, hypertriglyceridemia, and low HDL: that multiplicatively elevate risk of cardiovascular disease and type 2 diabetes.

Metallothionein : A cysteine-rich intracellular protein whose synthesis is strongly induced by excess zinc or copper, sequestering these metals to buffer their bioavailability and prevent oxidative toxicity.

Methionine synthase : A vitamin B12 (methylcobalamin)-dependent enzyme that catalyzes the remethylation of homocysteine to methionine using 5-methyltetrahydrofolate as the methyl donor, linking folate and B12 metabolism.

Methylcobalamin : The active, methyl-group-donating form of vitamin B12 that serves as an indispensable cofactor for methionine synthase in one-carbon metabolism and supports myelin synthesis and neurological function.

Micelles (mixed) : Amphipathic aggregates of bile salts, monoglycerides, free fatty acids, cholesterol, and fat-soluble vitamins that solubilize lipid digestion products in the aqueous intestinal lumen and ferry them to the enterocyte brush border.

Microbiome : The totality of microorganisms (bacteria, archaea, viruses, fungi) and their collective genomes inhabiting a specific body site, whose composition profoundly influences host metabolism, immunity, and disease susceptibility.

mTOR (mechanistic Target of Rapamycin) : A central nutrient-sensing serine/threonine kinase that stimulates anabolic processes (protein synthesis, cell growth) in the fed state and is inhibited during fasting to activate autophagy and AMPK signaling.

MTHFR (Methylenetetrahydrofolate reductase) : A riboflavin-dependent enzyme that converts 5,10-methyleneTHF to 5-methylTHF, the active folate form required for homocysteine remethylation to methionine; common MTHFR polymorphisms impair this reaction.

Myoglobin : An iron-porphyrin oxygen-binding protein in muscle fibers that stores oxygen at the site of use and releases it to mitochondria during high-intensity muscular activity.

NAD⁺/NADH : The oxidized and reduced forms of nicotinamide adenine dinucleotide, the central redox coenzyme pair that shuttles electrons through glycolysis, the TCA cycle, and the electron transport chain for ATP generation.

NADPH : The reduced form of NADP⁺, generated primarily in the pentose phosphate pathway, that provides reducing equivalents for biosynthetic reactions (fatty acid and cholesterol synthesis) and for regenerating reduced glutathione in antioxidant defense.

NaPi-IIb transporter : The primary sodium-coupled active phosphate transporter on intestinal enterocyte apical membranes responsible for dietary phosphate absorption, upregulated by calcitriol in states of phosphate deficiency.

Neural tube defects (NTDs) : Severe congenital malformations (anencephaly, spina bifida) arising from failure of embryonic neural tube closure, strongly and causally associated with periconceptional folate deficiency.

Neurogenesis : The generation of new functional neurons from neural stem and progenitor cells, occurring in the adult hippocampus and subventricular zone, regulated by folate, BDNF, and other nutritional factors.

NIS (Sodium-Iodide Symporter) : An active transport protein on thyroid follicular cell basolateral membranes that concentrates iodide from plasma into the thyroid for hormone synthesis; its expression is upregulated by TSH and down-regulated by excess iodide.

NMDA receptor : An ionotropic glutamate receptor and calcium channel in neurons whose voltage-dependent magnesium block must be relieved for activation, making it essential for synaptic plasticity, learning, and memory; downregulated by intraneuronal magnesium.

Non-exercise activity thermogenesis (NEAT) : The energy expended for all physical activity excluding structured exercise: encompassing standing, fidgeting, and incidental movement: which can vary by up to 2,000 kcal/day between individuals.

Norepinephrine : A catecholamine neurotransmitter and adrenal hormone synthesized from dopamine by the enzyme dopamine β -hydroxylase, which requires vitamin C as an electron-donating cofactor for its activity.

NPC1L1 : A sterol-selective apical transporter in intestinal enterocytes and hepatic canalicular membranes that mediates dietary cholesterol and fat-soluble vitamin uptake; the pharmacological target of ezetimibe.

Nutraceutical : A food-derived product or isolated bioactive food component providing health benefits beyond basic nutrition, regulated under food law rather than as a pharmaceutical, without pre-market efficacy proof requirements in most jurisdictions.

Osteocalcin : A vitamin K-dependent bone matrix protein secreted by osteoblasts that, when fully carboxylated, binds hydroxyapatite and, in its undercarboxylated form, acts as an endocrine hormone regulating insulin secretion and glucose metabolism.

Osteoblasts : Bone-forming cells derived from mesenchymal progenitors that synthesize the collagenous osteoid matrix and direct its mineralization with calcium phosphate during bone formation and remodeling.

Osteoclasts : Multinucleated bone-resorbing cells derived from monocyte/macrophage precursors that acidify the resorption lacuna and secrete cathepsin K to dissolve both the mineral and organic components of bone matrix.

Oxaloacetate : A four-carbon TCA cycle intermediate that condenses with acetyl-CoA to initiate each cycle turn; also a gluconeogenic substrate, an aspartate precursor, and a limiting factor whose depletion during fatty acid oxidation diverts acetyl-CoA to ketogenesis.

Oxidative phosphorylation : The mitochondrial process in which the electrochemical proton gradient established by the ETC drives ATP synthase (Complex V) to phosphorylate ADP, producing the majority of cellular ATP.

PARP (Poly(ADP-ribose) polymerase) : A family of NAD⁺-consuming nuclear enzymes that detect DNA strand breaks, recruit repair machinery, and regulate gene expression, apoptosis, and circadian rhythm; their activity depletes NAD⁺ under conditions of genotoxic stress.

Parathyroid hormone (PTH) : A polypeptide hormone secreted by parathyroid chief cells in response to hypocalcemia that raises serum calcium by stimulating osteoclastic bone resorption, renal calcium reabsorption, and renal calcitriol synthesis.

PEPCK (Phosphoenolpyruvate carboxykinase) : The rate-limiting gluconeogenic enzyme that converts oxaloacetate to phosphoenolpyruvate and CO₂, representing the committed step that bridges TCA cycle intermediates to de novo glucose synthesis.

Pepsinogen : The catalytically inactive zymogen precursor of pepsin, stored in gastric chief cell secretory granules and activated to pepsin by HCl or by autocatalytic pepsin cleavage below pH 2.

PEPT1 transporter : A proton-coupled dipeptide/tripeptide transporter on intestinal brush-border membranes that facilitates uptake of small peptides into enterocytes via secondary active transport driven by an inward H⁺ gradient.

Pentose phosphate pathway : A cytoplasmic metabolic branch from glycolysis that generates NADPH for reductive biosynthesis and antioxidant regeneration, and ribose-5-phosphate for nucleotide synthesis; its transketolase step requires thiamine (TDP) as a cofactor.

Phytate (phytic acid) : The principal phosphorus storage compound in plant seeds that chelates divalent minerals (iron, zinc, calcium, magnesium) in the intestinal lumen, substantially reducing their bioavailability in plant-based diets.

PLP (Pyridoxal 5-phosphate) : The biologically active coenzyme form of vitamin B6, covalently bound at the active site of over 150 enzymes catalyzing transamination, decarboxylation, racemization, and glycogen phosphorylase reactions.

Polyphenols : A structurally diverse class of plant-derived bioactive phenolic compounds with antioxidant, anti-inflammatory, and prebiotic properties that modulate gut microbiota composition, endothelial function, and chronic disease risk.

Prebiotics : Selectively fermented dietary substrates, typically non-digestible fibers such as inulin and fructooligosaccharides, that promote the growth and activity of health-beneficial gut microorganisms.

Probiotics : Live microorganisms that, when administered in adequate amounts, confer measurable physiological health benefits on the host, primarily by improving intestinal microbial composition and mucosal immunity.

Proteolysis : The hydrolytic cleavage of peptide bonds in proteins by endopeptidases or exopeptidases, reducing them to oligopeptides or free amino acids during digestion or regulated intracellular protein turnover.

Pyruvate : The three-carbon glycolytic end product that is decarboxylated to acetyl-CoA (aerobic, via PDH), reduced to lactate (anaerobic), or used as a gluconeogenic substrate, serving as the metabolic nexus between glycolysis and mitochondrial pathways.

Pyruvate dehydrogenase (PDH) : A thiamine diphosphate-dependent multienzyme mitochondrial complex that irreversibly converts pyruvate to acetyl-CoA and CO₂, linking glycolysis to the TCA cycle and committing carbon to aerobic oxidation.

Reactive oxygen species (ROS) : Chemically reactive oxygen-derived molecules: including superoxide anion, hydrogen peroxide, and hydroxyl radical: that cause oxidative damage to cellular macromolecules but also serve as physiological second messengers in immune and metabolic signaling.

Recommended Dietary Allowance (RDA) : The average daily dietary intake level sufficient to meet the nutrient requirements of 97.5% of healthy individuals in a given life stage and sex group, calculated as EAR + 2 standard deviations.

Renin-angiotensin-aldosterone system (RAAS) : A hormonal cascade triggered by low renal perfusion or sodium that produces vasoconstriction (angiotensin II) and renal sodium/water retention (aldosterone) to restore blood pressure and circulating volume.

Retinoic acid : The transcriptionally active vitamin A metabolite that binds nuclear receptors (RARs/RXRs) and regulates gene expression governing cell differentiation, embryonic organogenesis, immune function, and lipid metabolism.

Rhodopsin : The light-sensitive photopigment in retinal rod cells formed by conjugation of 11-cis-retinal (vitamin A aldehyde) with the protein opsin, enabling vision under scotopic (low-light) conditions.

SCFA (Short-Chain Fatty Acids) : Fermentation products: primarily butyrate, propionate, and acetate: produced by colonic bacteria from dietary fiber that nourish colonocytes, strengthen gut barrier integrity, modulate mucosal immunity, and influence systemic lipid and glucose metabolism.

Secretin : A hormone released by duodenal S-cells in response to luminal acid that stimulates bicarbonate-rich pancreatic secretion, neutralizing acidic chyme to create the pH-optimal environment for digestive enzyme activity.

Secretory IgA (sIgA) : The predominant immunoglobulin in mucosal secretions (saliva, breast milk, intestinal fluid) that agglutinates pathogens and prevents their adhesion to epithelial surfaces without eliciting an inflammatory response.

Sequential enzymatic synergy : The mechanistic cooperation in lipid digestion whereby diacylglycerols and free fatty acids produced by preduodenal lipases act as natural emulsifiers that enhance the subsequent activity of pancreatic lipase in the duodenum.

SGLT1 (Sodium-dependent Glucose Transporter-1) : The apical enterocyte co-transporter that couples glucose and galactose uptake to sodium influx, using the inward electrochemical Na^+ gradient maintained by the basolateral Na^+/K^+ -ATPase as the driving force.

Sirtuin signaling : NAD^+ -dependent deacetylase-driven regulatory pathways activated during caloric restriction and fasting that modify histone acetylation, regulate mitochondrial biogenesis, and promote stress resistance and metabolic adaptation.

SMVT (Sodium-dependent Multivitamin Transporter) : An active intestinal transport protein that absorbs biotin, pantothenic acid (B5), and lipoate against concentration gradients using the inward sodium electrochemical gradient; also active in the proximal colon.

Superoxide dismutase (SOD) : A metalloenzyme (Cu/Zn-SOD in cytoplasm; Mn-SOD in mitochondria) that catalyzes the dismutation of superoxide radicals to hydrogen peroxide and oxygen, constituting the first enzymatic line of defense against oxidative damage.

TDP (Thiamine Diphosphate) : The biologically active coenzyme form of vitamin B1, required as a cofactor by pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, and transketolase for carbohydrate oxidation and pentose phosphate pathway activity.

Tetrahydrofolate (THF) : The reduced, active coenzyme form of folate that accepts and donates one-carbon units in reactions essential for purine and thymidylate synthesis, thereby supporting DNA replication and cellular division.

Thermic effect of food (TEF) : The increment in energy expenditure above basal levels attributable to the digestion, absorption, transport, and storage of ingested nutrients, representing approximately 5–15% of total daily energy expenditure.

Thyroglobulin (TG) : The large (~660 kDa) iodinated glycoprotein synthesized by thyroid follicular cells that serves as both the molecular scaffold for thyroid hormone biosynthesis and the intrafollicular storage reservoir for T3 and T4.

Thyroid peroxidase (TPO) : The heme-containing thyroid enzyme that catalyzes iodide oxidation and its subsequent organification into tyrosine residues of thyroglobulin, forming the iodinated tyrosine precursors that couple to yield T4 and T3.

Thyroid-stimulating hormone (TSH) : A pituitary glycoprotein that binds thyroid TSH receptors to stimulate NIS expression, iodide uptake, thyroglobulin iodination, and thyroid hormone synthesis and secretion.

Tolerable Upper Intake Level (UL) : The highest daily nutrient intake level that is unlikely to cause adverse health effects in virtually all healthy individuals; intakes exceeding the UL carry progressively increasing risk of toxicity.

Total energy expenditure (TEE) : The total daily caloric energy converted by an organism, comprising basal metabolic rate (~60%), the thermic effect of food (~10%), and activity energy expenditure (~15–30%), which together determine caloric requirements for energy balance.

Transcobalamin II (TCII) : The plasma vitamin B12-binding protein that binds newly absorbed cobalamin in portal blood and delivers it to peripheral tissues and bone marrow via specific cellular receptors; physiologically essential for tissue vitamin B12 delivery.

Transamination : A reversible PLP-dependent enzymatic reaction in which an amino group is transferred from a donor amino acid to an alpha-keto acid, simultaneously generating a new amino acid and converting the donor to its corresponding keto acid.

Transferrin : A plasma beta-globulin glycoprotein that binds up to two Fe^{3+} ions and transports them through the bloodstream, delivering iron to cells via receptor-mediated endocytosis of the transferrin-receptor complex.

Universal Salt Iodisation (USI) : A public health strategy that mandates the addition of iodine to all salt intended for human and livestock consumption as the most cost-effective intervention to eliminate iodine deficiency disorders population-wide.

Urea cycle : A five-step hepatic metabolic pathway that detoxifies ammonia generated from amino acid catabolism by incorporating it into urea for safe excretion in urine, preventing hyperammonemia.

VDR (Vitamin D receptor) : A nuclear receptor expressed in most tissues throughout the body that binds calcitriol to regulate the transcription of approximately 3% of the human genome, influencing calcium homeostasis, immune regulation, and cellular differentiation.

Villous atrophy : The pathological flattening and loss of small intestinal villi and their associated absorptive surface, the defining mucosal lesion of untreated celiac disease, causing broad malabsorption of macronutrients and micronutrients.

VLDL (Very Low-Density Lipoprotein) : A triglyceride-rich lipoprotein secreted by the liver that distributes endogenously synthesized fatty acids to peripheral tissues, progressively converted to IDL and then LDL as triglycerides are hydrolyzed by lipoprotein lipase.

Zinc fingers : Protein structural domains in which one or more zinc ions are tetrahedrally coordinated by cysteine and histidine residues, stabilizing a compact fold that enables transcription factors and other regulatory proteins to bind specific DNA or RNA sequences.

ZIP4 : The primary zinc-selective transporter on the apical membrane of duodenal and jejunal enterocytes that mediates dietary zinc absorption; loss-of-function mutations cause acrodermatitis enteropathica, a severe hereditary zinc deficiency disorder.

Zymogens : Catalytically inactive enzyme precursor proteins (e.g., pepsinogen, trypsinogen, chymotrypsinogen, procolipase) secreted by glands and activated only at their intended site of action to prevent autodigestion of secretory tissue.